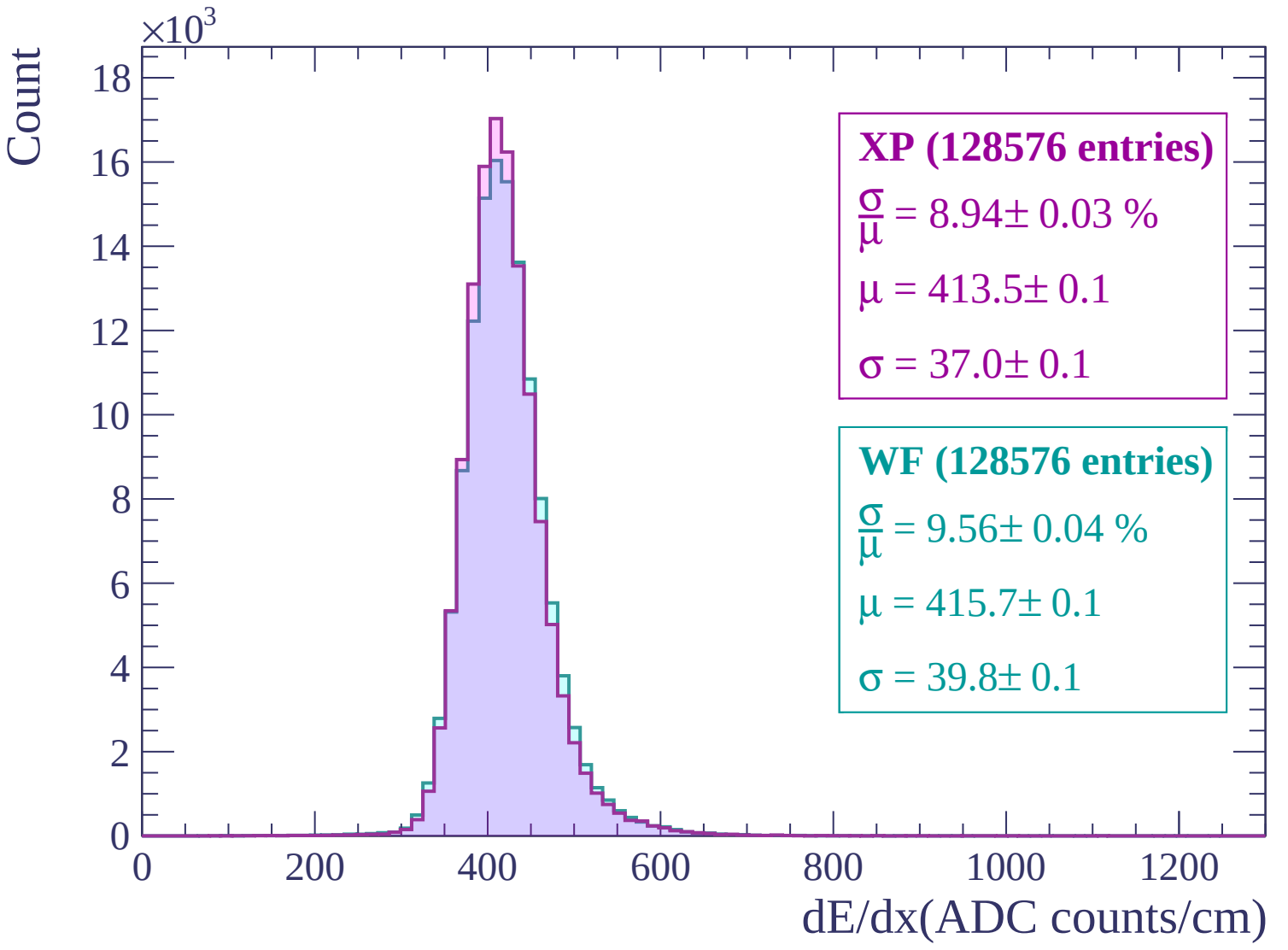
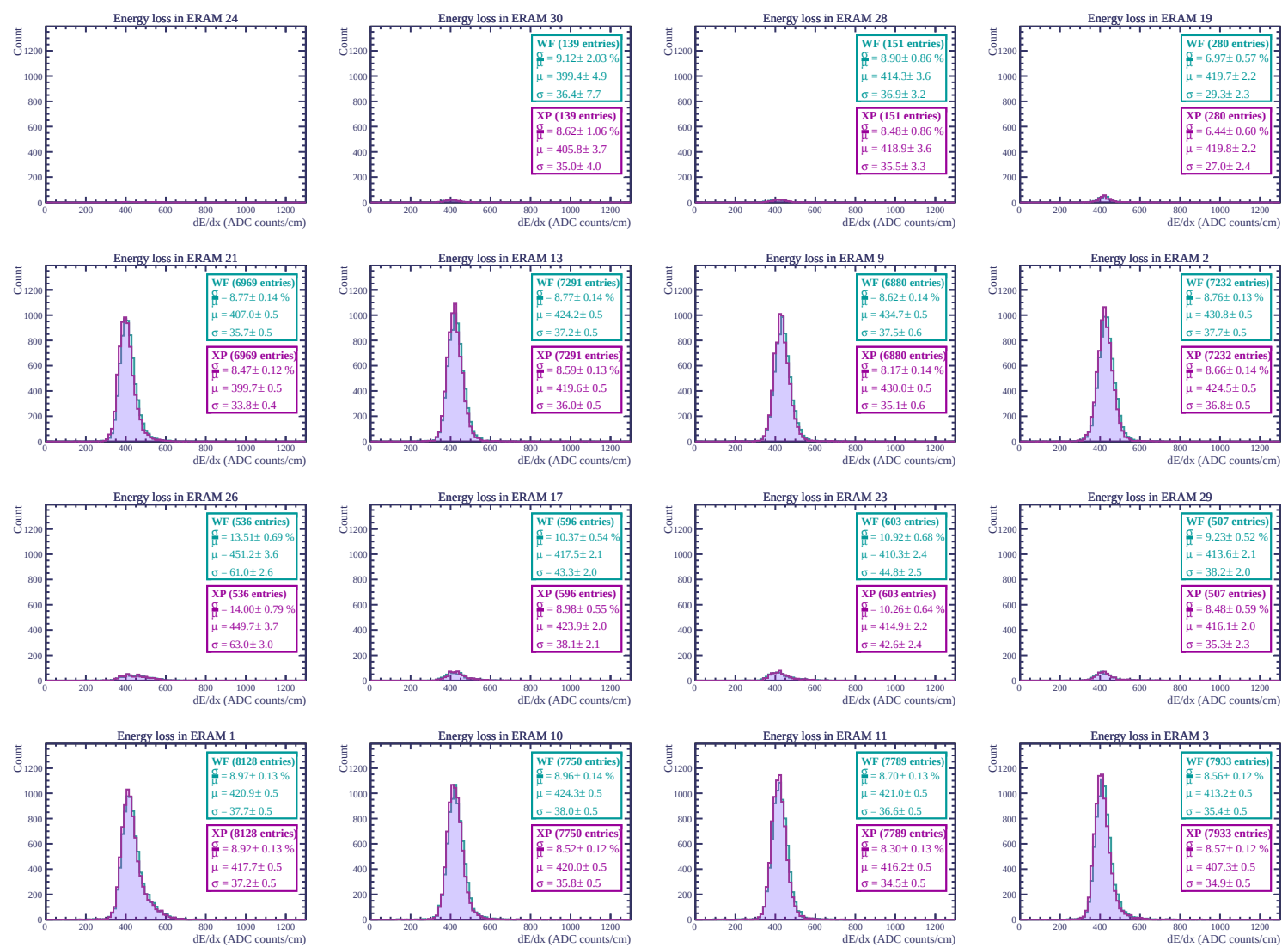
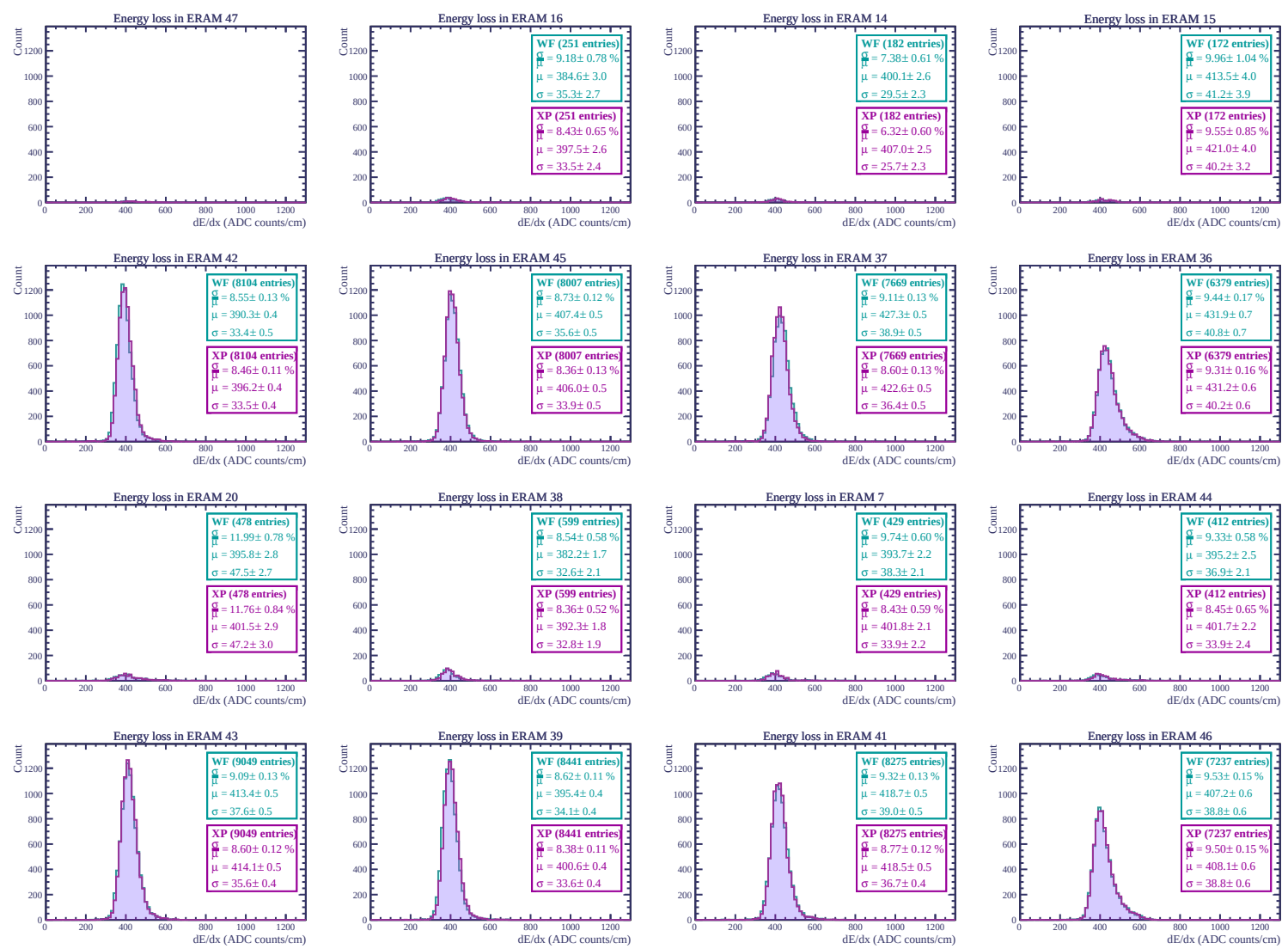
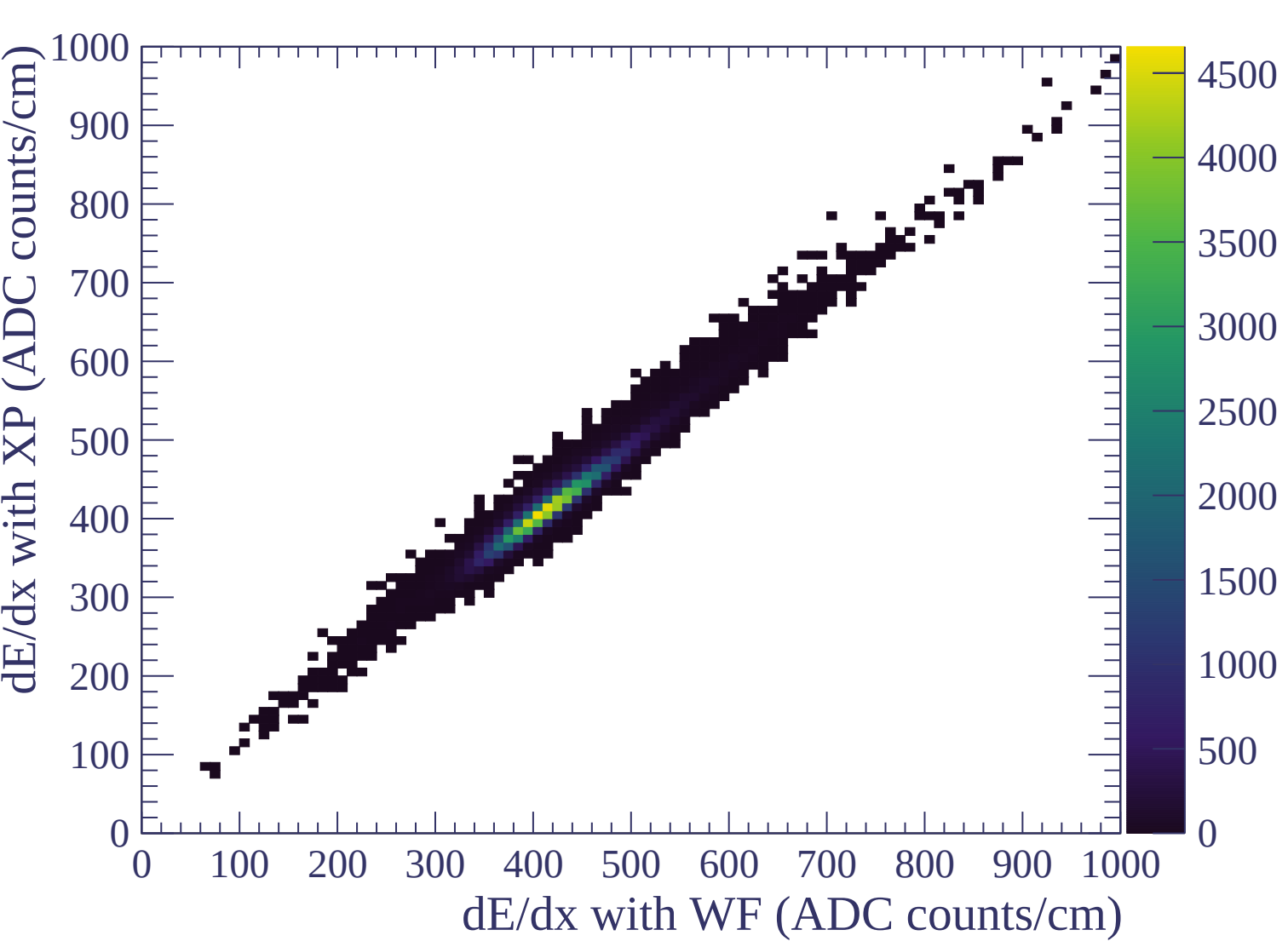
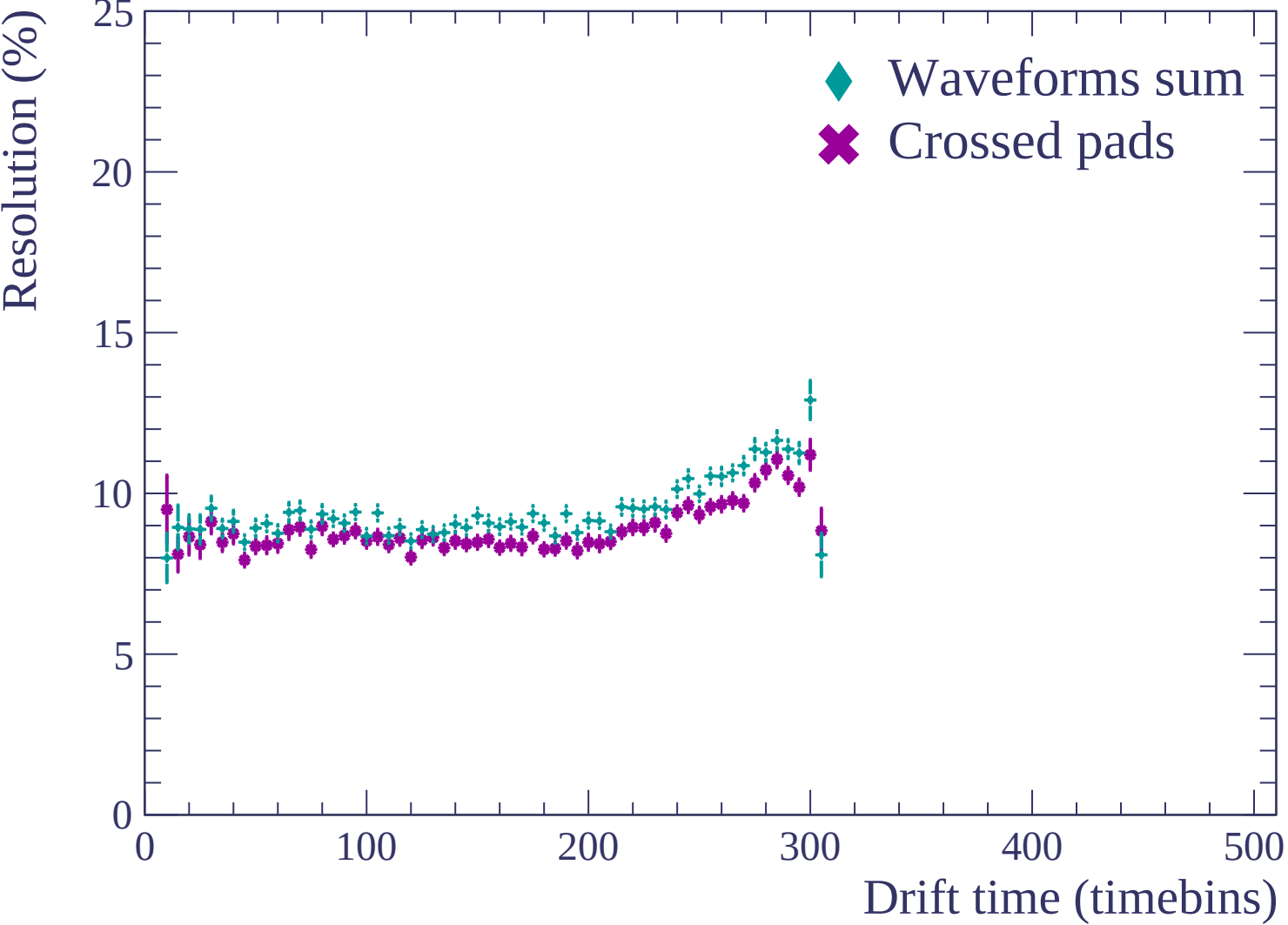


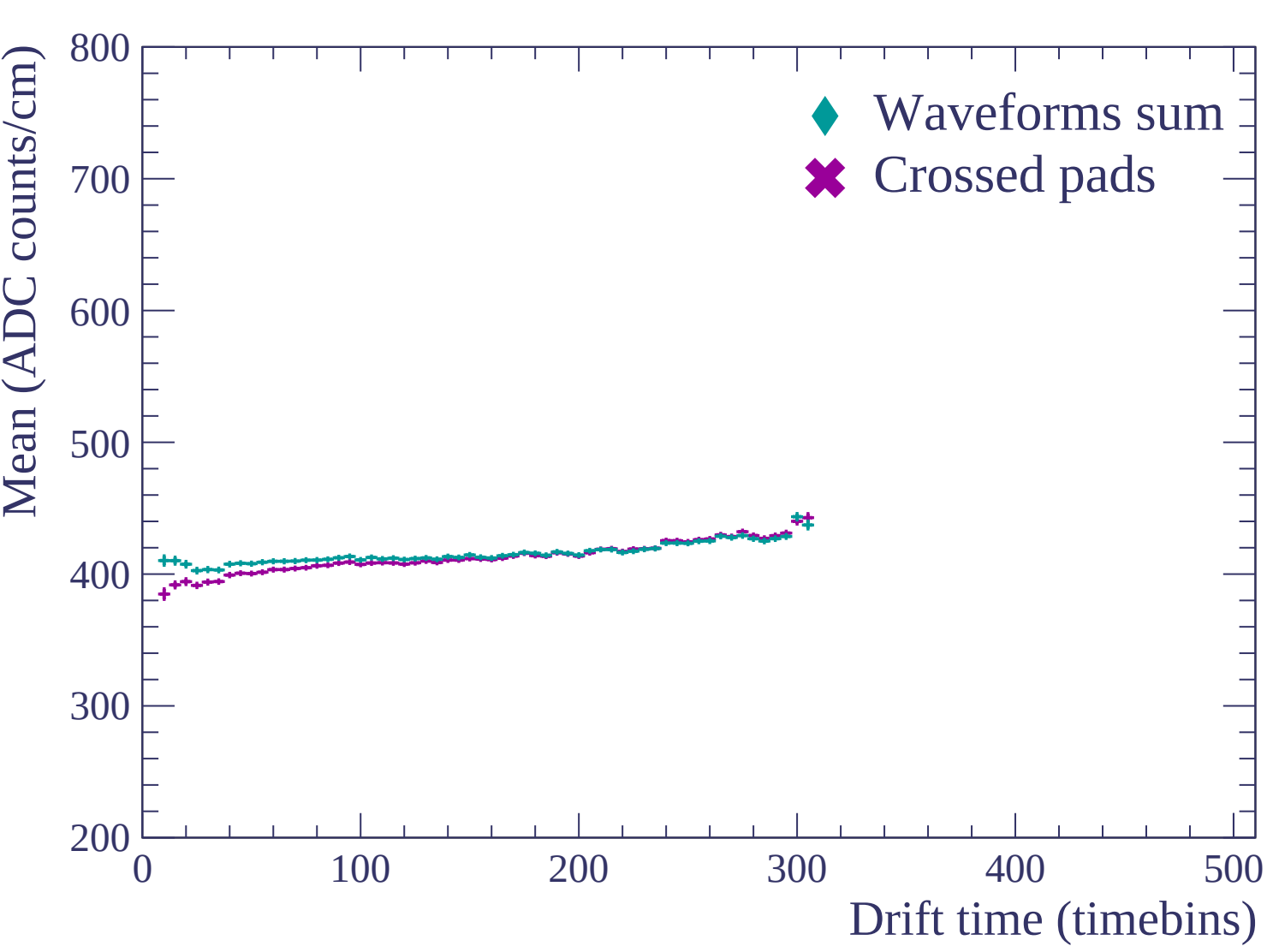


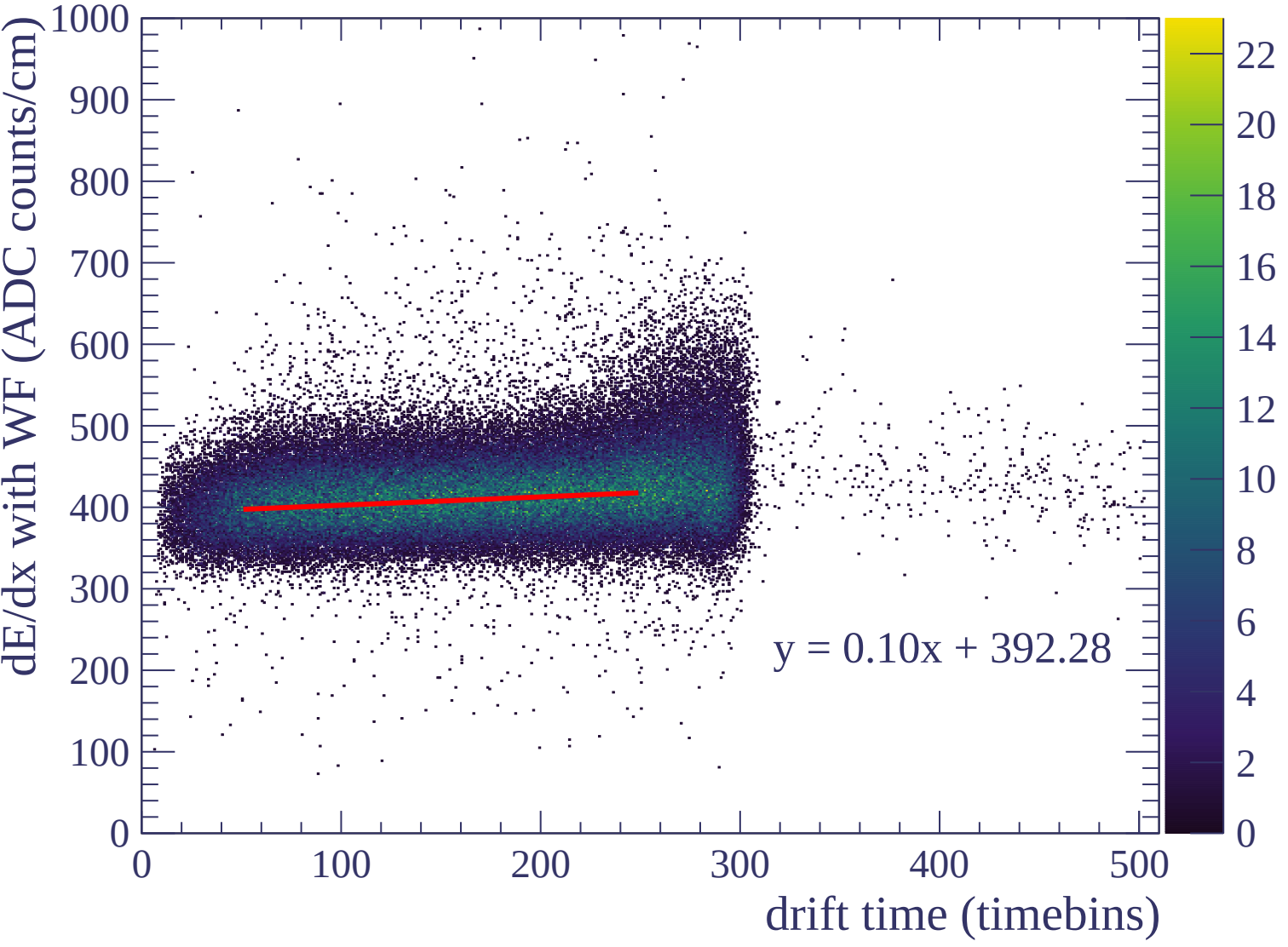


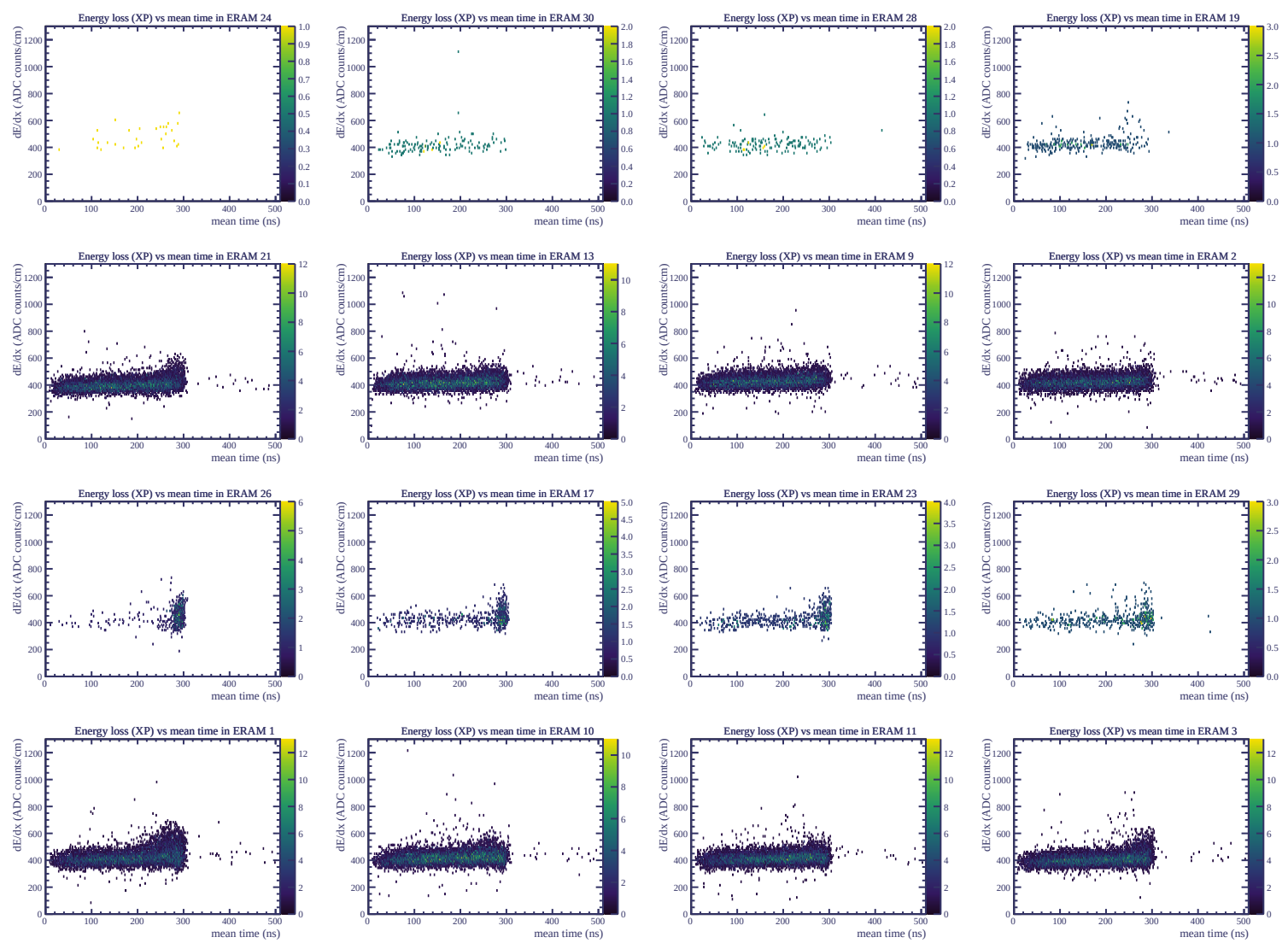


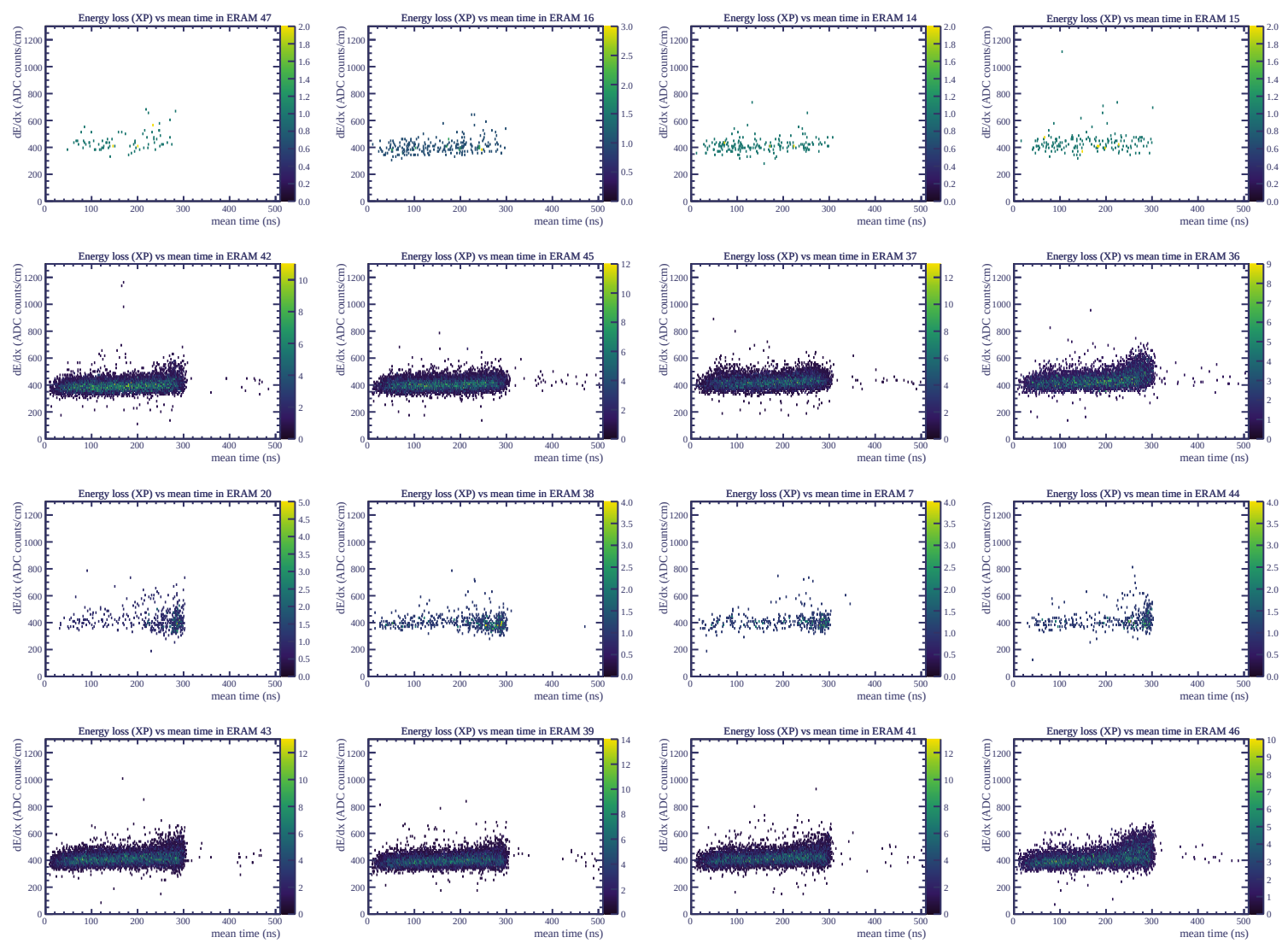


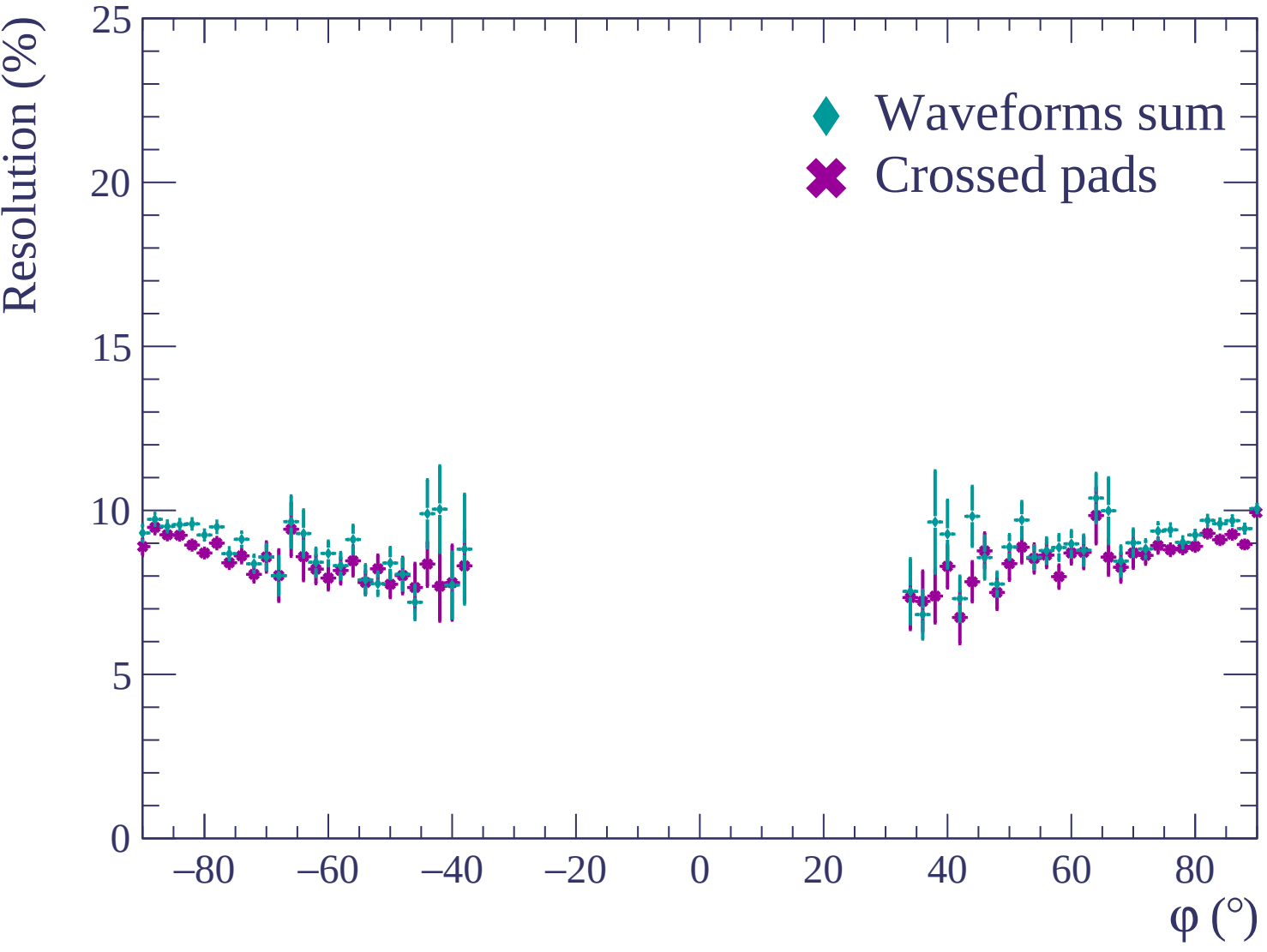


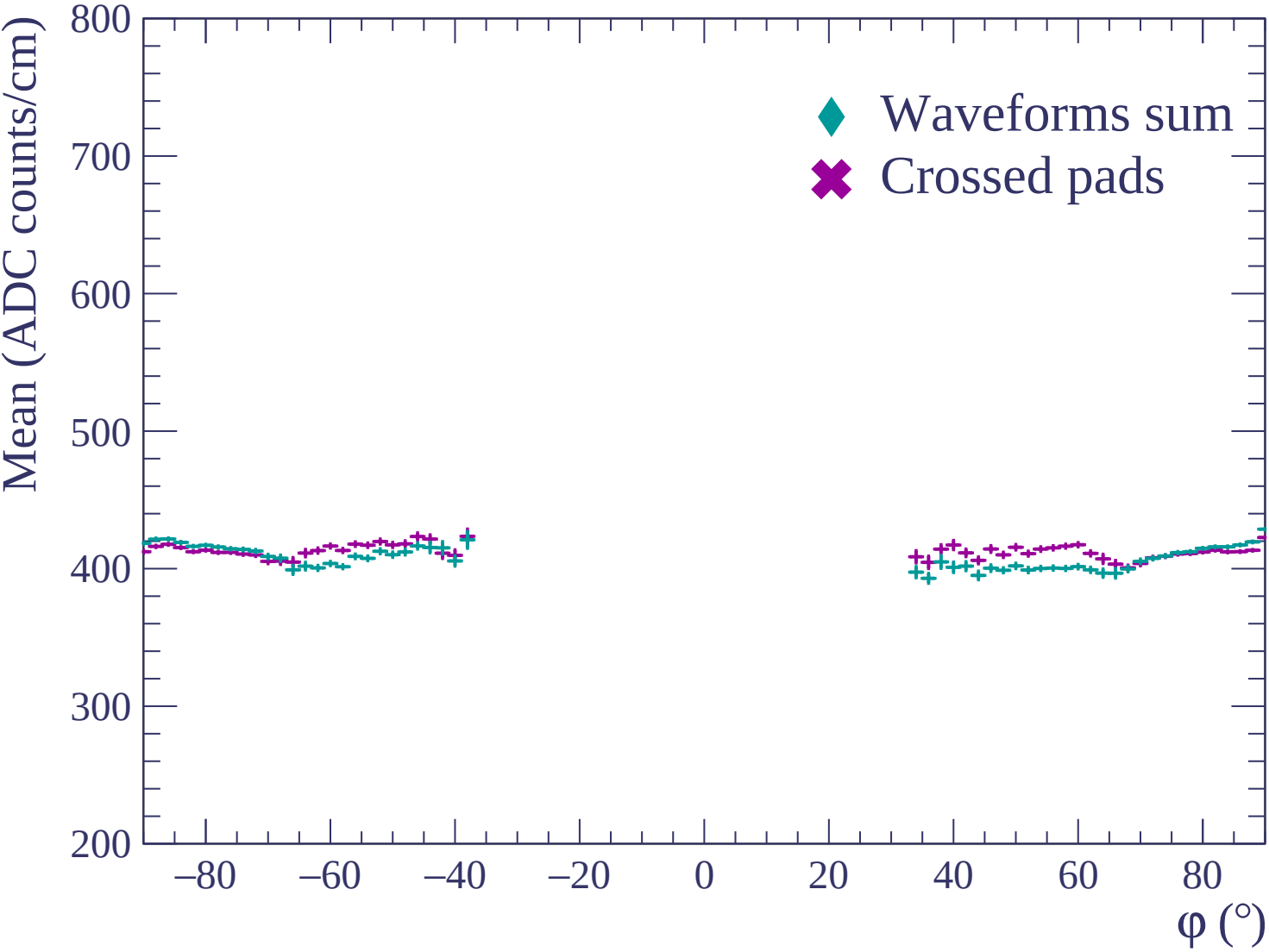


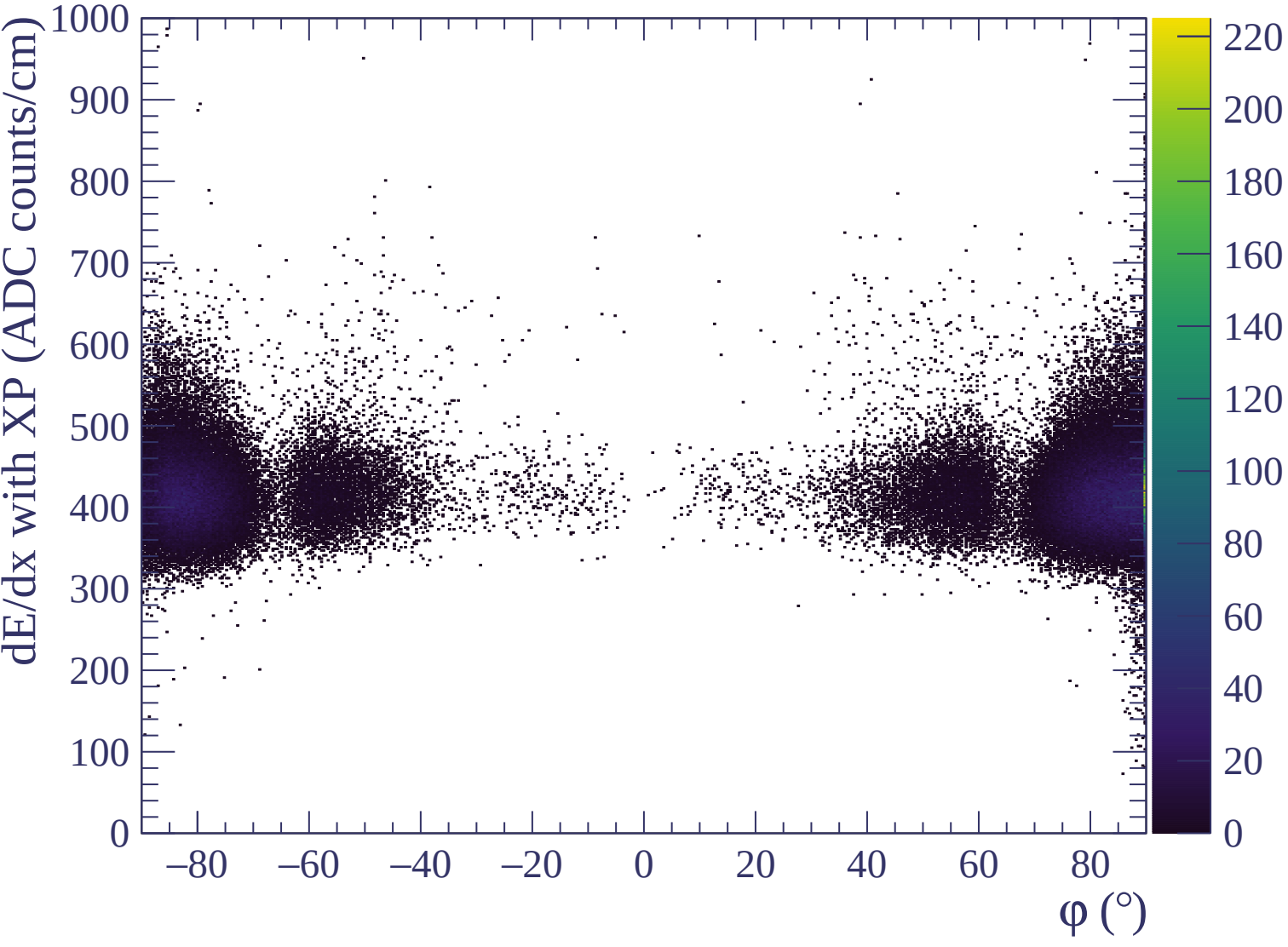


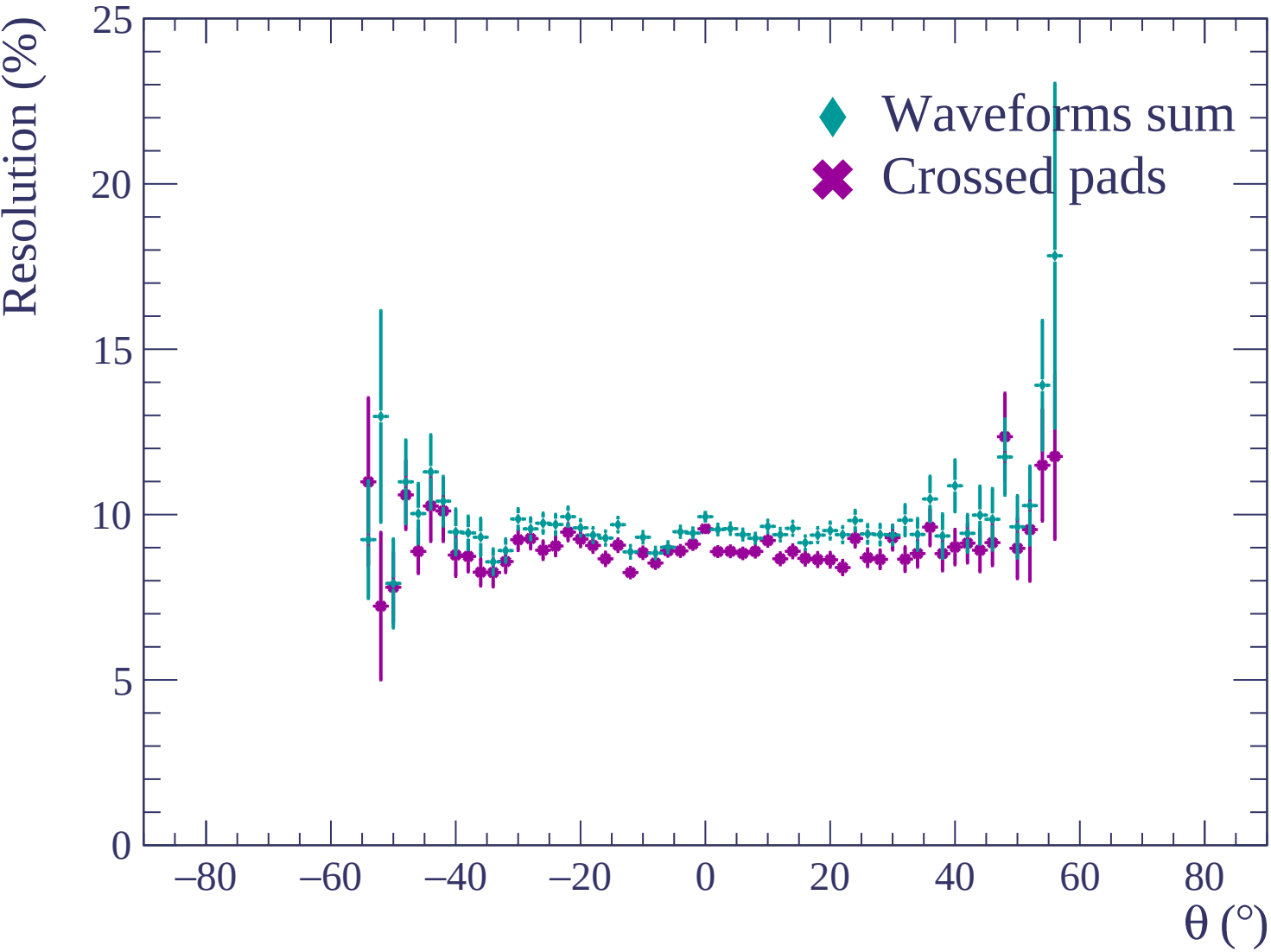


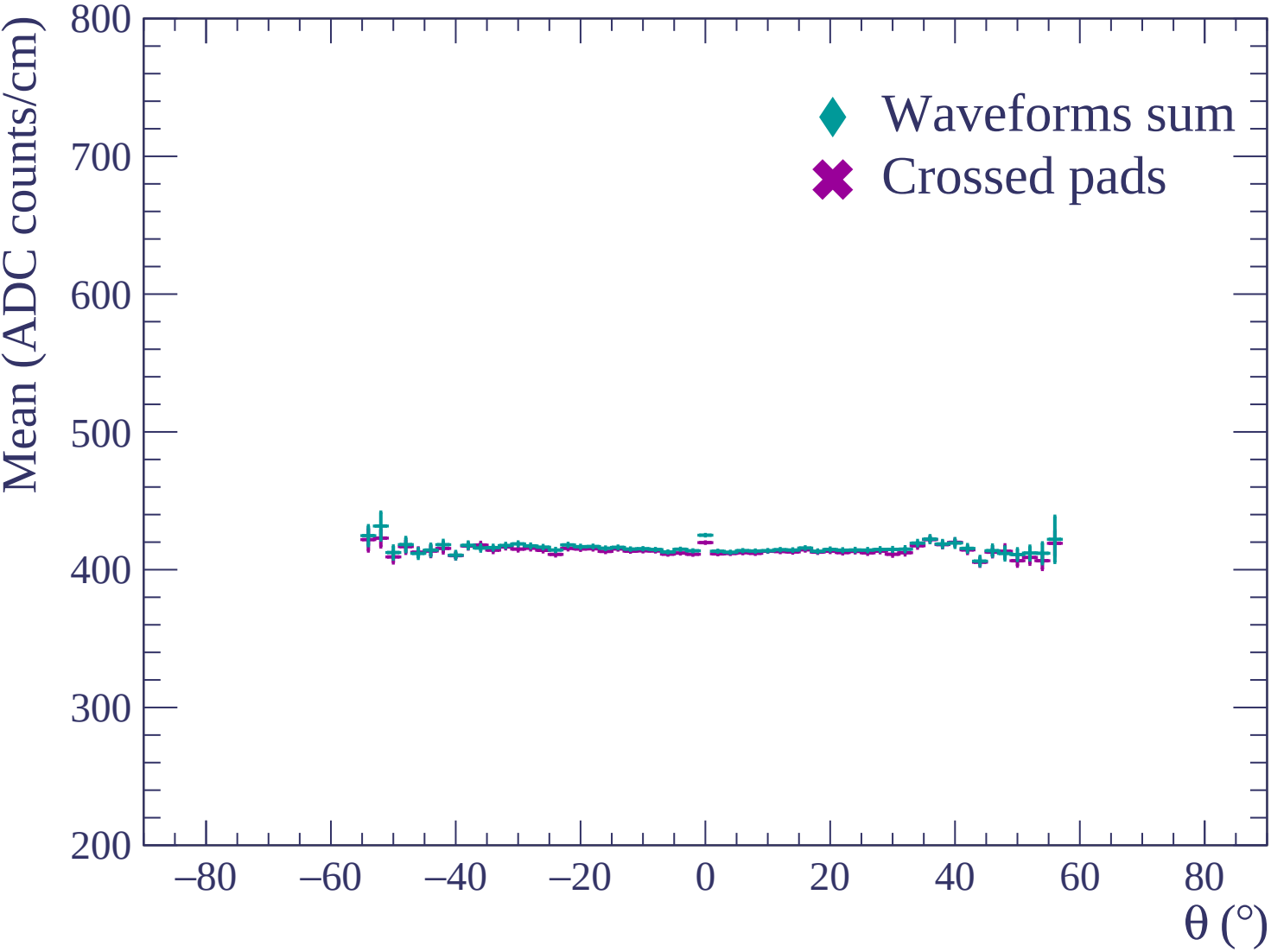


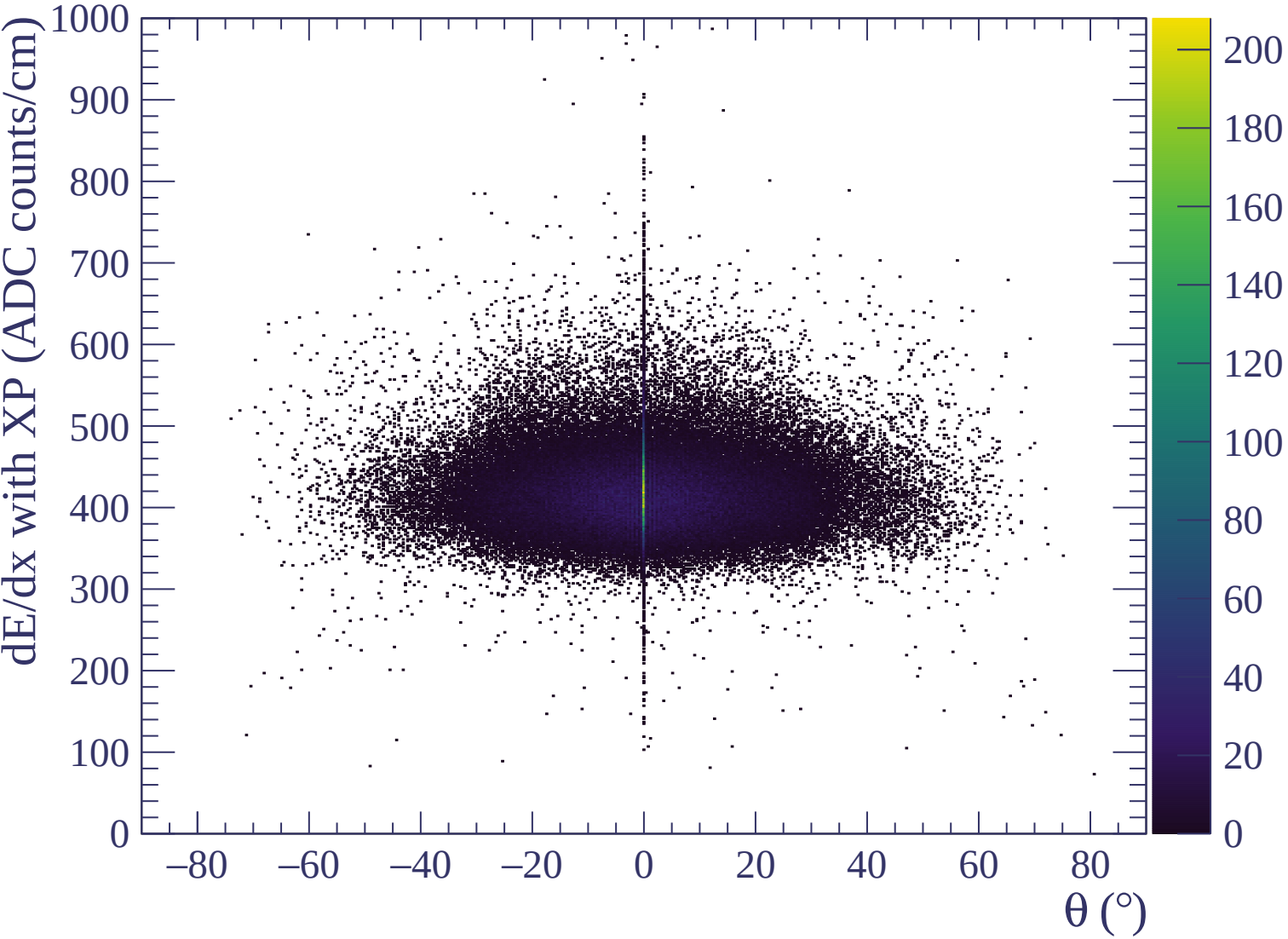


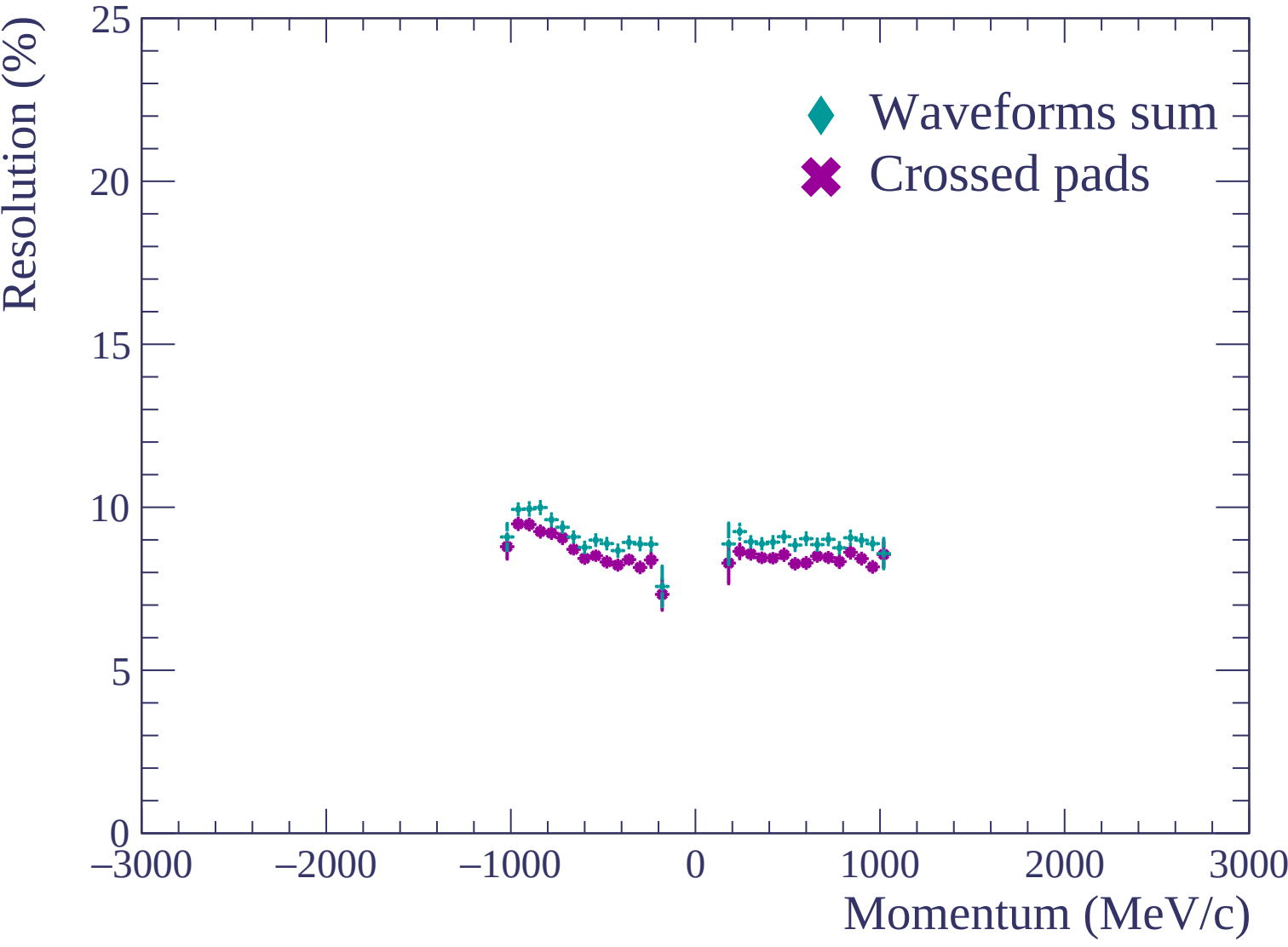


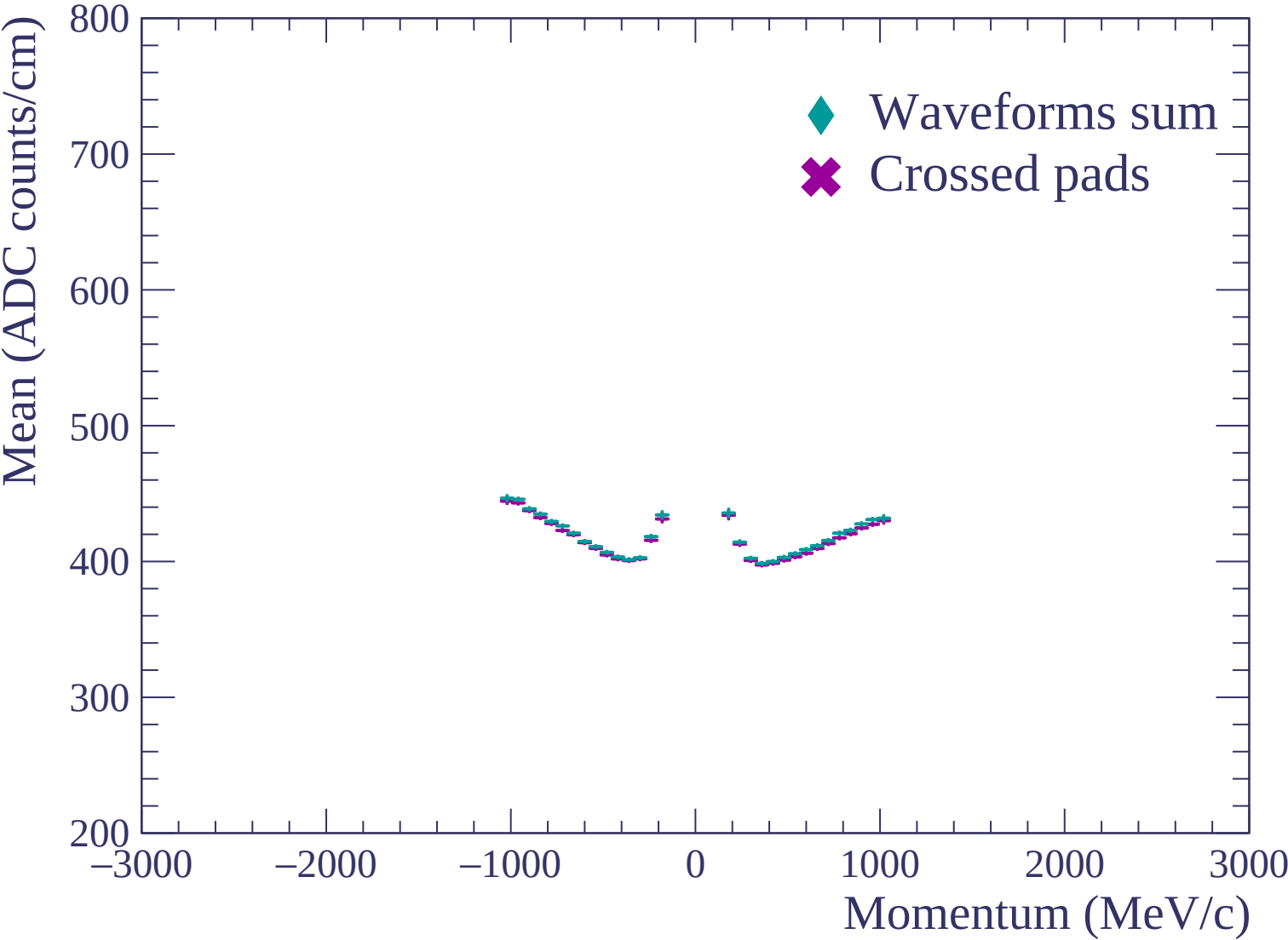


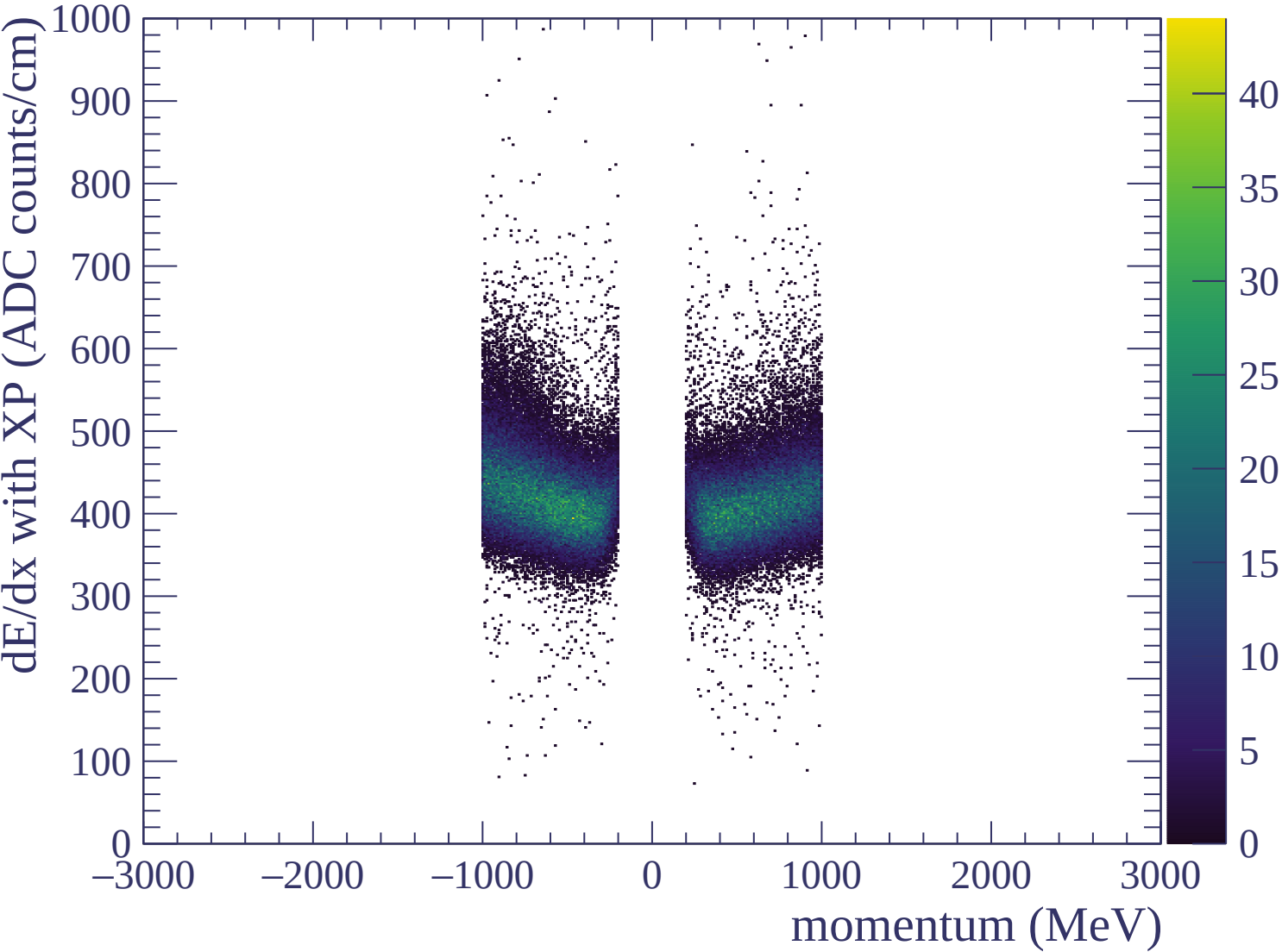


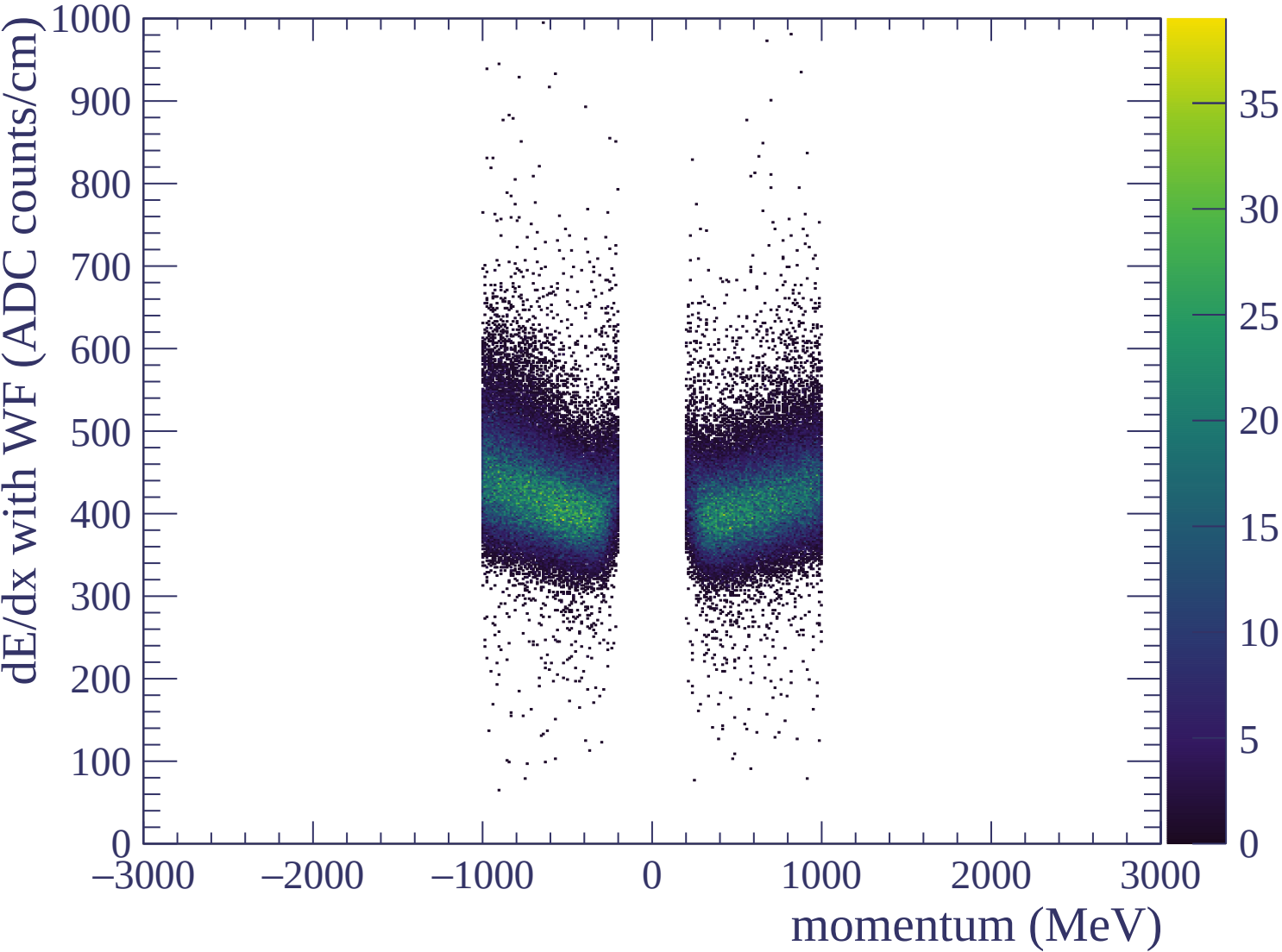


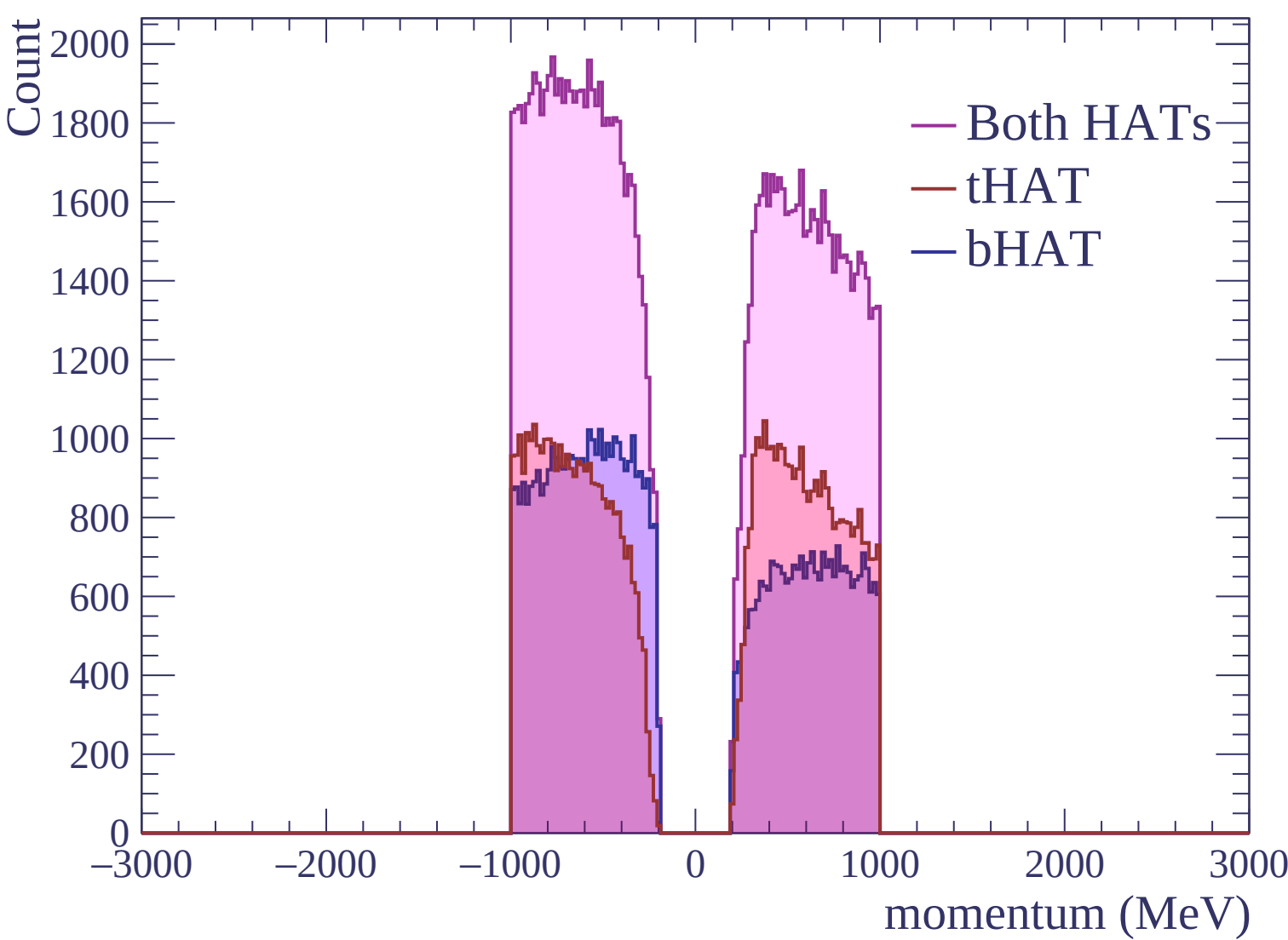


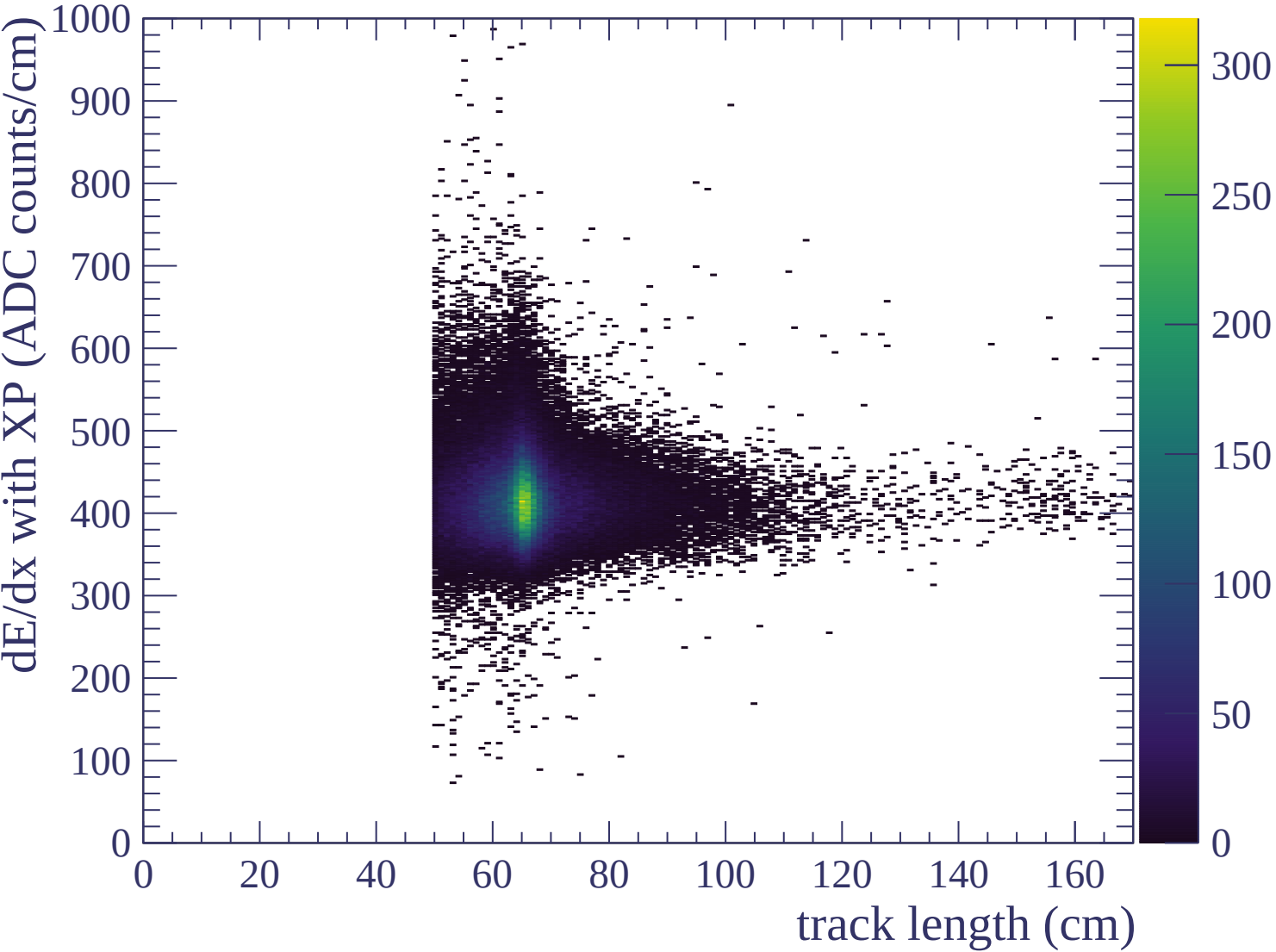


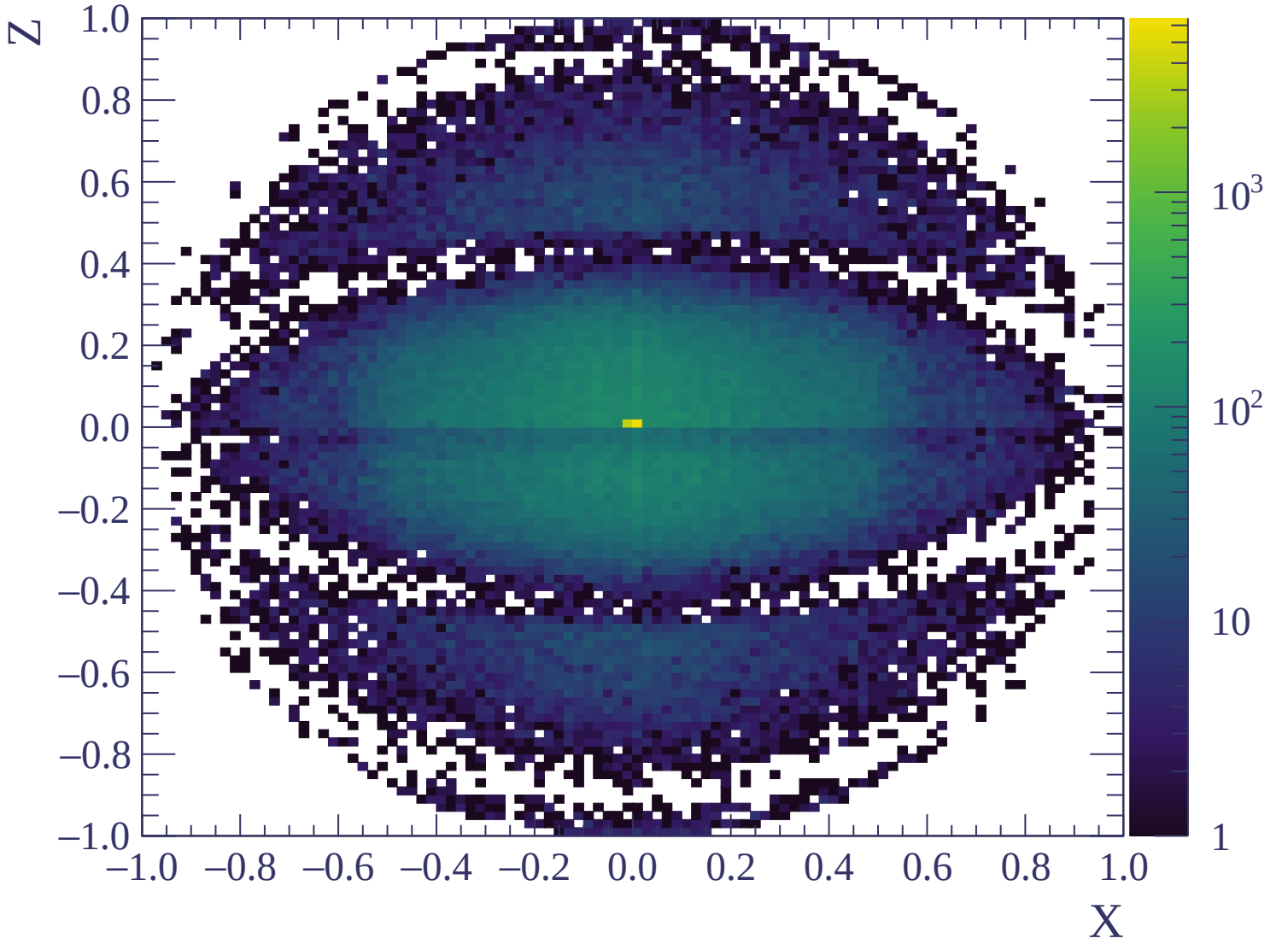


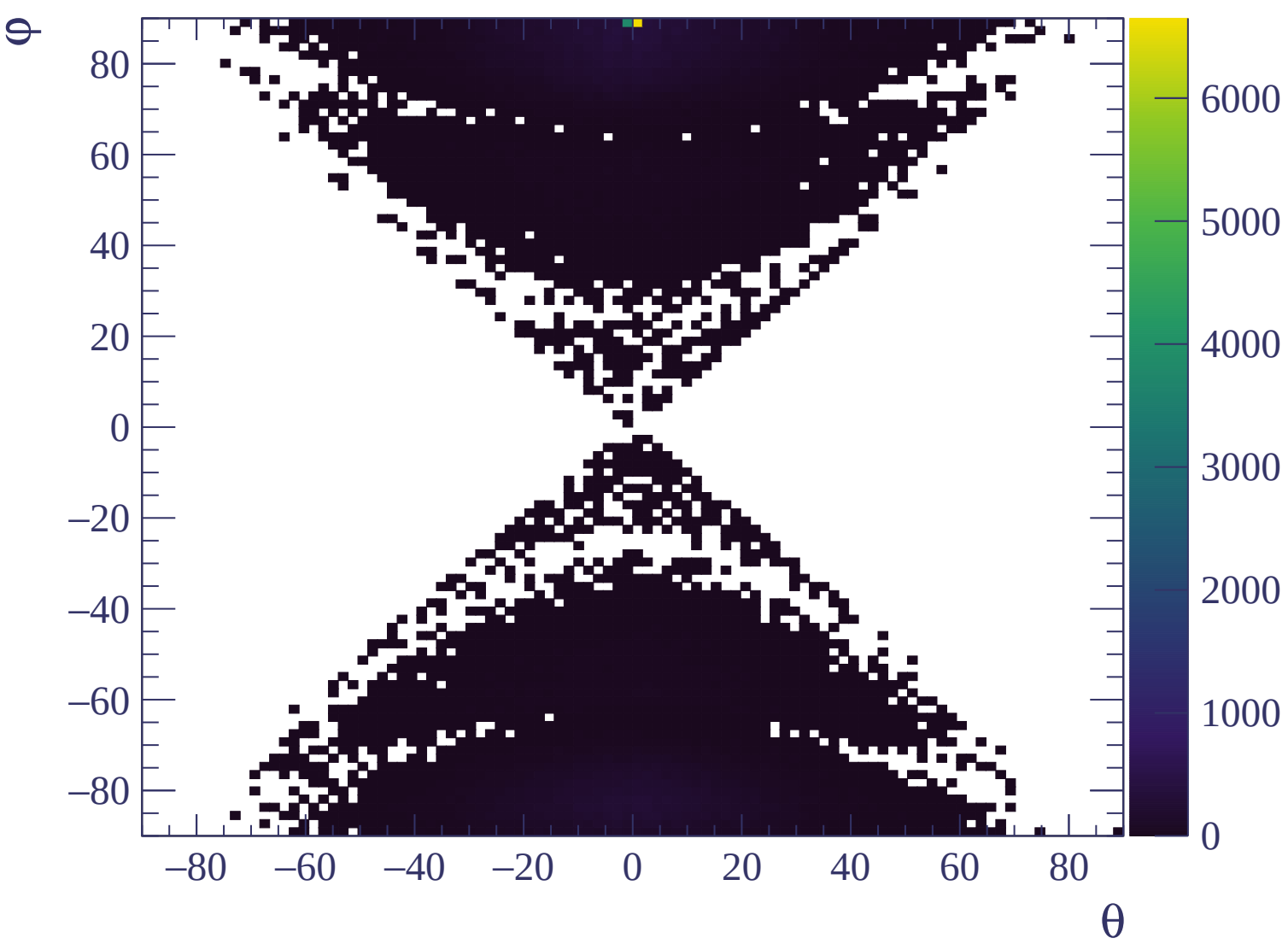


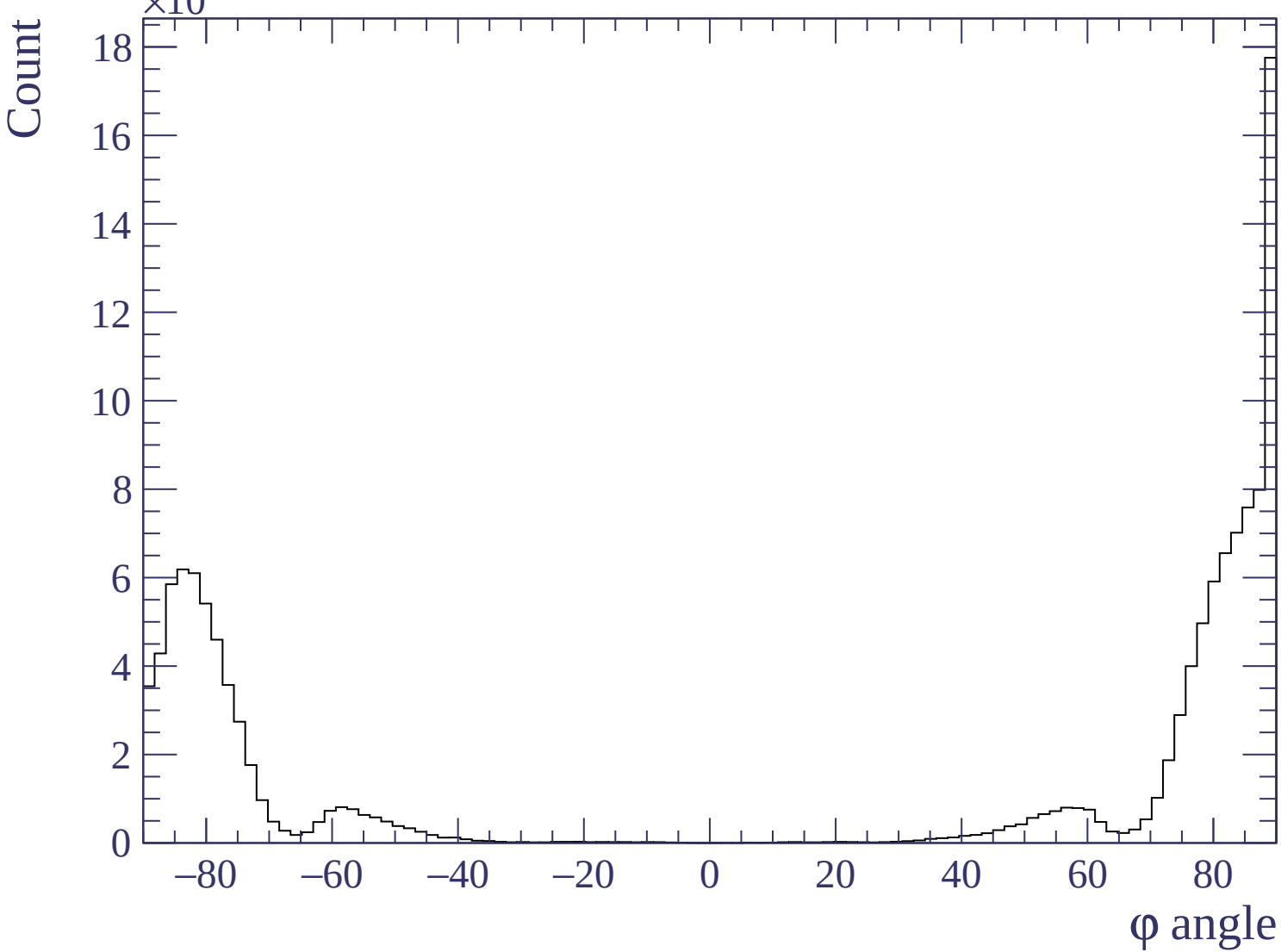


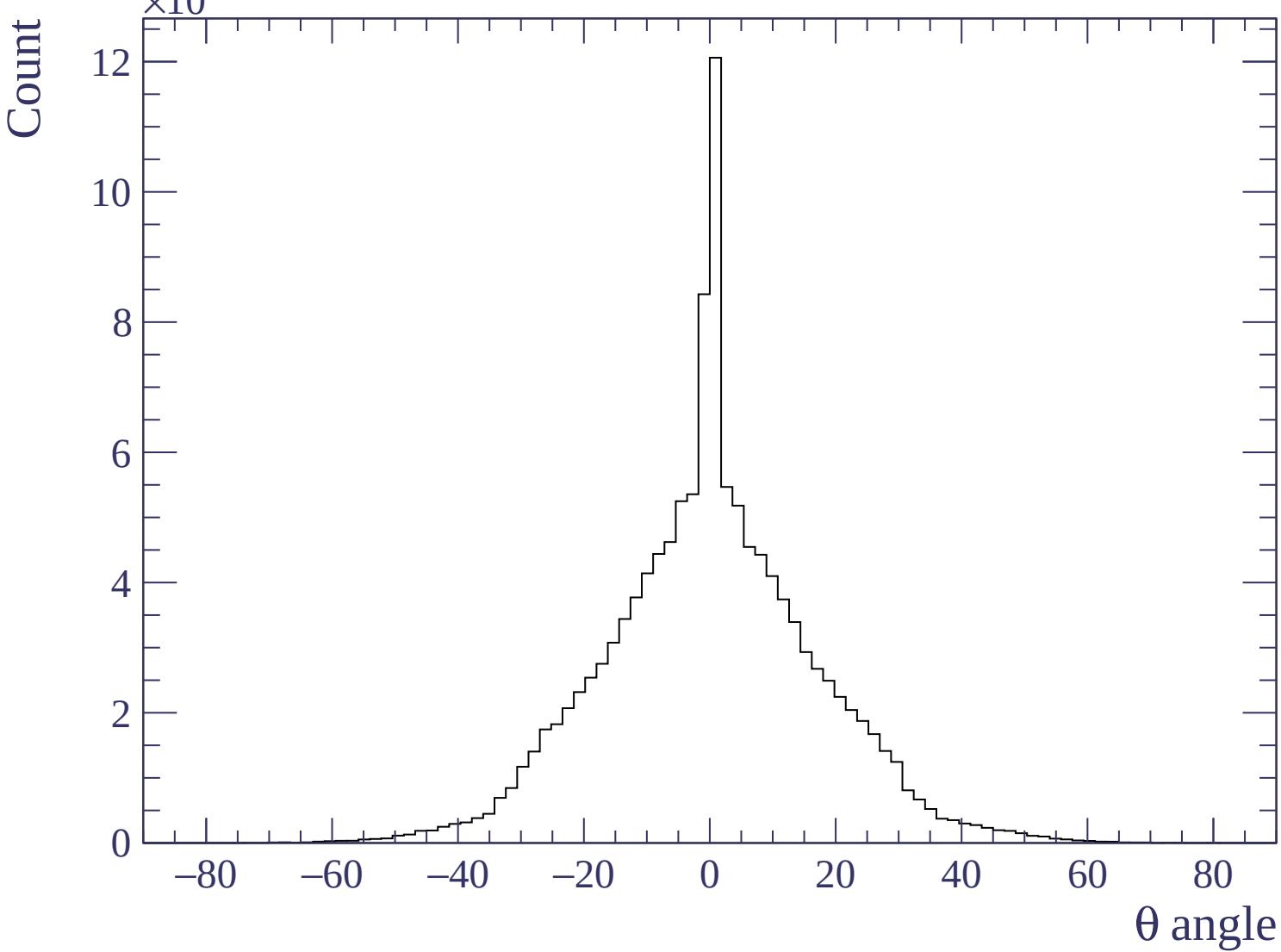


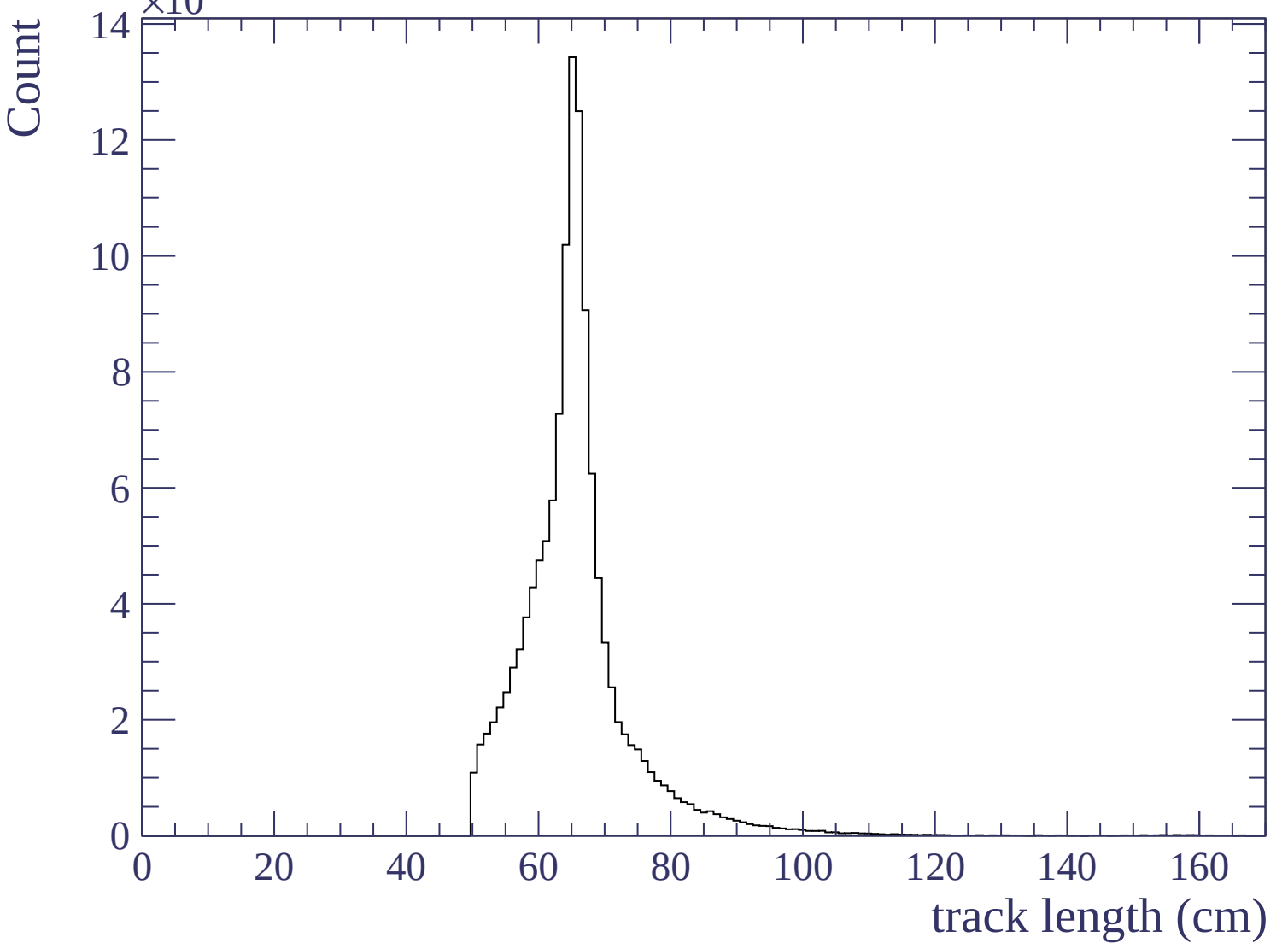


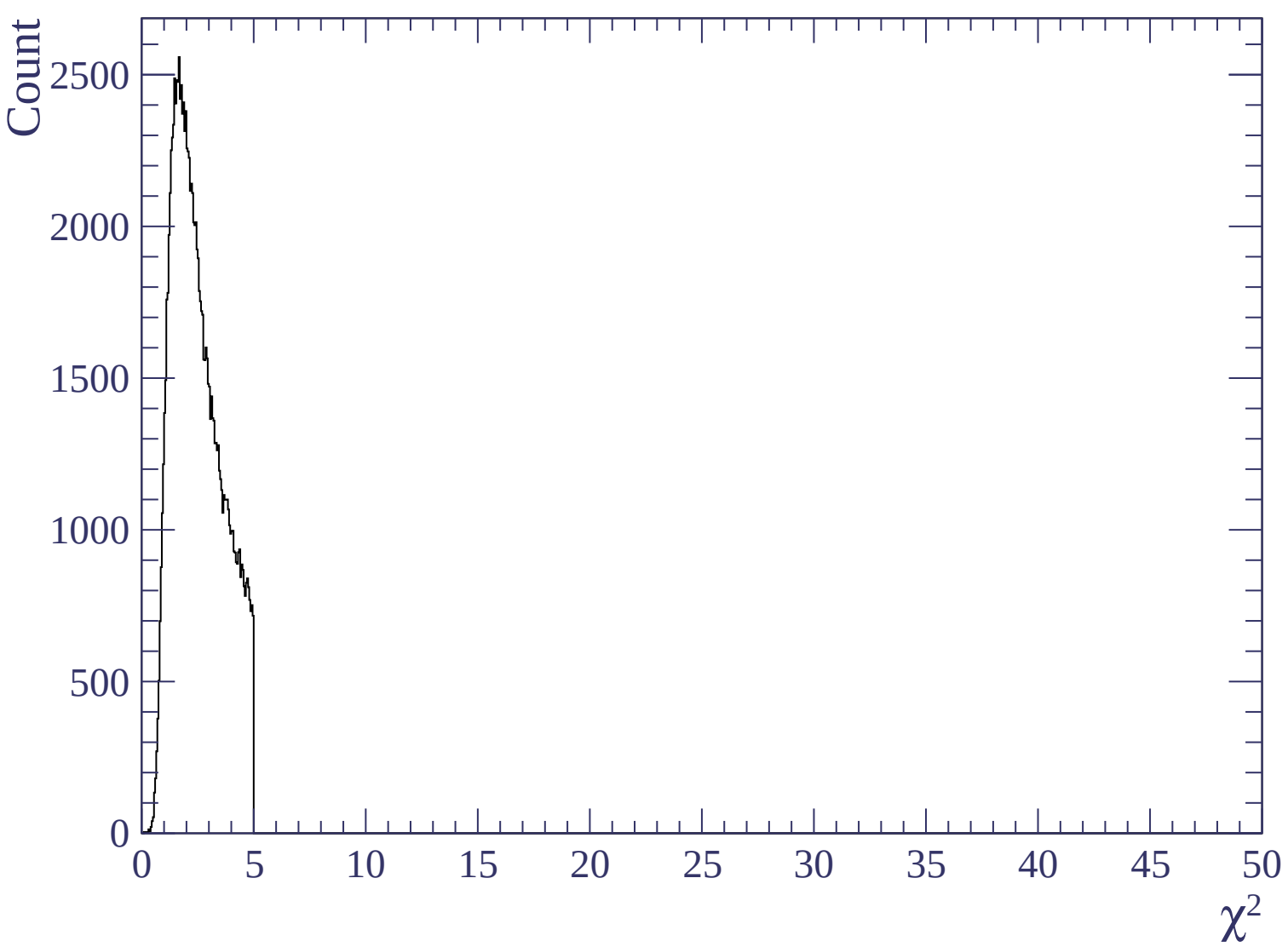


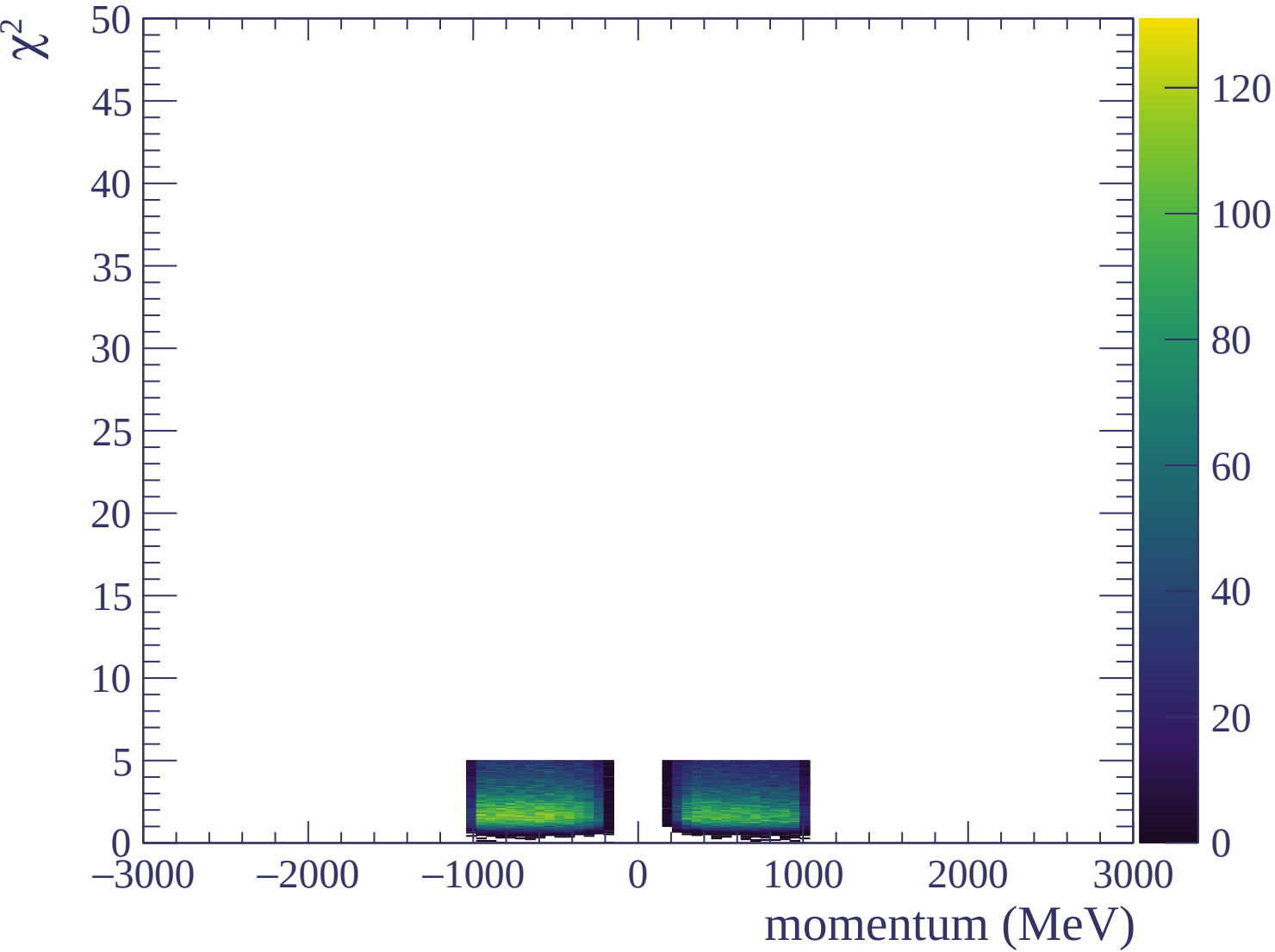


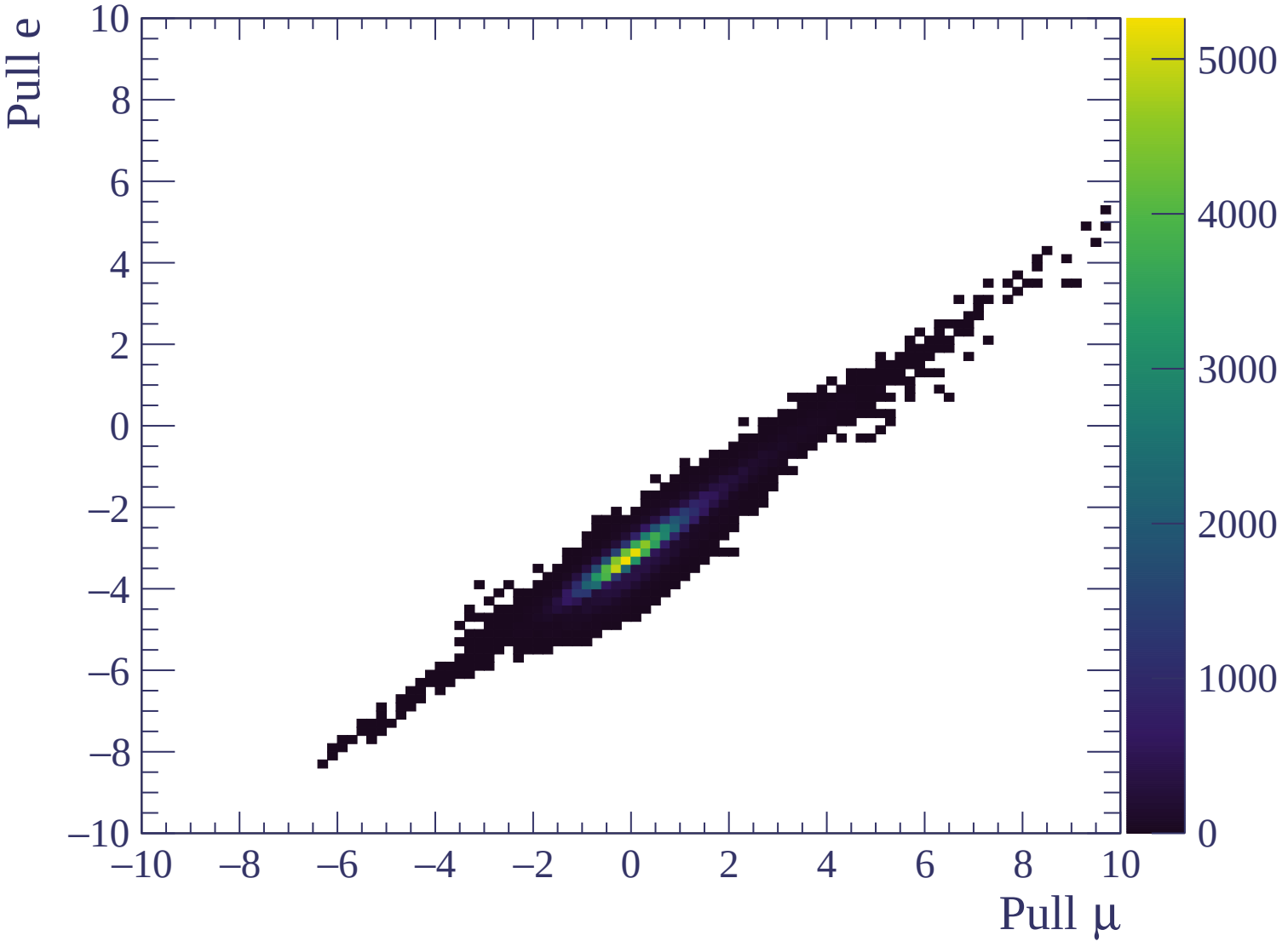


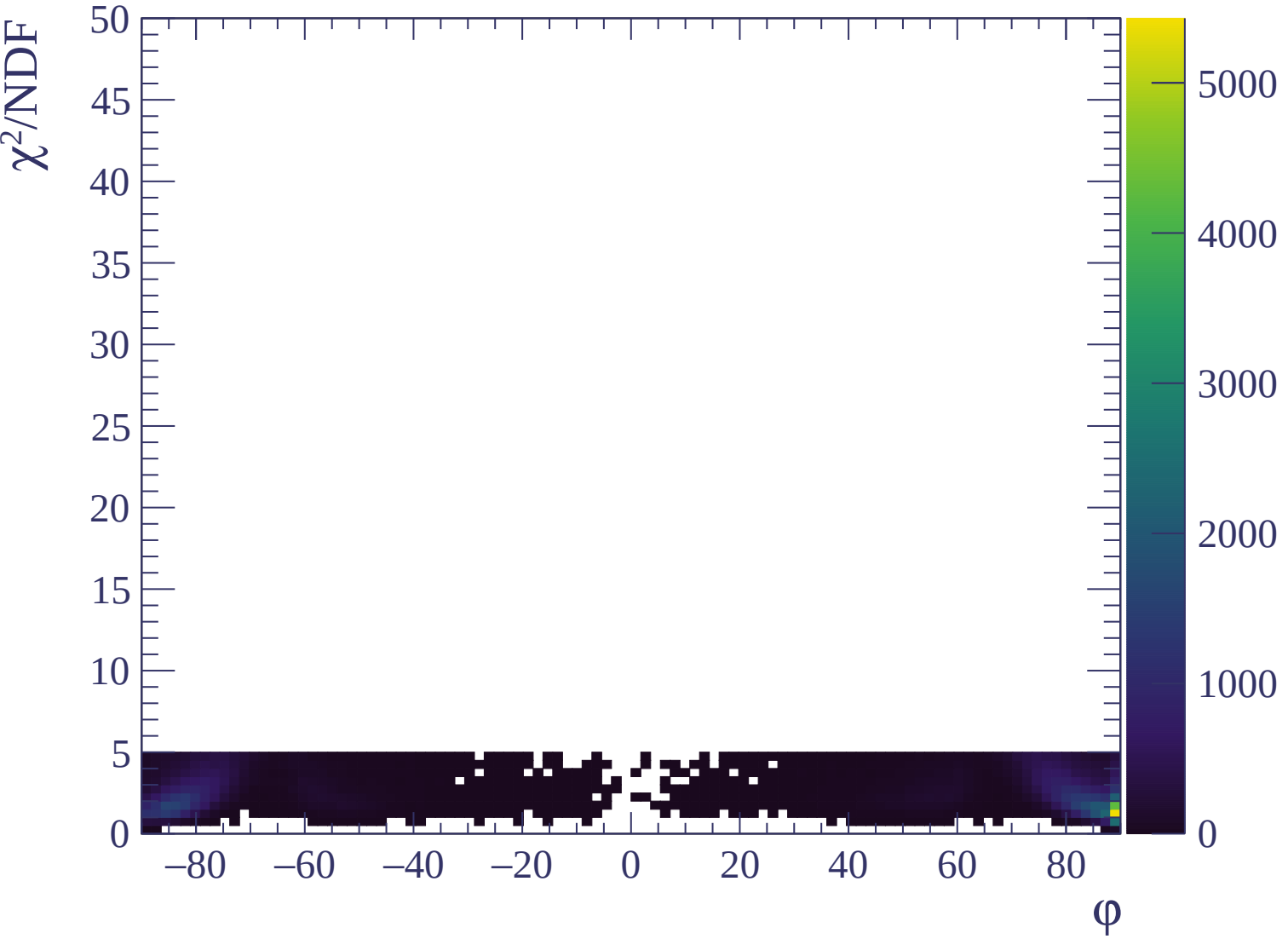


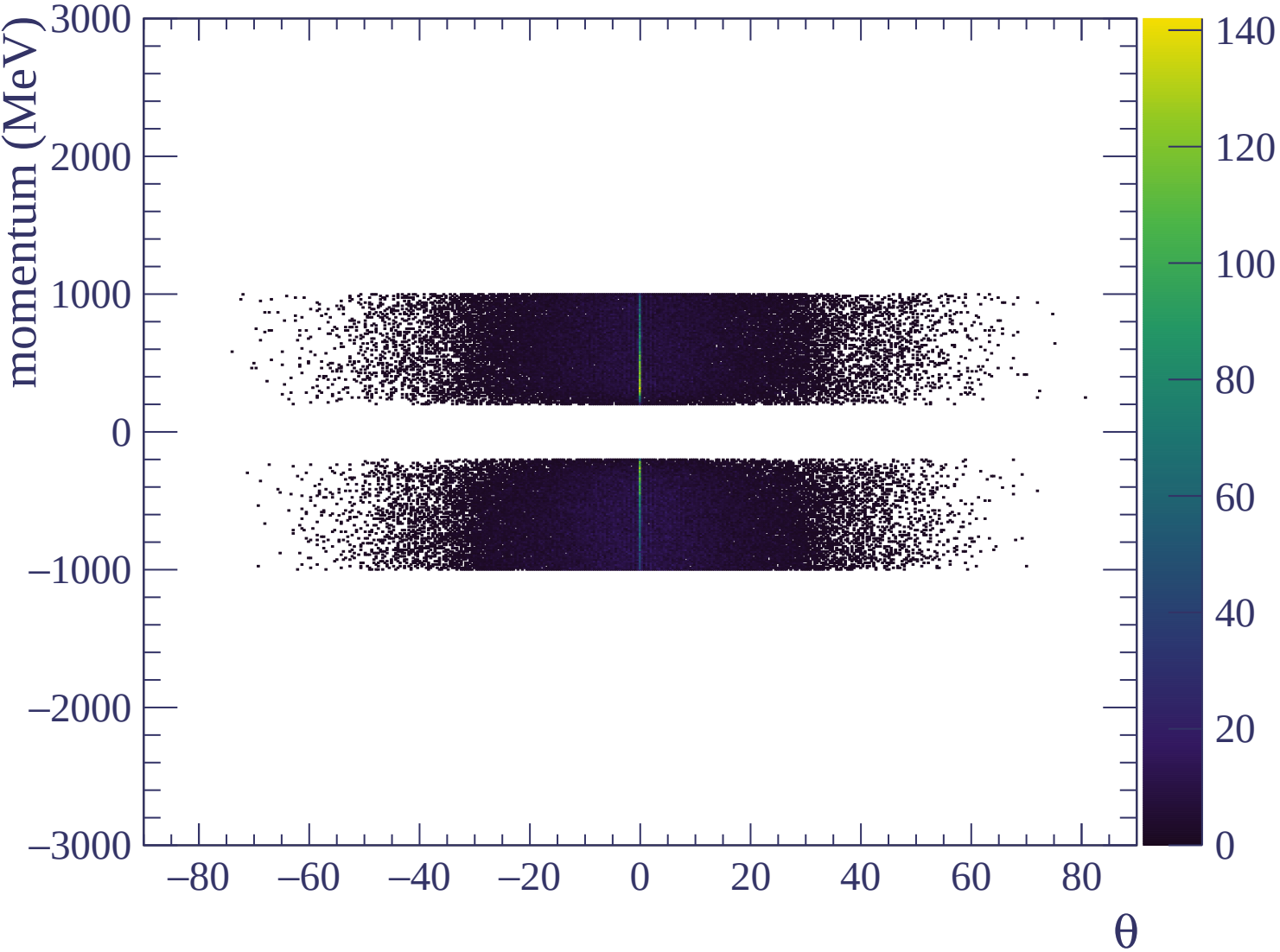


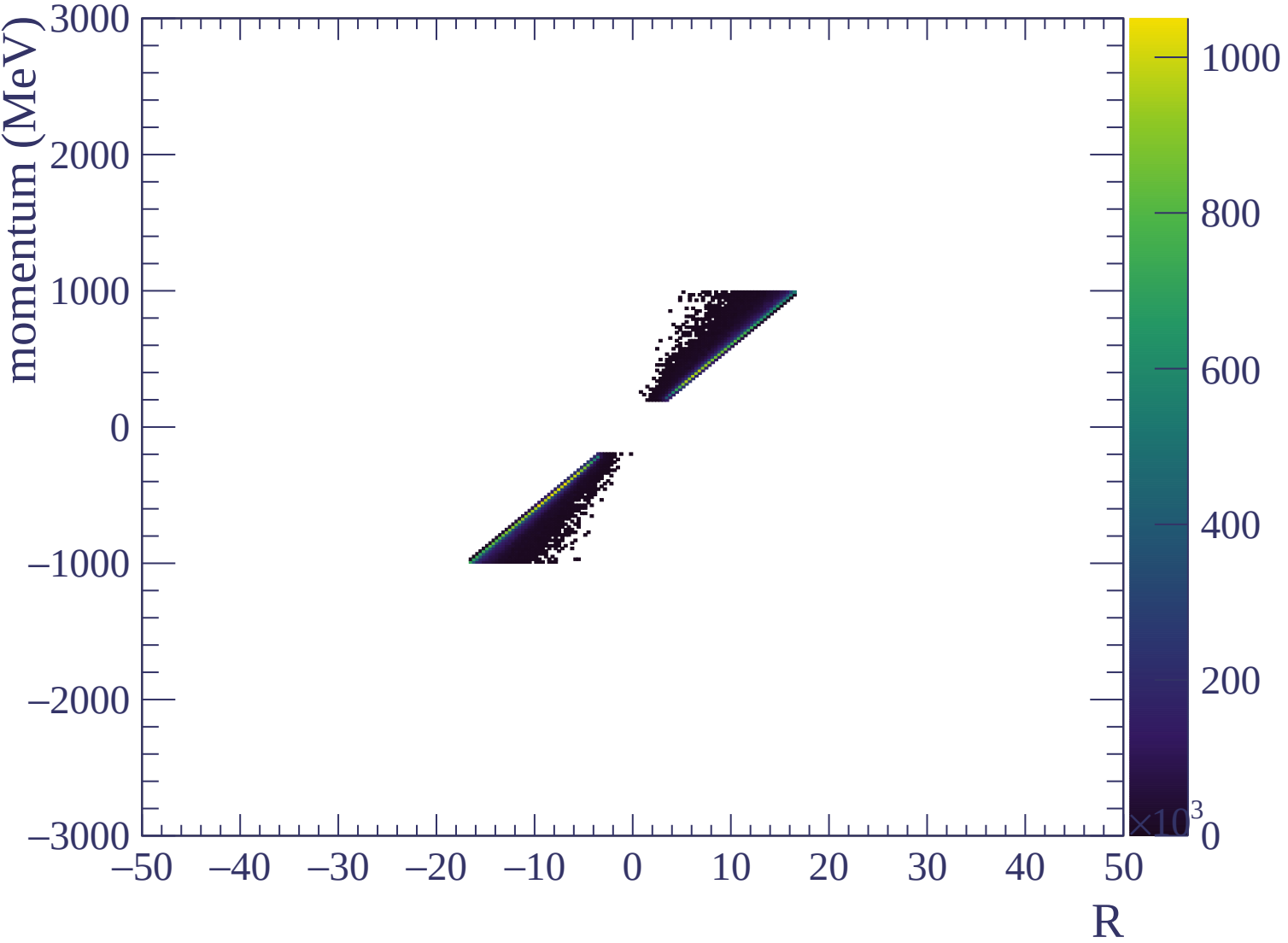


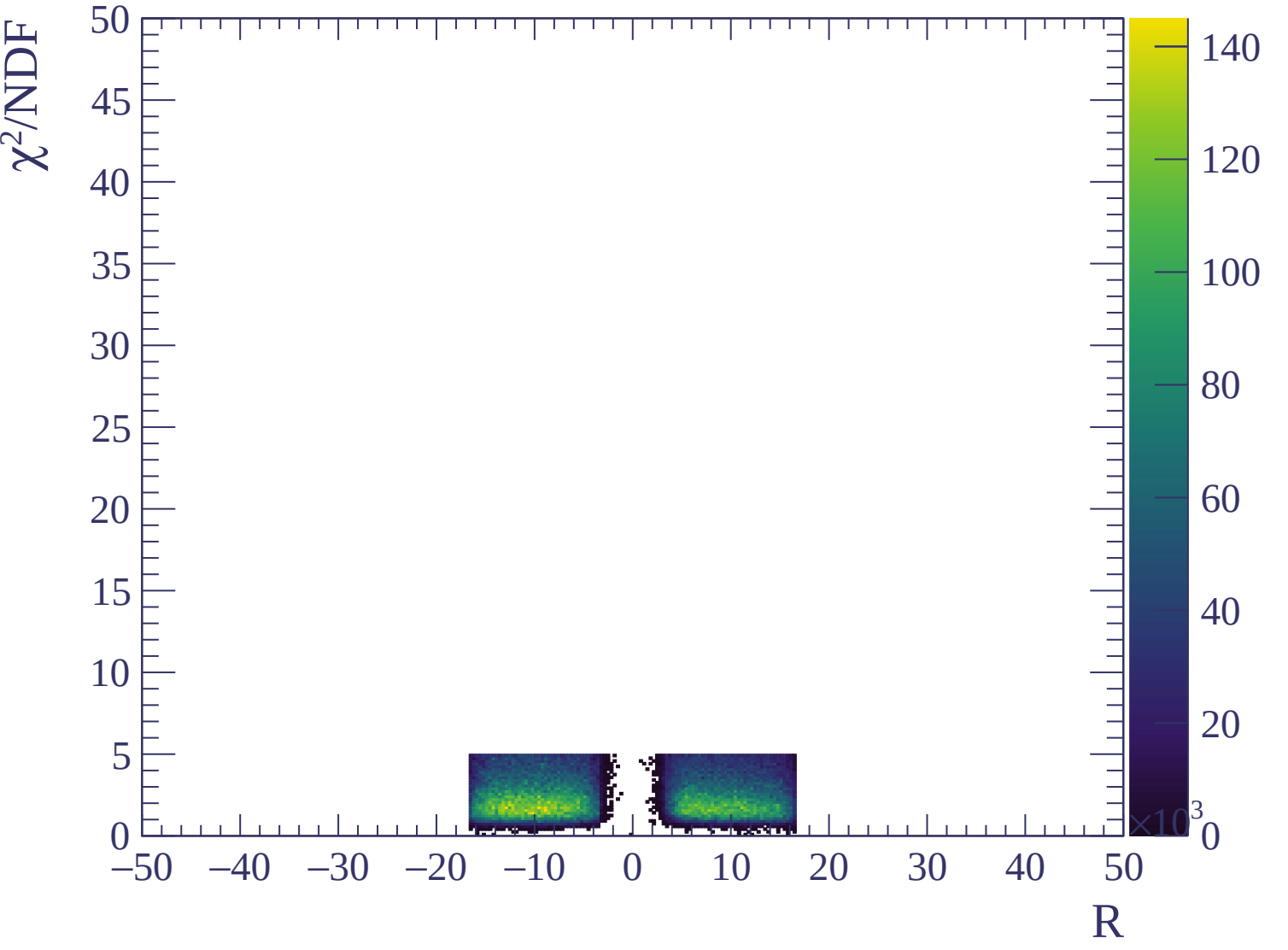


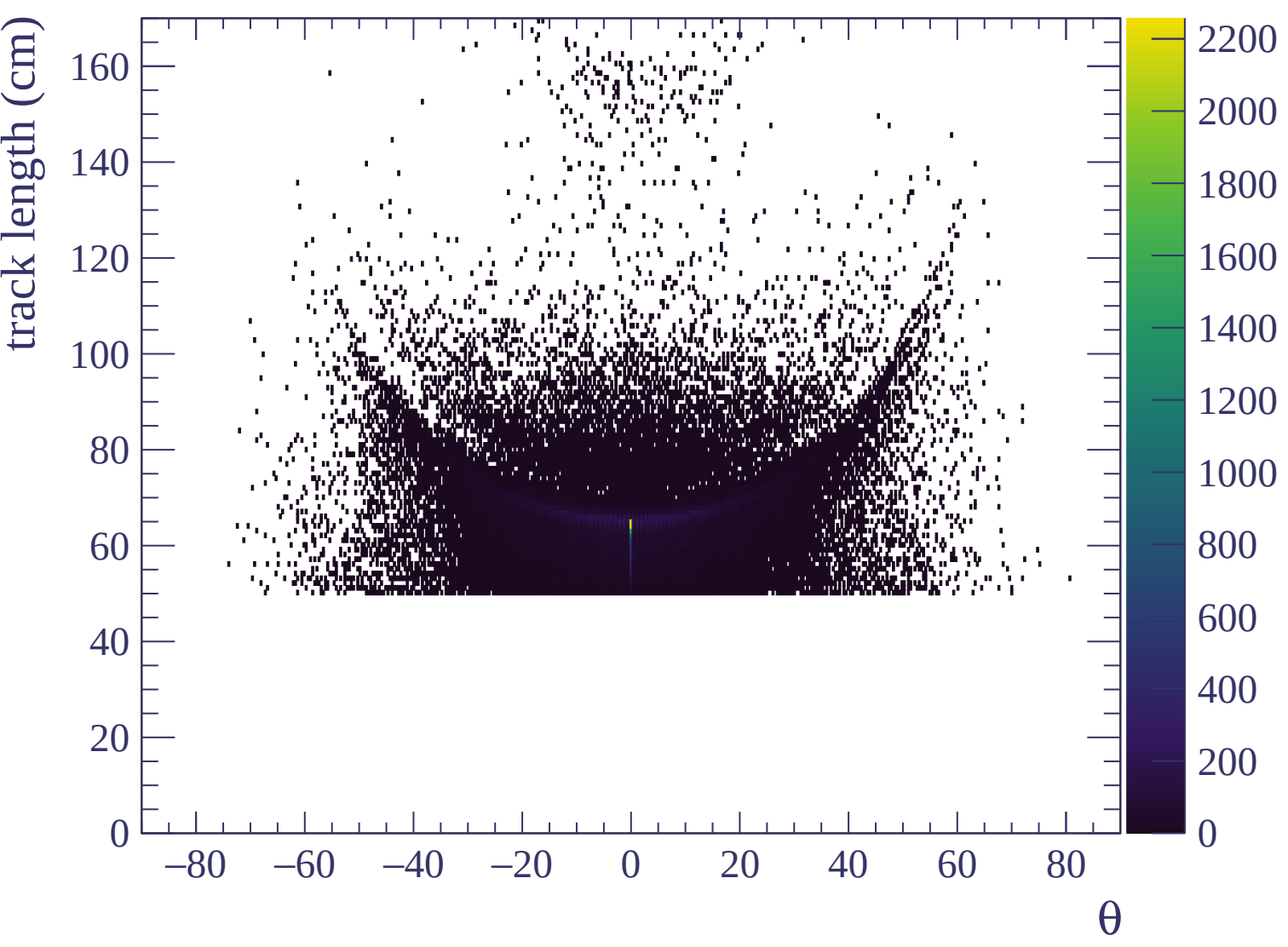


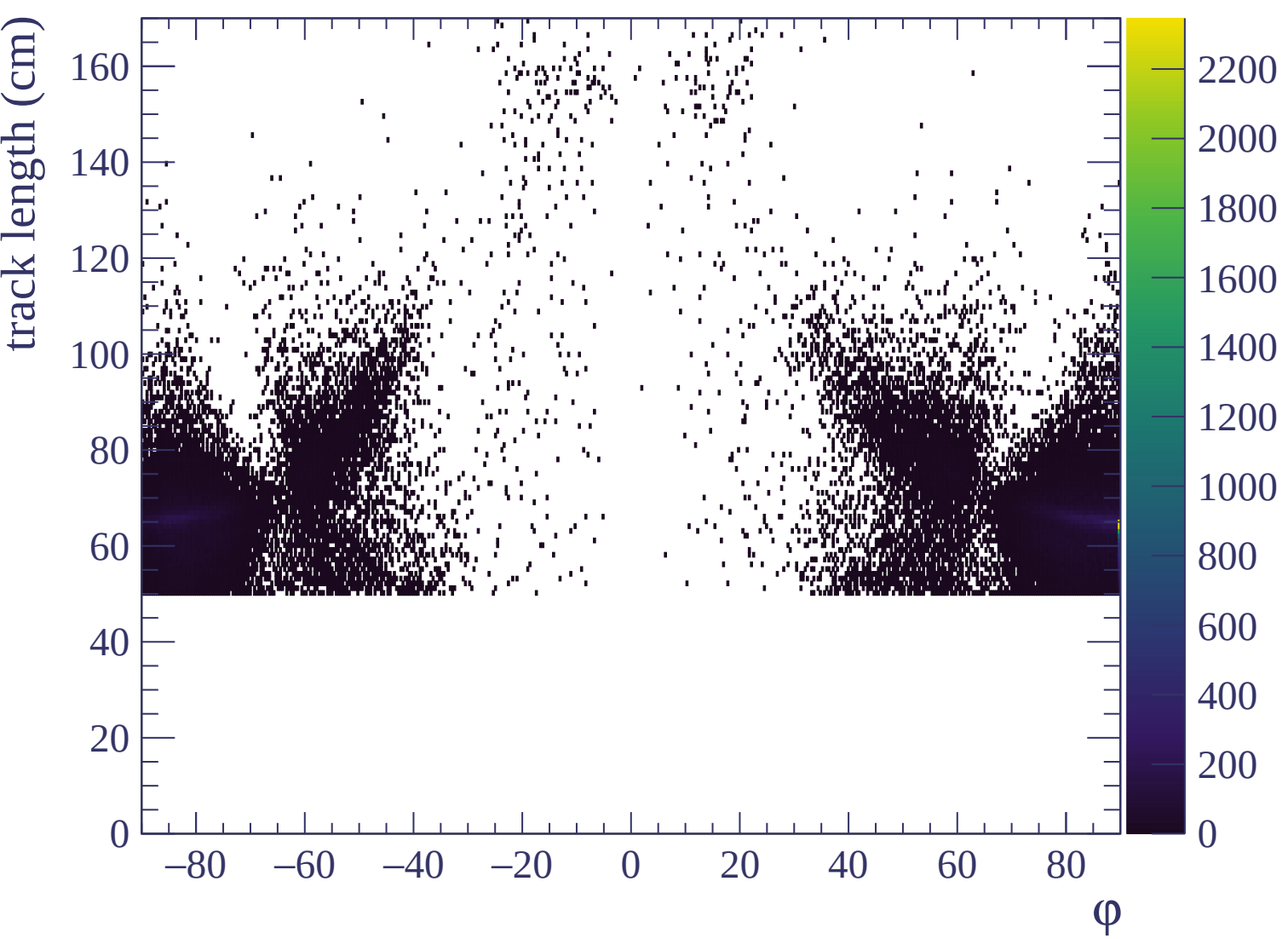












Energy loss | $-3000 < p < -2940$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2940 < p < -2880

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2880 < p < -2820

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2820 < p < -2760$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-2760 < p < -2700$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2700 < p < -2640

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-2640 < p < -2580$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2580 < p < -2520

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-2520 < p < -2460$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2460 < p < -2400$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2400 < p < -2340

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2340 < p < -2280$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2280 < p < -2220

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2220 < p < -2160$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2160 < p < -2100$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -2100 < p < -2040

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-2040 < p < -1980$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1980 < p < -1920$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1920 < p < -1860

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1860 < p < -1800

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1740$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1740 < p < -1680

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1680 < p < -1620

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1620 < p < -1560

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

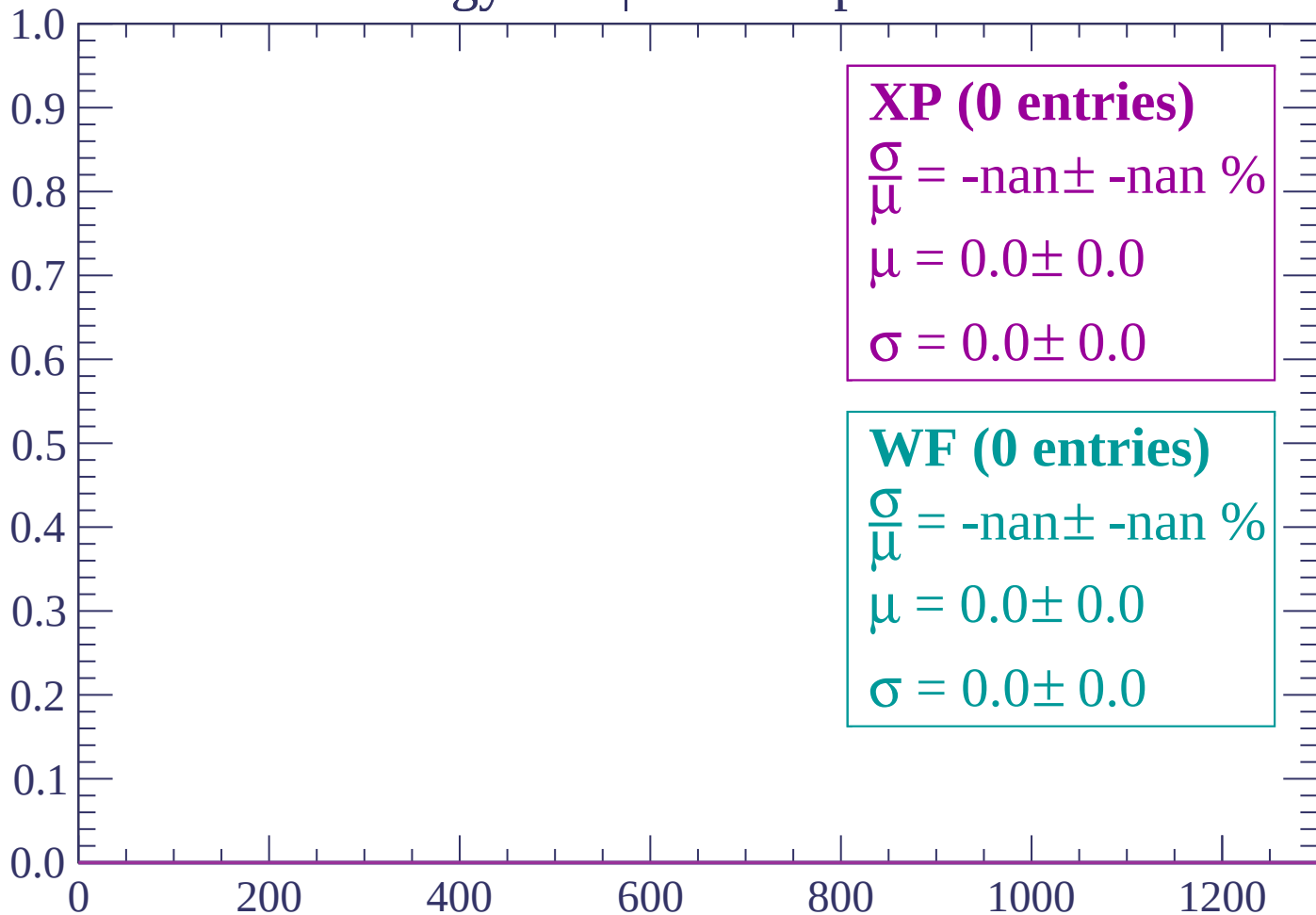
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1500$

Count



dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1440 < p < -1380

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1380 < p < -1320

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1260$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1260 < p < -1200$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1200 < p < -1140

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1140 < p < -1080

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | -1080 < p < -1020

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

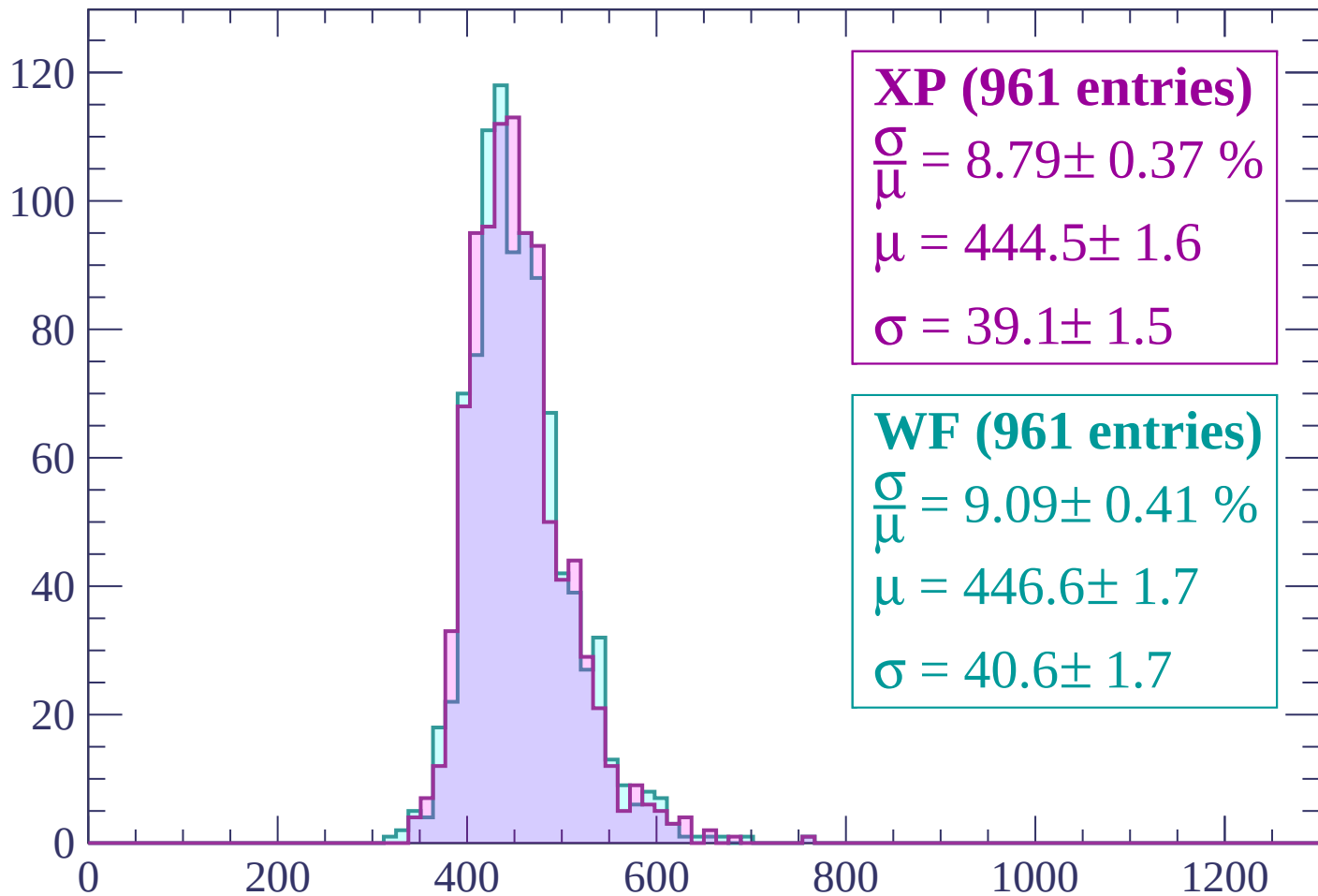
$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-1020 < p < -960$

Count



XP (961 entries)

$$\frac{\sigma}{\mu} = 8.79 \pm 0.37 \%$$

$$\mu = 444.5 \pm 1.6$$

$$\sigma = 39.1 \pm 1.5$$

WF (961 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.41 \%$$

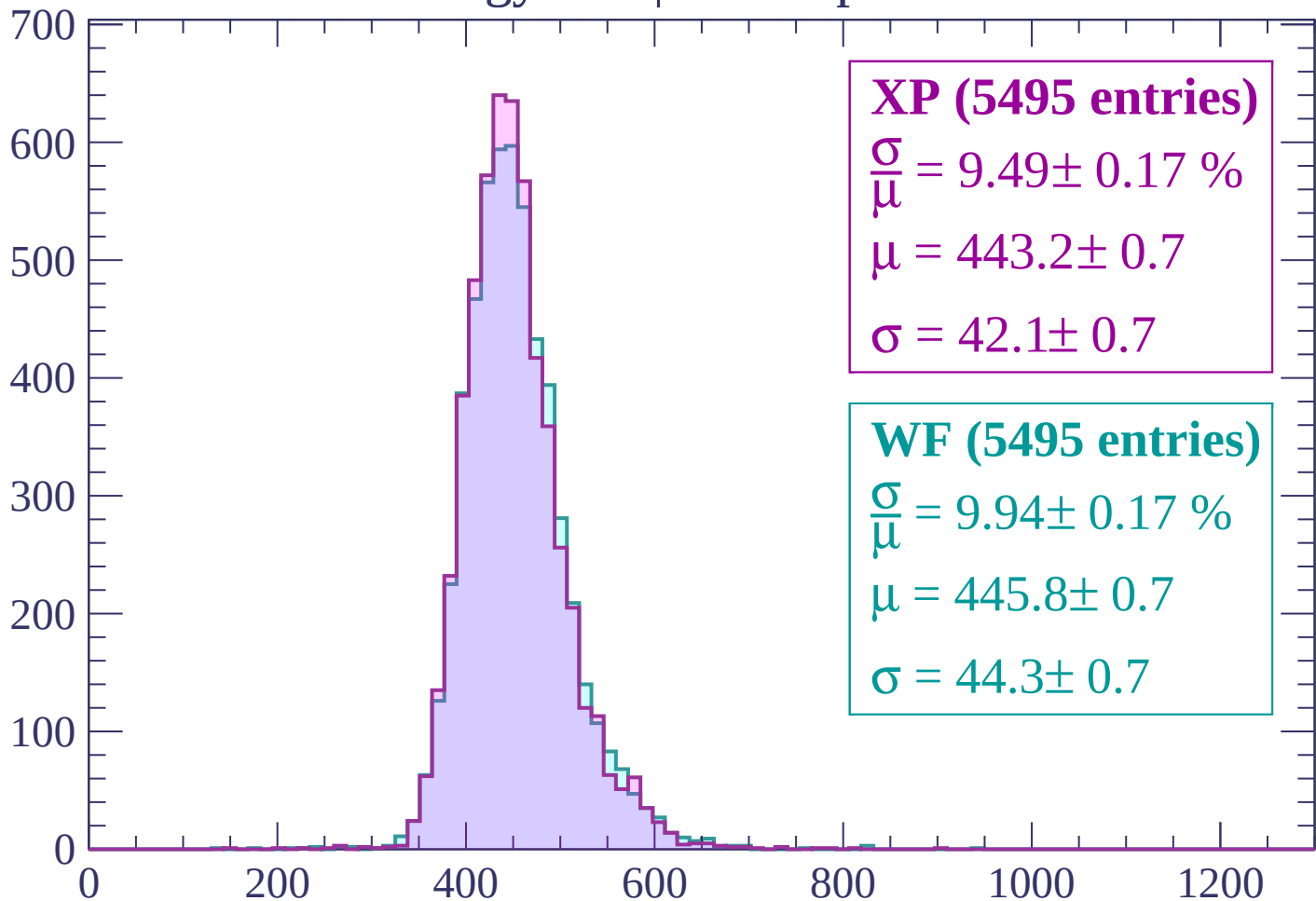
$$\mu = 446.6 \pm 1.7$$

$$\sigma = 40.6 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (5495 entries)

$$\frac{\sigma}{\mu} = 9.49 \pm 0.17 \%$$

$$\mu = 443.2 \pm 0.7$$

$$\sigma = 42.1 \pm 0.7$$

WF (5495 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

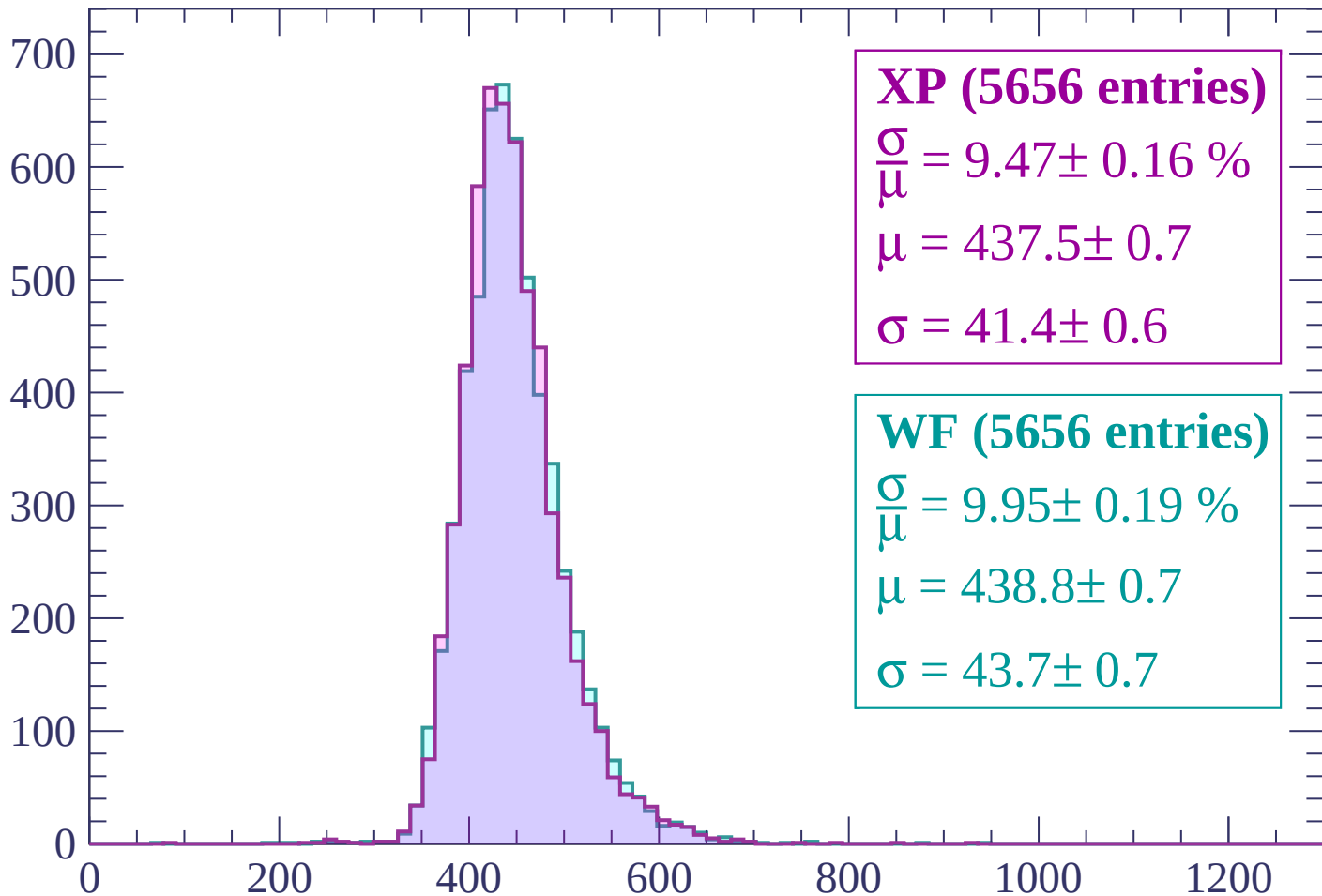
$$\mu = 445.8 \pm 0.7$$

$$\sigma = 44.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (5656 entries)

$$\frac{\sigma}{\mu} = 9.47 \pm 0.16 \%$$

$$\mu = 437.5 \pm 0.7$$

$$\sigma = 41.4 \pm 0.6$$

WF (5656 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.19 \%$$

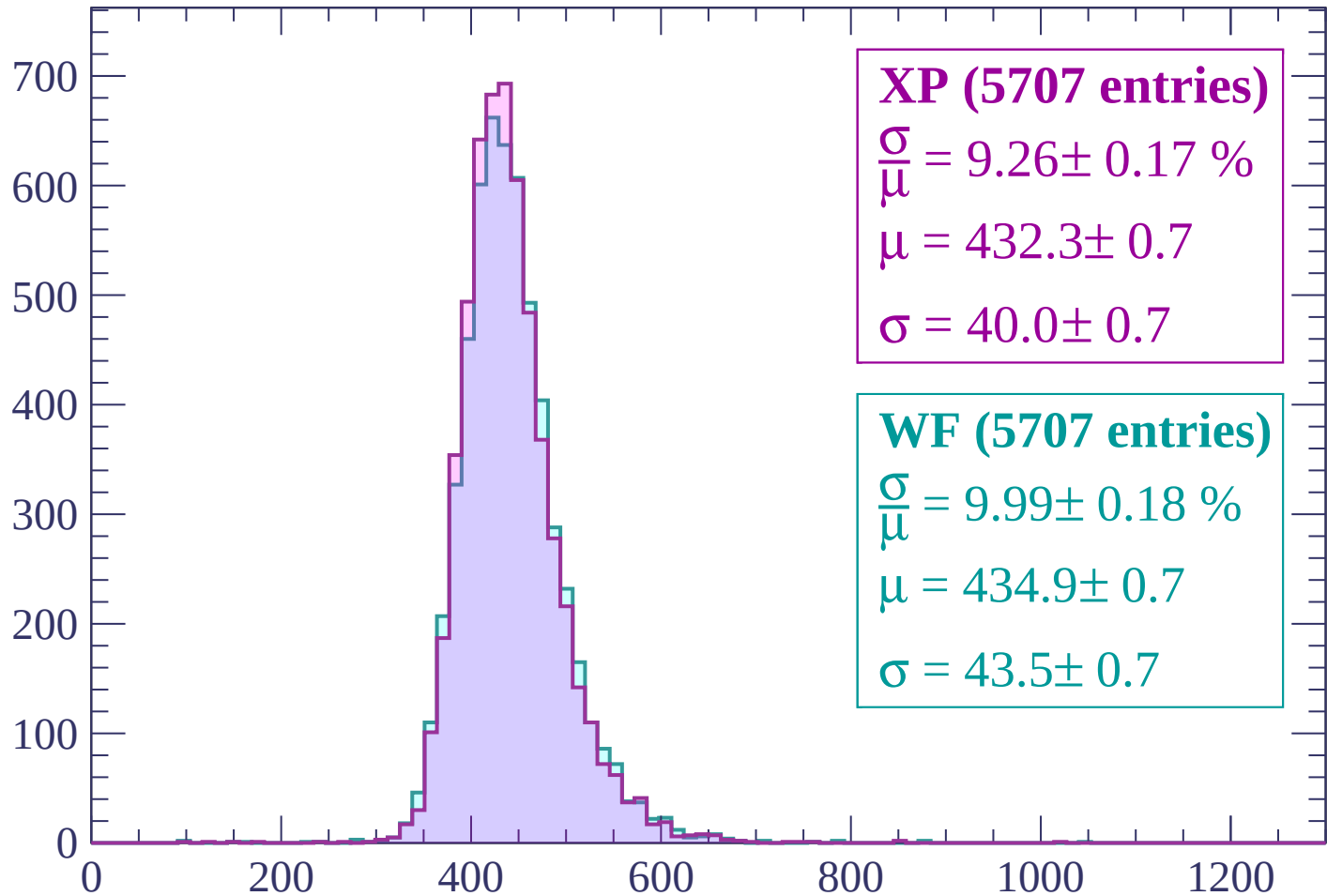
$$\mu = 438.8 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (5707 entries)

$$\frac{\sigma}{\mu} = 9.26 \pm 0.17 \%$$

$$\mu = 432.3 \pm 0.7$$

$$\sigma = 40.0 \pm 0.7$$

WF (5707 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.18 \%$$

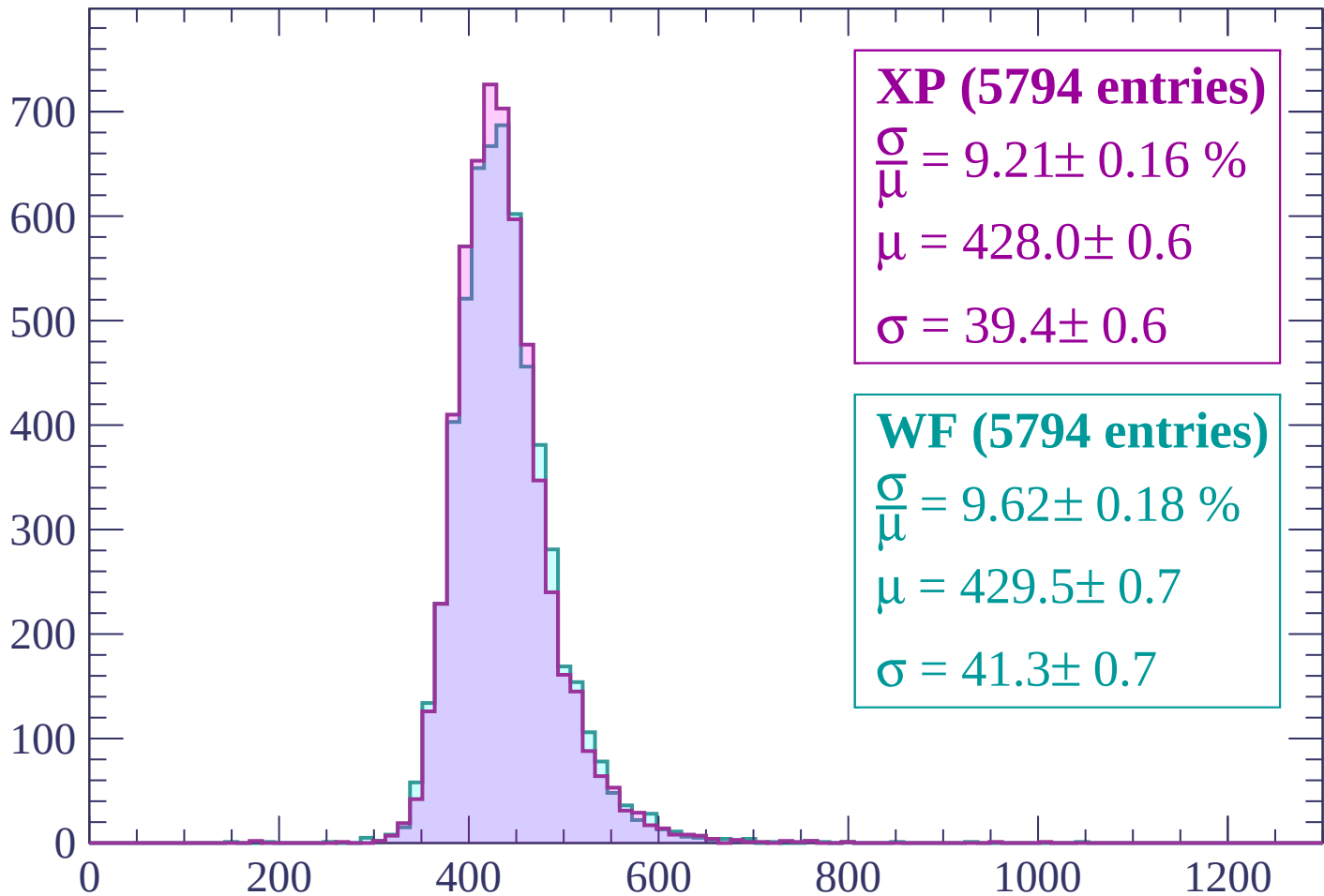
$$\mu = 434.9 \pm 0.7$$

$$\sigma = 43.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (5794 entries)

$$\frac{\sigma}{\mu} = 9.21 \pm 0.16 \%$$

$$\mu = 428.0 \pm 0.6$$

$$\sigma = 39.4 \pm 0.6$$

WF (5794 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

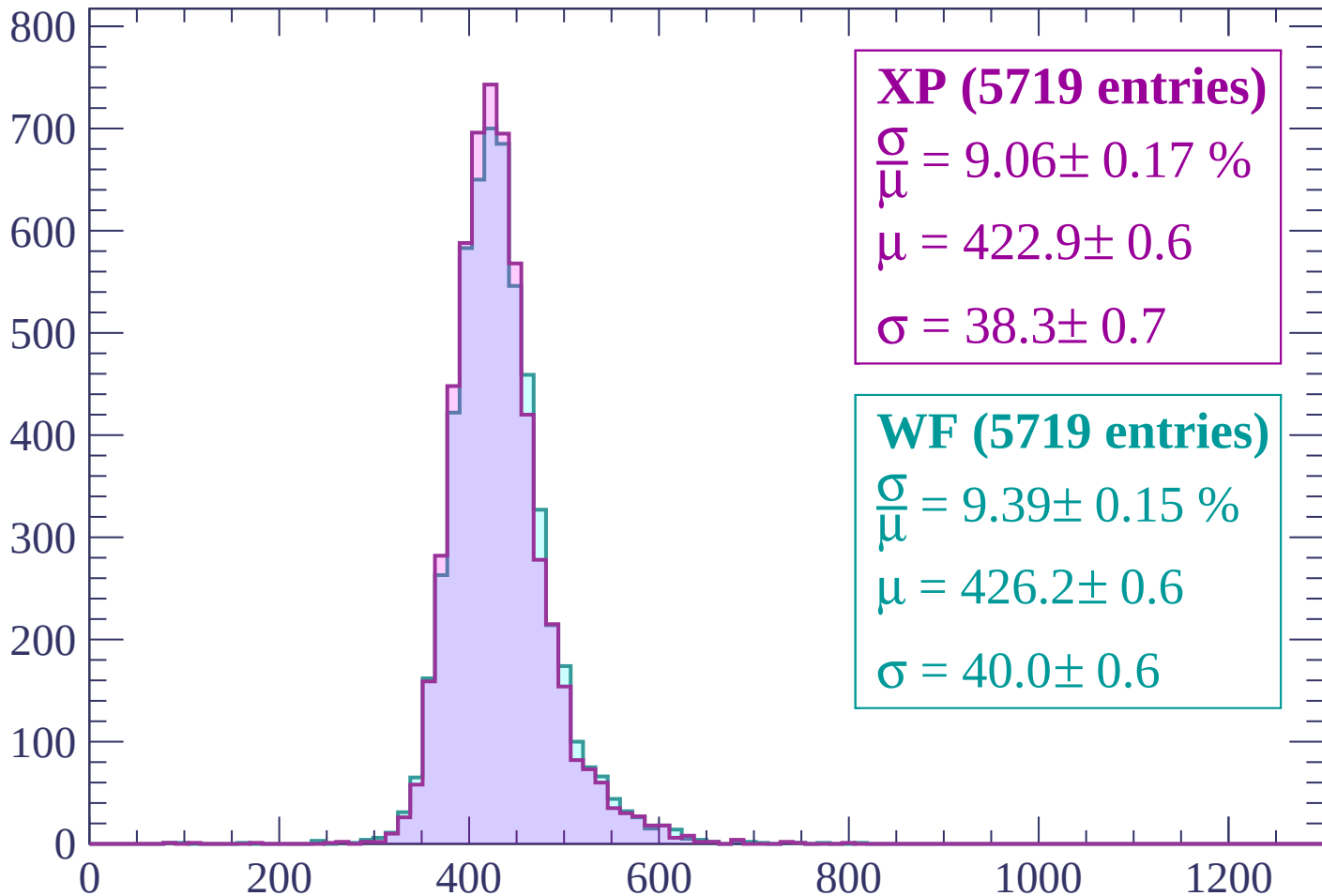
$$\mu = 429.5 \pm 0.7$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (5719 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.17 \%$$

$$\mu = 422.9 \pm 0.6$$

$$\sigma = 38.3 \pm 0.7$$

WF (5719 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.15 \%$$

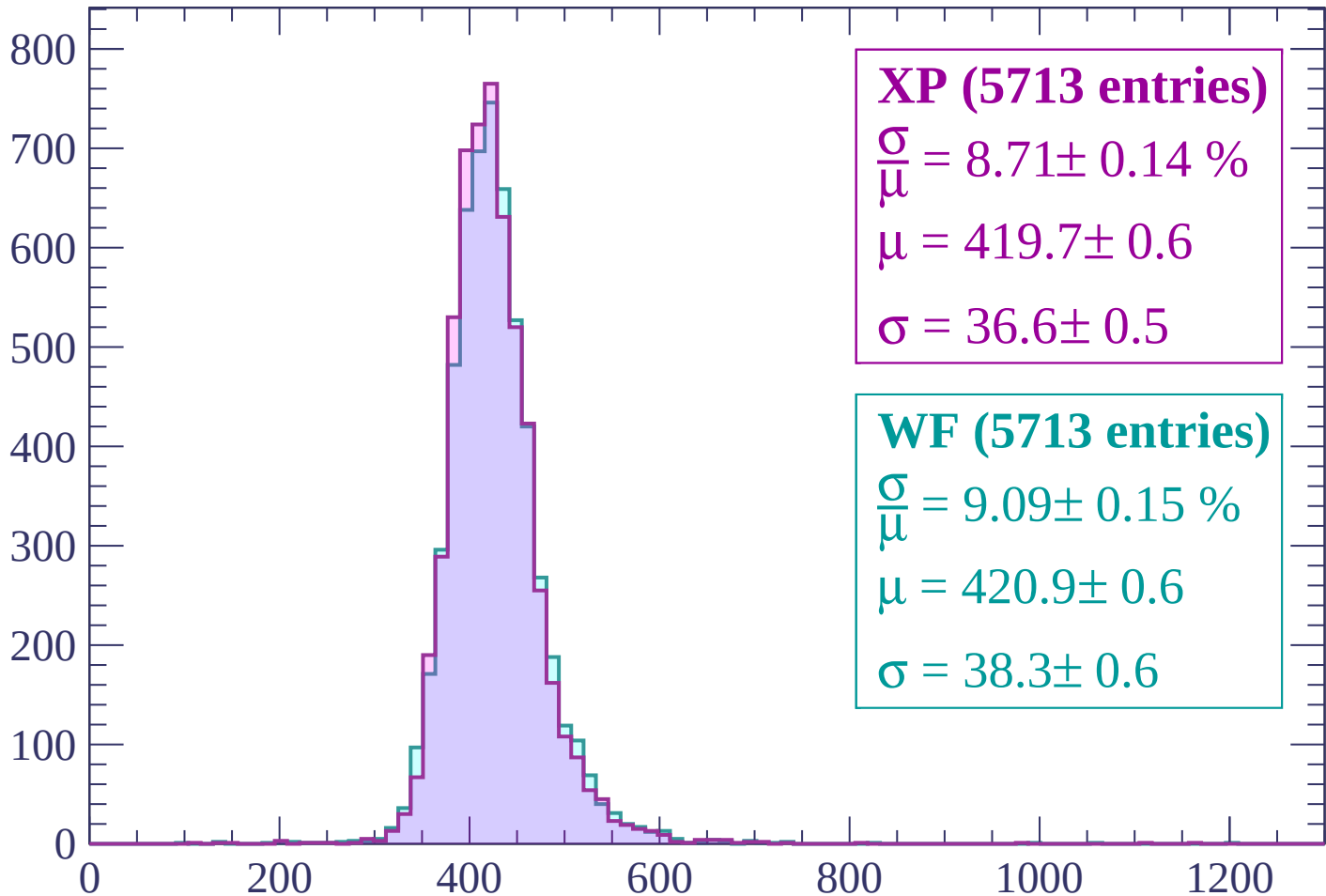
$$\mu = 426.2 \pm 0.6$$

$$\sigma = 40.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (5713 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.14 \%$$

$$\mu = 419.7 \pm 0.6$$

$$\sigma = 36.6 \pm 0.5$$

WF (5713 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.15 \%$$

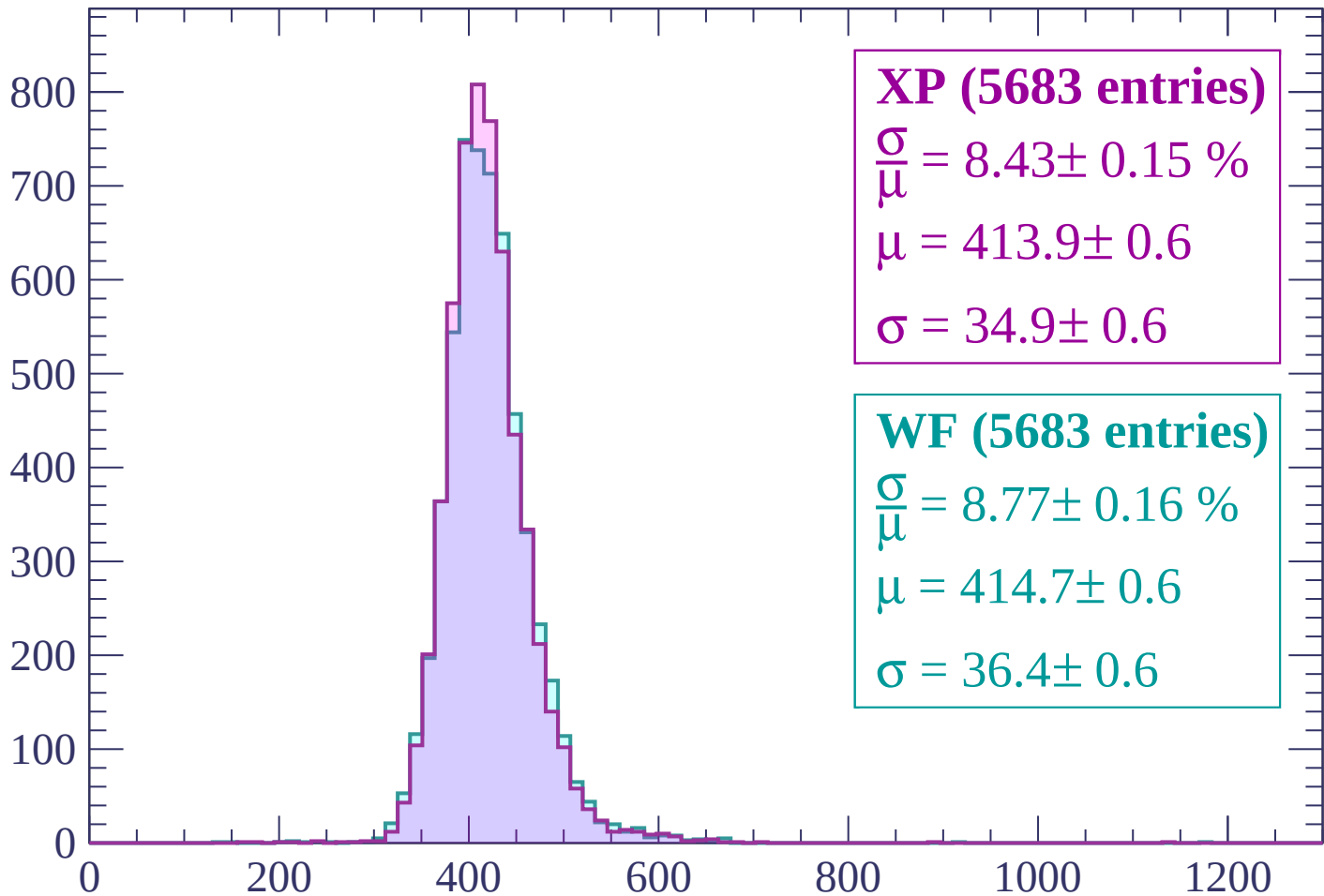
$$\mu = 420.9 \pm 0.6$$

$$\sigma = 38.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (5683 entries)

$\frac{\sigma}{\mu} = 8.43 \pm 0.15 \%$

$\mu = 413.9 \pm 0.6$

$\sigma = 34.9 \pm 0.6$

WF (5683 entries)

$\frac{\sigma}{\mu} = 8.77 \pm 0.16 \%$

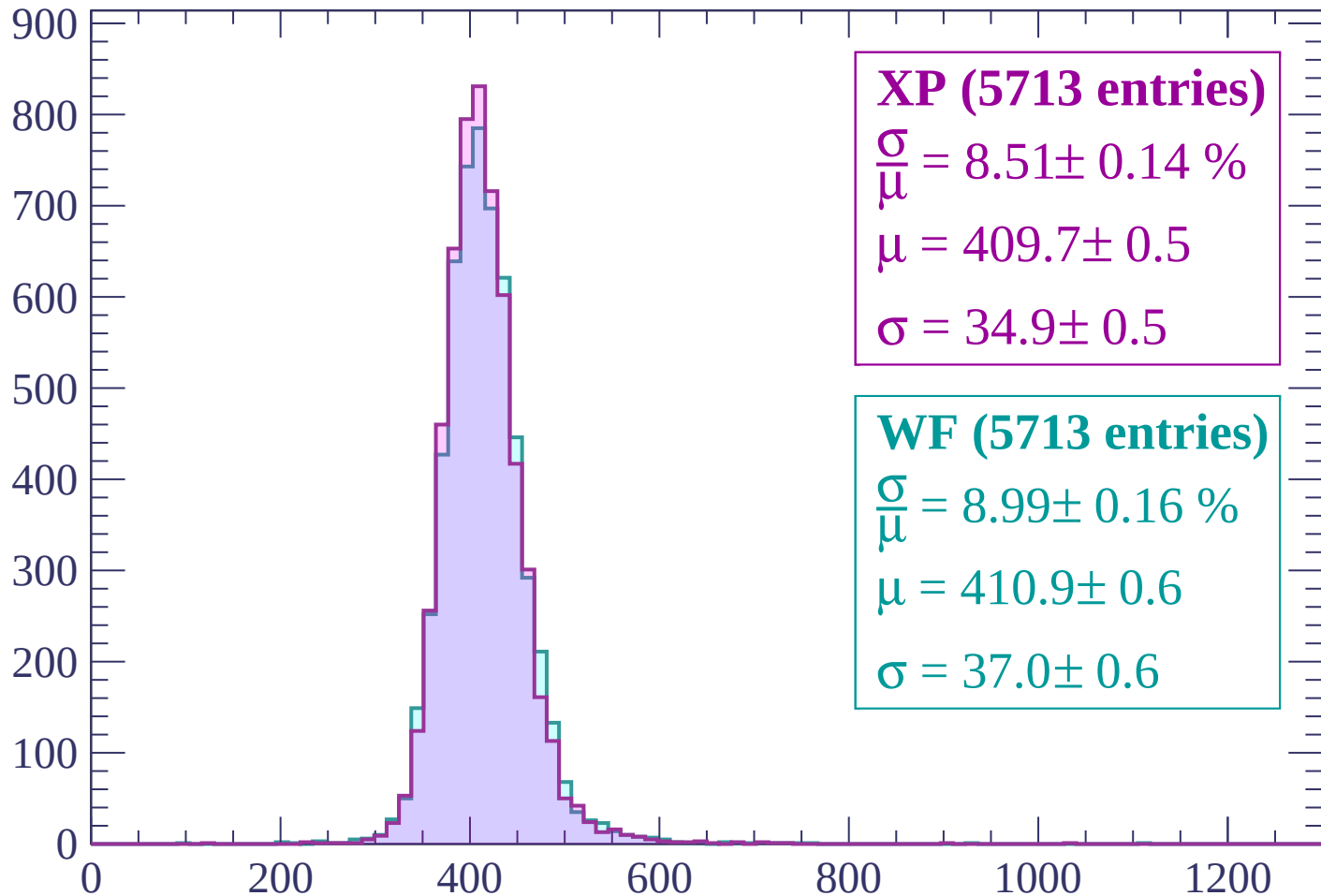
$\mu = 414.7 \pm 0.6$

$\sigma = 36.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (5713 entries)

$$\frac{\sigma}{\mu} = 8.51 \pm 0.14 \%$$

$$\mu = 409.7 \pm 0.5$$

$$\sigma = 34.9 \pm 0.5$$

WF (5713 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$$

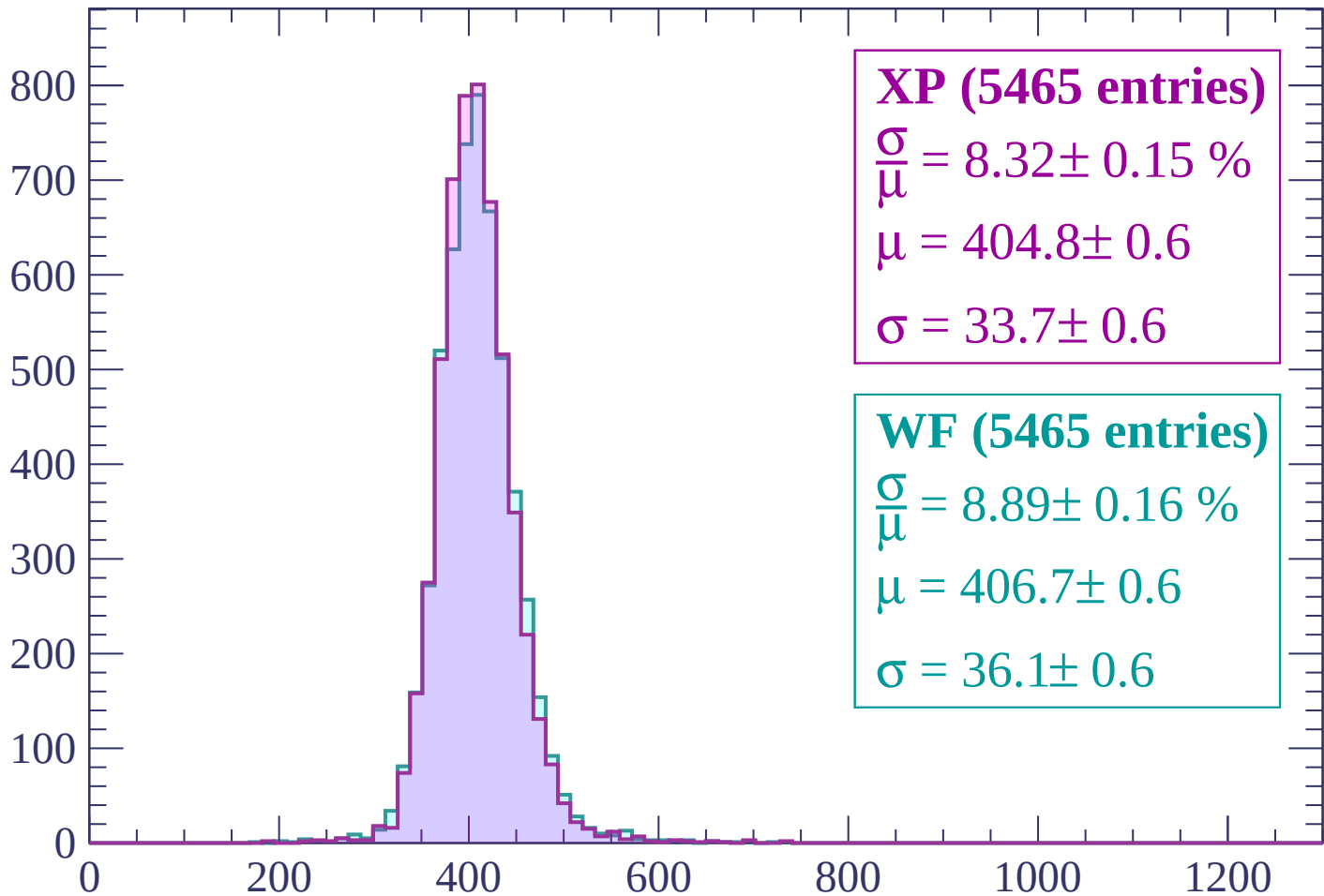
$$\mu = 410.9 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (5465 entries)

$$\frac{\sigma}{\mu} = 8.32 \pm 0.15 \%$$

$$\mu = 404.8 \pm 0.6$$

$$\sigma = 33.7 \pm 0.6$$

WF (5465 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

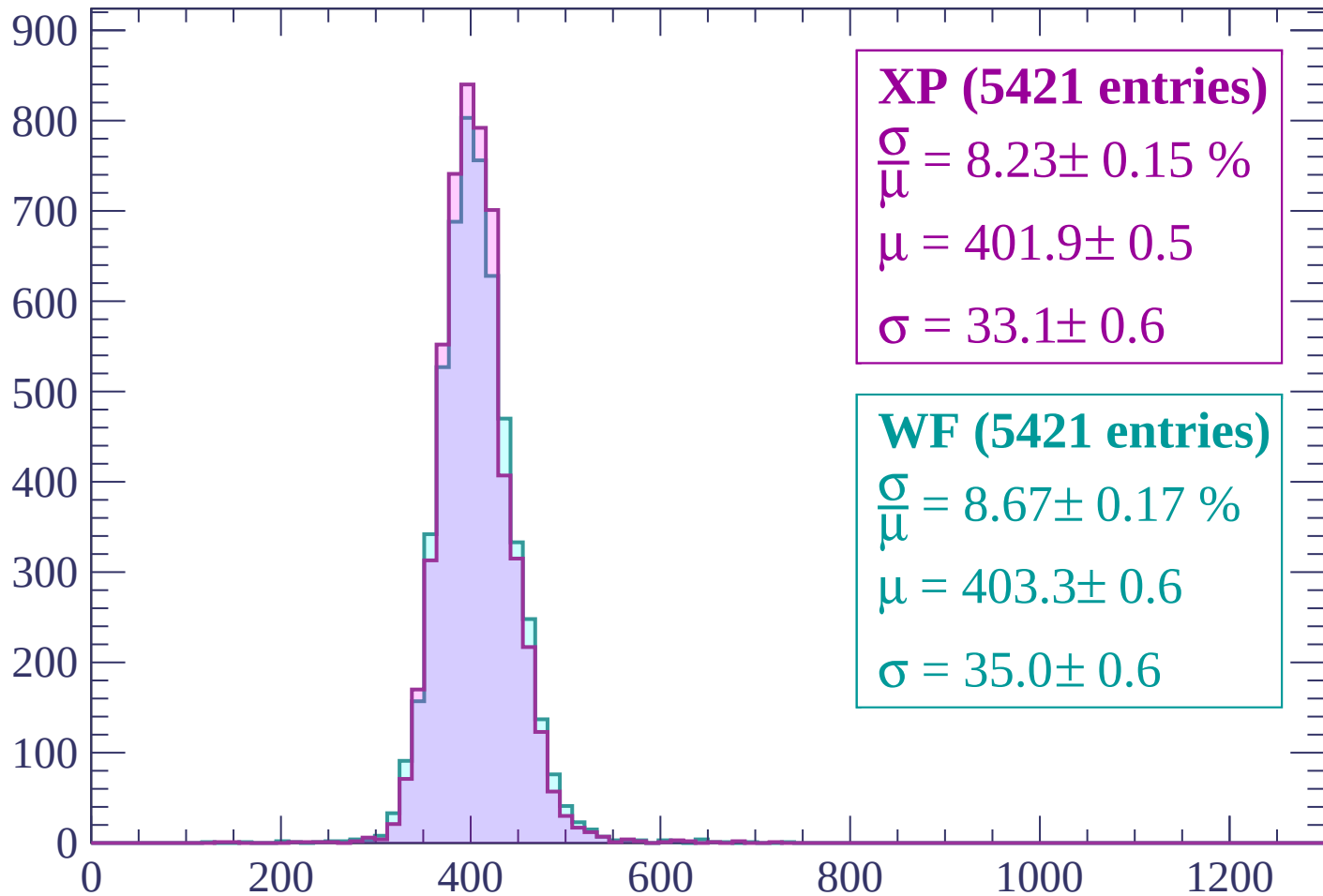
$$\mu = 406.7 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (5421 entries)

$$\frac{\sigma}{\mu} = 8.23 \pm 0.15 \%$$

$$\mu = 401.9 \pm 0.5$$

$$\sigma = 33.1 \pm 0.6$$

WF (5421 entries)

$$\frac{\sigma}{\mu} = 8.67 \pm 0.17 \%$$

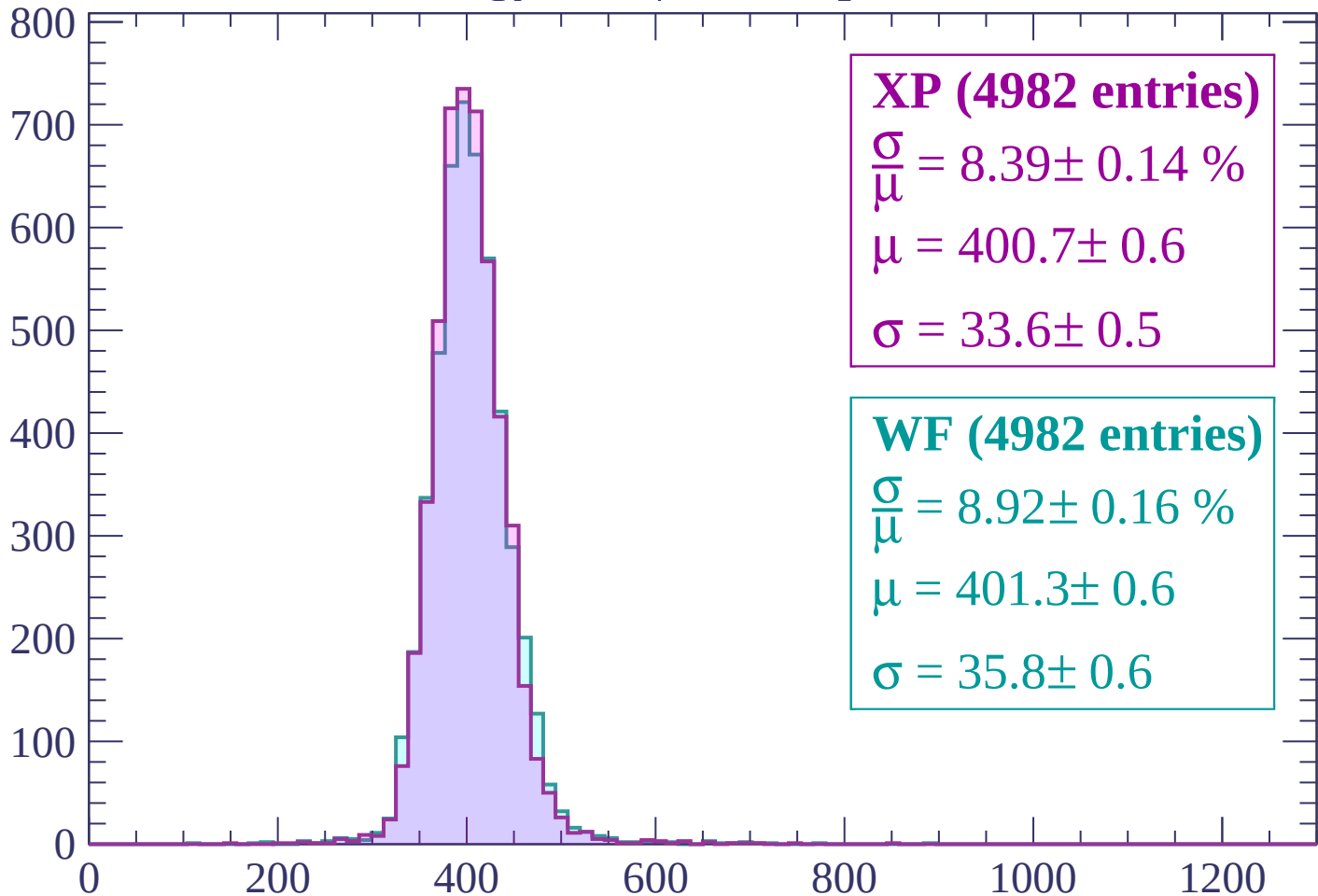
$$\mu = 403.3 \pm 0.6$$

$$\sigma = 35.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | -360 < p < -300

Count



XP (4982 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.14 \%$$

$$\mu = 400.7 \pm 0.6$$

$$\sigma = 33.6 \pm 0.5$$

WF (4982 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

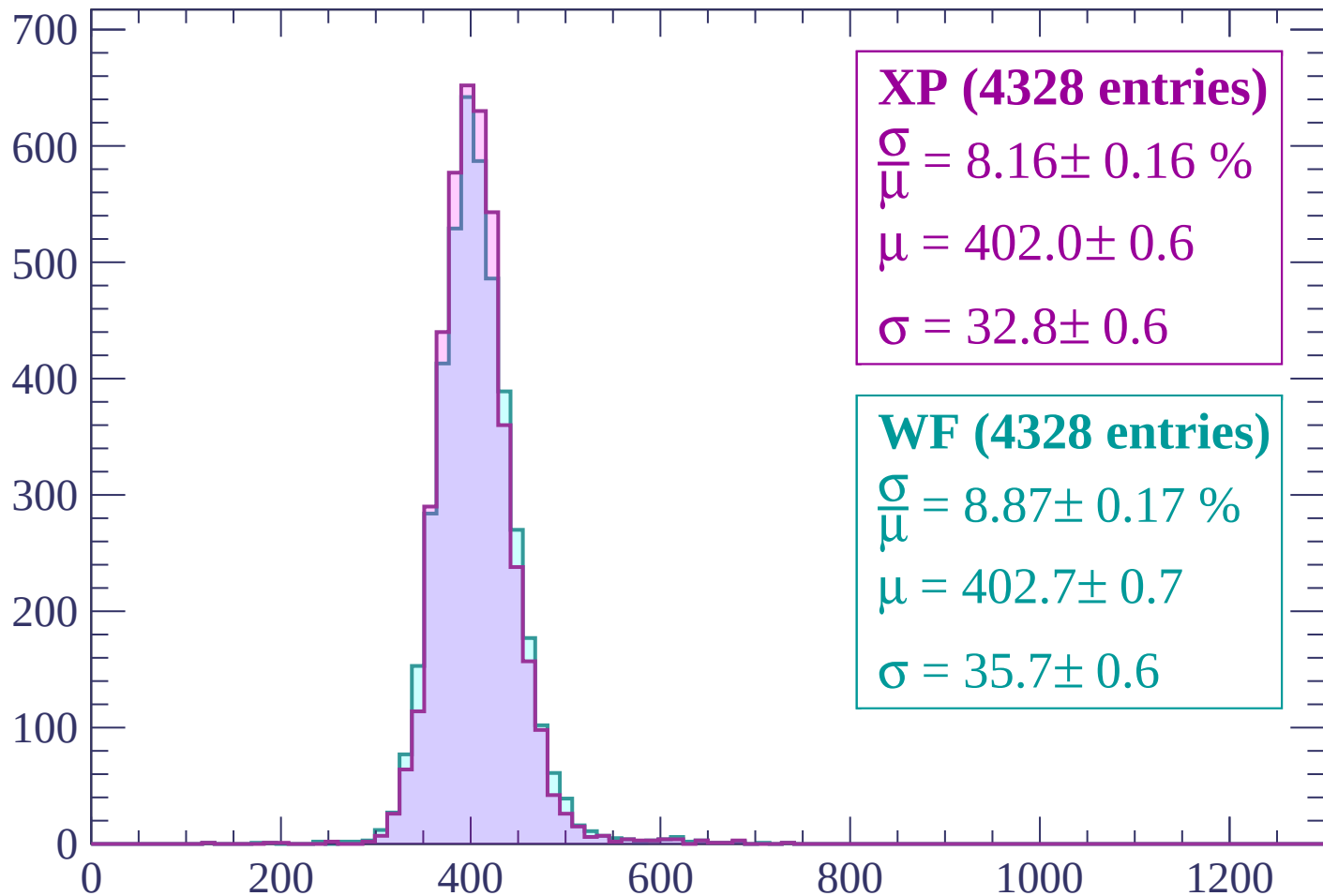
$$\mu = 401.3 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (4328 entries)

$$\frac{\sigma}{\mu} = 8.16 \pm 0.16 \%$$

$$\mu = 402.0 \pm 0.6$$

$$\sigma = 32.8 \pm 0.6$$

WF (4328 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.17 \%$$

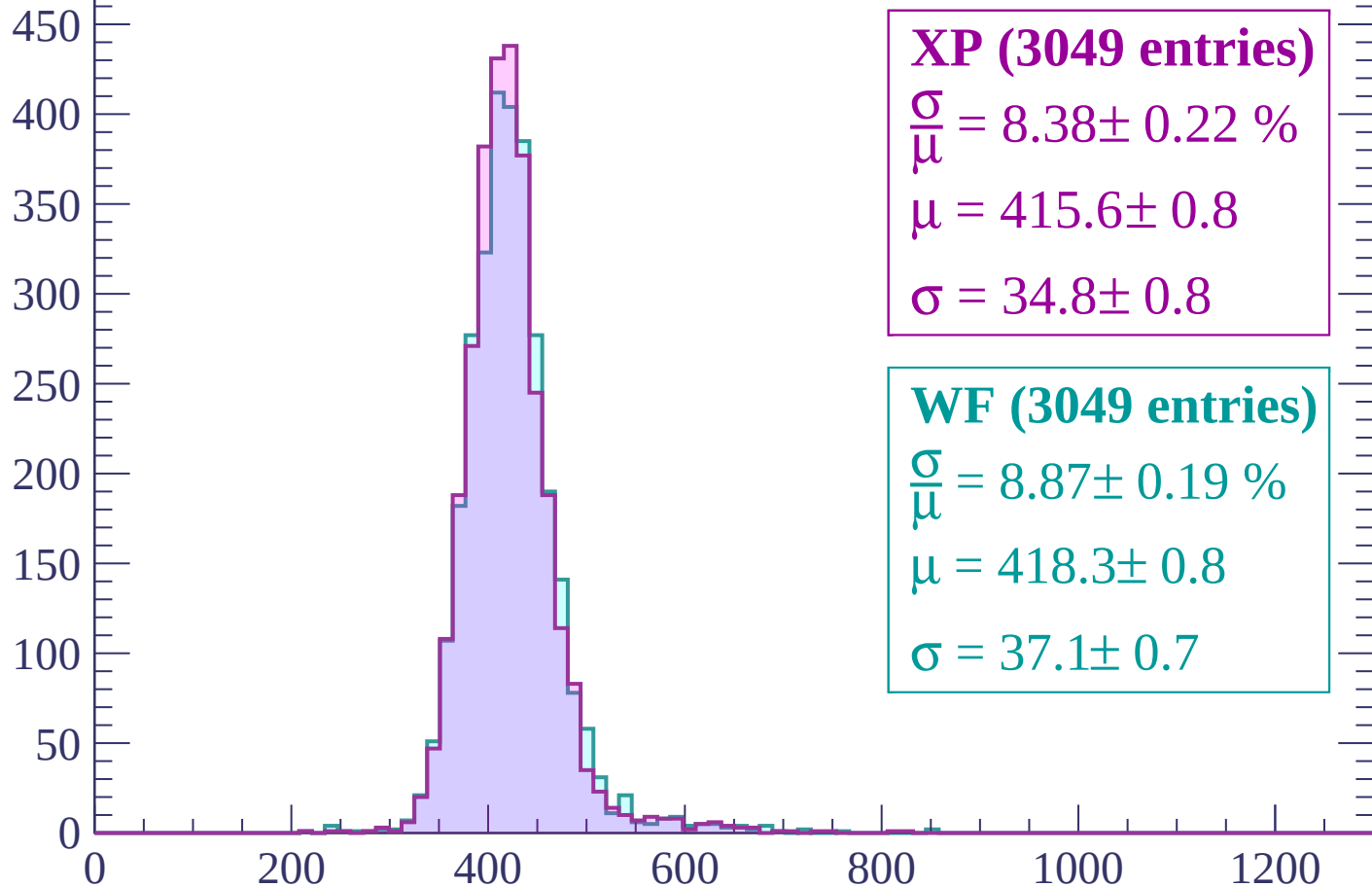
$$\mu = 402.7 \pm 0.7$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (3049 entries)

$$\frac{\sigma}{\mu} = 8.38 \pm 0.22 \%$$

$$\mu = 415.6 \pm 0.8$$

$$\sigma = 34.8 \pm 0.8$$

WF (3049 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.19 \%$$

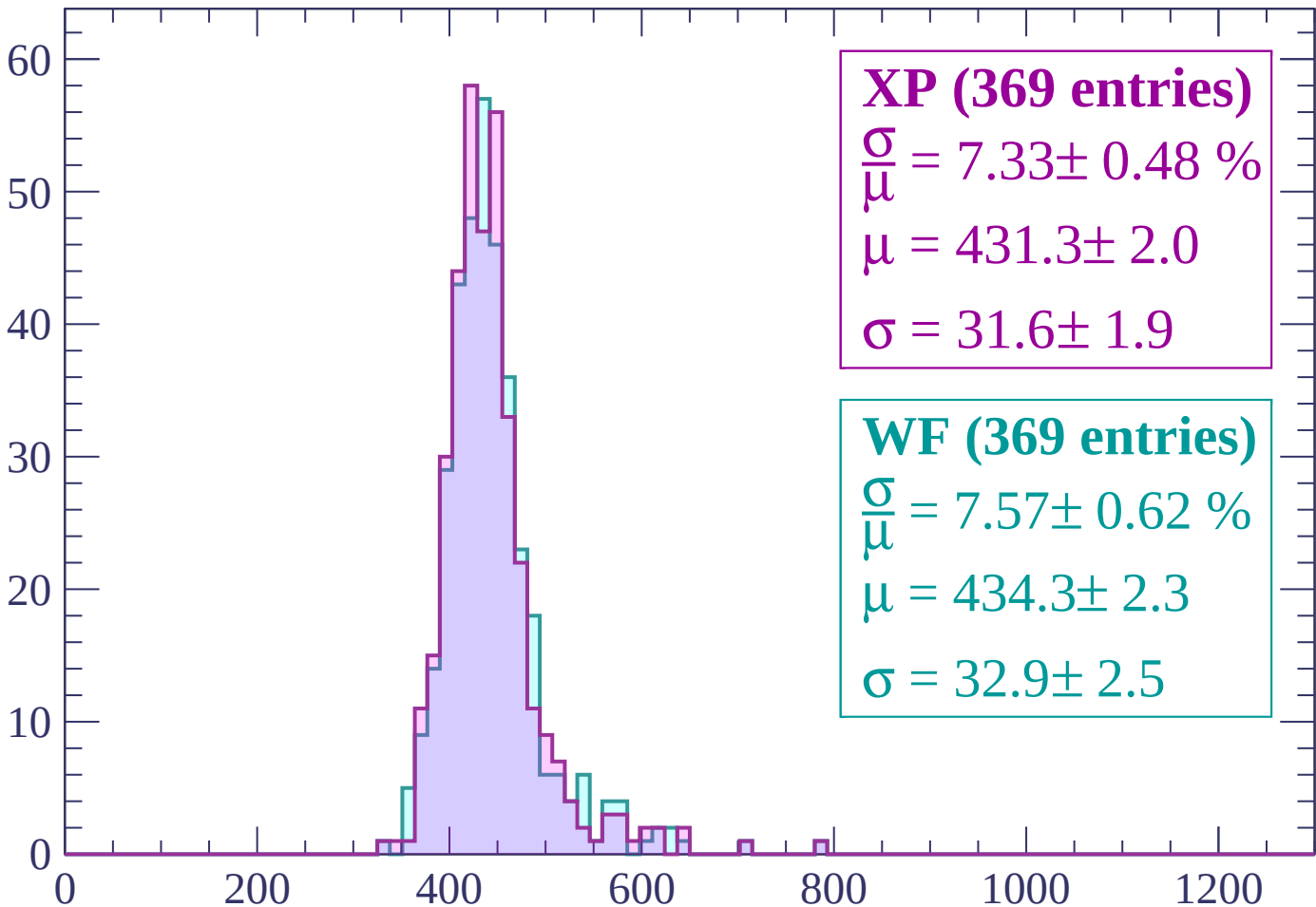
$$\mu = 418.3 \pm 0.8$$

$$\sigma = 37.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-180 < p < -120$

Count



XP (369 entries)

$\frac{\sigma}{\mu} = 7.33 \pm 0.48 \%$

$\mu = 431.3 \pm 2.0$

$\sigma = 31.6 \pm 1.9$

WF (369 entries)

$\frac{\sigma}{\mu} = 7.57 \pm 0.62 \%$

$\mu = 434.3 \pm 2.3$

$\sigma = 32.9 \pm 2.5$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

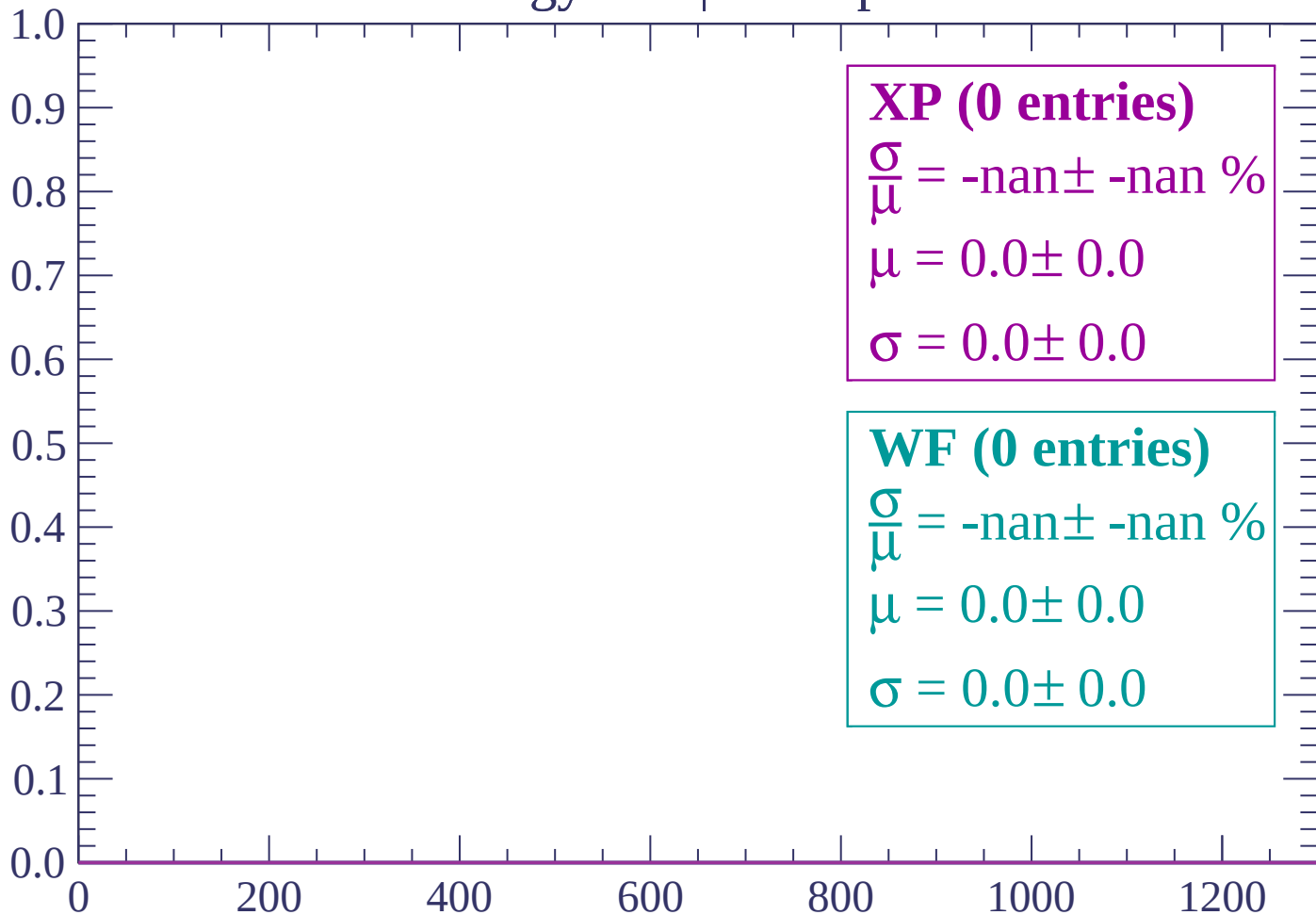
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-60 < p < 0$

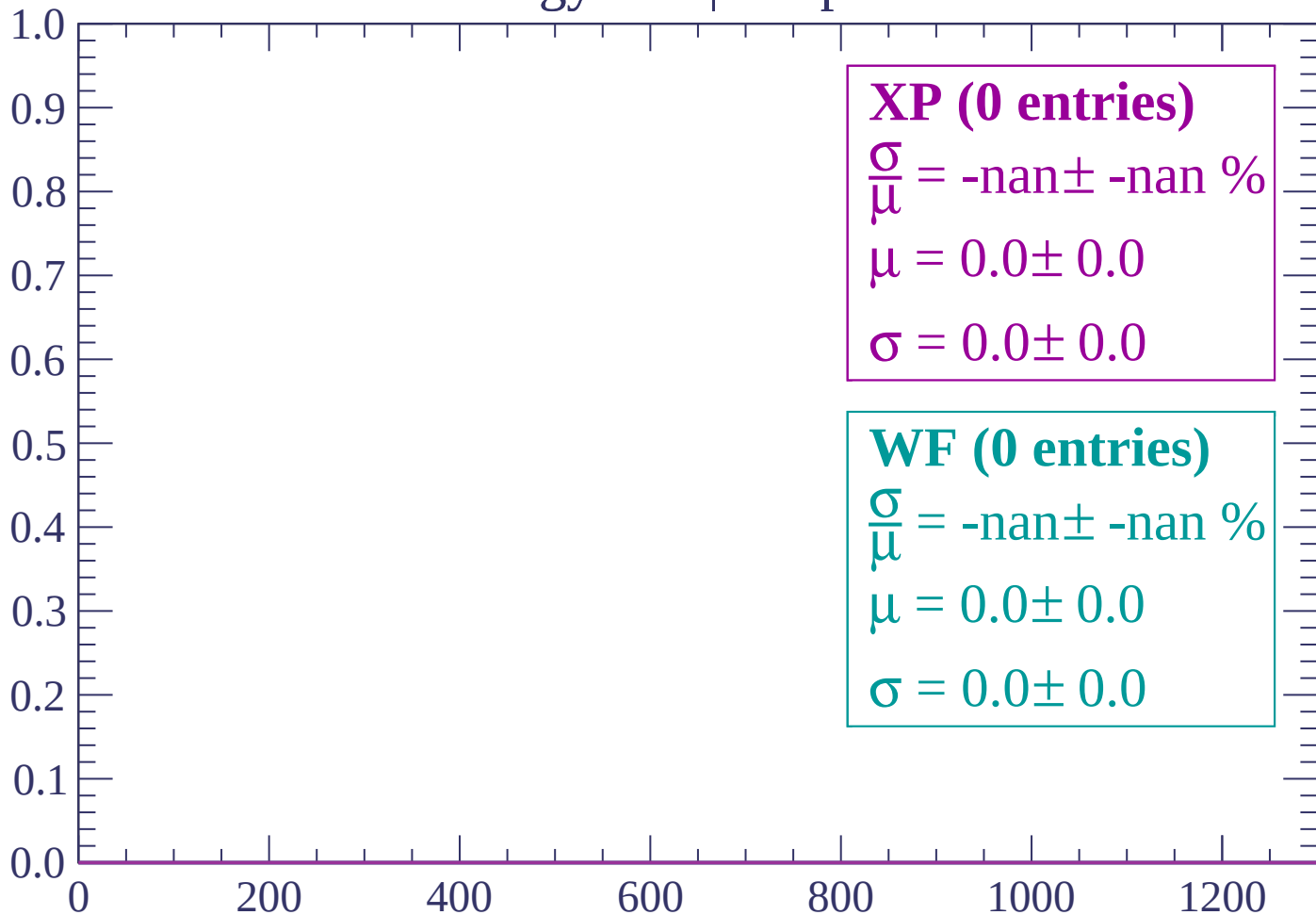
Count



dE/dx (ADC counts/cm)

Energy loss | $0 < p < 60$

Count



dE/dx (ADC counts/cm)

Energy loss | $60 < p < 120$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $120 < p < 180$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

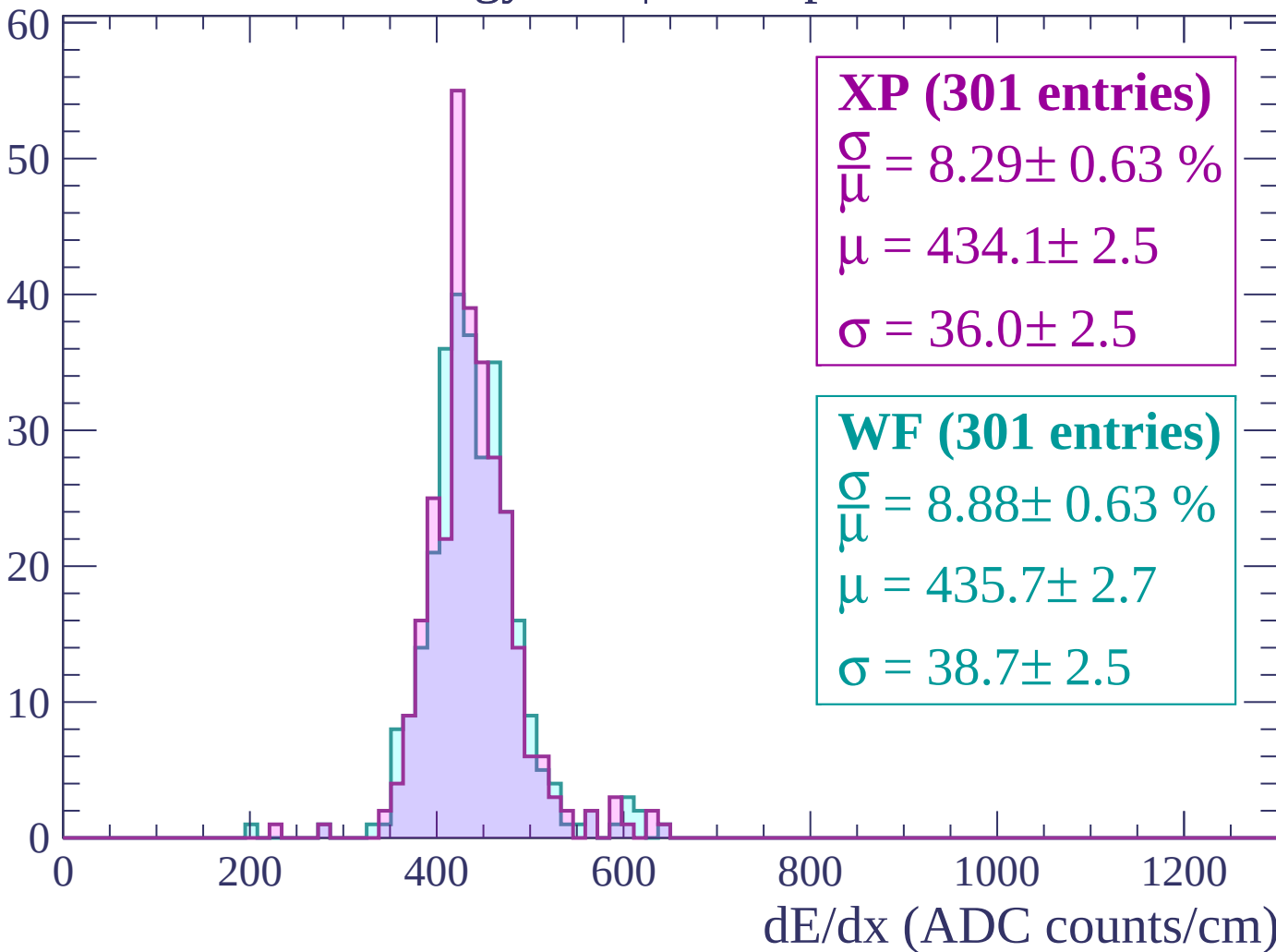
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

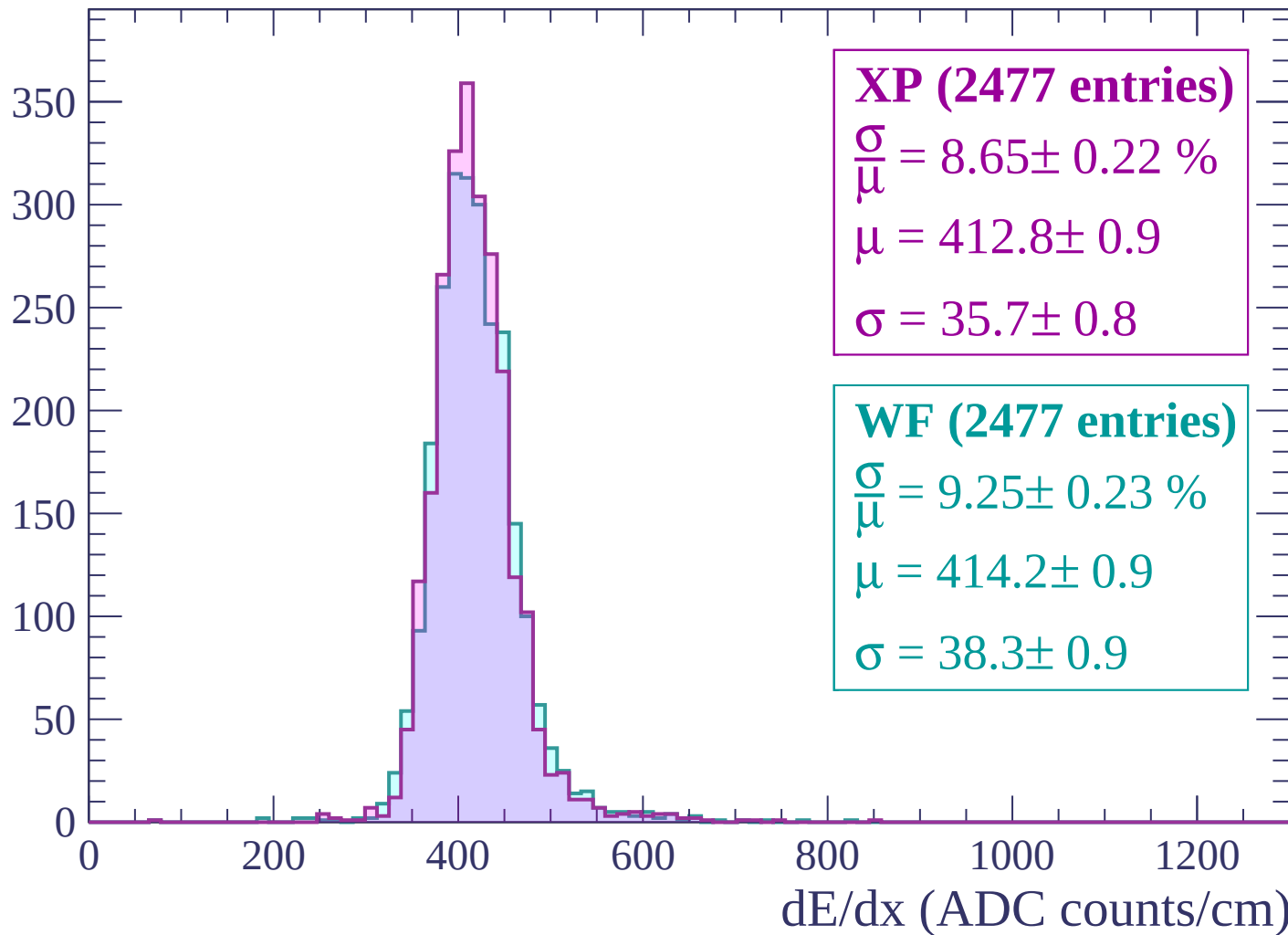
Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count



Energy loss | $300 < p < 360$

Count

600
500
400
300
200
100
0

XP (4221 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.15 \%$$

$$\mu = 400.8 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

WF (4221 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.15 \%$$

$$\mu = 402.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

0

200

400

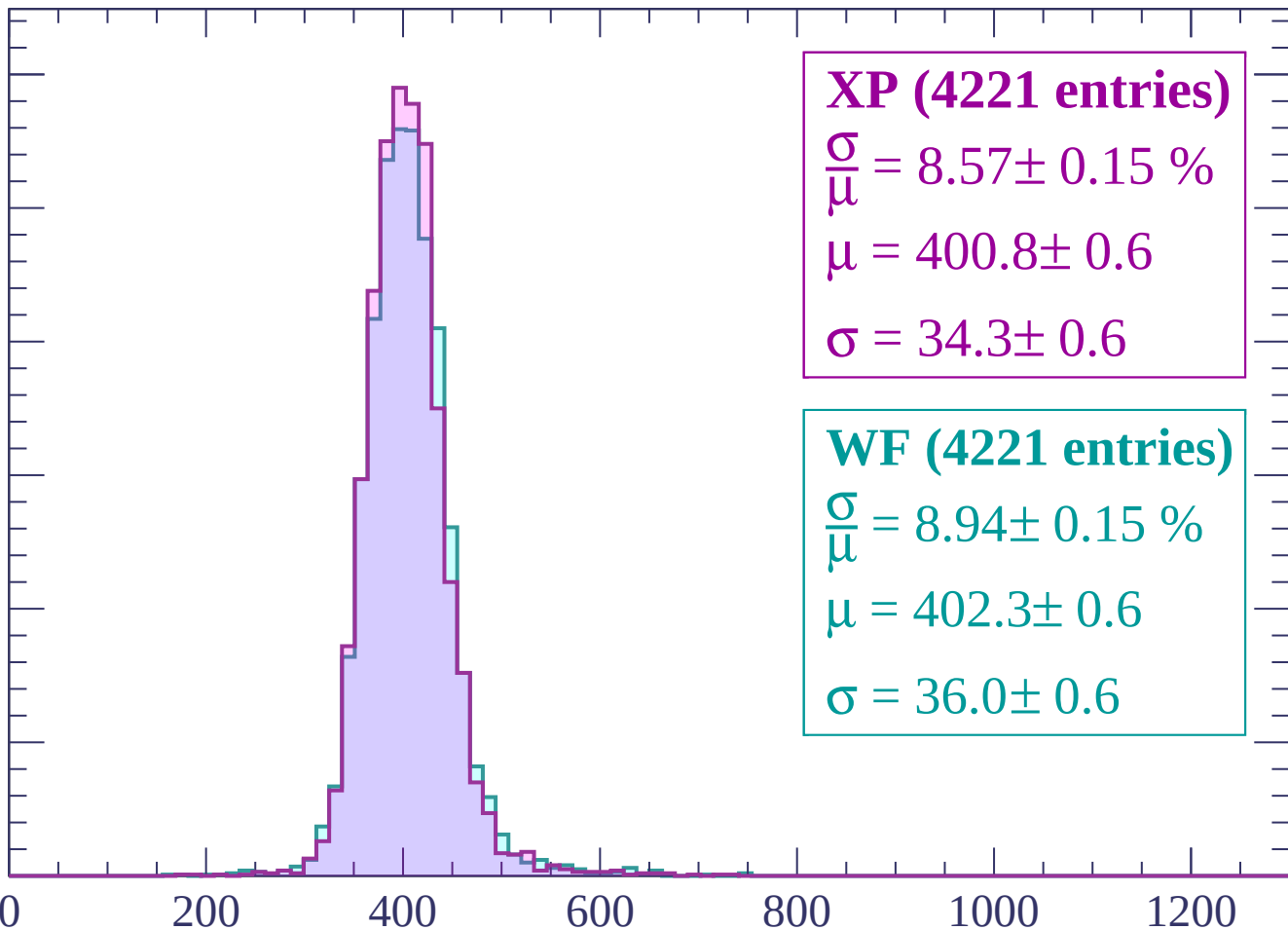
600

800

1000

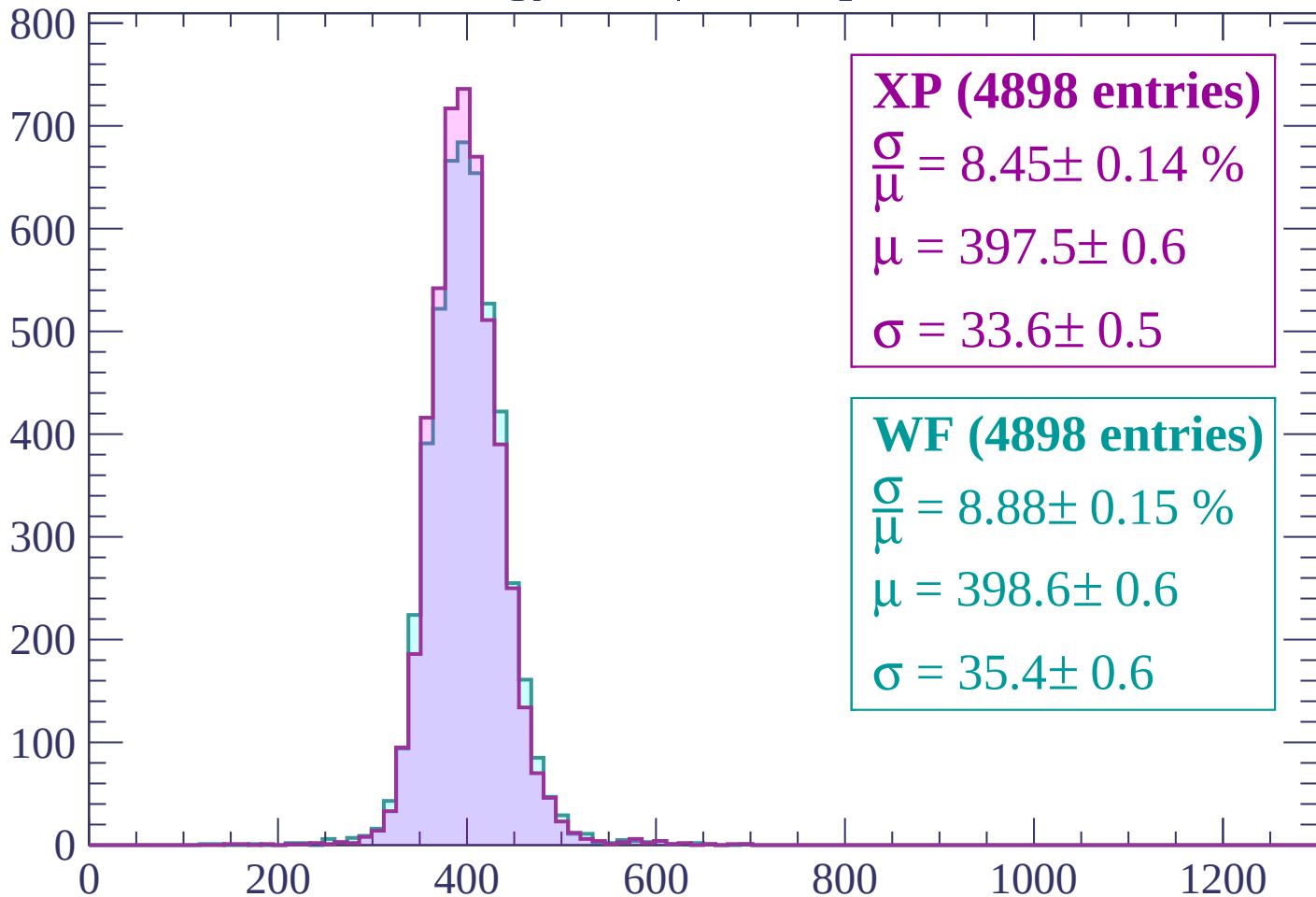
1200

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$

Count



XP (4898 entries)

$$\frac{\sigma}{\mu} = 8.45 \pm 0.14 \%$$

$$\mu = 397.5 \pm 0.6$$

$$\sigma = 33.6 \pm 0.5$$

WF (4898 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.15 \%$$

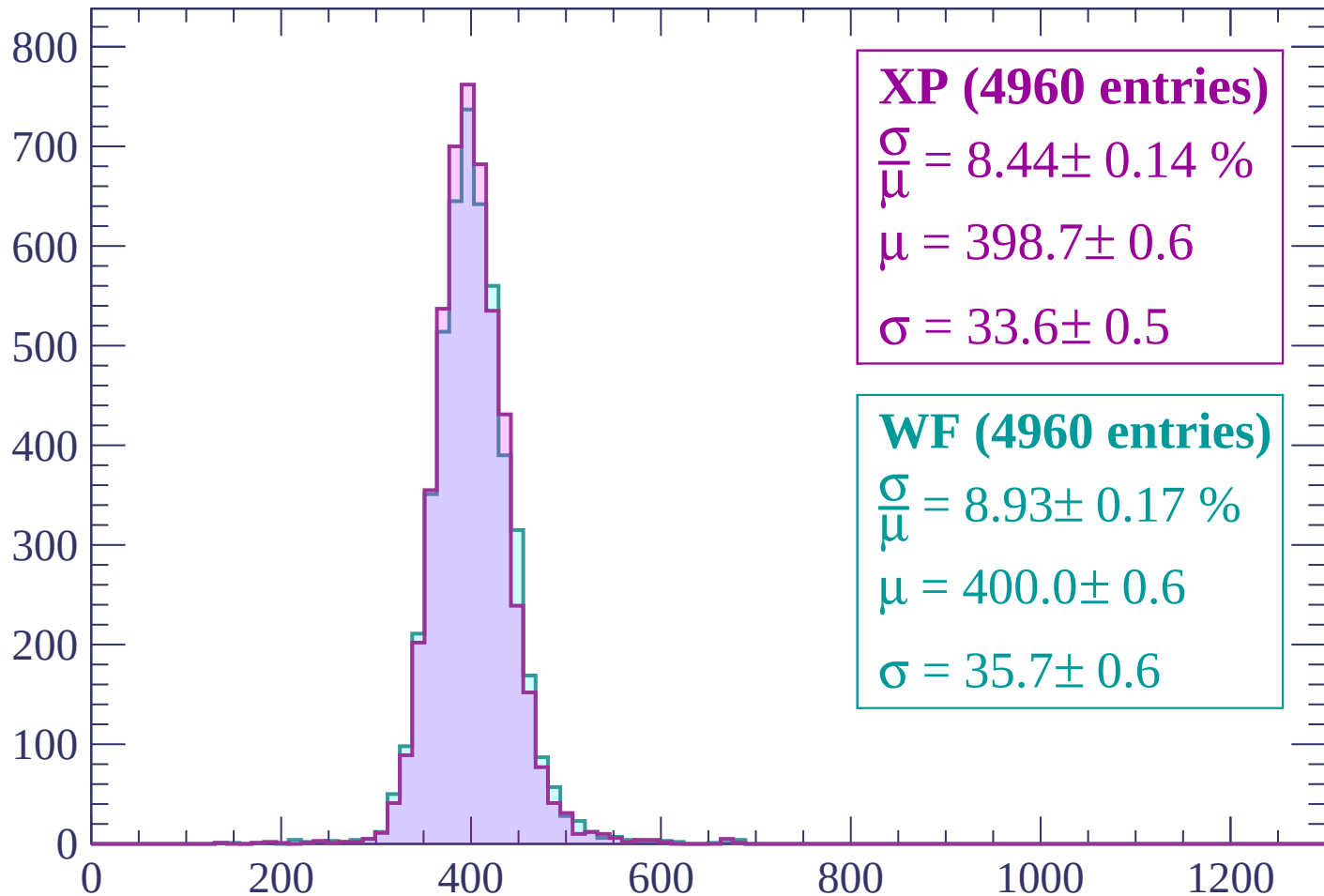
$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (4960 entries)

$$\frac{\sigma}{\mu} = 8.44 \pm 0.14 \%$$

$$\mu = 398.7 \pm 0.6$$

$$\sigma = 33.6 \pm 0.5$$

WF (4960 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.17 \%$$

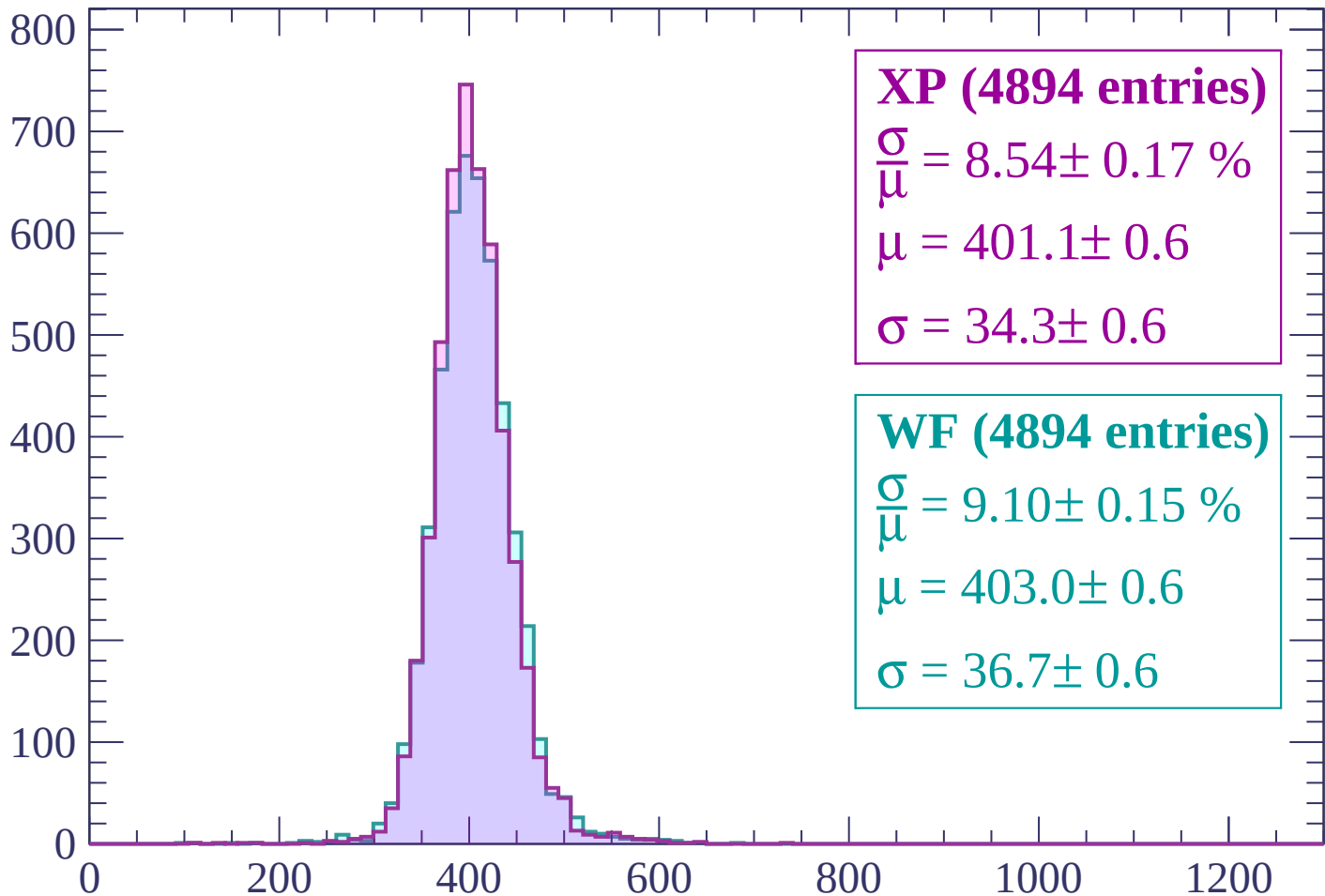
$$\mu = 400.0 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (4894 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.17 \%$$

$$\mu = 401.1 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

WF (4894 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.15 \%$$

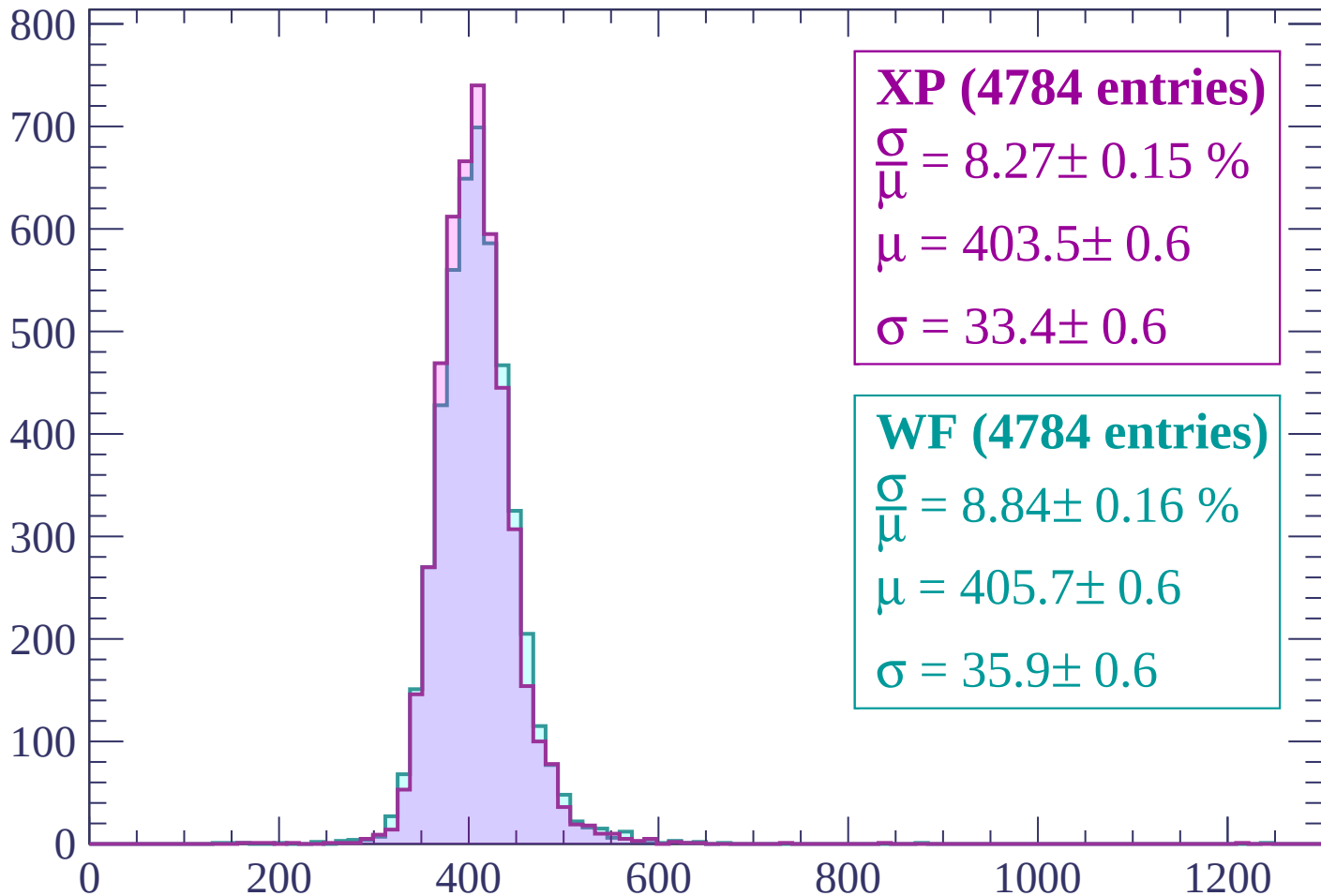
$$\mu = 403.0 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (4784 entries)

$\frac{\sigma}{\mu} = 8.27 \pm 0.15 \%$

$\mu = 403.5 \pm 0.6$

$\sigma = 33.4 \pm 0.6$

WF (4784 entries)

$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$

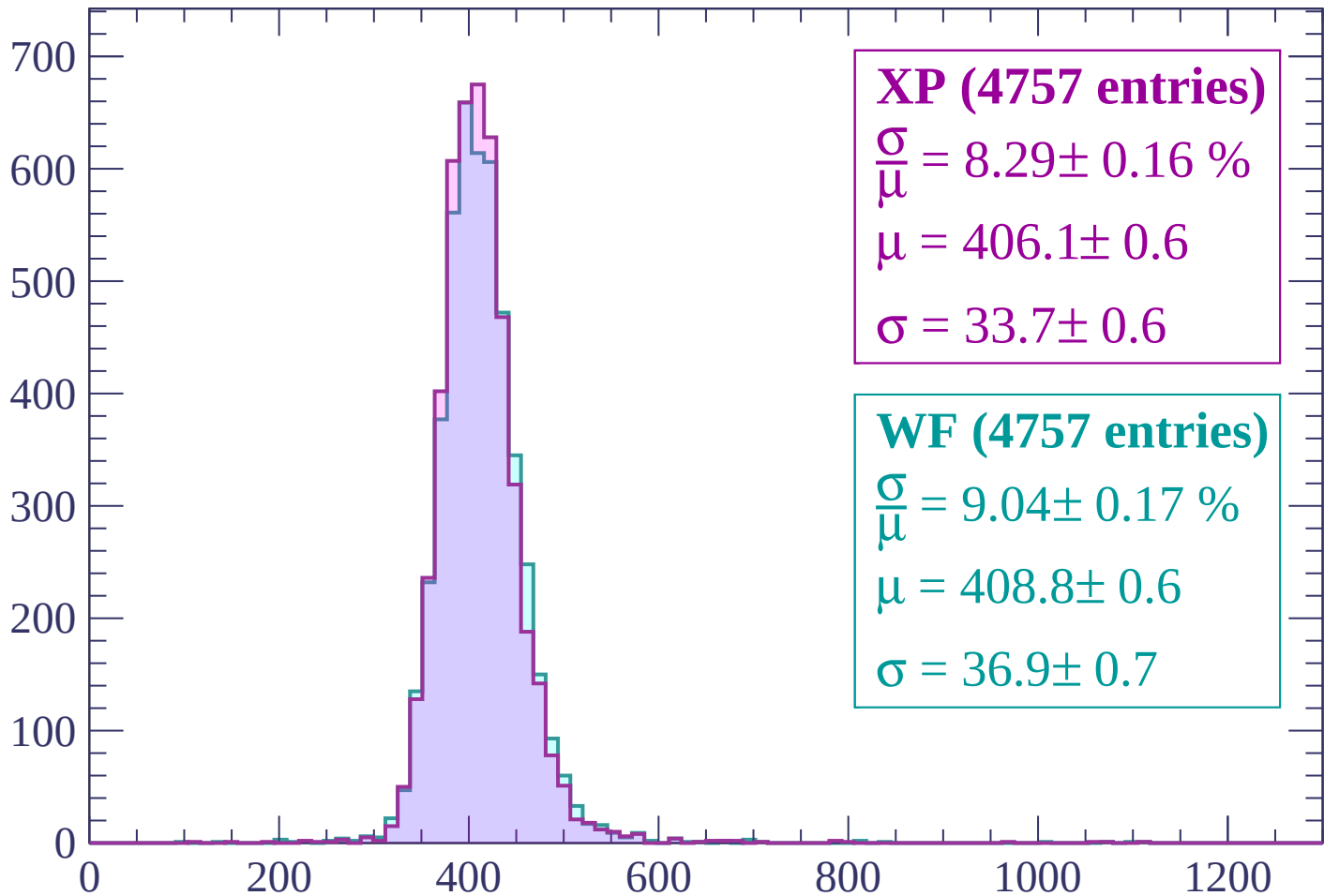
$\mu = 405.7 \pm 0.6$

$\sigma = 35.9 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (4757 entries)

$$\frac{\sigma}{\mu} = 8.29 \pm 0.16 \%$$

$$\mu = 406.1 \pm 0.6$$

$$\sigma = 33.7 \pm 0.6$$

WF (4757 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.17 \%$$

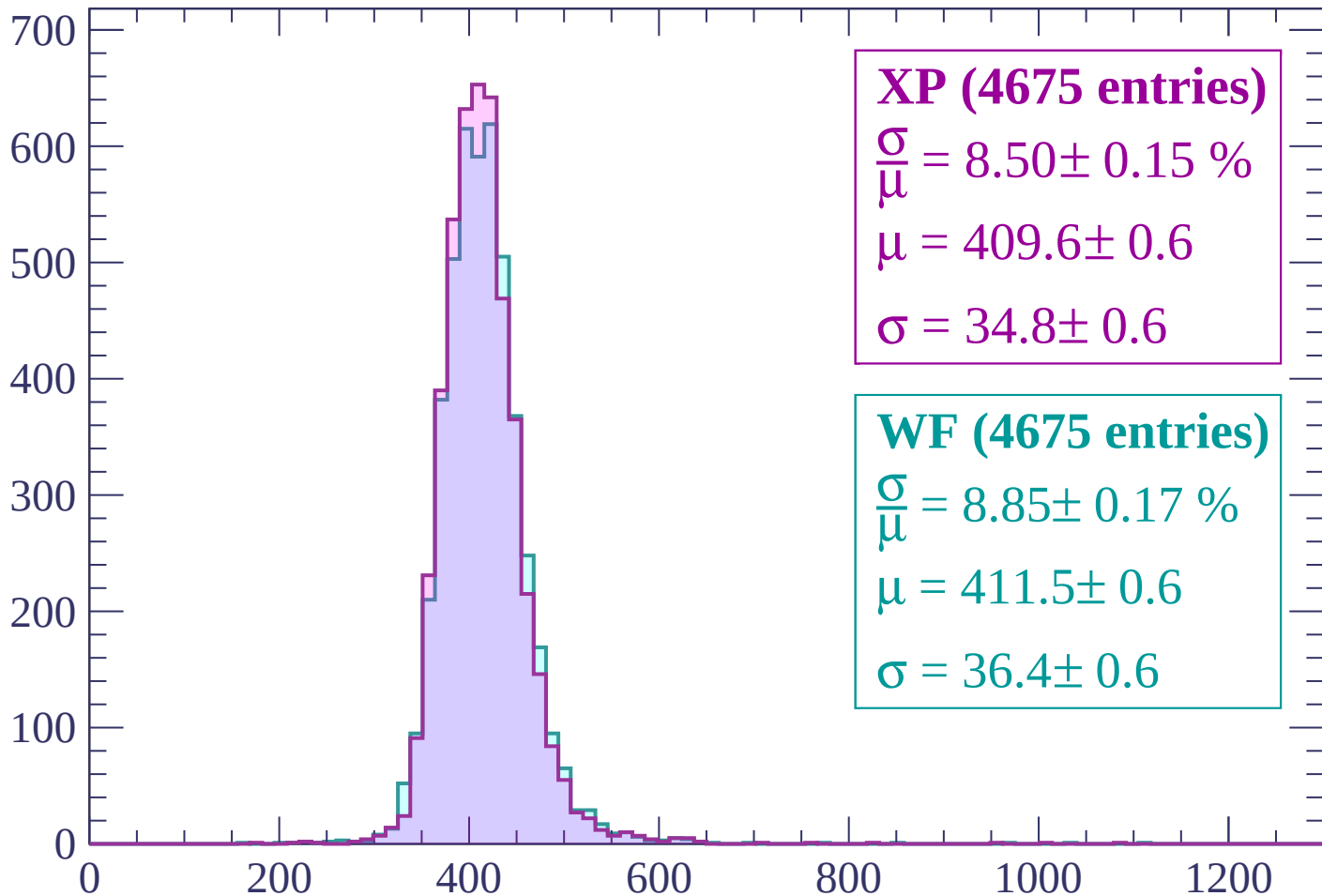
$$\mu = 408.8 \pm 0.6$$

$$\sigma = 36.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (4675 entries)

$$\frac{\sigma}{\mu} = 8.50 \pm 0.15 \%$$

$$\mu = 409.6 \pm 0.6$$

$$\sigma = 34.8 \pm 0.6$$

WF (4675 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

$$\mu = 411.5 \pm 0.6$$

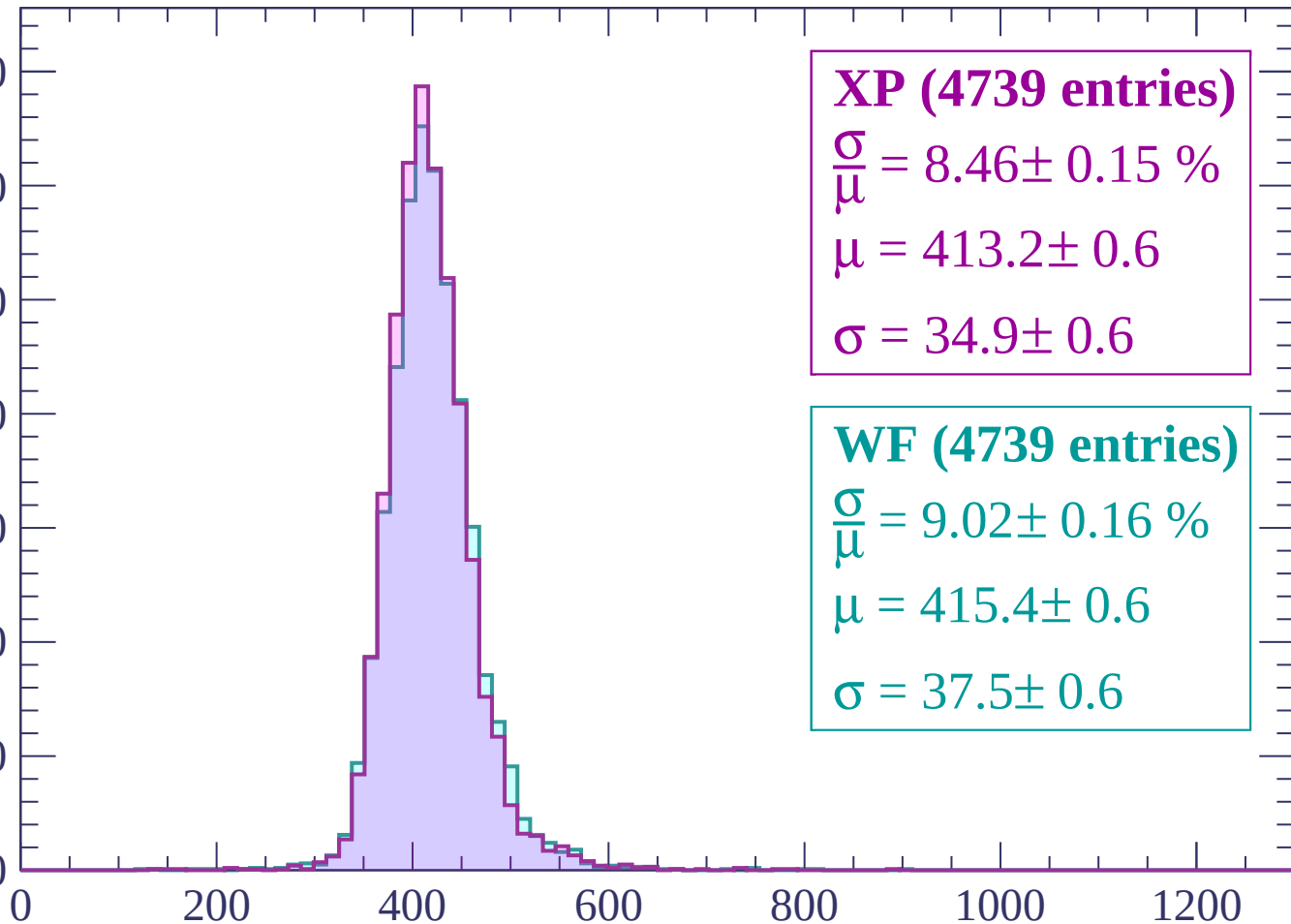
$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count

700
600
500
400
300
200
100
0



XP (4739 entries)

$\frac{\sigma}{\mu} = 8.46 \pm 0.15 \%$

$\mu = 413.2 \pm 0.6$

$\sigma = 34.9 \pm 0.6$

WF (4739 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$

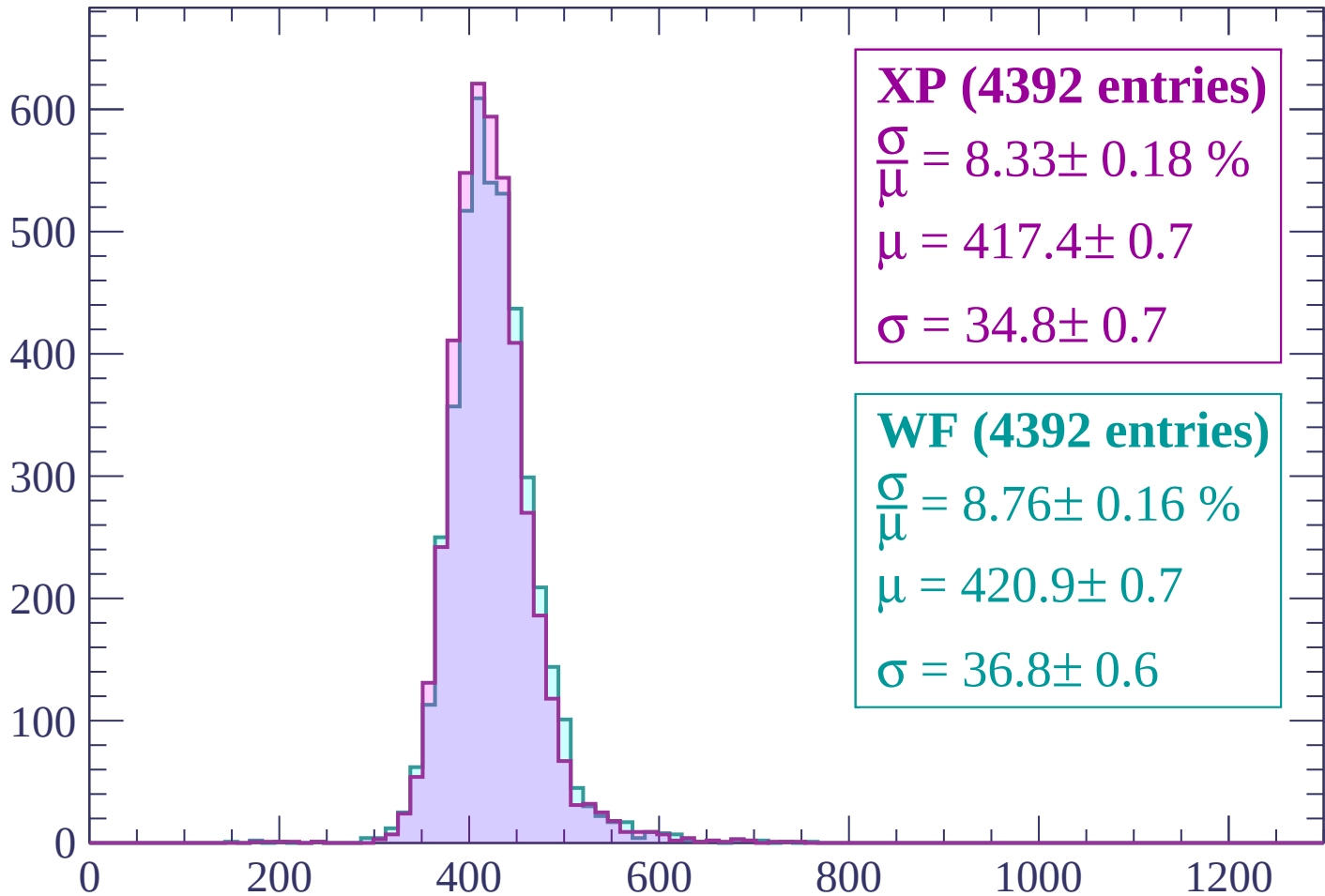
$\mu = 415.4 \pm 0.6$

$\sigma = 37.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $780 < p < 840$

Count



XP (4392 entries)

$$\frac{\sigma}{\mu} = 8.33 \pm 0.18 \%$$

$$\mu = 417.4 \pm 0.7$$

$$\sigma = 34.8 \pm 0.7$$

WF (4392 entries)

$$\frac{\sigma}{\mu} = 8.76 \pm 0.16 \%$$

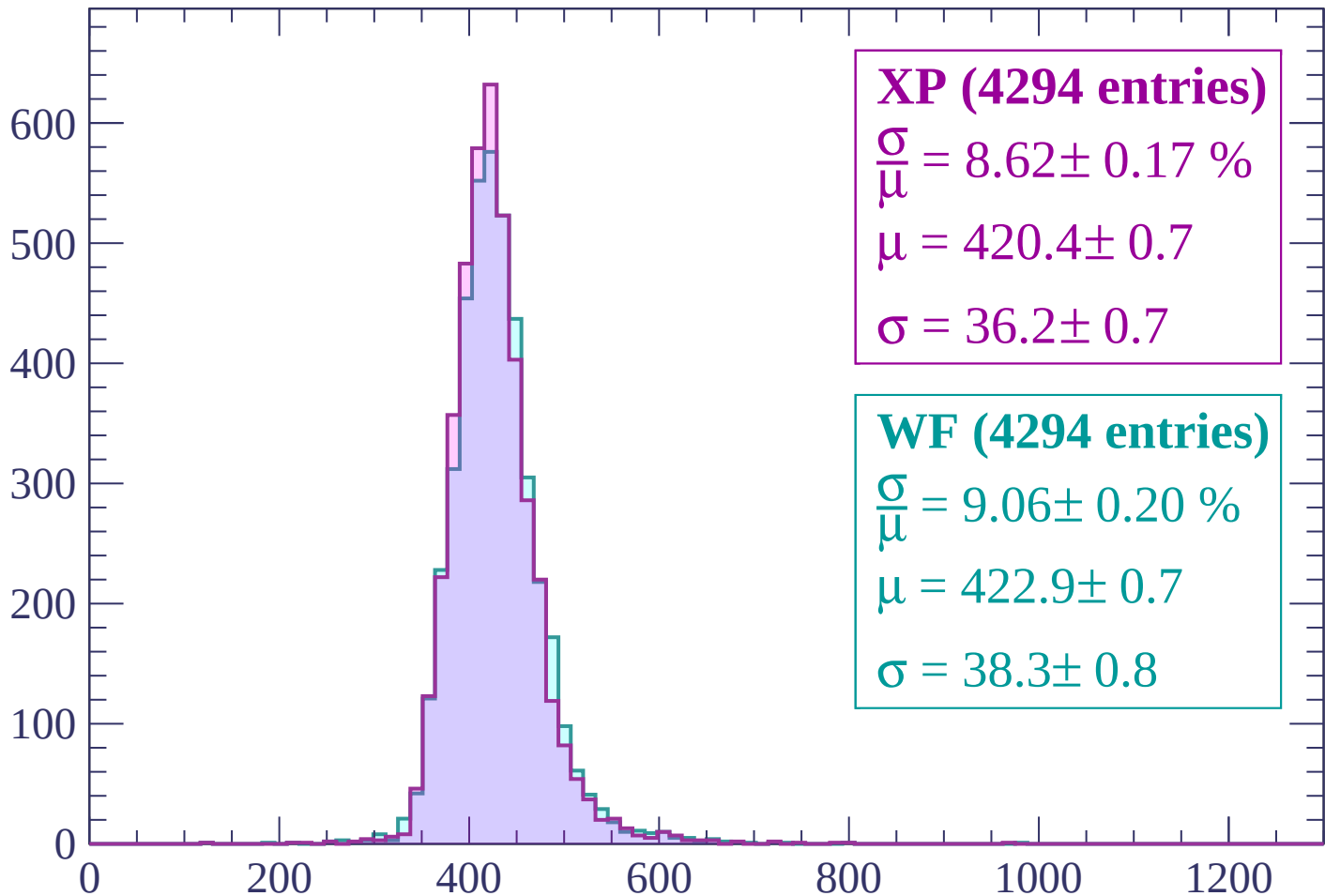
$$\mu = 420.9 \pm 0.7$$

$$\sigma = 36.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $840 < p < 900$

Count



XP (4294 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.17 \%$$

$$\mu = 420.4 \pm 0.7$$

$$\sigma = 36.2 \pm 0.7$$

WF (4294 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.20 \%$$

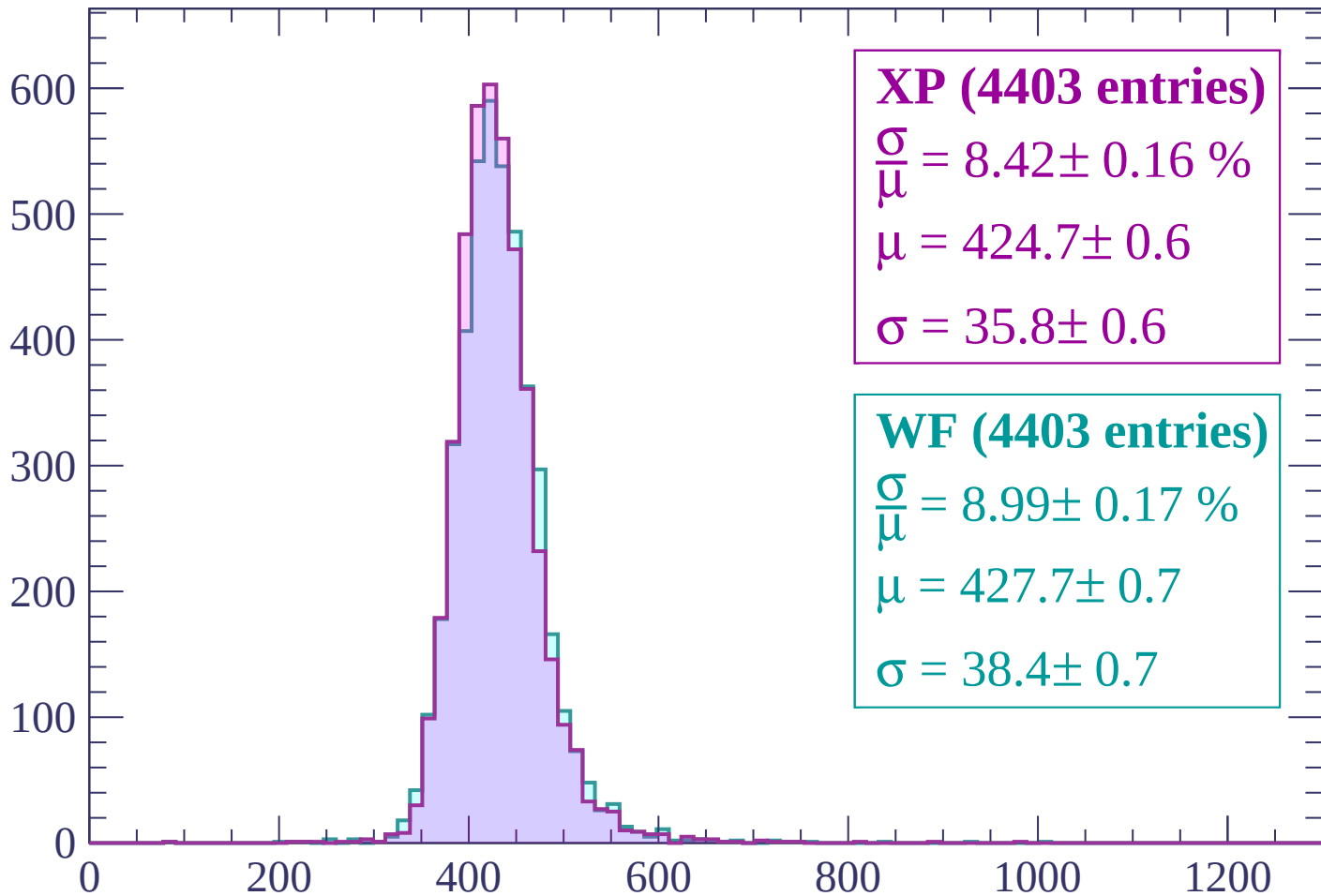
$$\mu = 422.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (4403 entries)

$$\frac{\sigma}{\mu} = 8.42 \pm 0.16 \%$$

$$\mu = 424.7 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

WF (4403 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.17 \%$$

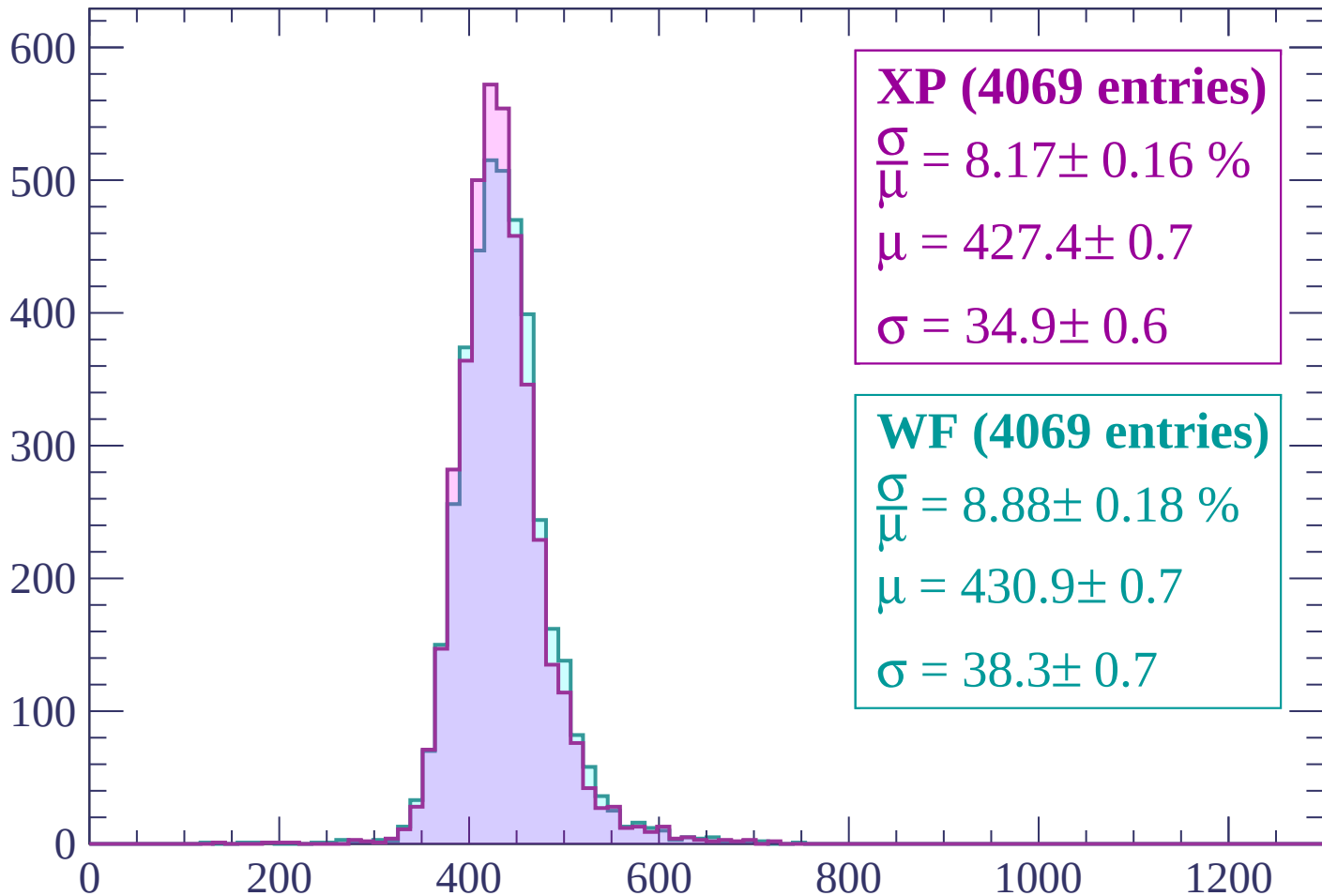
$$\mu = 427.7 \pm 0.7$$

$$\sigma = 38.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (4069 entries)

$$\frac{\sigma}{\mu} = 8.17 \pm 0.16 \%$$

$$\mu = 427.4 \pm 0.7$$

$$\sigma = 34.9 \pm 0.6$$

WF (4069 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.18 \%$$

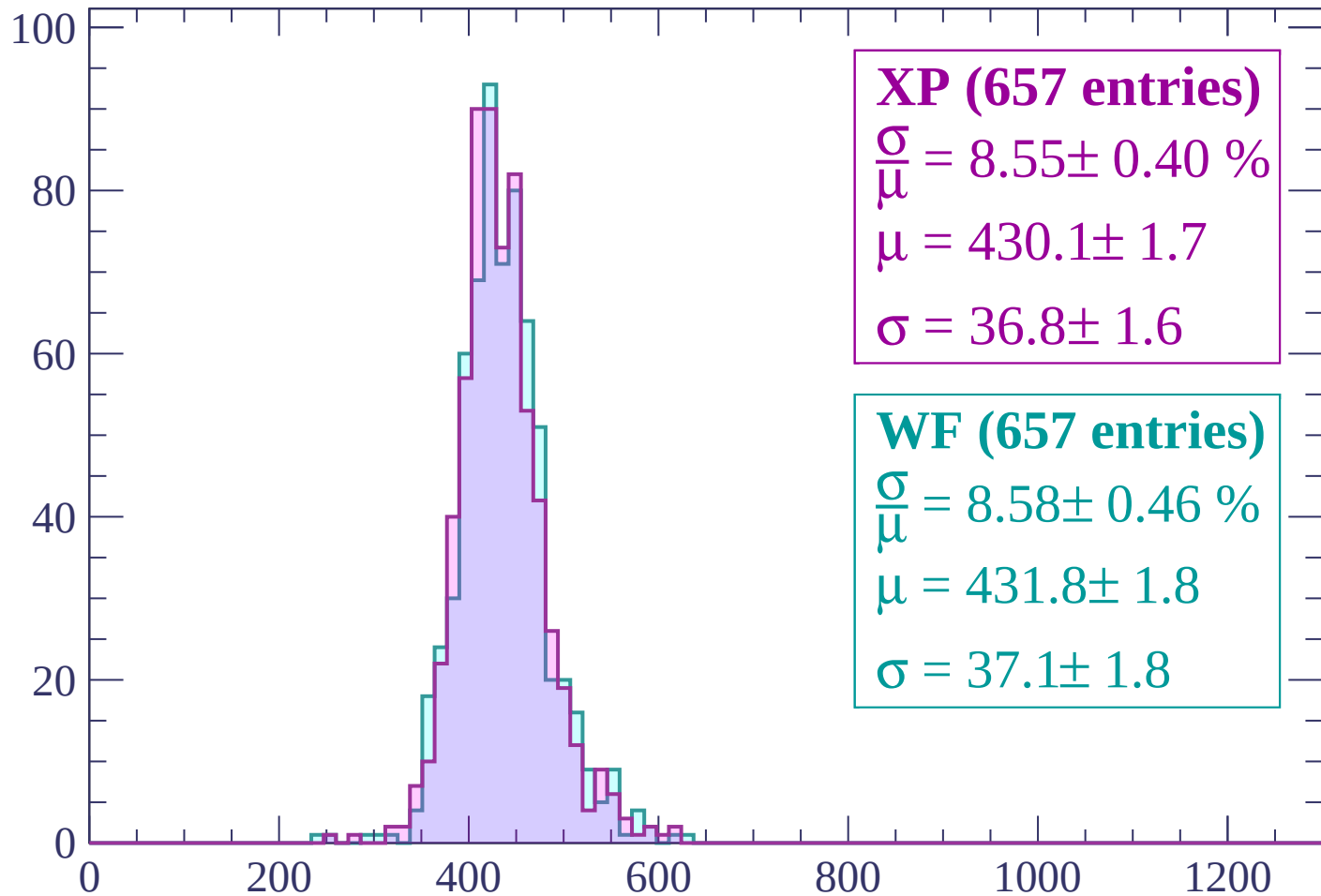
$$\mu = 430.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (657 entries)

$\frac{\sigma}{\mu} = 8.55 \pm 0.40 \%$

$\mu = 430.1 \pm 1.7$

$\sigma = 36.8 \pm 1.6$

WF (657 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.46 \%$

$\mu = 431.8 \pm 1.8$

$\sigma = 37.1 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1140 < p < 1200

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1200 < p < 1260$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1260 < p < 1320$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1320 < p < 1380$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

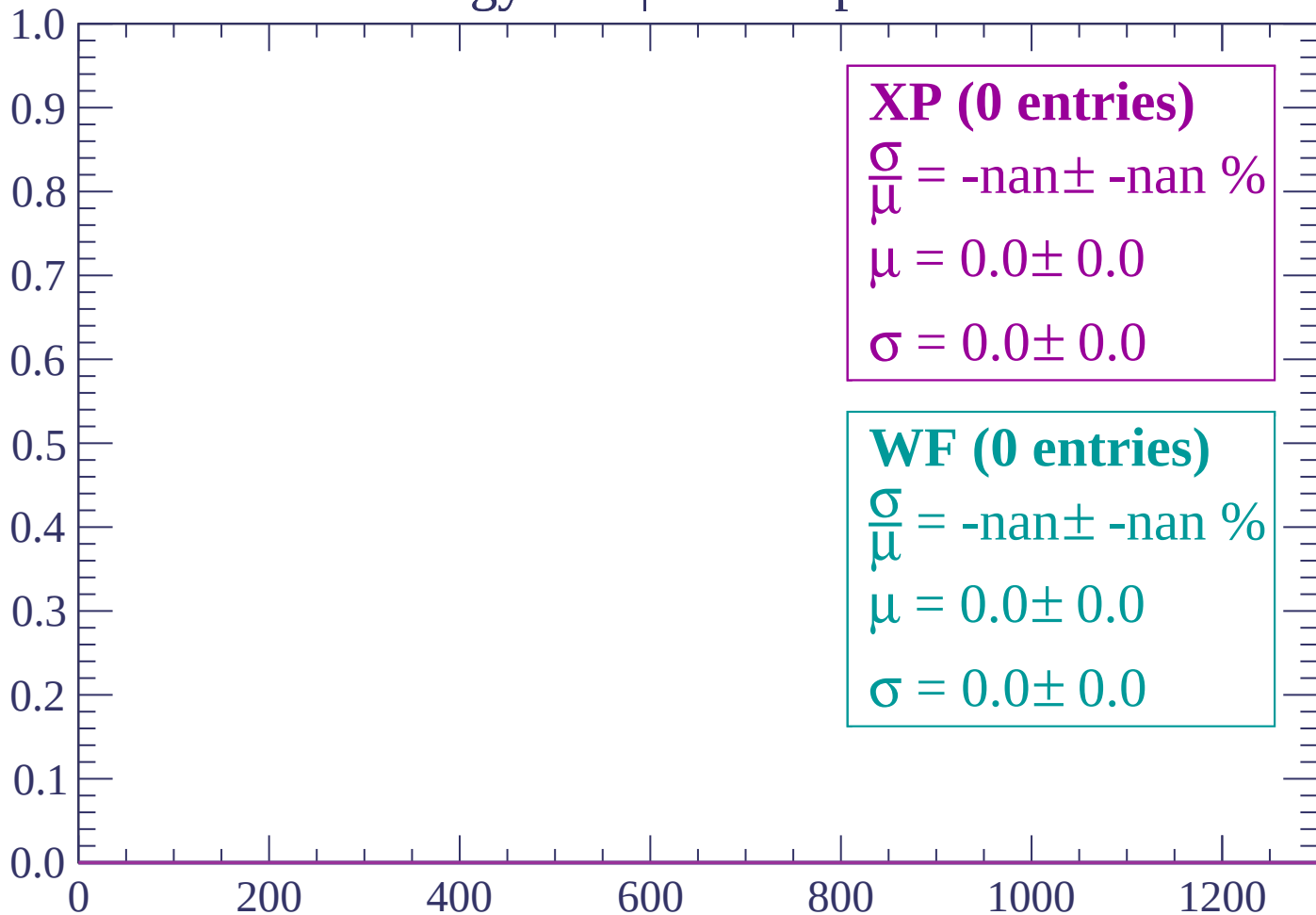
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1380 < p < 1440$

Count



dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1500 < p < 1560$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1560 < p < 1620

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $1620 < p < 1680$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1680 < p < 1740

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1800 < p < 1860

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1860 < p < 1920

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1920 < p < 1980

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 1980 < p < 2040

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2040 < p < 2100$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2100 < p < 2160

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2160 < p < 2220

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2220 < p < 2280

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2280 < p < 2340

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2340 < p < 2400$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $2400 < p < 2460$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | 2460 < p < 2520

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | 2520 < p < 2580

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | 2580 < p < 2640

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | 2640 < p < 2700

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2700 < p < 2760

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2760 < p < 2820

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | 2820 < p < 2880

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2880 < p < 2940

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | 2940 < p < 3000

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $3000 < p < 3060$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-78 < \theta < -76$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-74 < \theta < -72$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

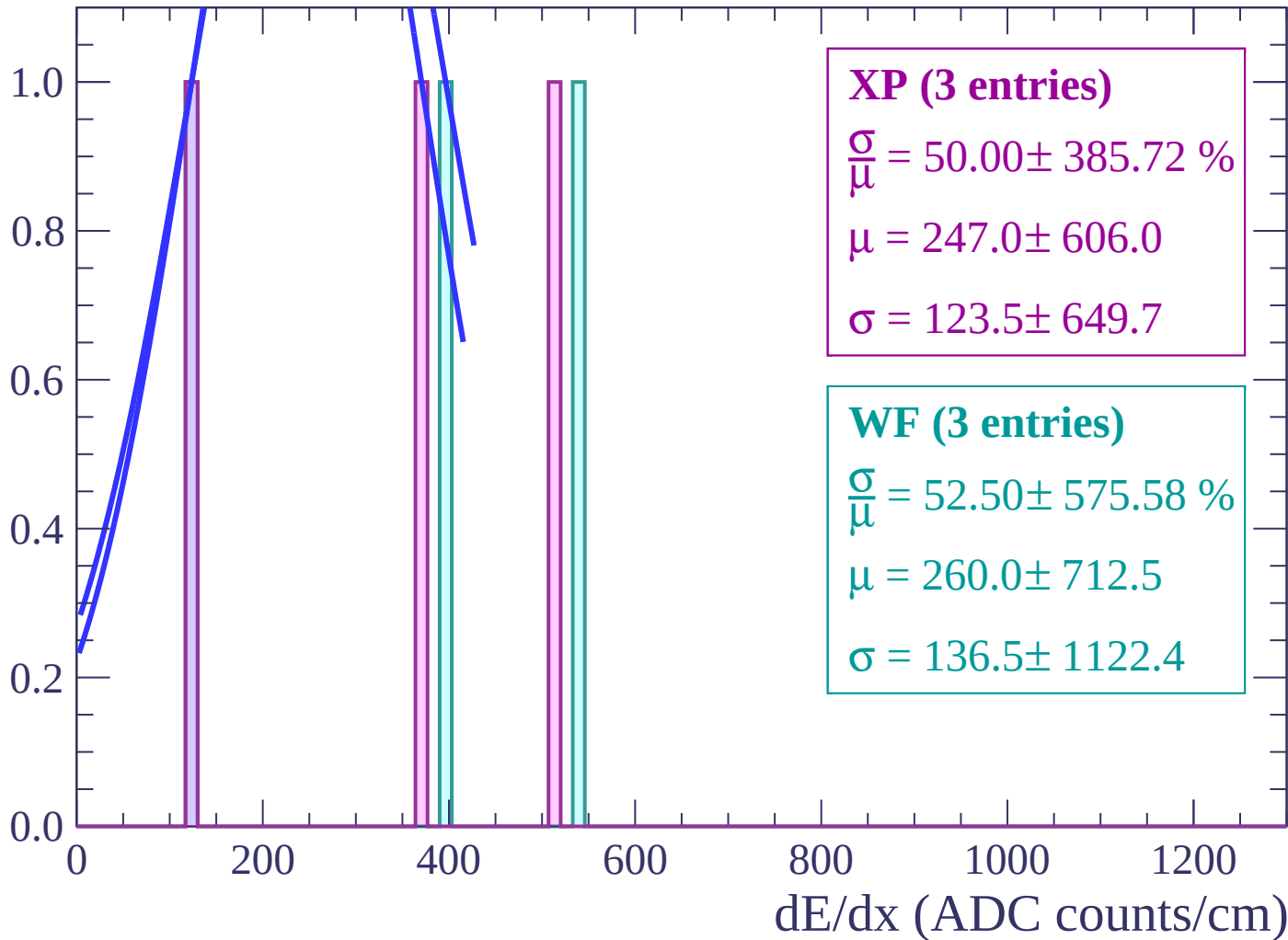
$\frac{\sigma}{\bar{\mu}} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

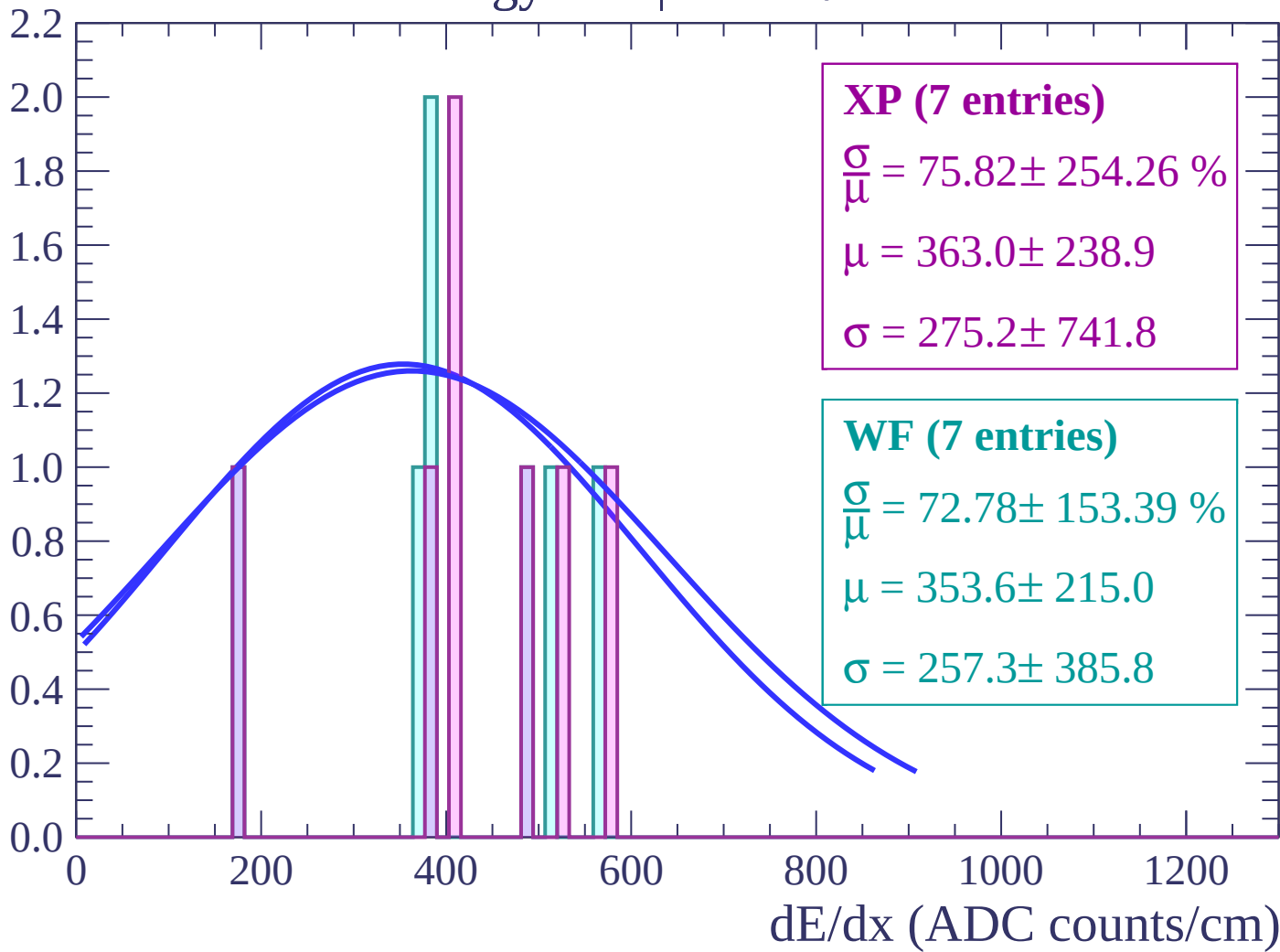
Energy loss | $-72 < \theta < -70$

Count



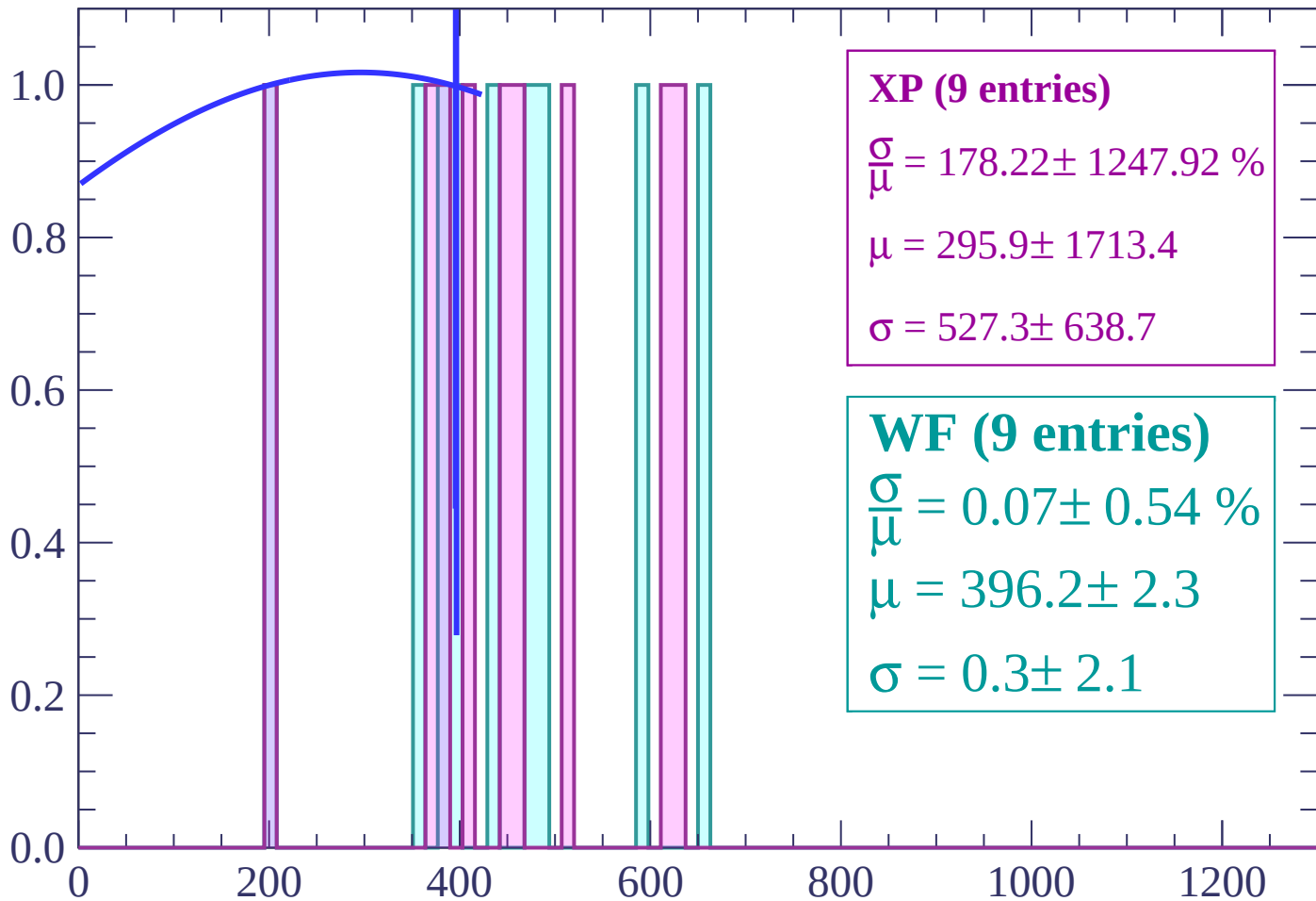
Energy loss | $-70 < \theta < -68$

Count



Energy loss | $-68 < \theta < -66$

Count



XP (9 entries)

$$\frac{\sigma}{\mu} = 178.22 \pm 1247.92 \%$$

$$\mu = 295.9 \pm 1713.4$$

$$\sigma = 527.3 \pm 638.7$$

WF (9 entries)

$$\frac{\sigma}{\mu} = 0.07 \pm 0.54 \%$$

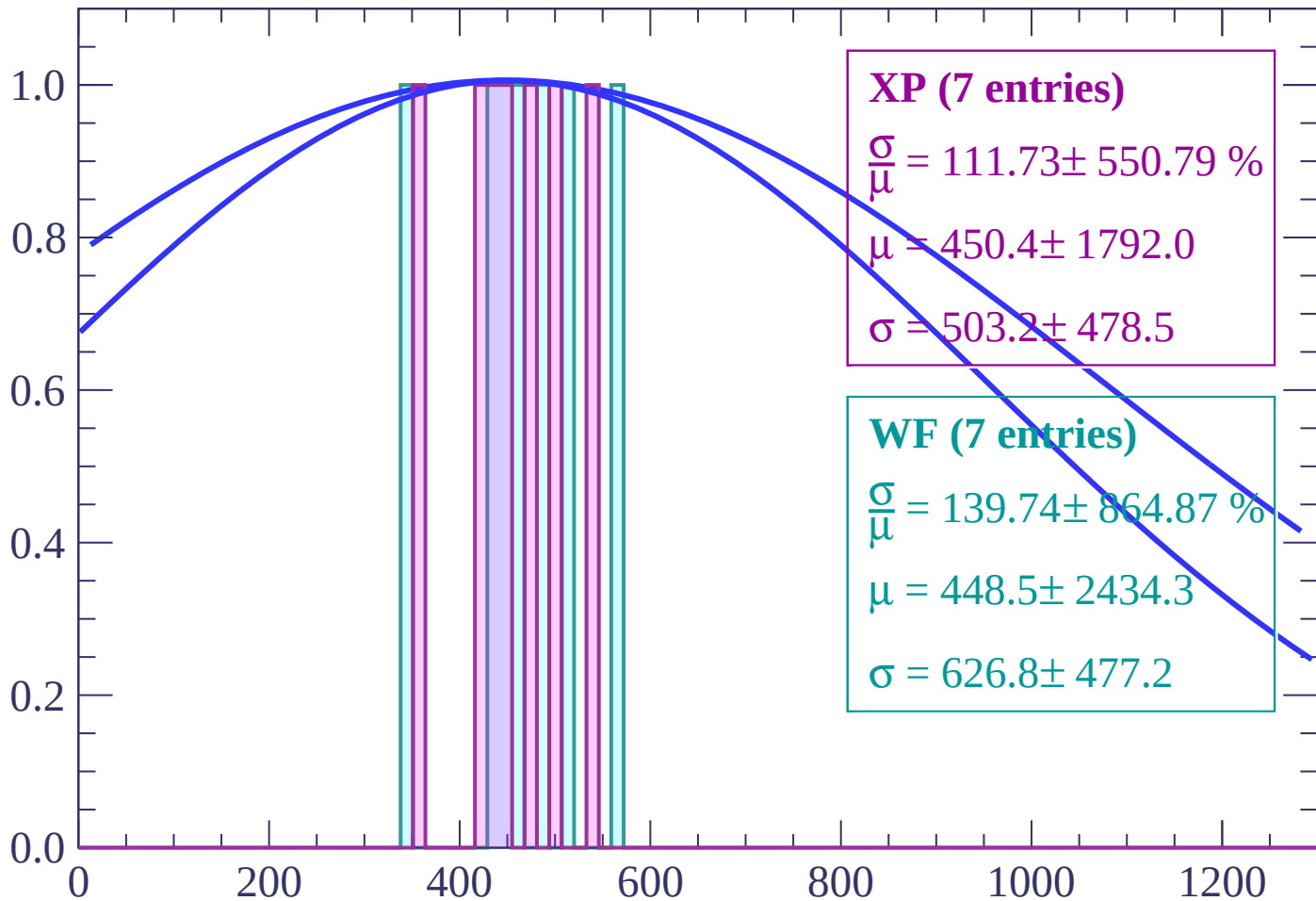
$$\mu = 396.2 \pm 2.3$$

$$\sigma = 0.3 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -64$

Count



XP (7 entries)

$$\frac{\sigma}{\mu} = 111.73 \pm 550.79 \%$$

$$\mu = 450.4 \pm 1792.0$$

$$\sigma = 503.2 \pm 478.5$$

WF (7 entries)

$$\frac{\sigma}{\mu} = 139.74 \pm 864.87 \%$$

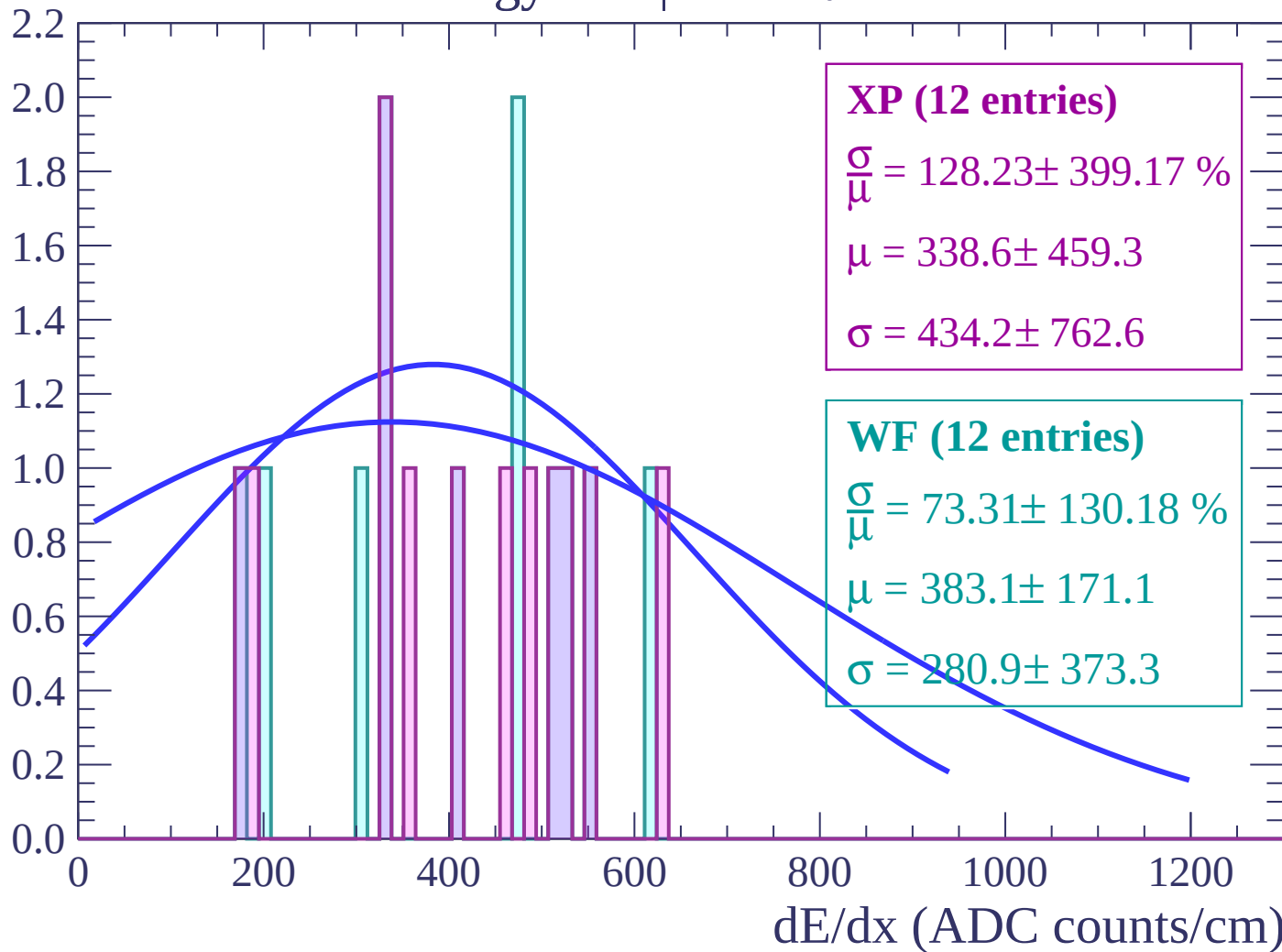
$$\mu = 448.5 \pm 2434.3$$

$$\sigma = 626.8 \pm 477.2$$

dE/dx (ADC counts/cm)

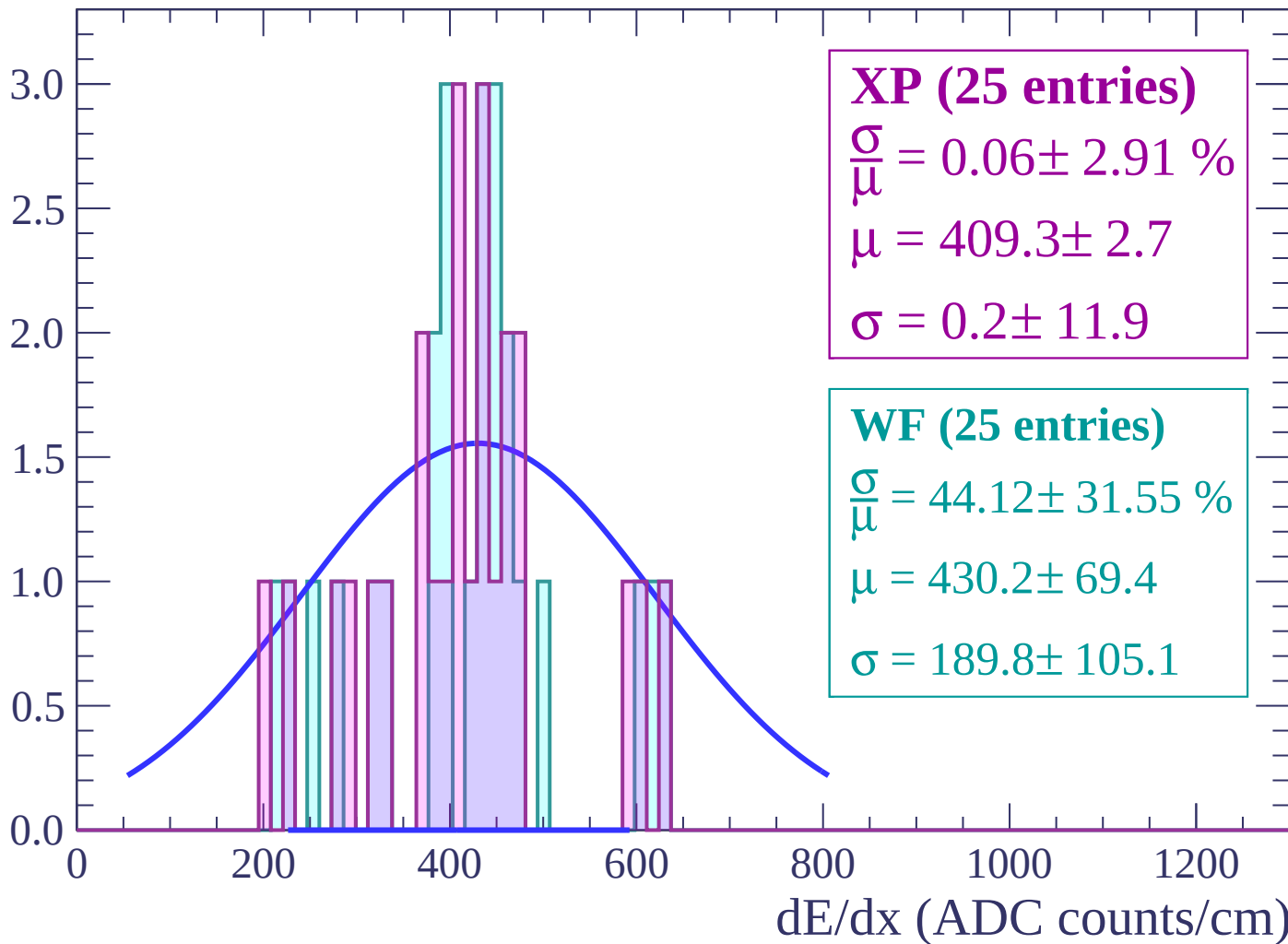
Energy loss | $-64 < \theta < -62$

Count



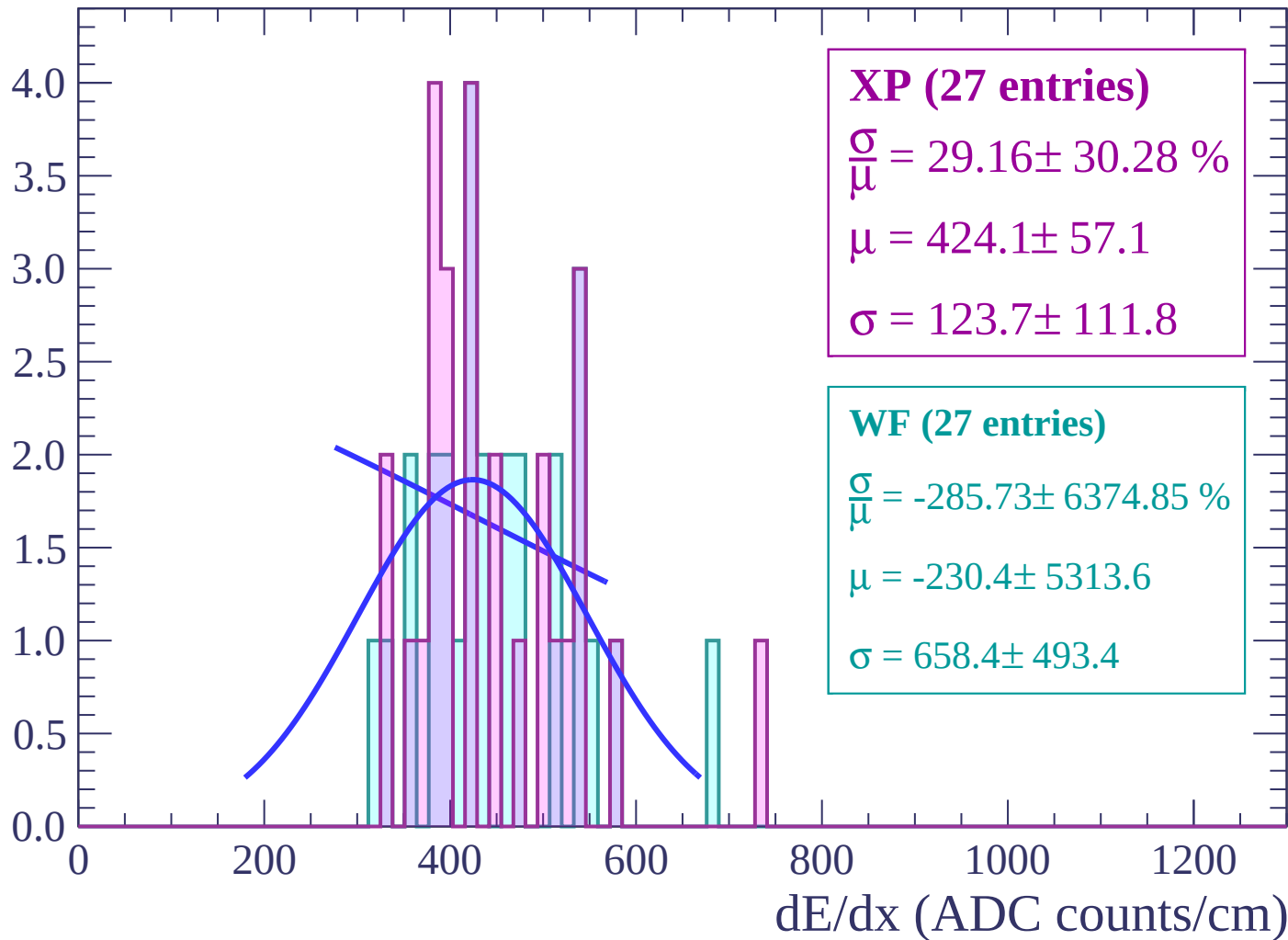
Energy loss | $-62 < \theta < -60$

Count



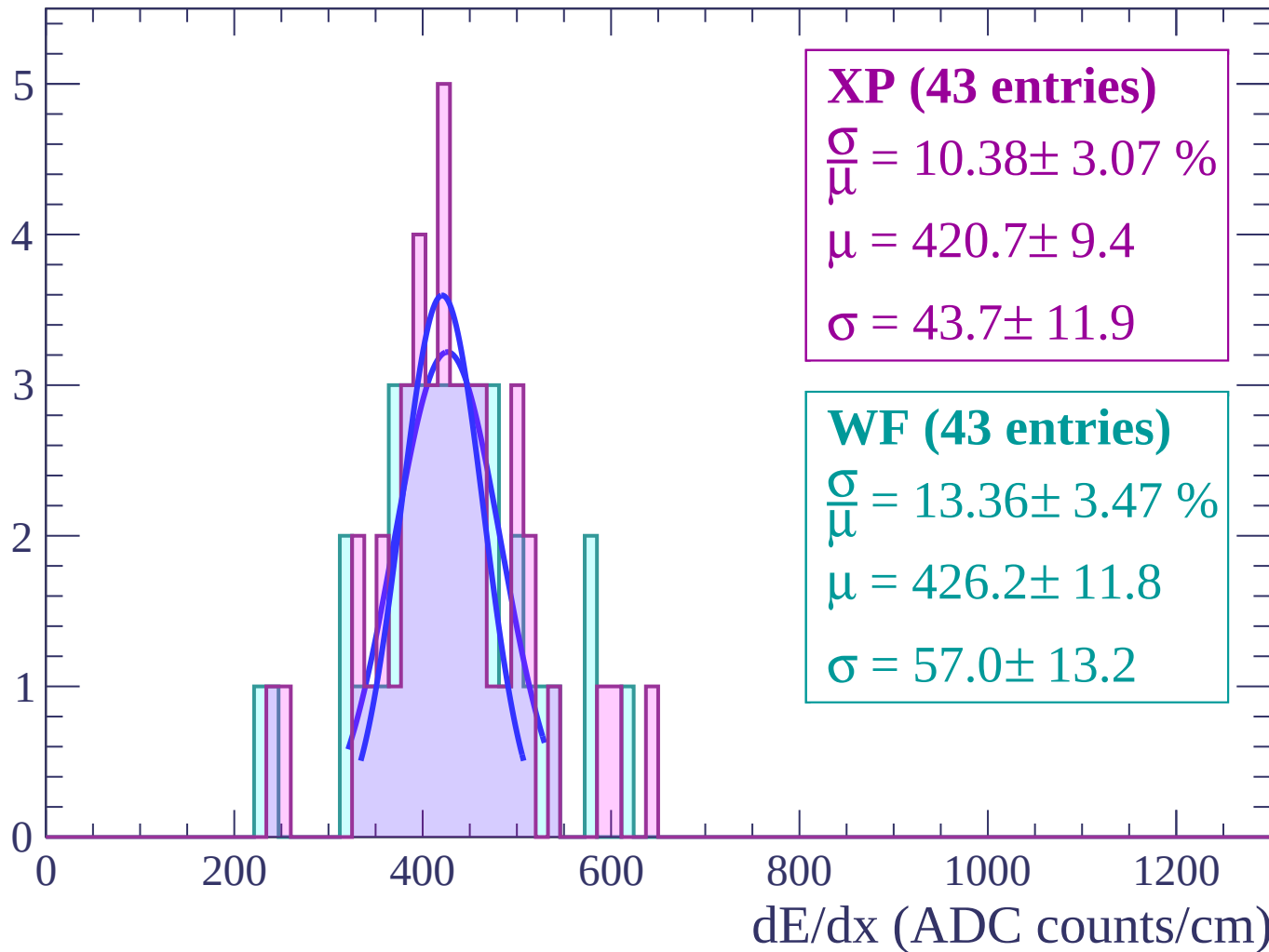
Energy loss | $-60 < \theta < -58$

Count



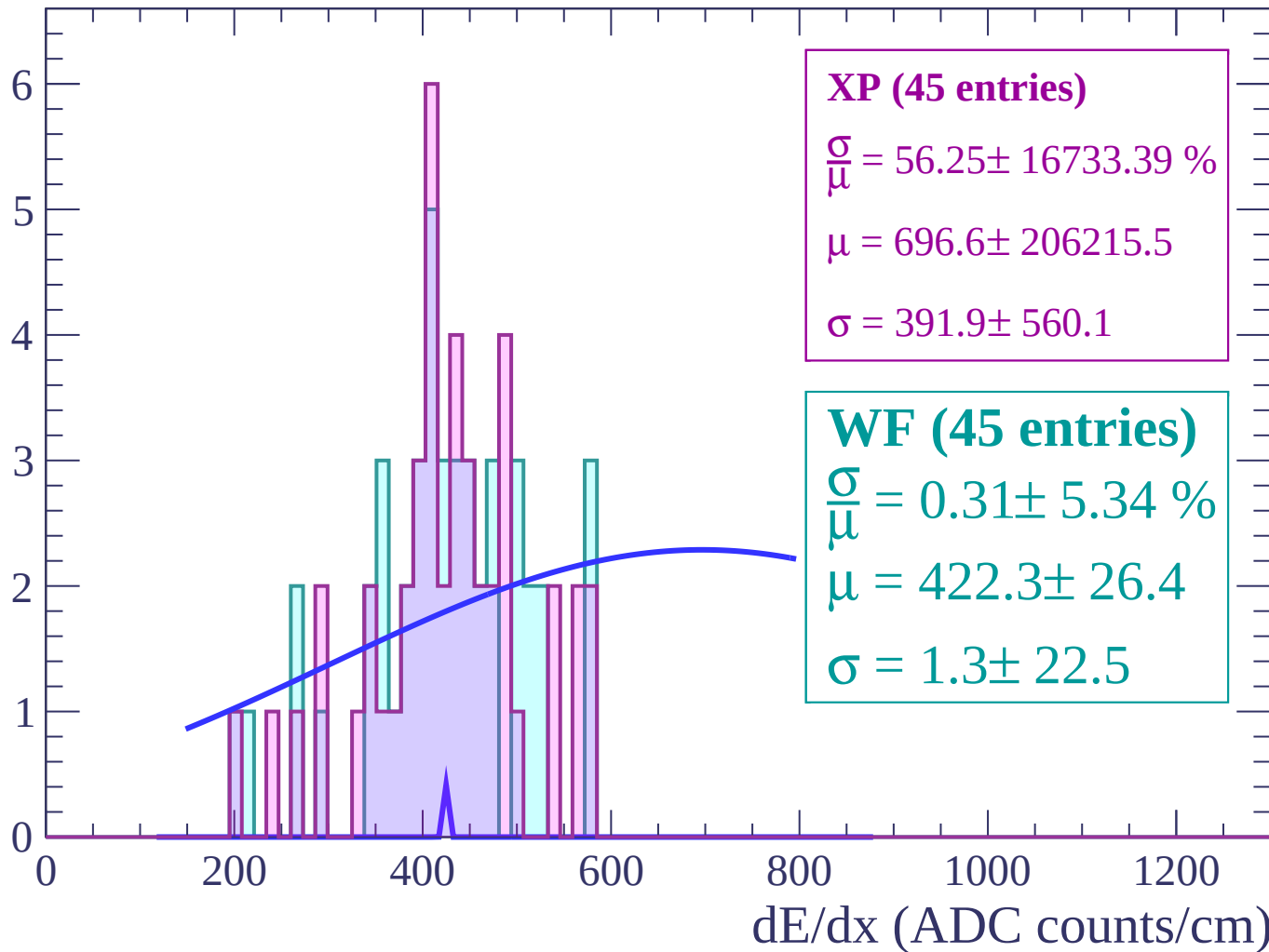
Energy loss | $-58 < \theta < -56$

Count



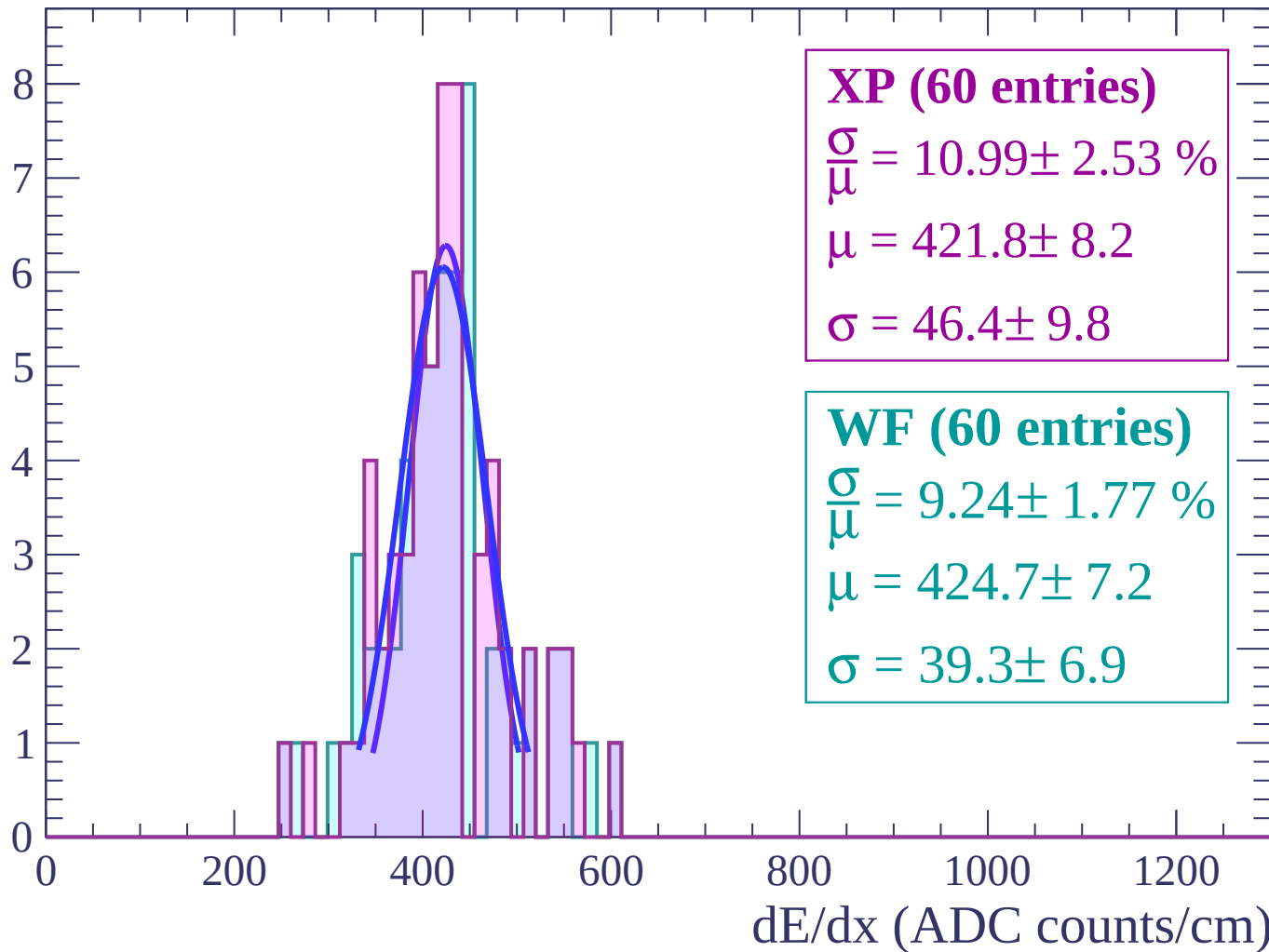
Energy loss | $-56 < \theta < -54$

Count



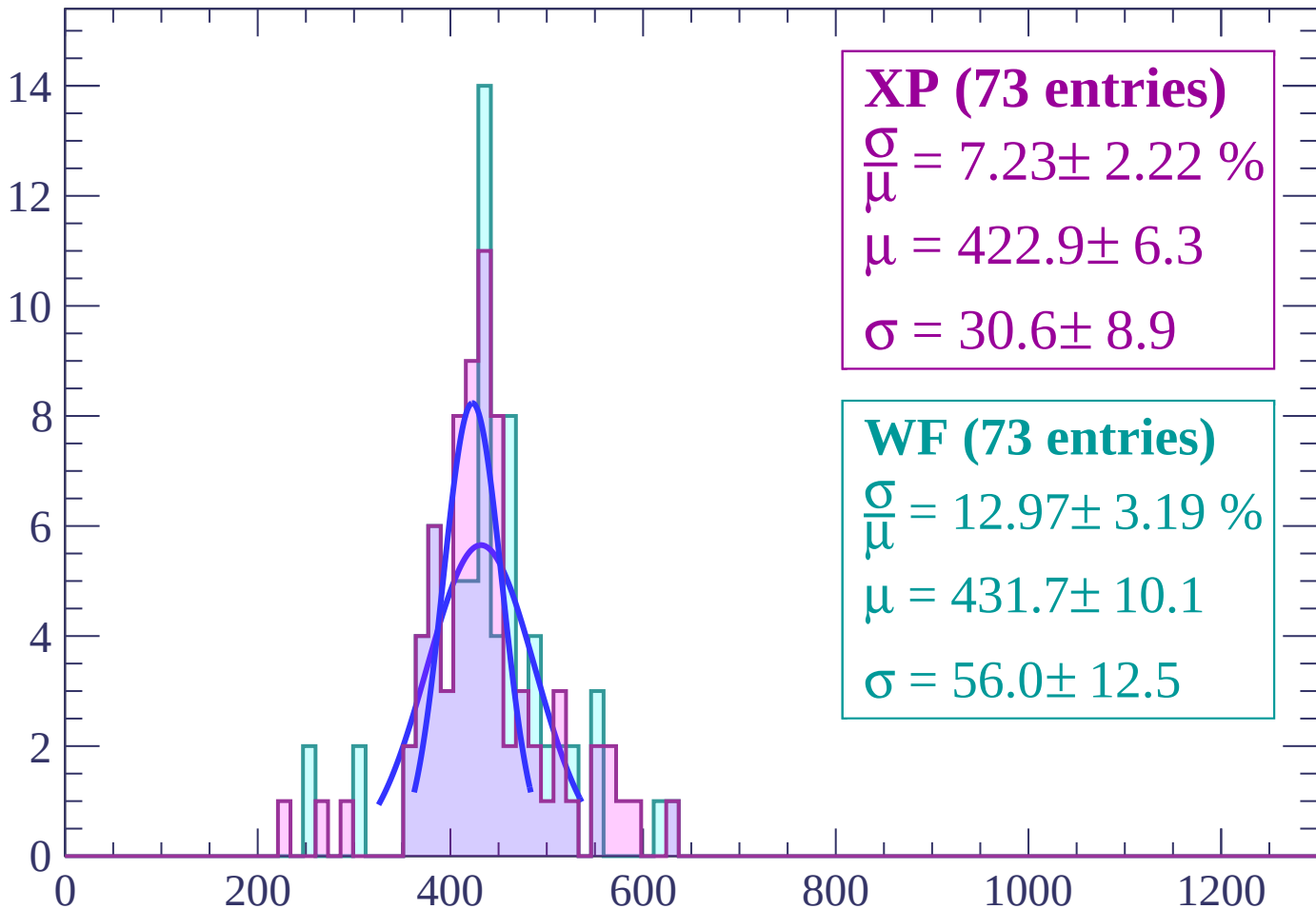
Energy loss | $-54 < \theta < -52$

Count



Energy loss | $-52 < \theta < -50$

Count



XP (73 entries)

$$\frac{\sigma}{\mu} = 7.23 \pm 2.22 \%$$

$$\mu = 422.9 \pm 6.3$$

$$\sigma = 30.6 \pm 8.9$$

WF (73 entries)

$$\frac{\sigma}{\mu} = 12.97 \pm 3.19 \%$$

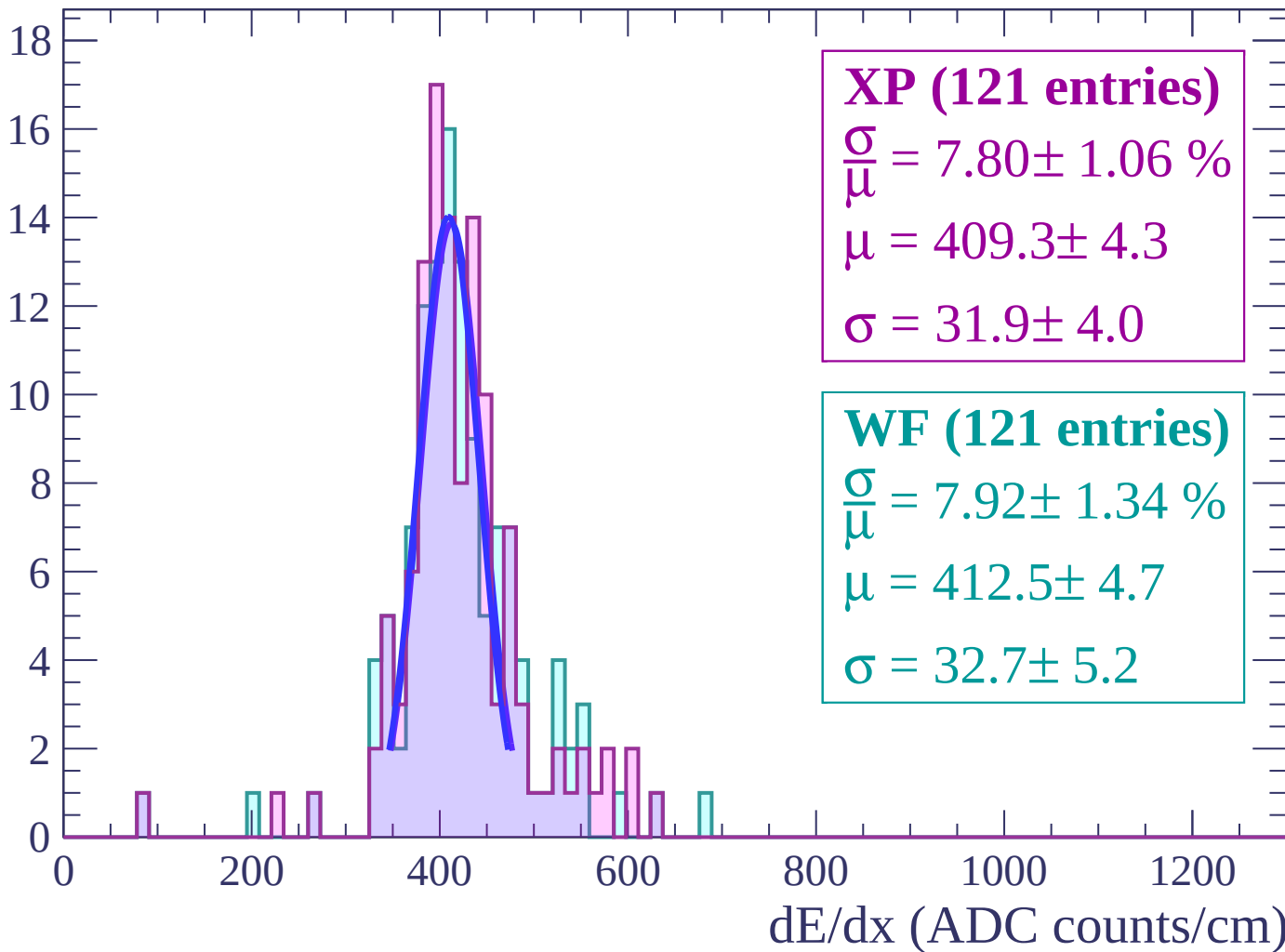
$$\mu = 431.7 \pm 10.1$$

$$\sigma = 56.0 \pm 12.5$$

dE/dx (ADC counts/cm)

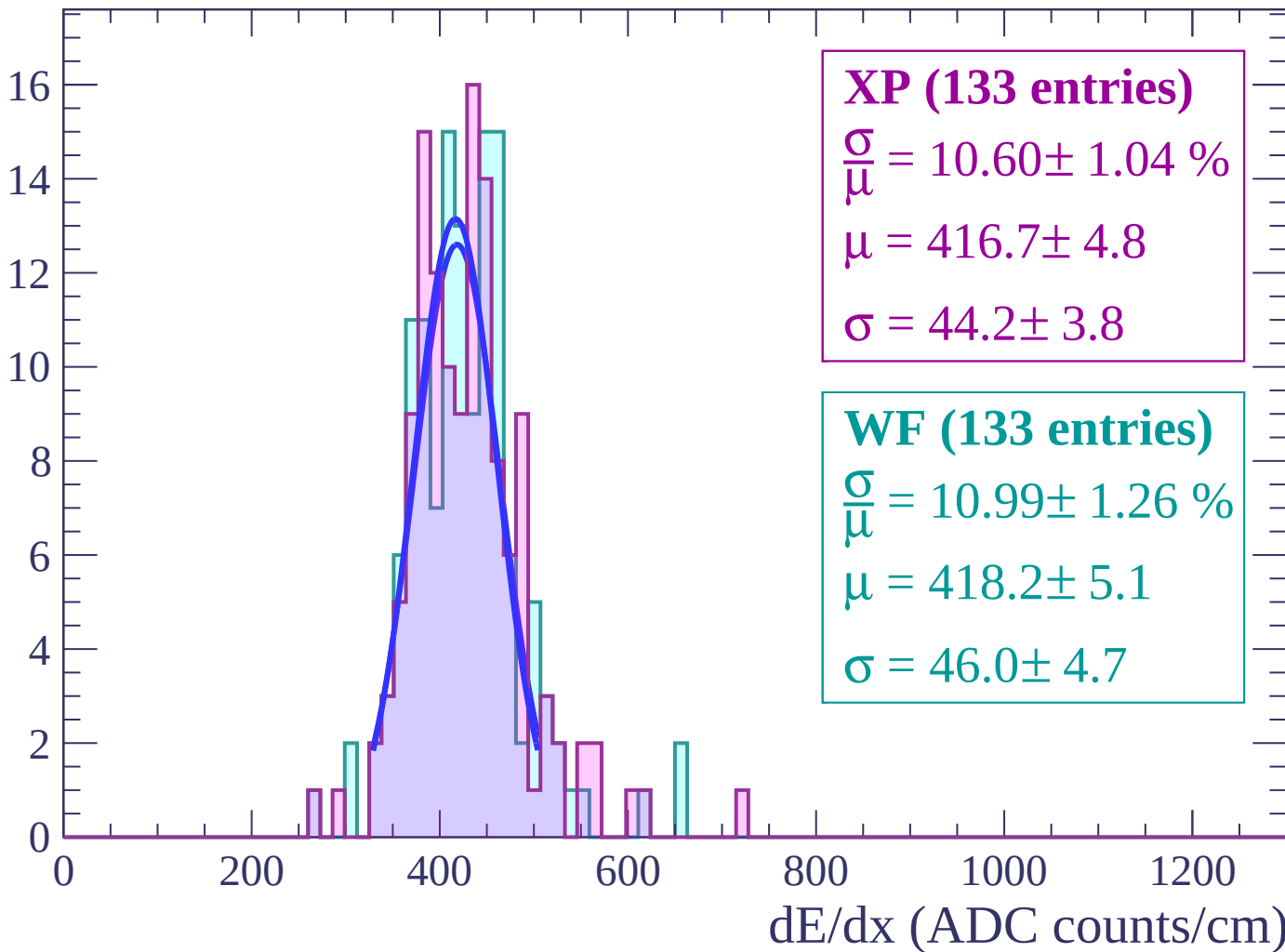
Energy loss | $-50 < \theta < -48$

Count



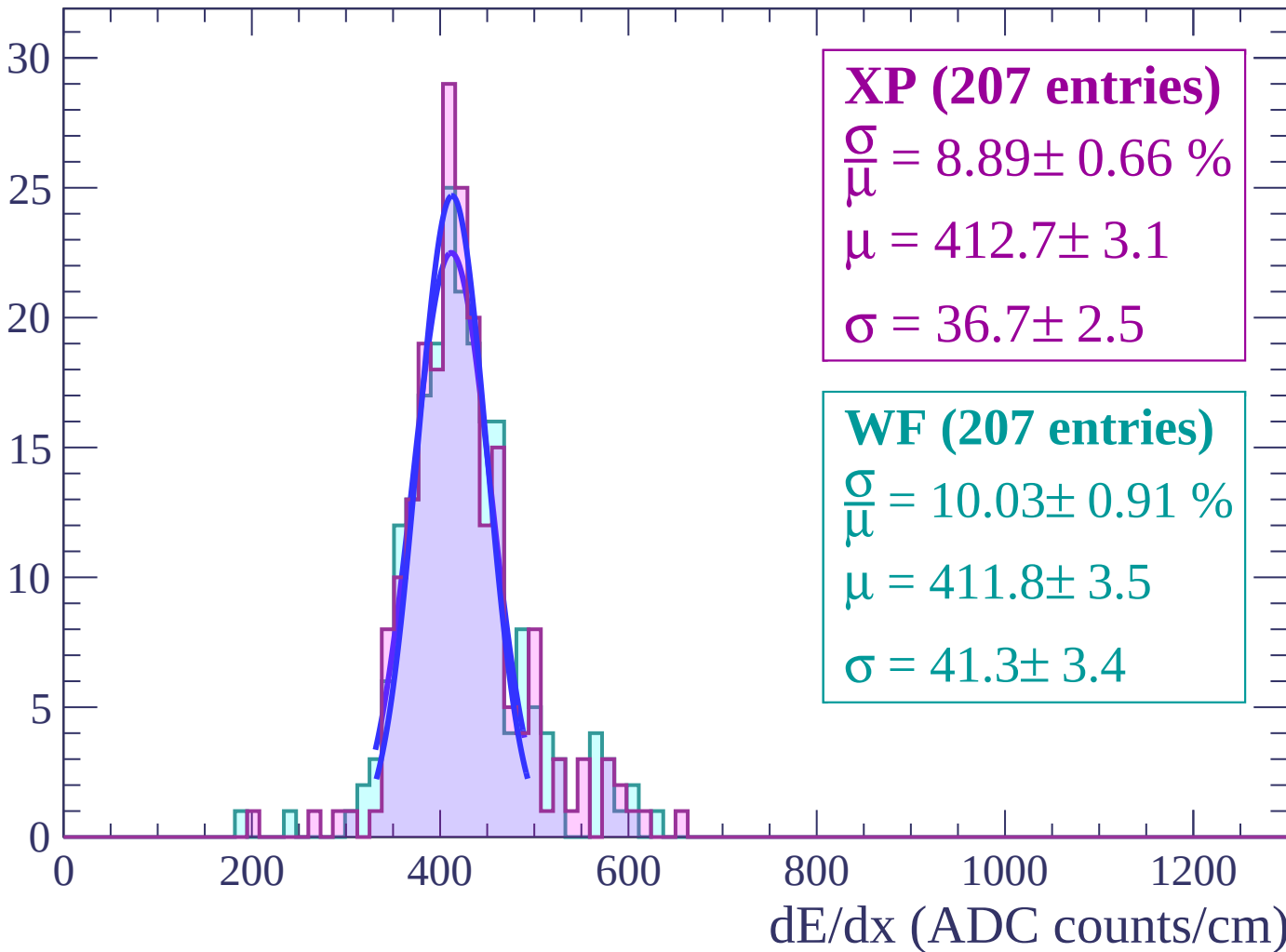
Energy loss | $-48 < \theta < -46$

Count



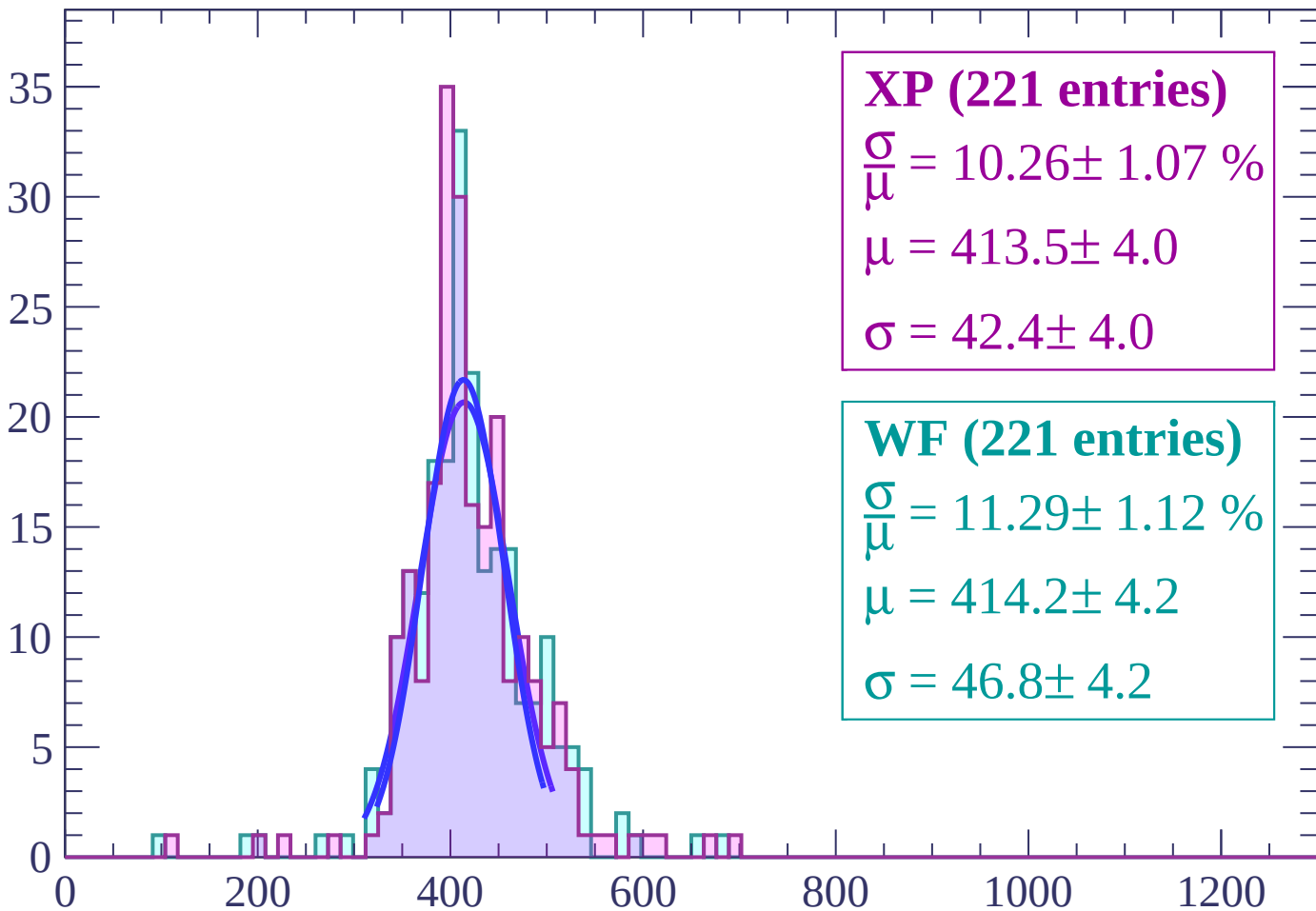
Energy loss | $-46 < \theta < -44$

Count



Energy loss | $-44 < \theta < -42$

Count



XP (221 entries)

$$\frac{\sigma}{\mu} = 10.26 \pm 1.07 \%$$

$$\mu = 413.5 \pm 4.0$$

$$\sigma = 42.4 \pm 4.0$$

WF (221 entries)

$$\frac{\sigma}{\mu} = 11.29 \pm 1.12 \%$$

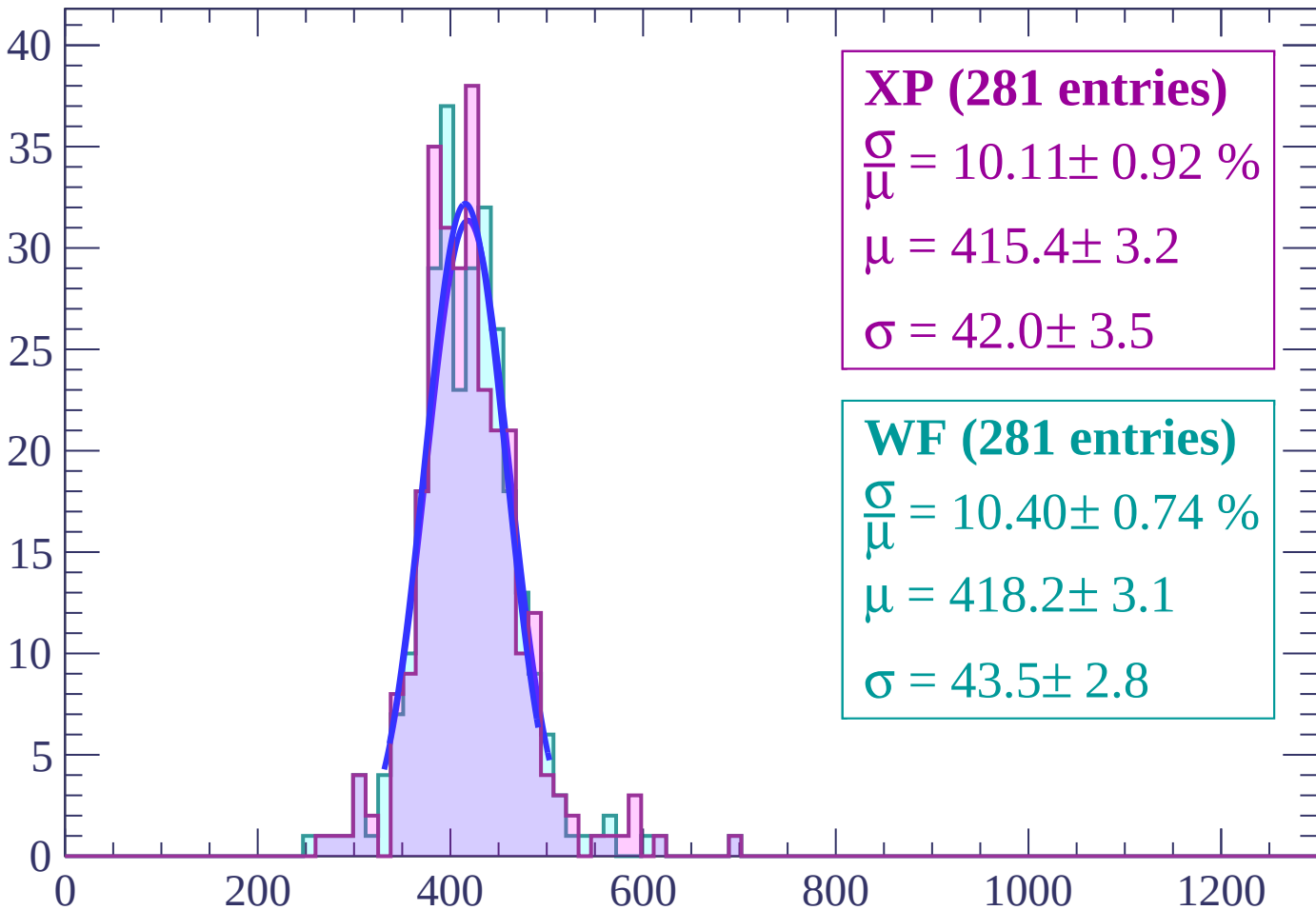
$$\mu = 414.2 \pm 4.2$$

$$\sigma = 46.8 \pm 4.2$$

dE/dx (ADC counts/cm)

Energy loss | $-42 < \theta < -40$

Count



XP (281 entries)

$$\frac{\sigma}{\mu} = 10.11 \pm 0.92 \%$$

$$\mu = 415.4 \pm 3.2$$

$$\sigma = 42.0 \pm 3.5$$

WF (281 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.74 \%$$

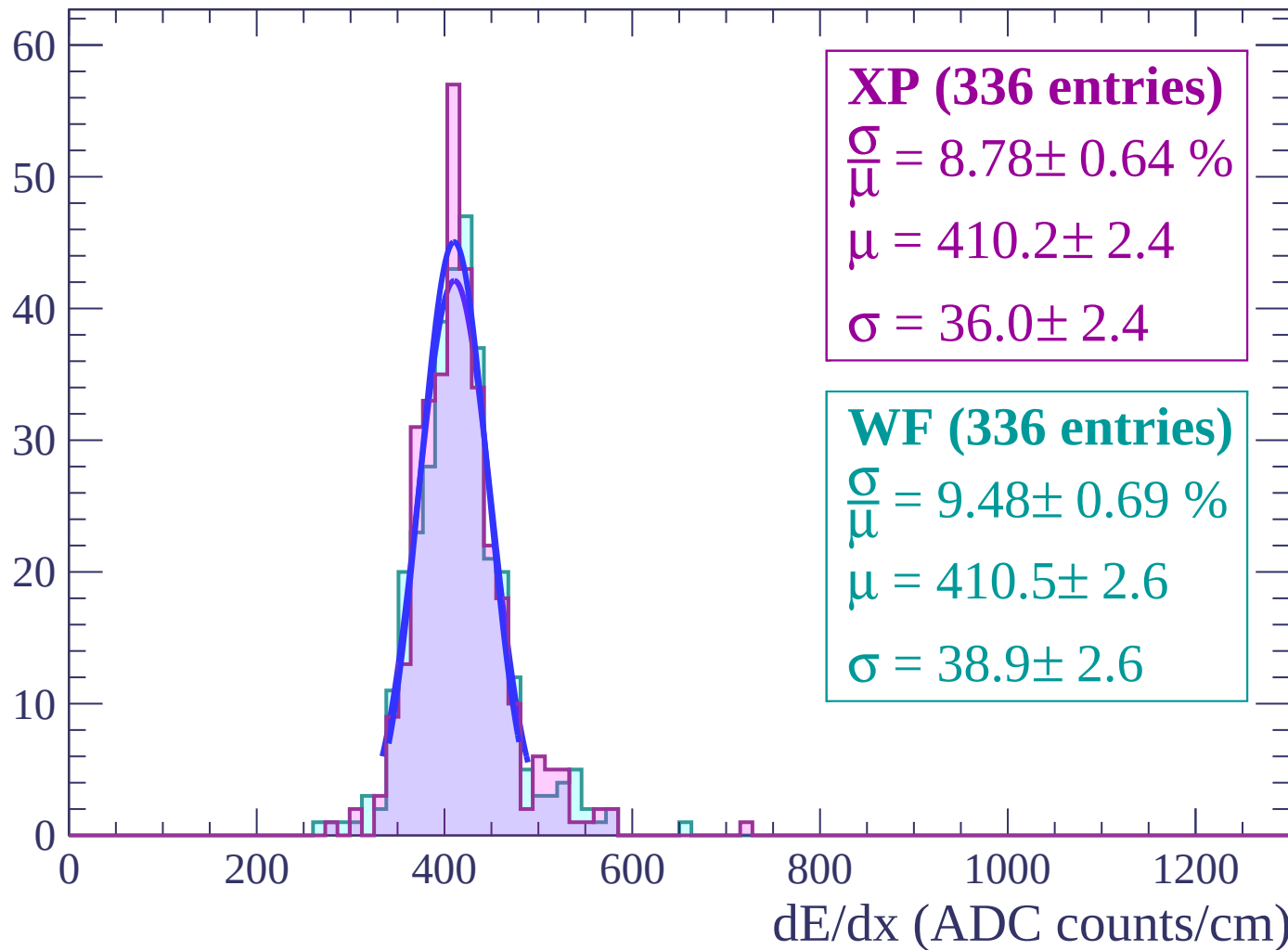
$$\mu = 418.2 \pm 3.1$$

$$\sigma = 43.5 \pm 2.8$$

dE/dx (ADC counts/cm)

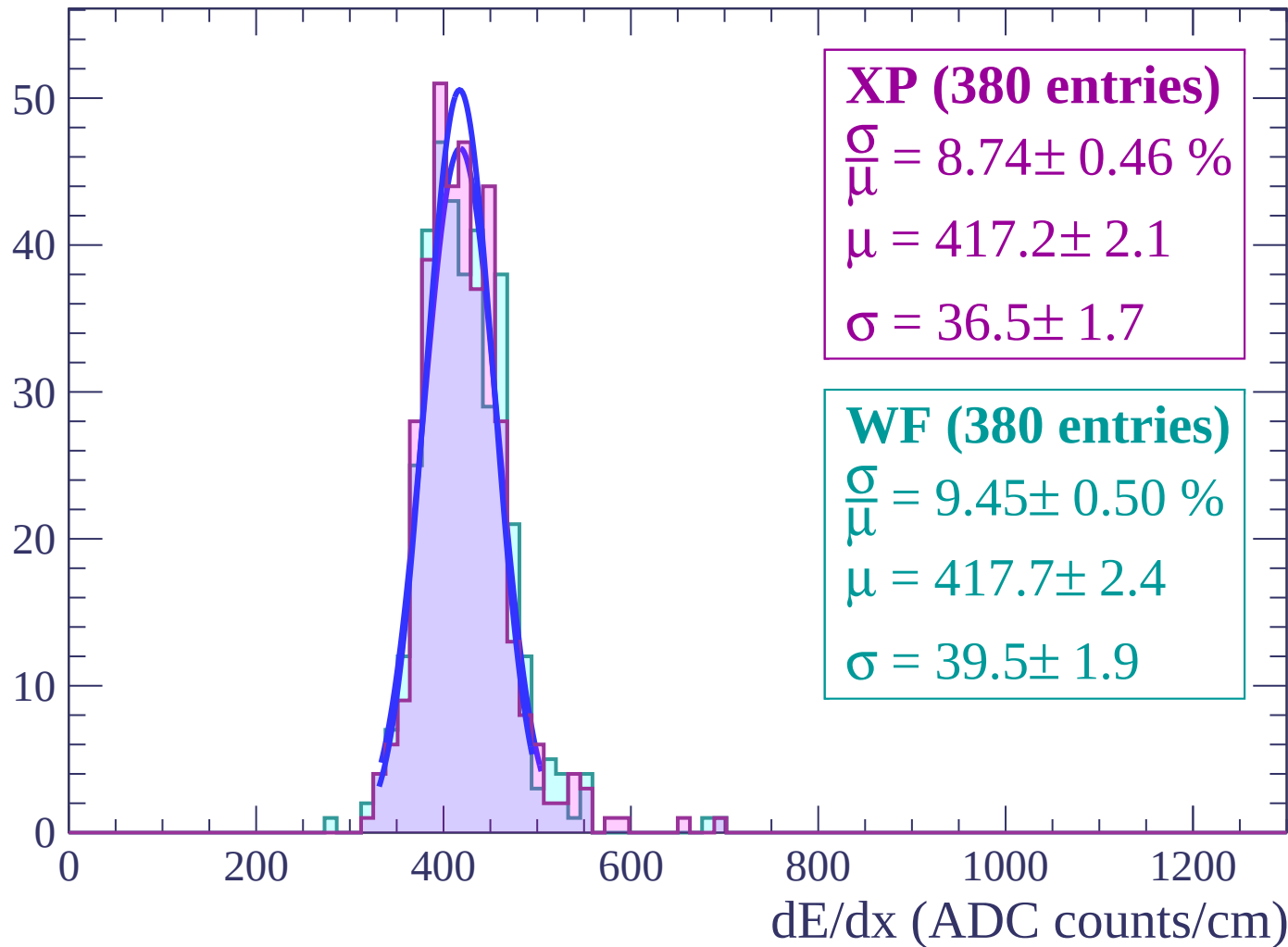
Energy loss | $-40 < \theta < -38$

Count



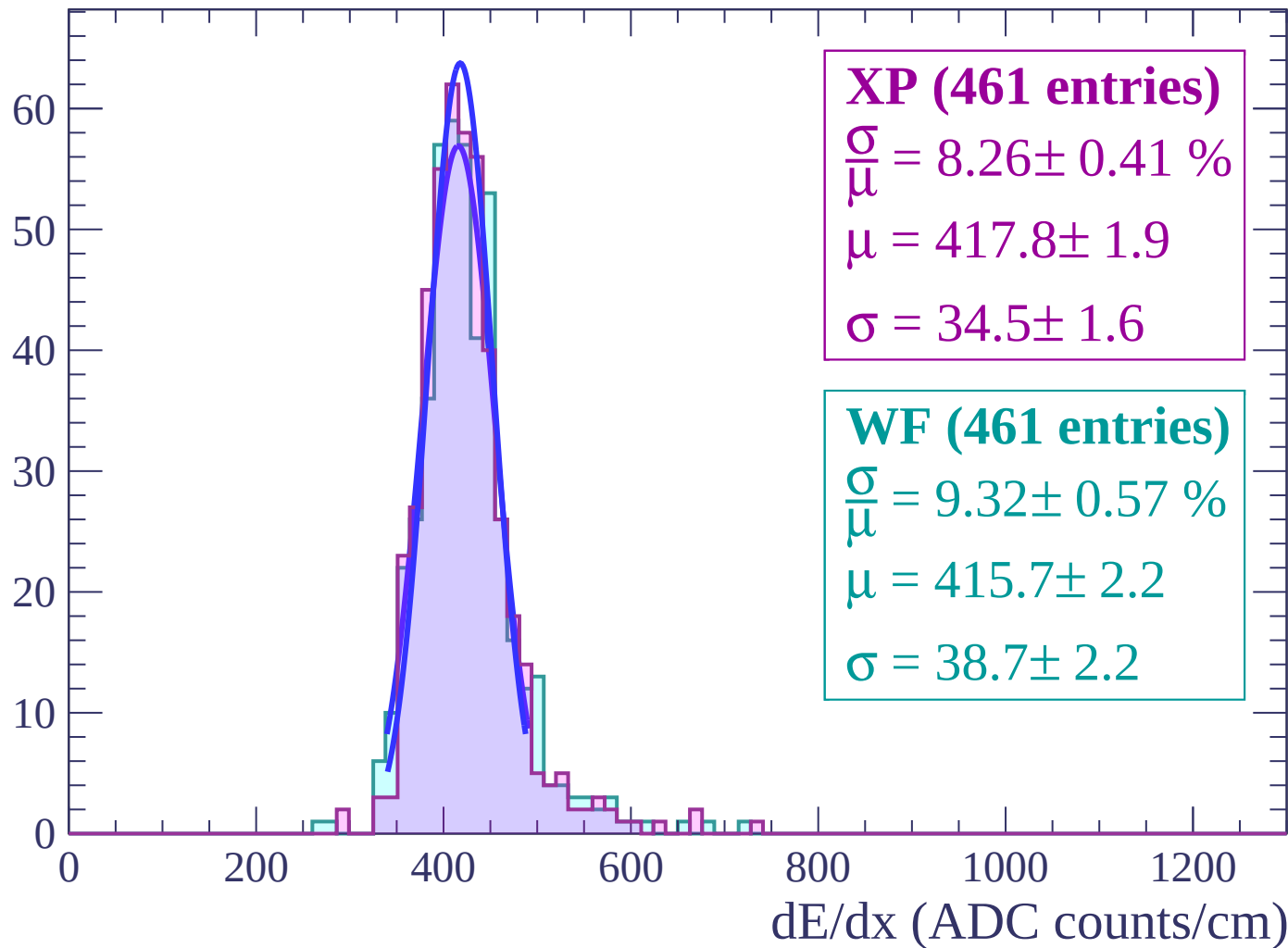
Energy loss | $-38 < \theta < -36$

Count



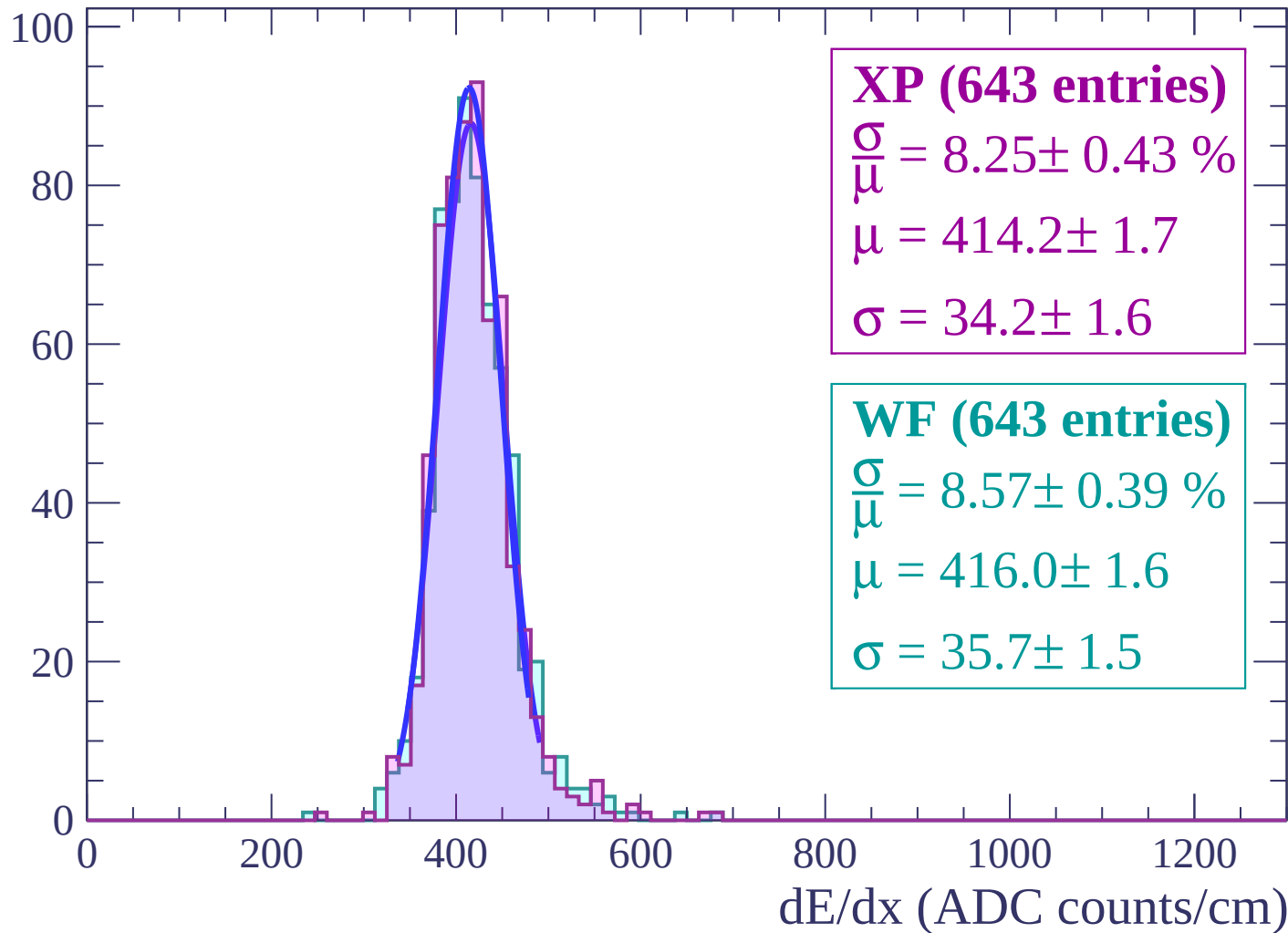
Energy loss | $-36 < \theta < -34$

Count



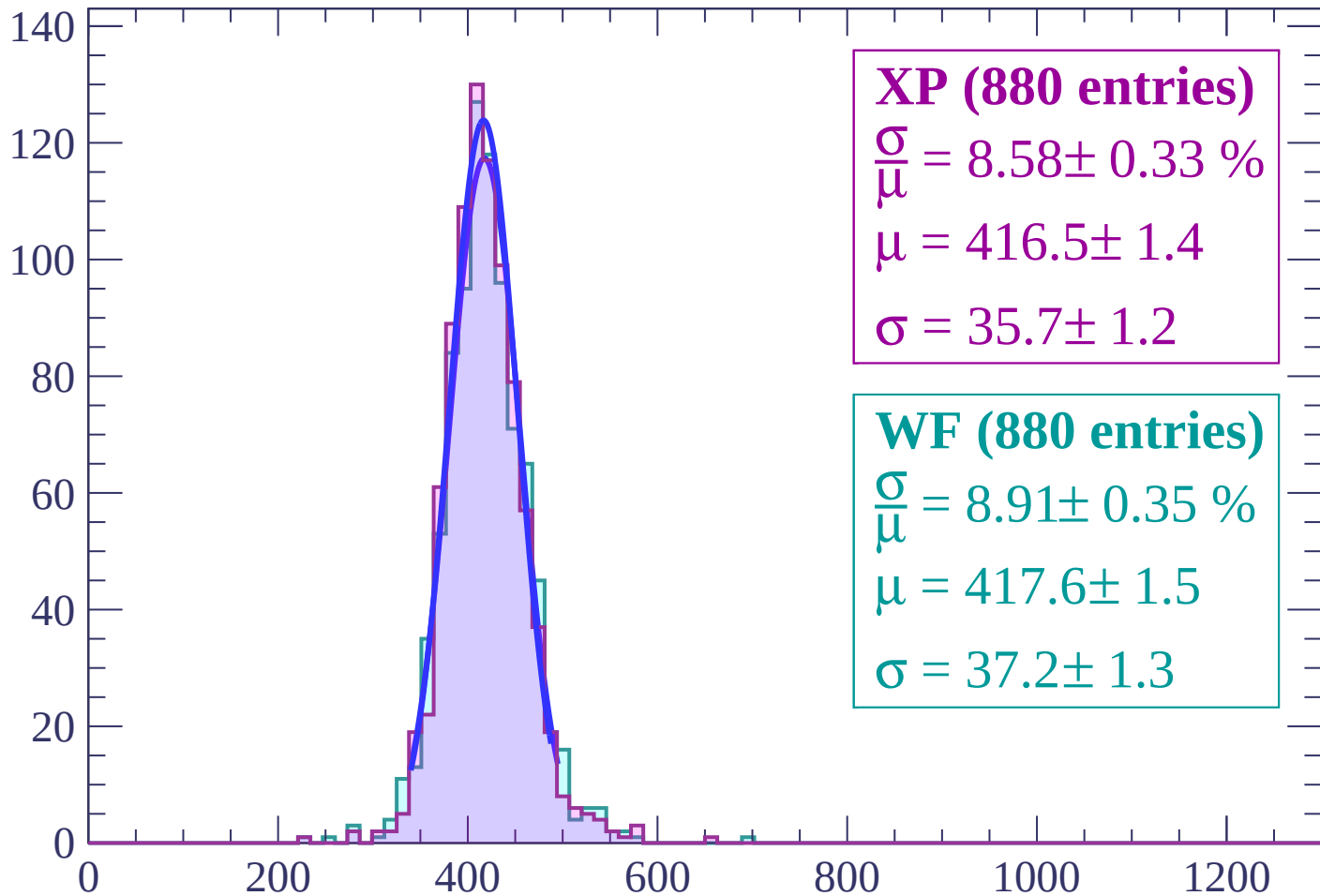
Energy loss | $-34 < \theta < -32$

Count



Energy loss | $-32 < \theta < -30$

Count



XP (880 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.33 \%$

$\mu = 416.5 \pm 1.4$

$\sigma = 35.7 \pm 1.2$

WF (880 entries)

$\frac{\sigma}{\mu} = 8.91 \pm 0.35 \%$

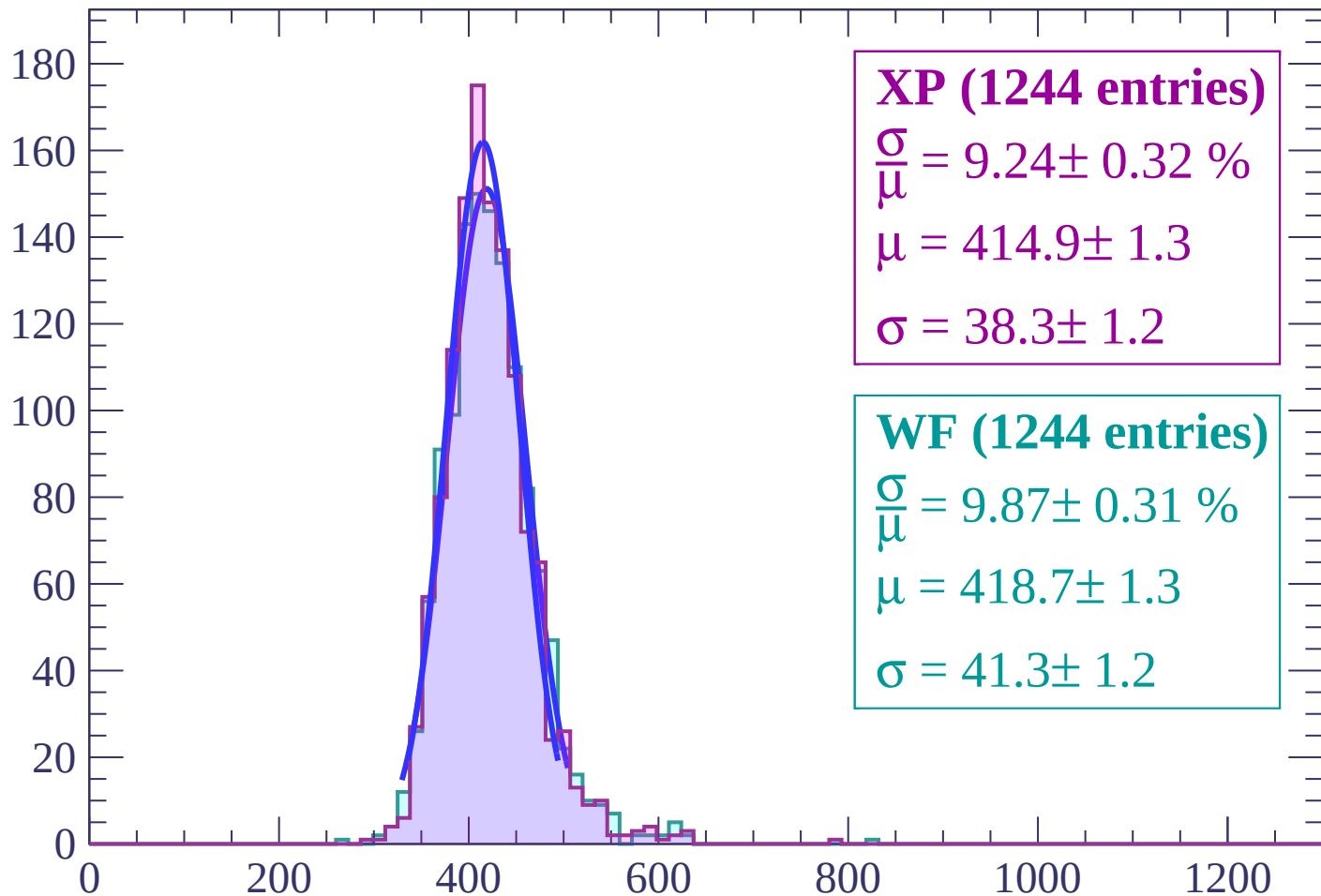
$\mu = 417.6 \pm 1.5$

$\sigma = 37.2 \pm 1.3$

dE/dx (ADC counts/cm)

Energy loss | $-30 < \theta < -28$

Count



XP (1244 entries)

$\frac{\sigma}{\mu} = 9.24 \pm 0.32 \%$

$\mu = 414.9 \pm 1.3$

$\sigma = 38.3 \pm 1.2$

WF (1244 entries)

$\frac{\sigma}{\mu} = 9.87 \pm 0.31 \%$

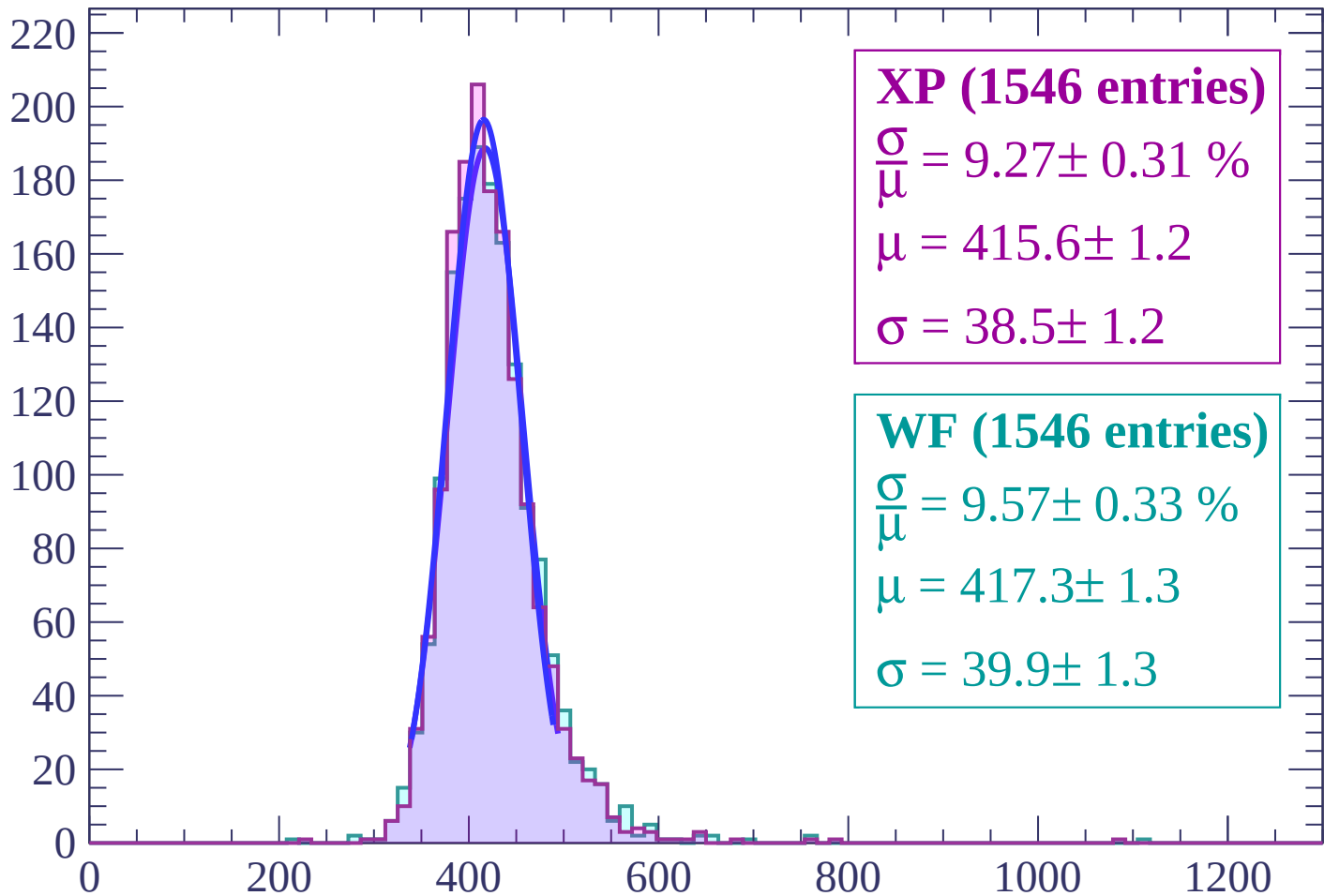
$\mu = 418.7 \pm 1.3$

$\sigma = 41.3 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $-28 < \theta < -26$

Count



XP (1546 entries)

$$\frac{\sigma}{\mu} = 9.27 \pm 0.31 \%$$

$$\mu = 415.6 \pm 1.2$$

$$\sigma = 38.5 \pm 1.2$$

WF (1546 entries)

$$\frac{\sigma}{\mu} = 9.57 \pm 0.33 \%$$

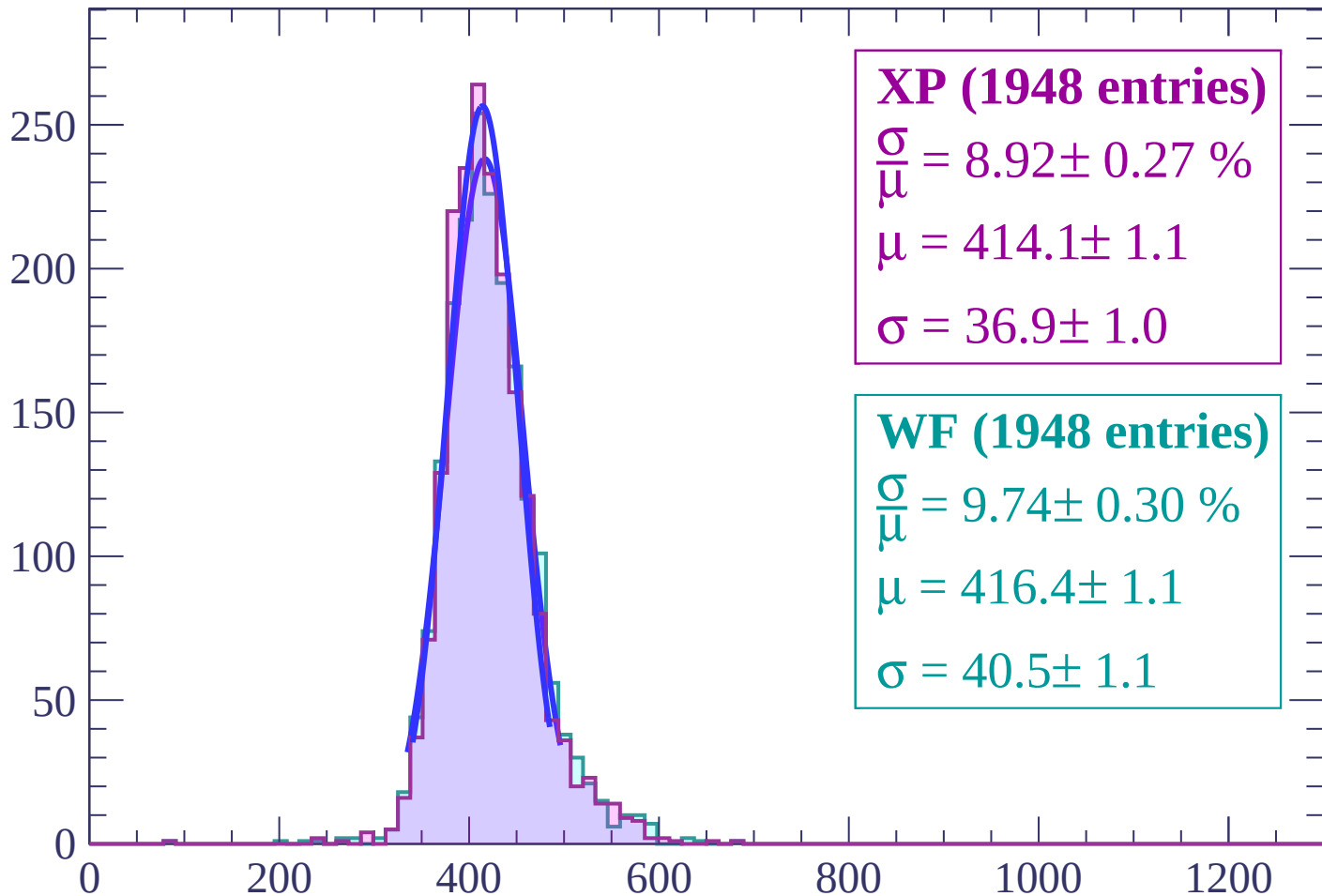
$$\mu = 417.3 \pm 1.3$$

$$\sigma = 39.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-26 < \theta < -24$

Count



XP (1948 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.27 \%$$

$$\mu = 414.1 \pm 1.1$$

$$\sigma = 36.9 \pm 1.0$$

WF (1948 entries)

$$\frac{\sigma}{\mu} = 9.74 \pm 0.30 \%$$

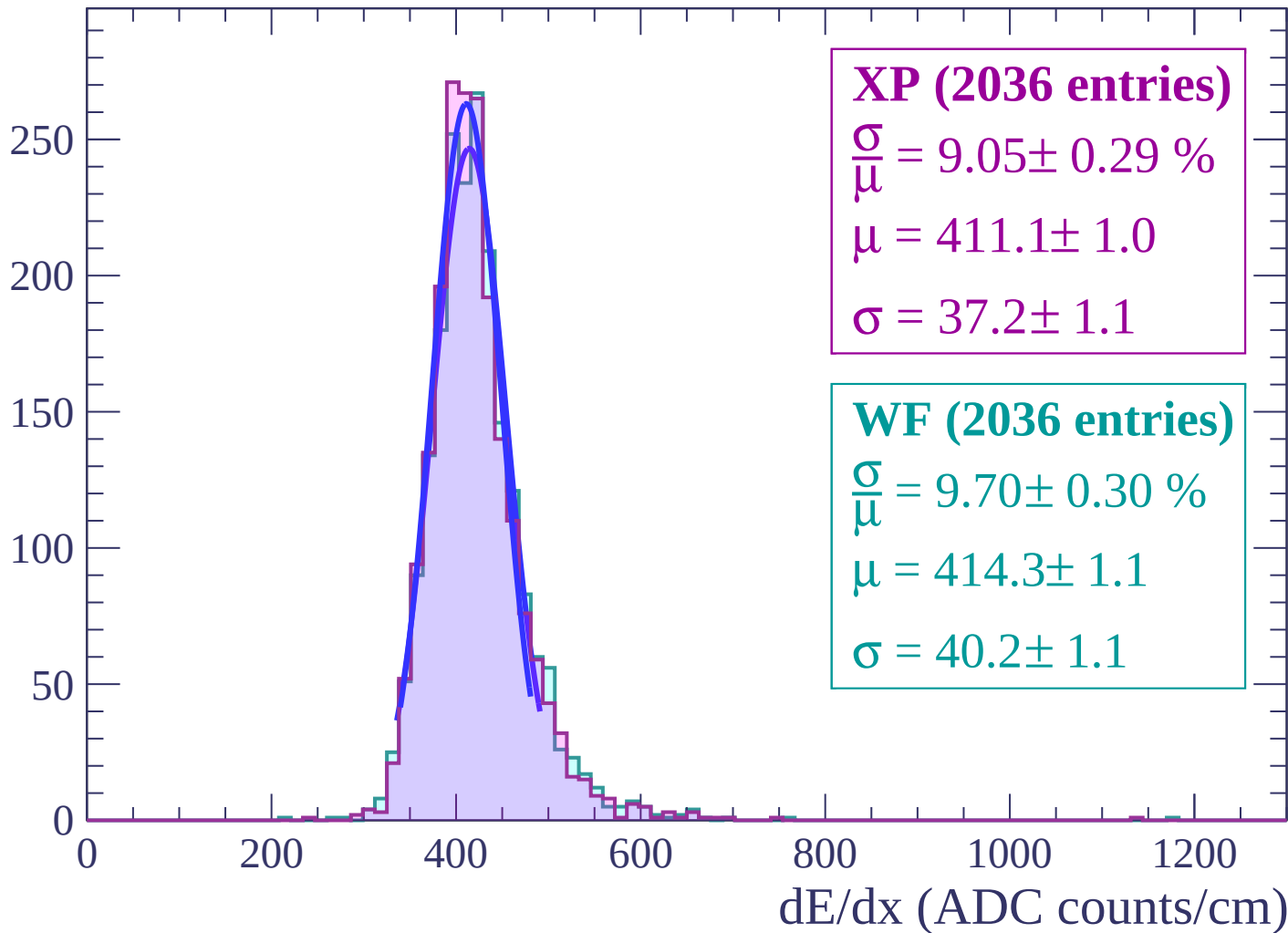
$$\mu = 416.4 \pm 1.1$$

$$\sigma = 40.5 \pm 1.1$$

dE/dx (ADC counts/cm)

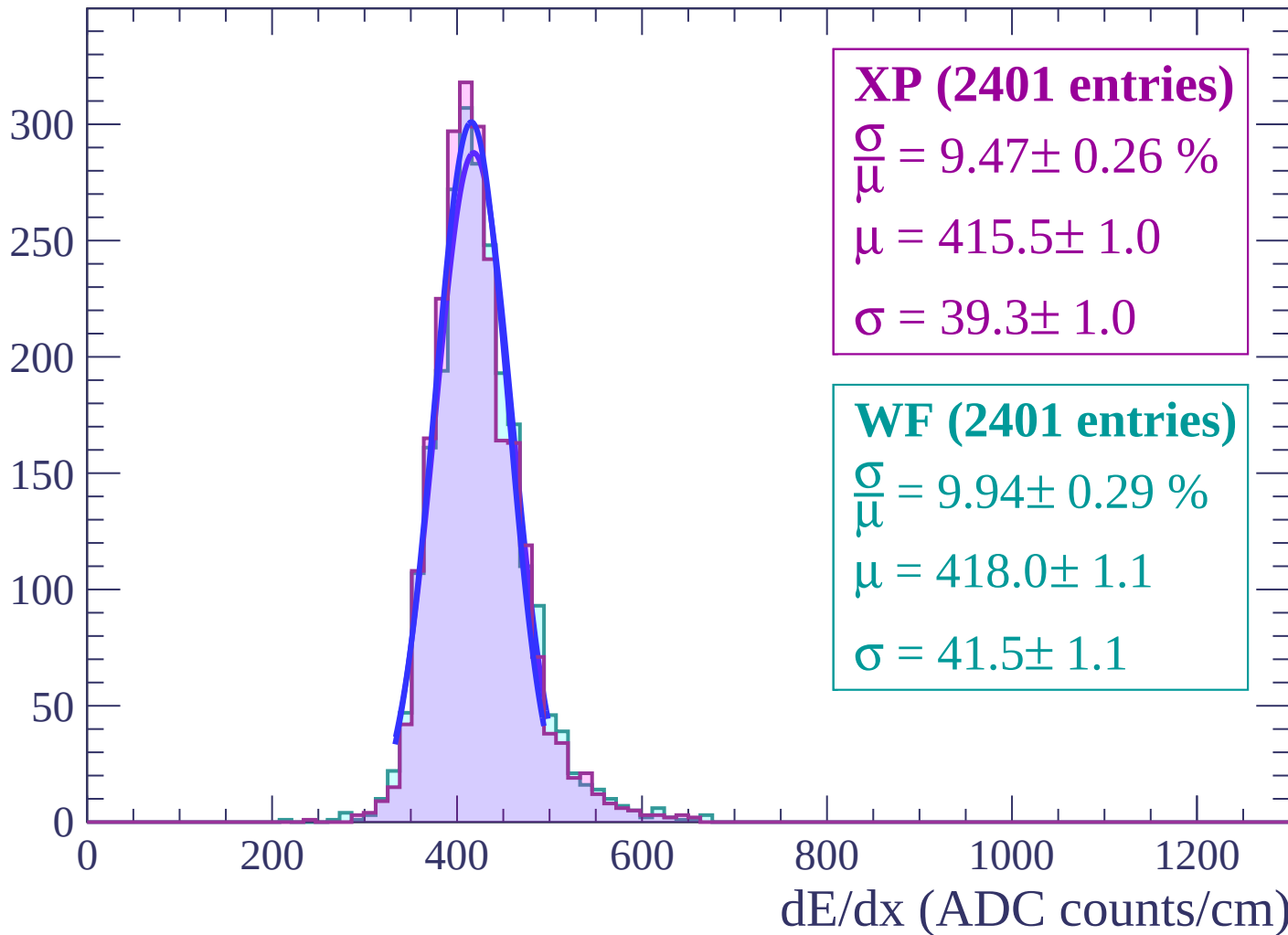
Energy loss | $-24 < \theta < -22$

Count



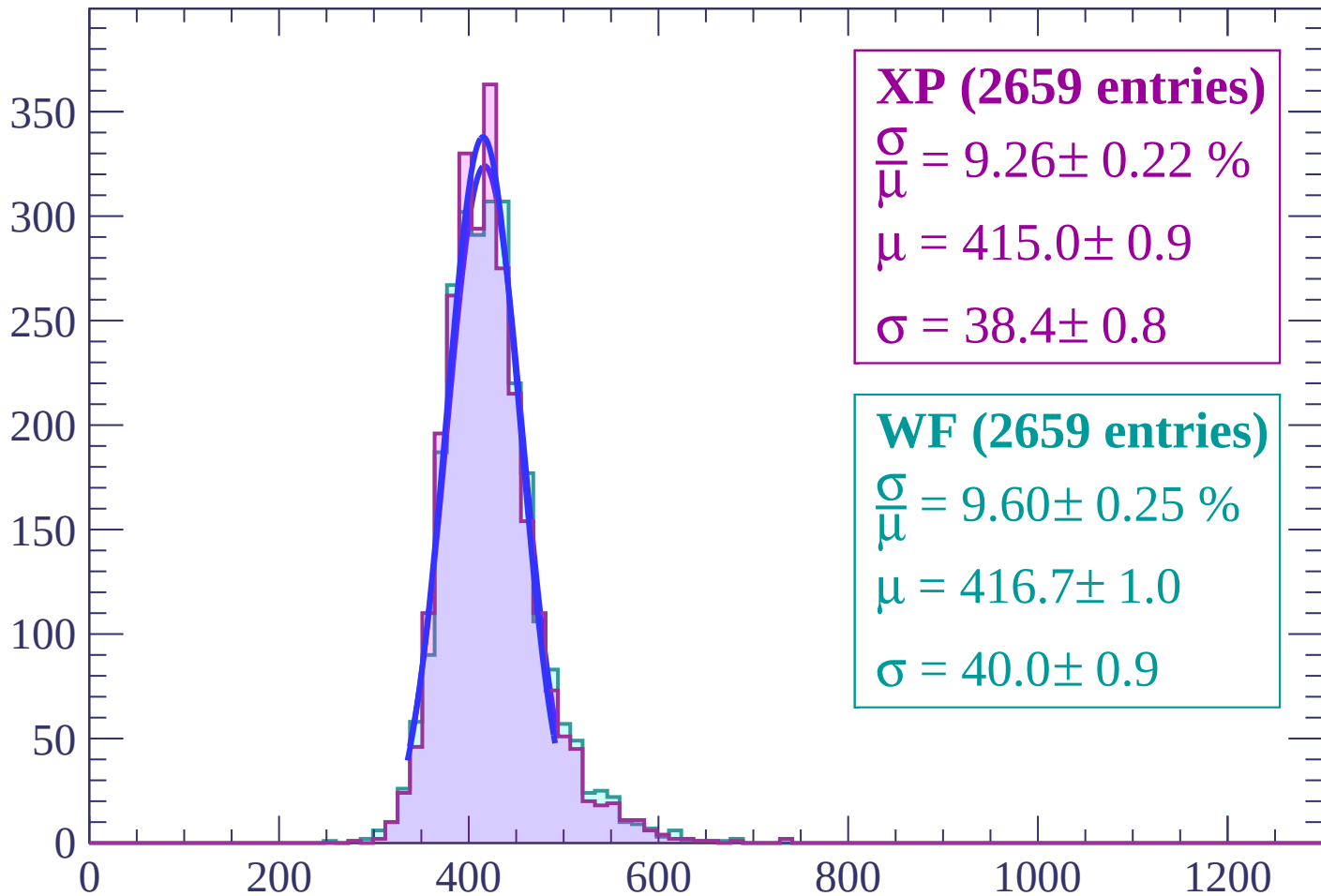
Energy loss | $-22 < \theta < -20$

Count



Energy loss | $-20 < \theta < -18$

Count



XP (2659 entries)

$\frac{\sigma}{\mu} = 9.26 \pm 0.22 \%$

$\mu = 415.0 \pm 0.9$

$\sigma = 38.4 \pm 0.8$

WF (2659 entries)

$\frac{\sigma}{\mu} = 9.60 \pm 0.25 \%$

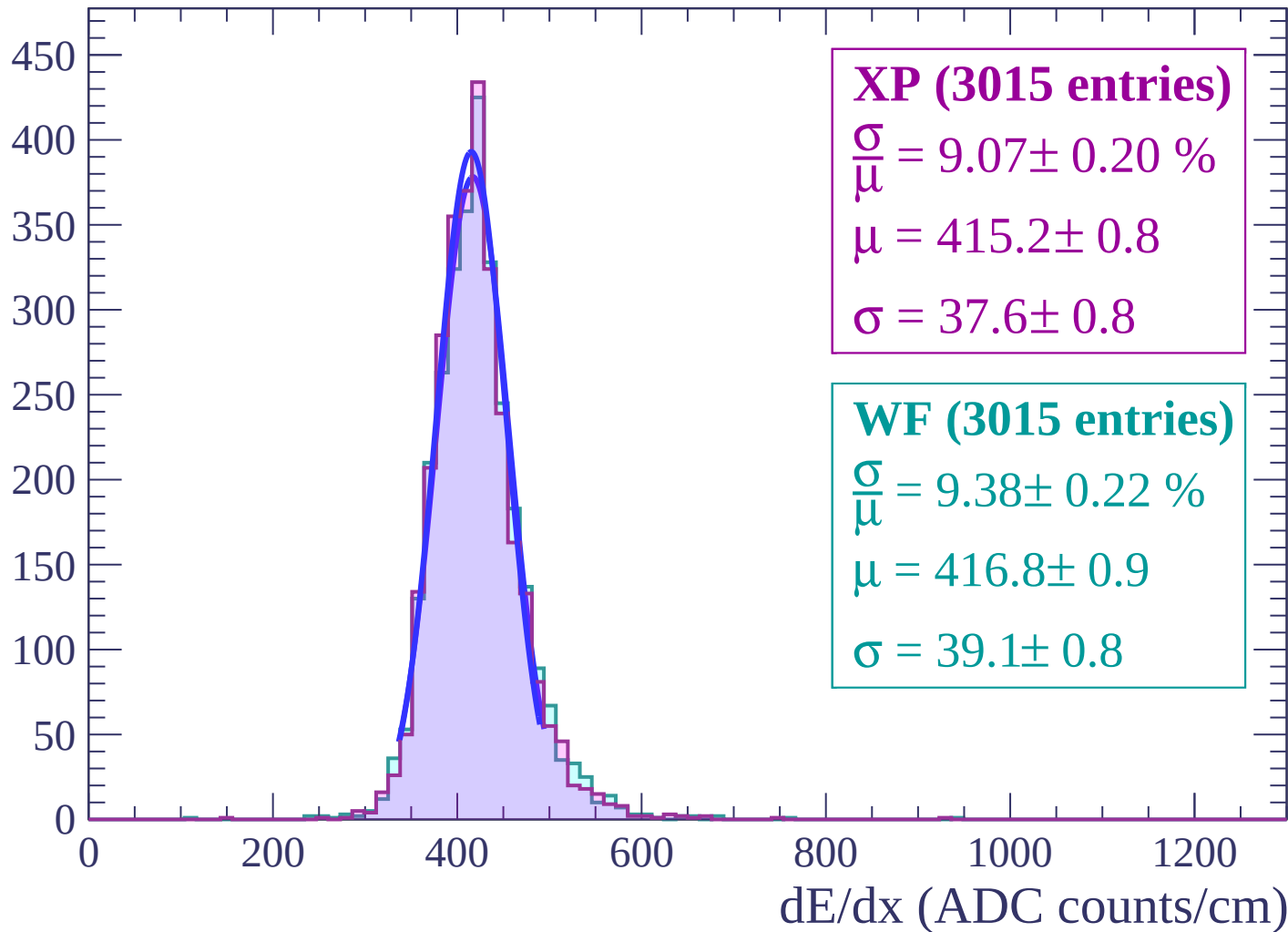
$\mu = 416.7 \pm 1.0$

$\sigma = 40.0 \pm 0.9$

dE/dx (ADC counts/cm)

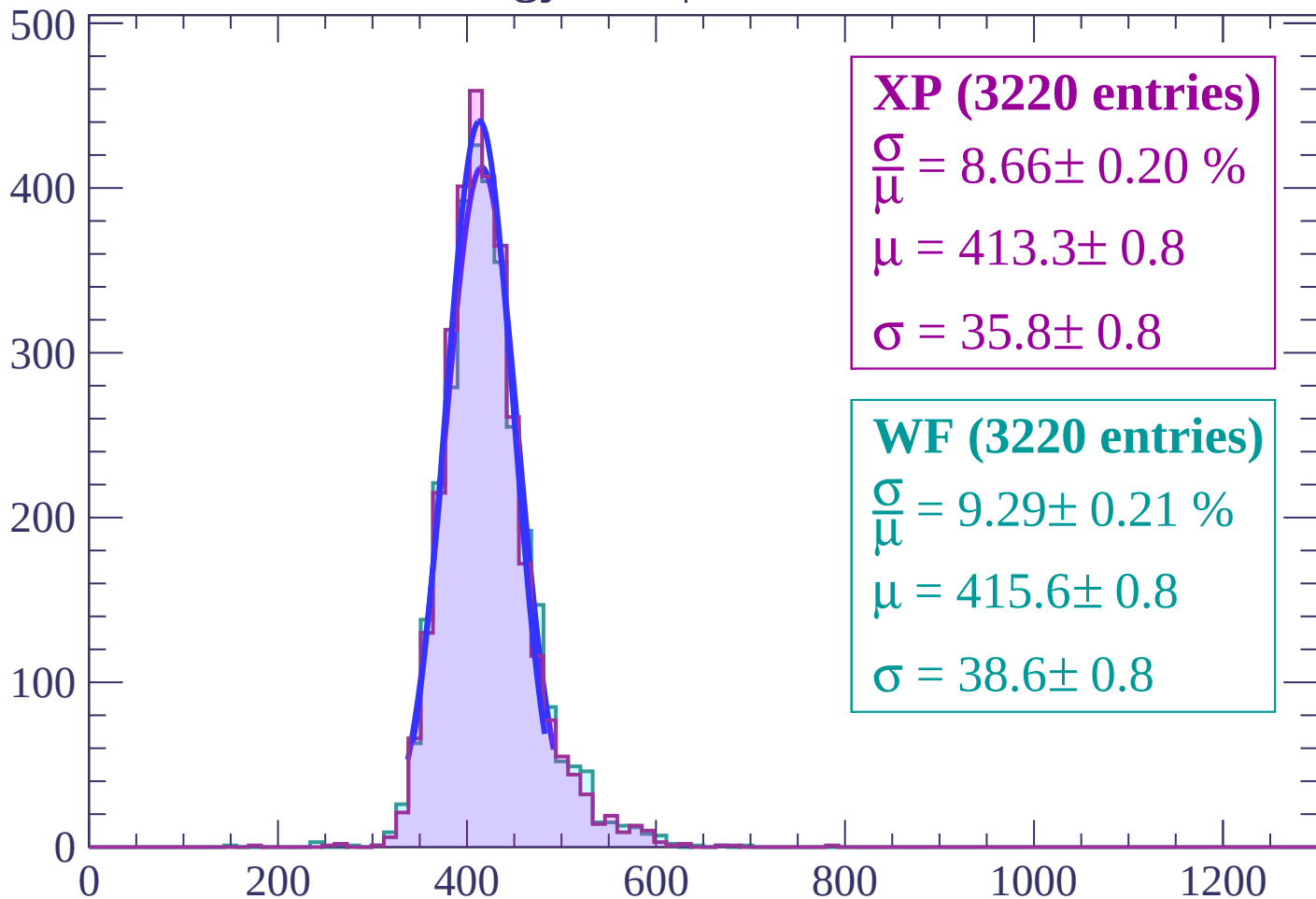
Energy loss | $-18 < \theta < -16$

Count



Energy loss | $-16 < \theta < -14$

Count



XP (3220 entries)

$\frac{\sigma}{\mu} = 8.66 \pm 0.20 \%$

$\mu = 413.3 \pm 0.8$

$\sigma = 35.8 \pm 0.8$

WF (3220 entries)

$\frac{\sigma}{\mu} = 9.29 \pm 0.21 \%$

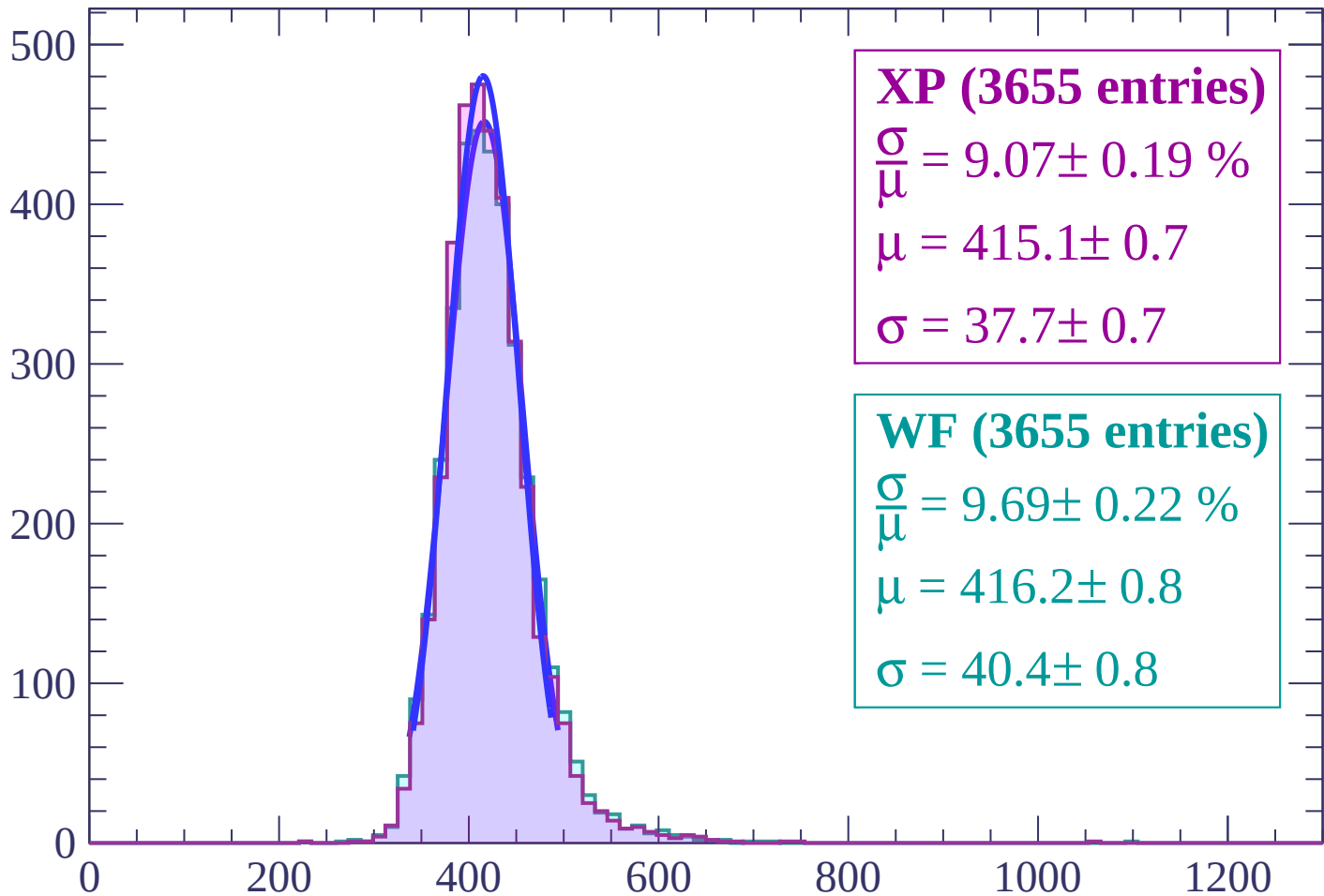
$\mu = 415.6 \pm 0.8$

$\sigma = 38.6 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-14 < \theta < -12$

Count



XP (3655 entries)

$\frac{\sigma}{\mu} = 9.07 \pm 0.19 \%$

$\mu = 415.1 \pm 0.7$

$\sigma = 37.7 \pm 0.7$

WF (3655 entries)

$\frac{\sigma}{\mu} = 9.69 \pm 0.22 \%$

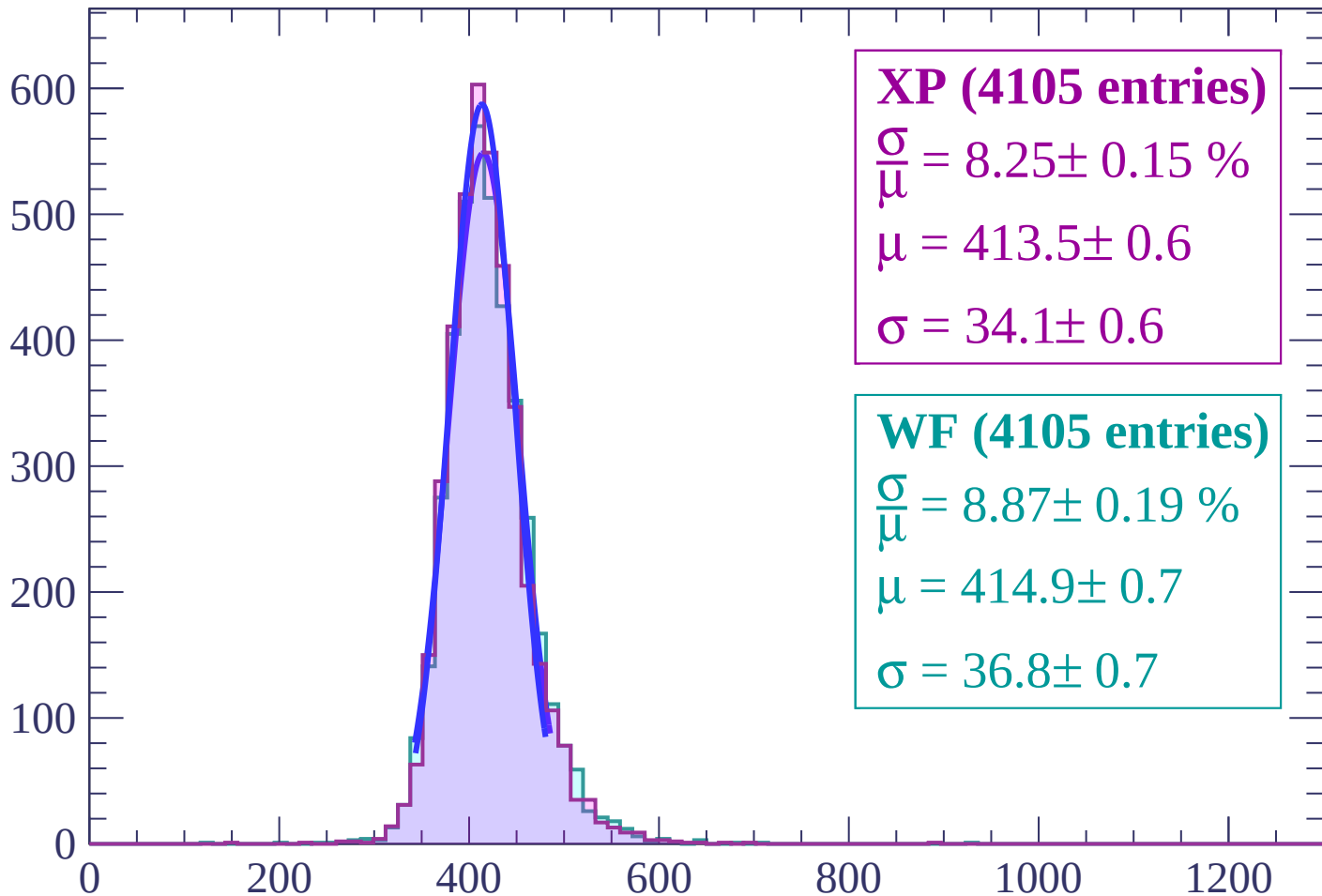
$\mu = 416.2 \pm 0.8$

$\sigma = 40.4 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $-12 < \theta < -10$

Count



XP (4105 entries)

$$\frac{\sigma}{\mu} = 8.25 \pm 0.15 \%$$

$$\mu = 413.5 \pm 0.6$$

$$\sigma = 34.1 \pm 0.6$$

WF (4105 entries)

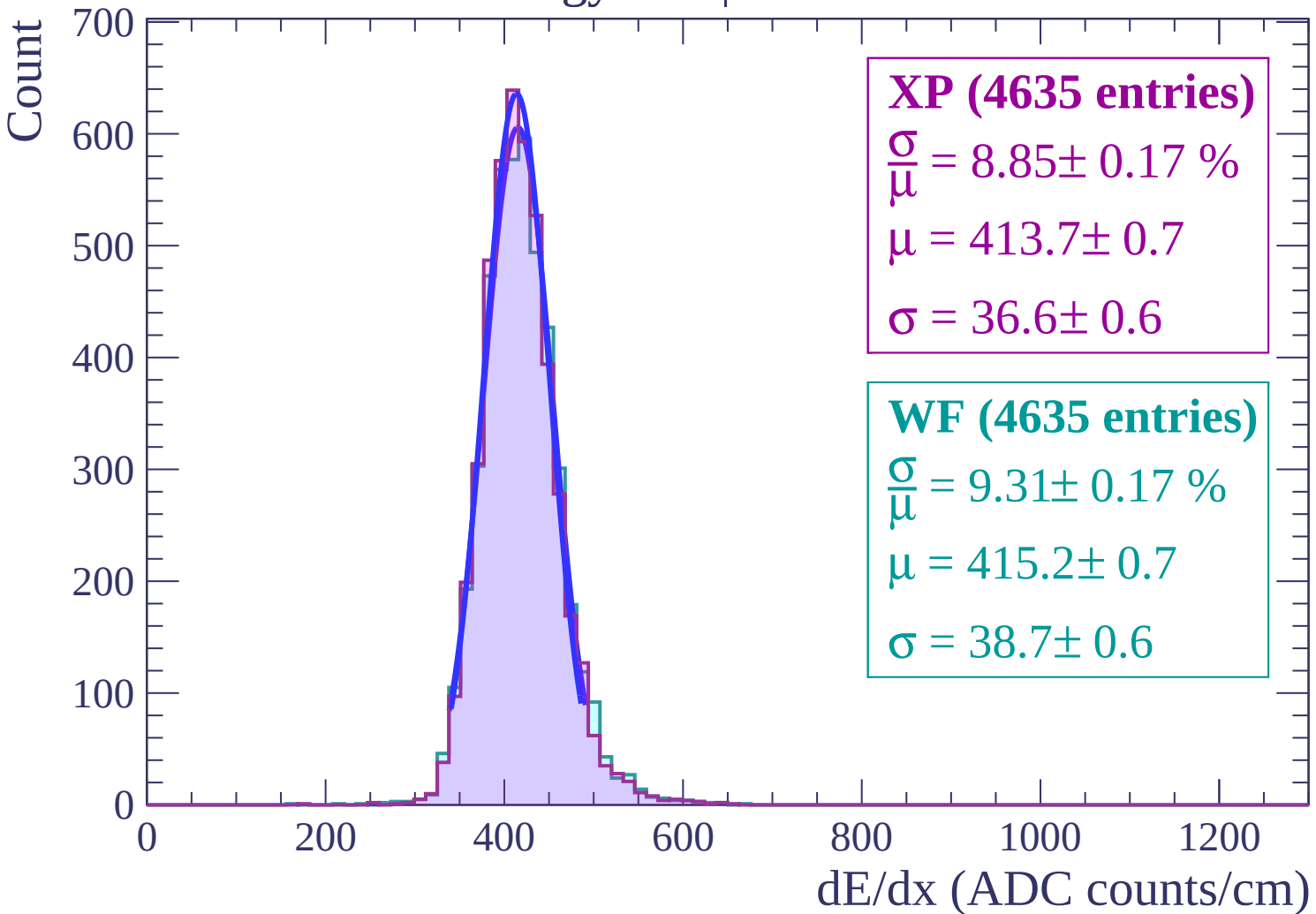
$$\frac{\sigma}{\mu} = 8.87 \pm 0.19 \%$$

$$\mu = 414.9 \pm 0.7$$

$$\sigma = 36.8 \pm 0.7$$

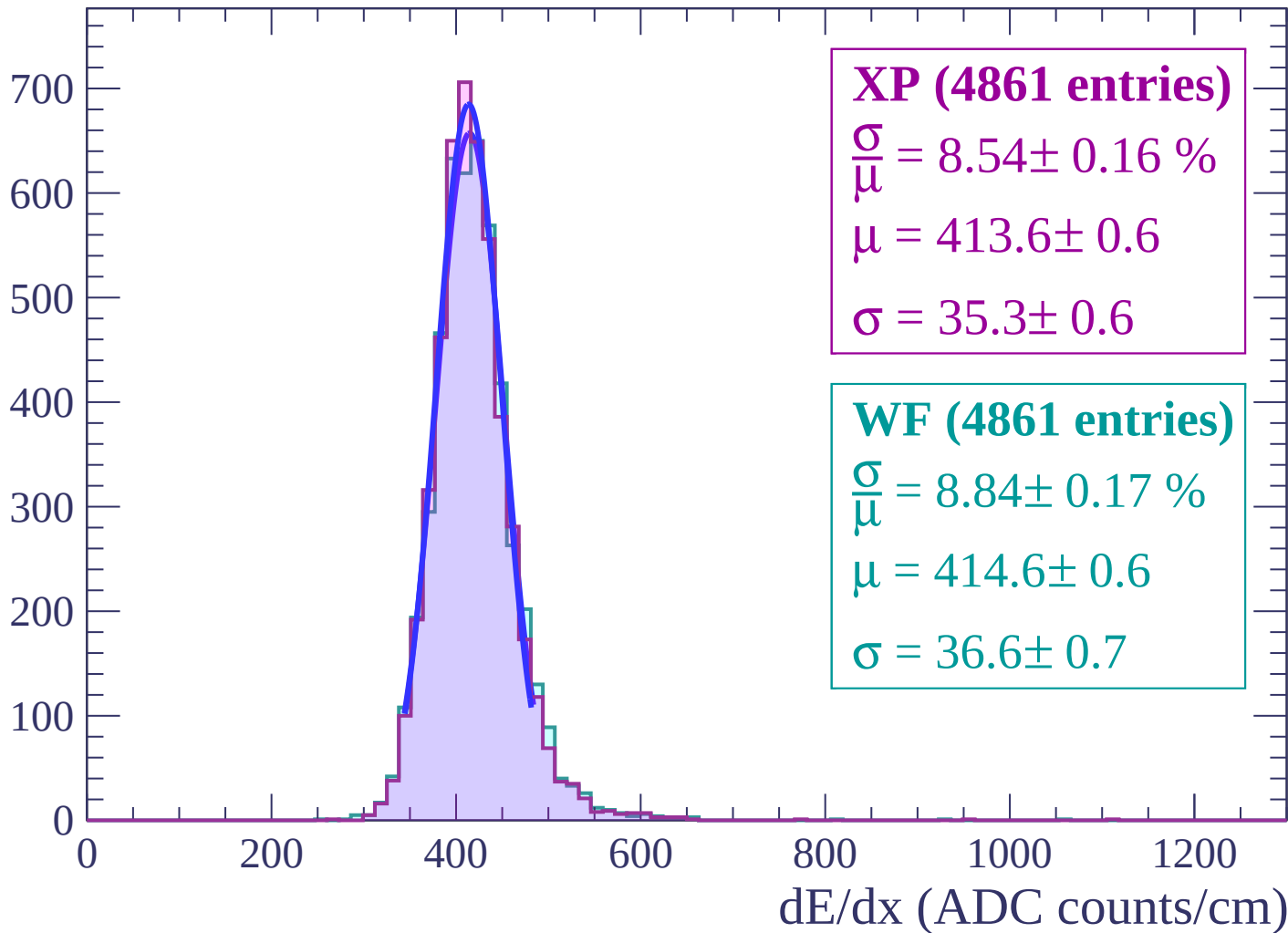
dE/dx (ADC counts/cm)

Energy loss | $-10 < \theta < -8$



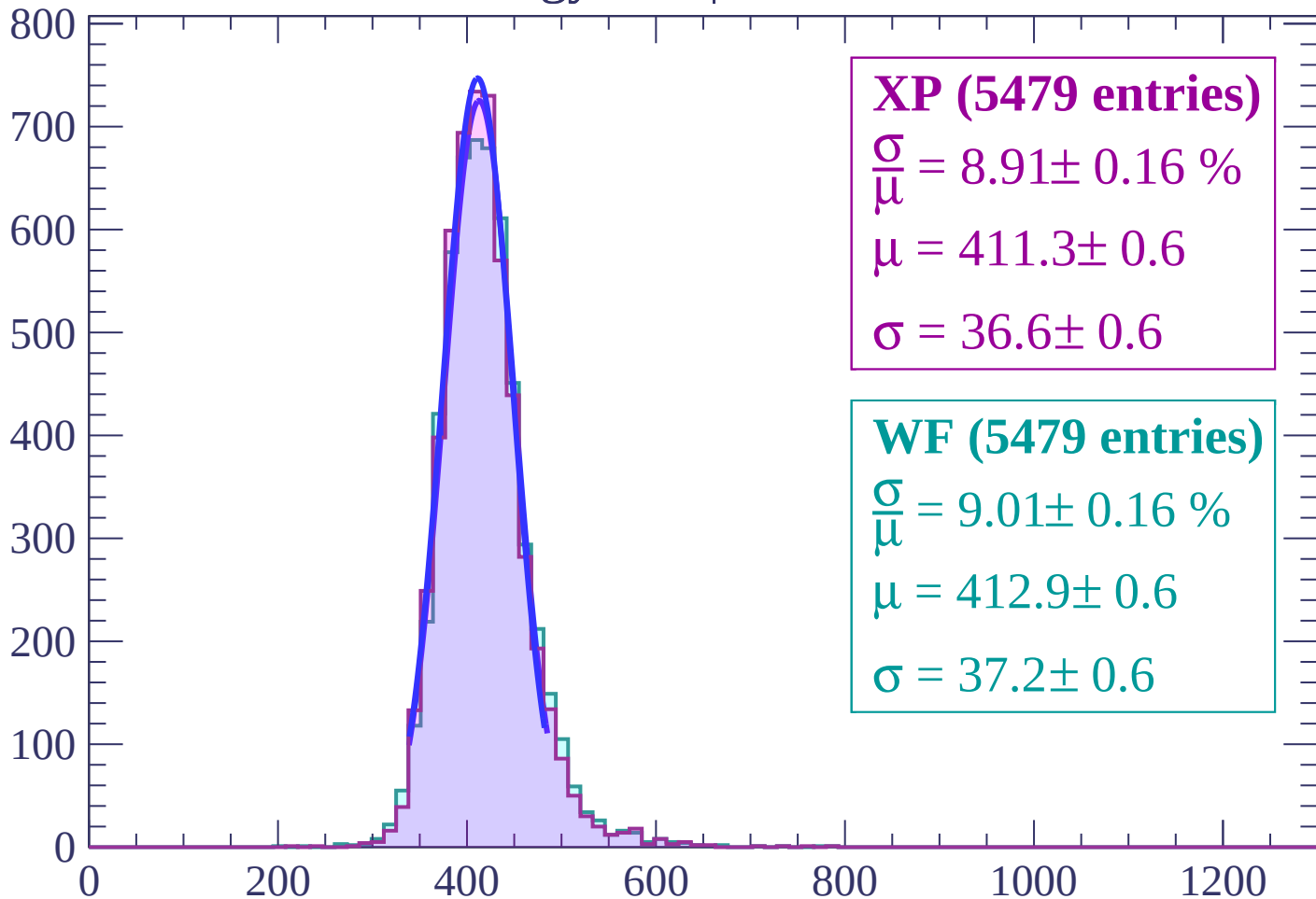
Energy loss | $-8 < \theta < -6$

Count



Energy loss | $-6 < \theta < -4$

Count



XP (5479 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.16 \%$$

$$\mu = 411.3 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

WF (5479 entries)

$$\frac{\sigma}{\mu} = 9.01 \pm 0.16 \%$$

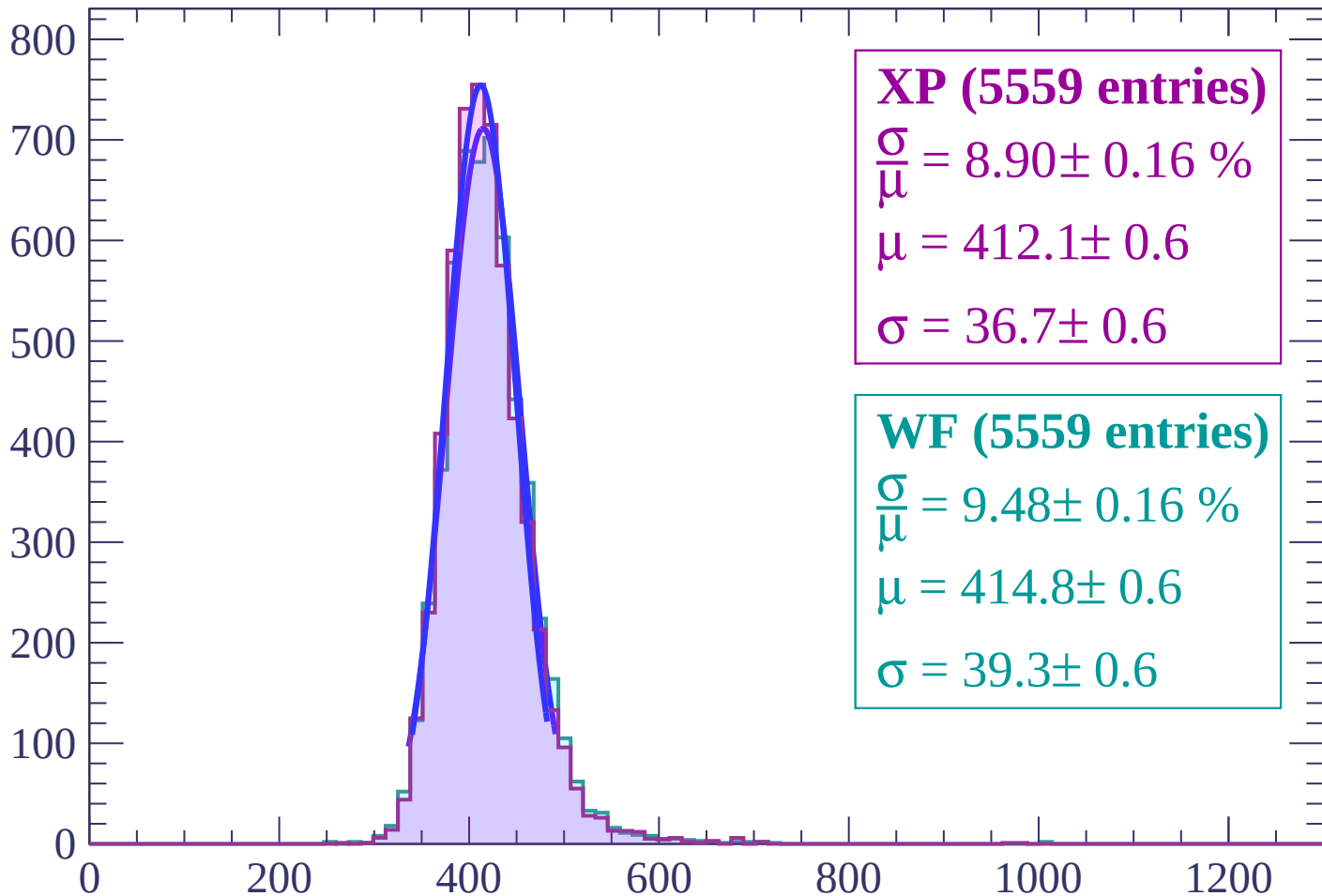
$$\mu = 412.9 \pm 0.6$$

$$\sigma = 37.2 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-4 < \theta < -2$

Count



XP (5559 entries)

$\frac{\sigma}{\mu} = 8.90 \pm 0.16 \%$

$\mu = 412.1 \pm 0.6$

$\sigma = 36.7 \pm 0.6$

WF (5559 entries)

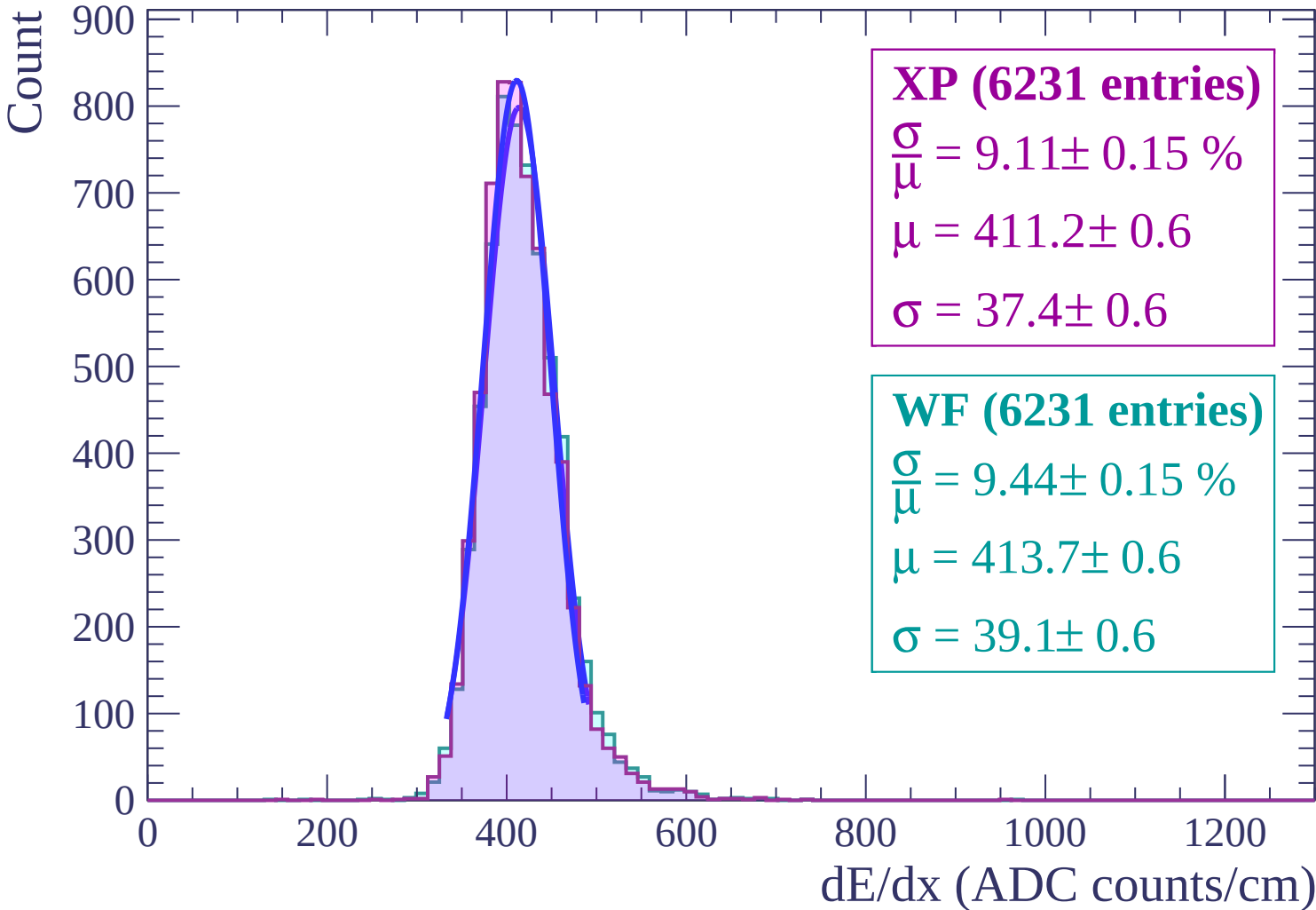
$\frac{\sigma}{\mu} = 9.48 \pm 0.16 \%$

$\mu = 414.8 \pm 0.6$

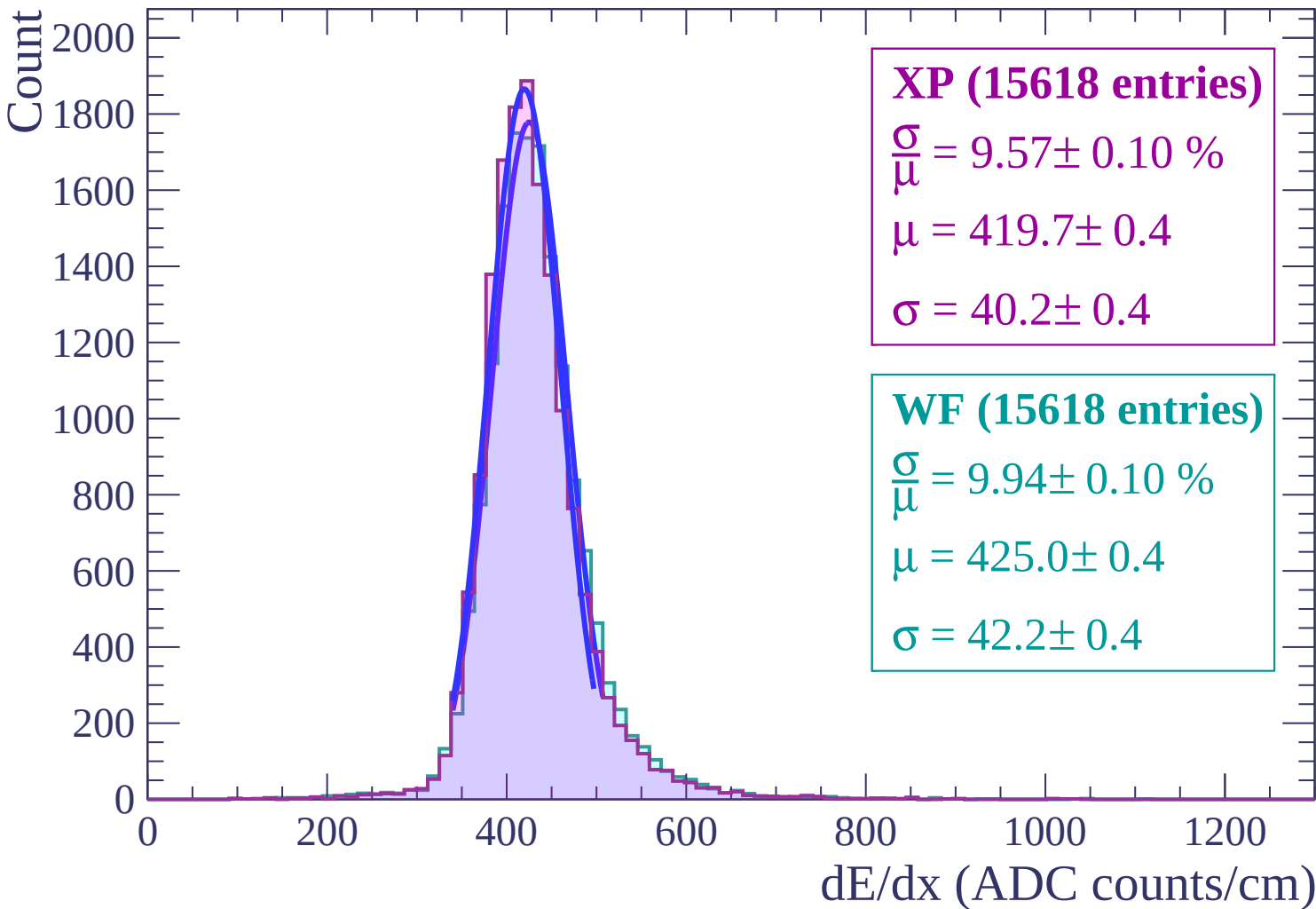
$\sigma = 39.3 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss $-2 < \theta < 0$

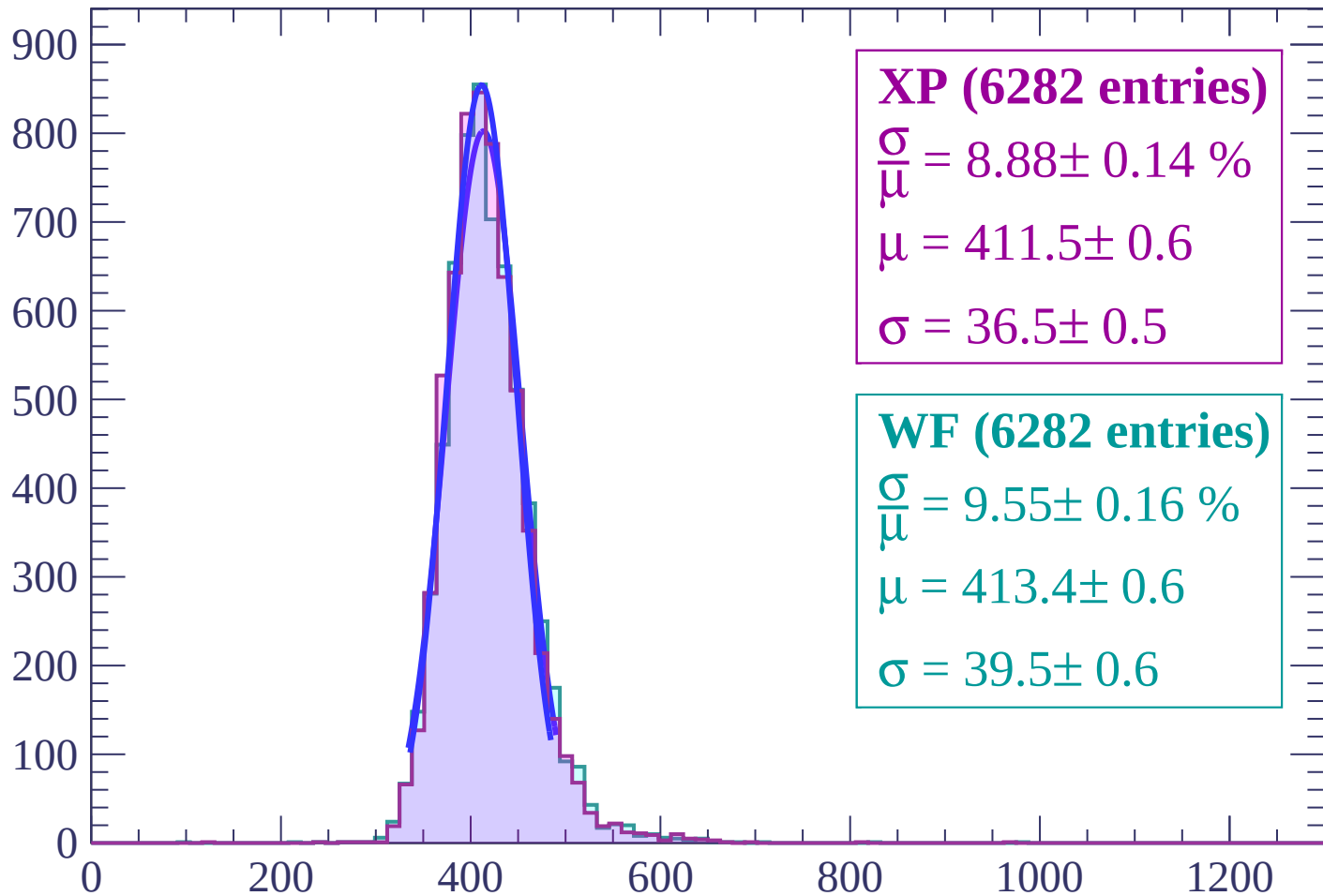


Energy loss | $0 < \theta < 2$



Energy loss | $2 < \theta < 4$

Count



XP (6282 entries)

$\frac{\sigma}{\mu} = 8.88 \pm 0.14 \%$

$\mu = 411.5 \pm 0.6$

$\sigma = 36.5 \pm 0.5$

WF (6282 entries)

$\frac{\sigma}{\mu} = 9.55 \pm 0.16 \%$

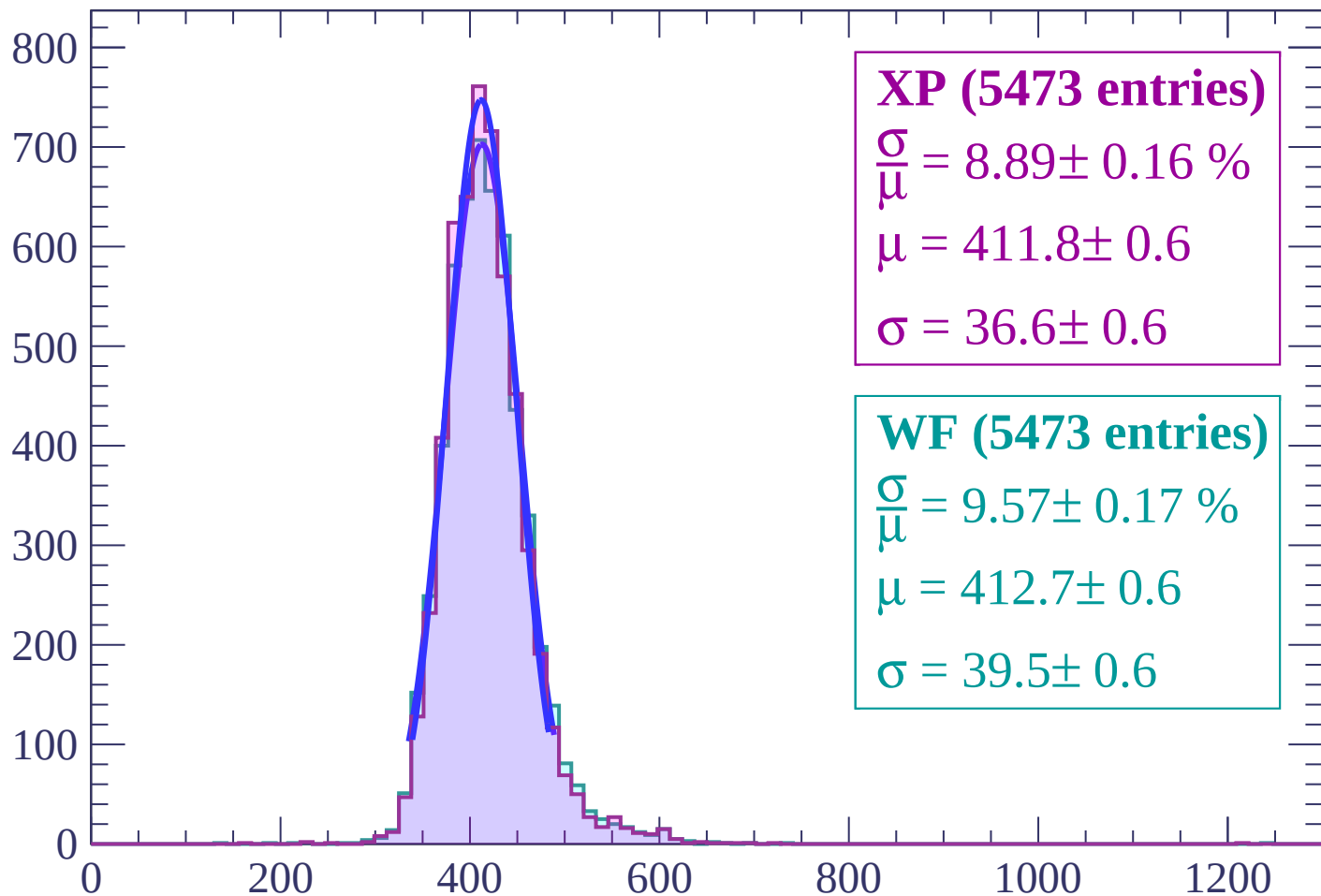
$\mu = 413.4 \pm 0.6$

$\sigma = 39.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $4 < \theta < 6$

Count



XP (5473 entries)

$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$

$\mu = 411.8 \pm 0.6$

$\sigma = 36.6 \pm 0.6$

WF (5473 entries)

$\frac{\sigma}{\mu} = 9.57 \pm 0.17 \%$

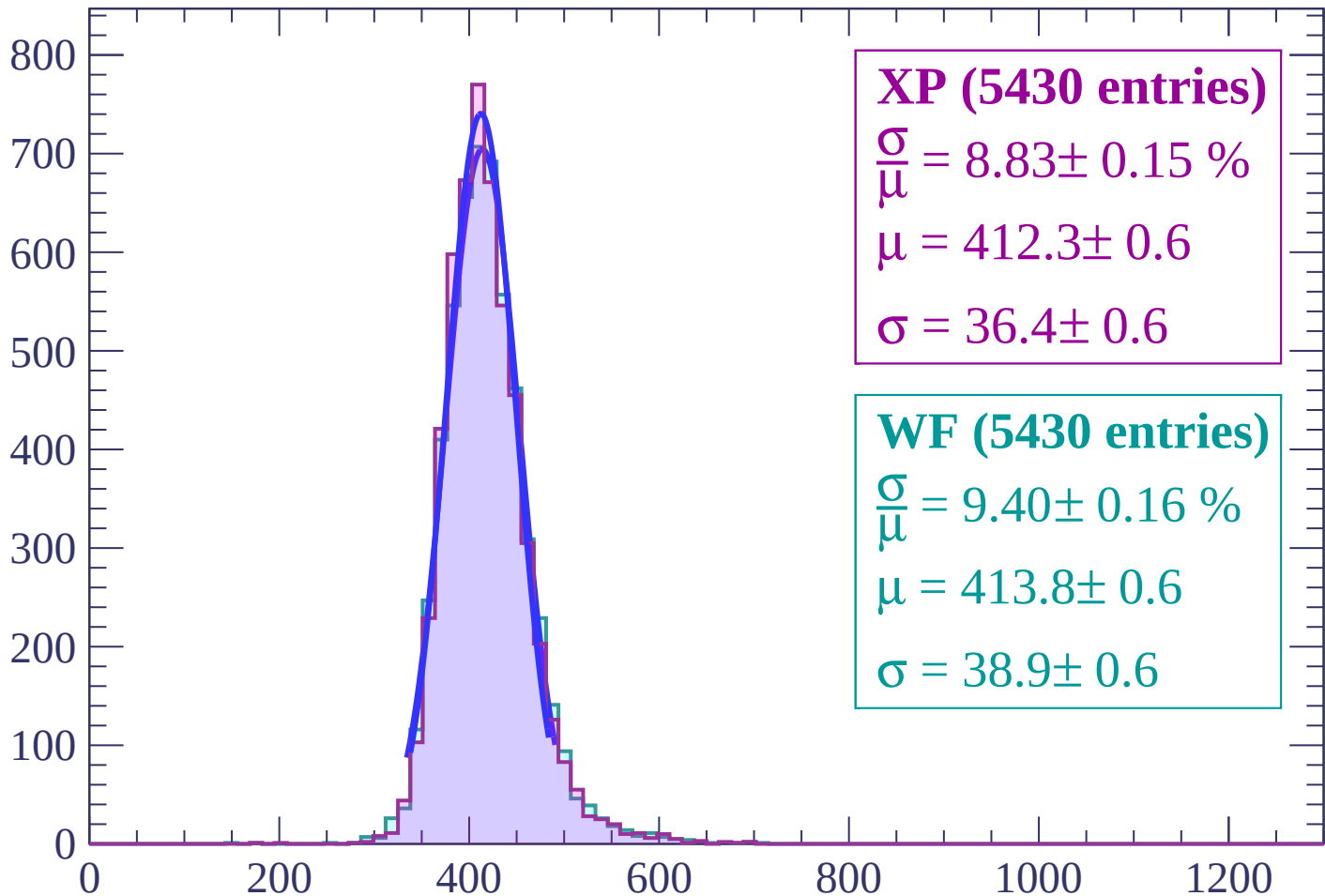
$\mu = 412.7 \pm 0.6$

$\sigma = 39.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

Count



XP (5430 entries)

$\frac{\sigma}{\mu} = 8.83 \pm 0.15 \%$

$\mu = 412.3 \pm 0.6$

$\sigma = 36.4 \pm 0.6$

WF (5430 entries)

$\frac{\sigma}{\mu} = 9.40 \pm 0.16 \%$

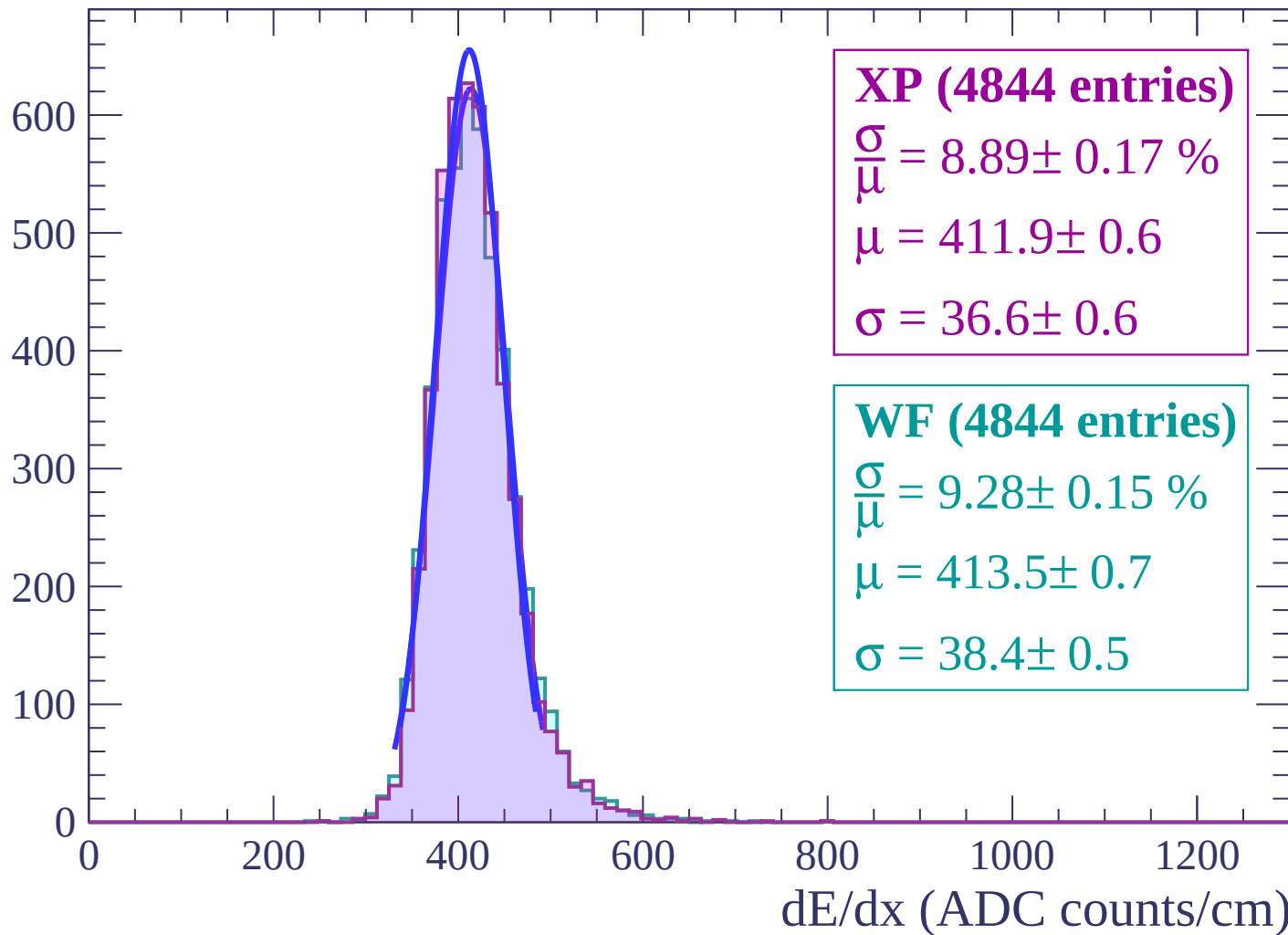
$\mu = 413.8 \pm 0.6$

$\sigma = 38.9 \pm 0.6$

dE/dx (ADC counts/cm)

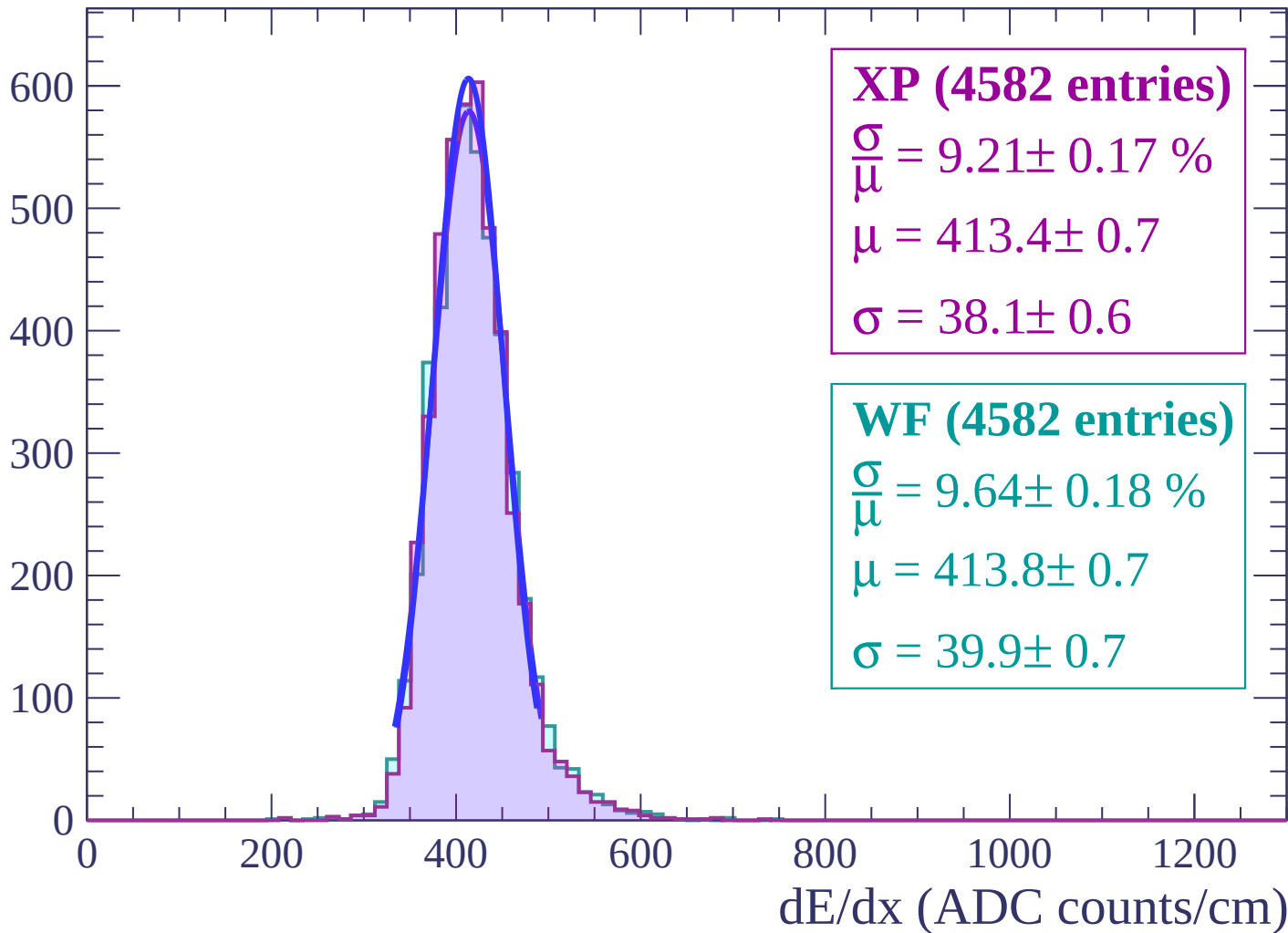
Energy loss | $8 < \theta < 10$

Count



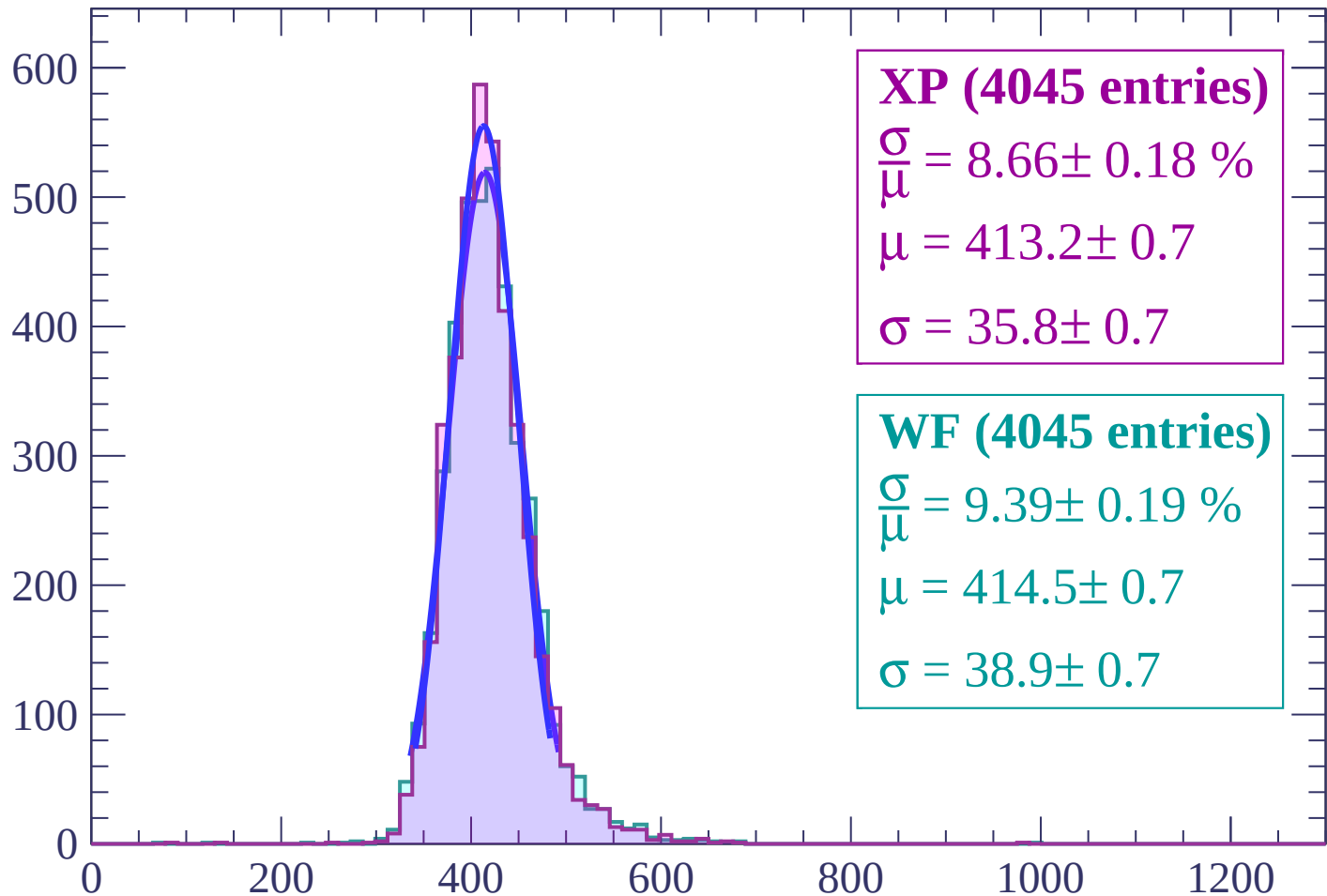
Energy loss | $10 < \theta < 12$

Count



Energy loss | $12 < \theta < 14$

Count



XP (4045 entries)

$$\frac{\sigma}{\mu} = 8.66 \pm 0.18 \%$$

$$\mu = 413.2 \pm 0.7$$

$$\sigma = 35.8 \pm 0.7$$

WF (4045 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.19 \%$$

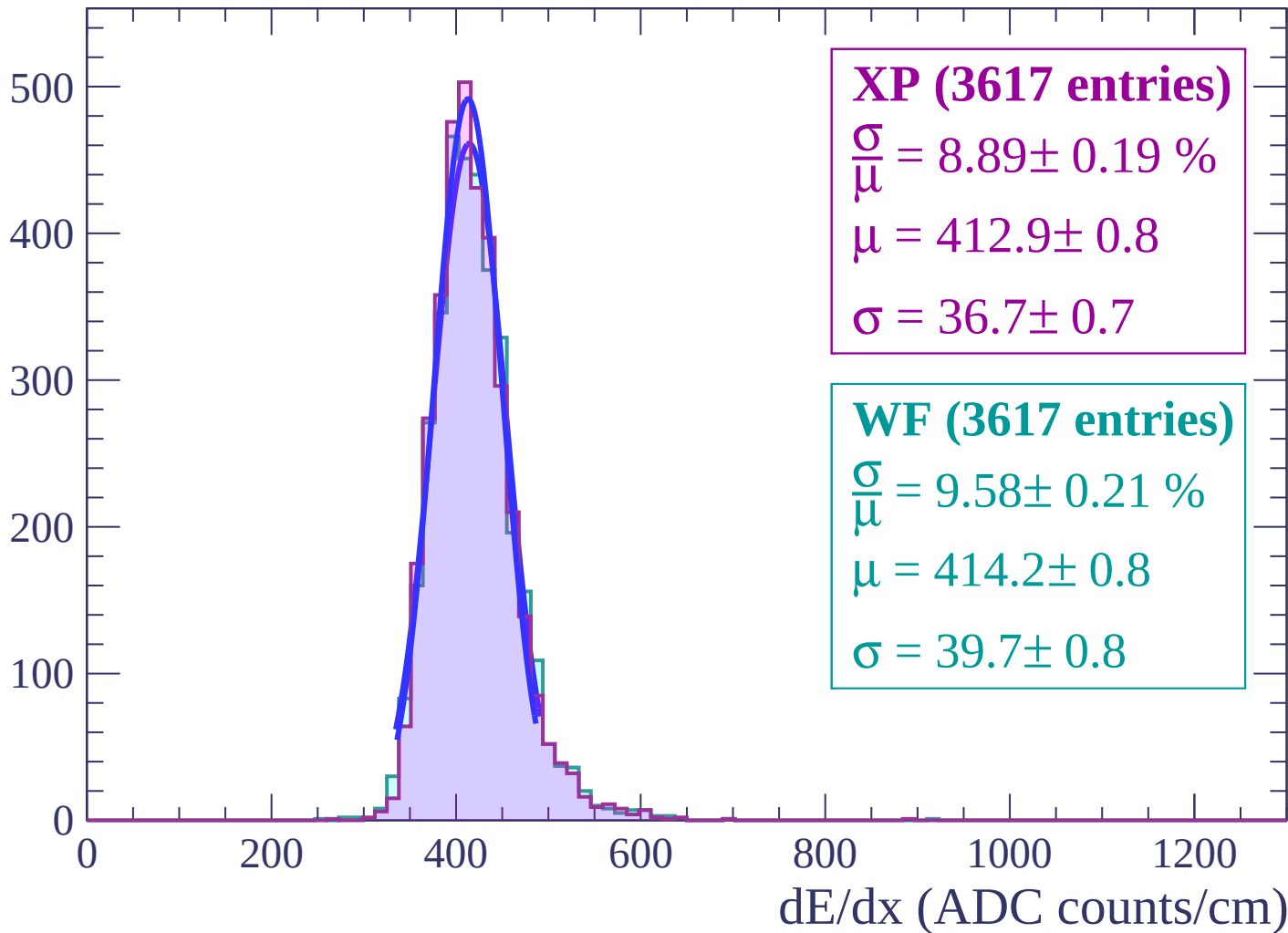
$$\mu = 414.5 \pm 0.7$$

$$\sigma = 38.9 \pm 0.7$$

dE/dx (ADC counts/cm)

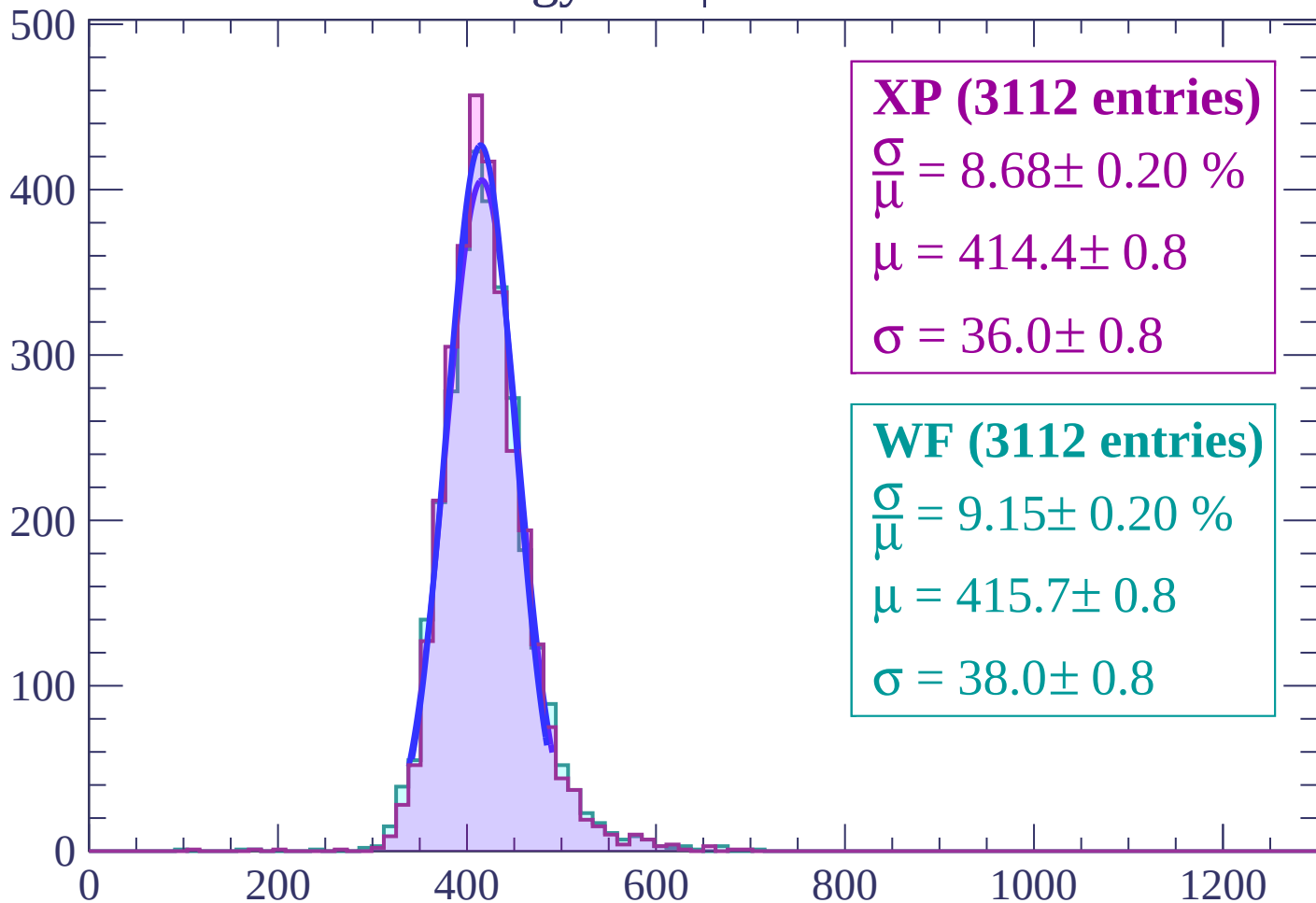
Energy loss | $14 < \theta < 16$

Count



Energy loss | $16 < \theta < 18$

Count



XP (3112 entries)

$\frac{\sigma}{\mu} = 8.68 \pm 0.20 \%$

$\mu = 414.4 \pm 0.8$

$\sigma = 36.0 \pm 0.8$

WF (3112 entries)

$\frac{\sigma}{\mu} = 9.15 \pm 0.20 \%$

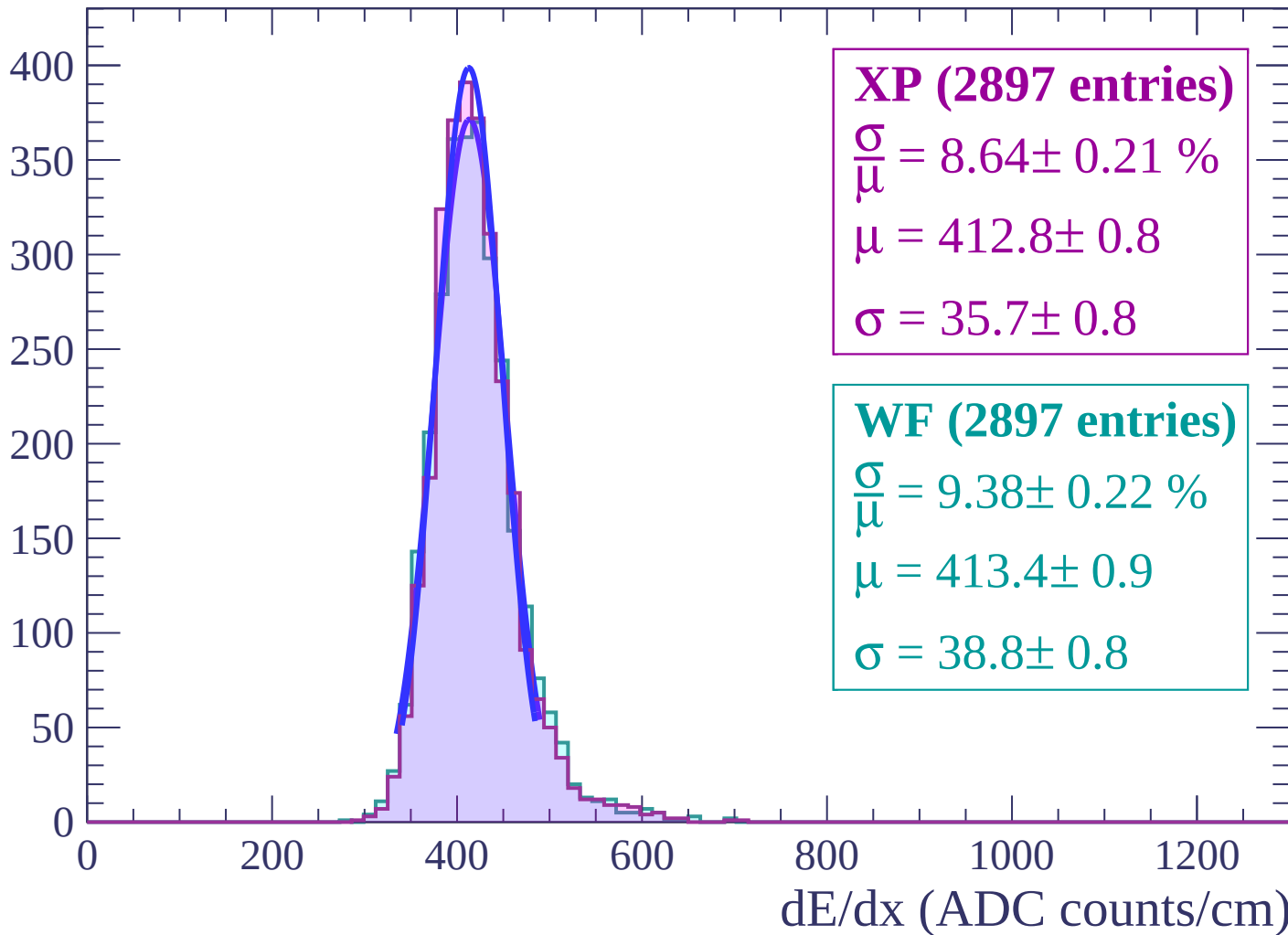
$\mu = 415.7 \pm 0.8$

$\sigma = 38.0 \pm 0.8$

dE/dx (ADC counts/cm)

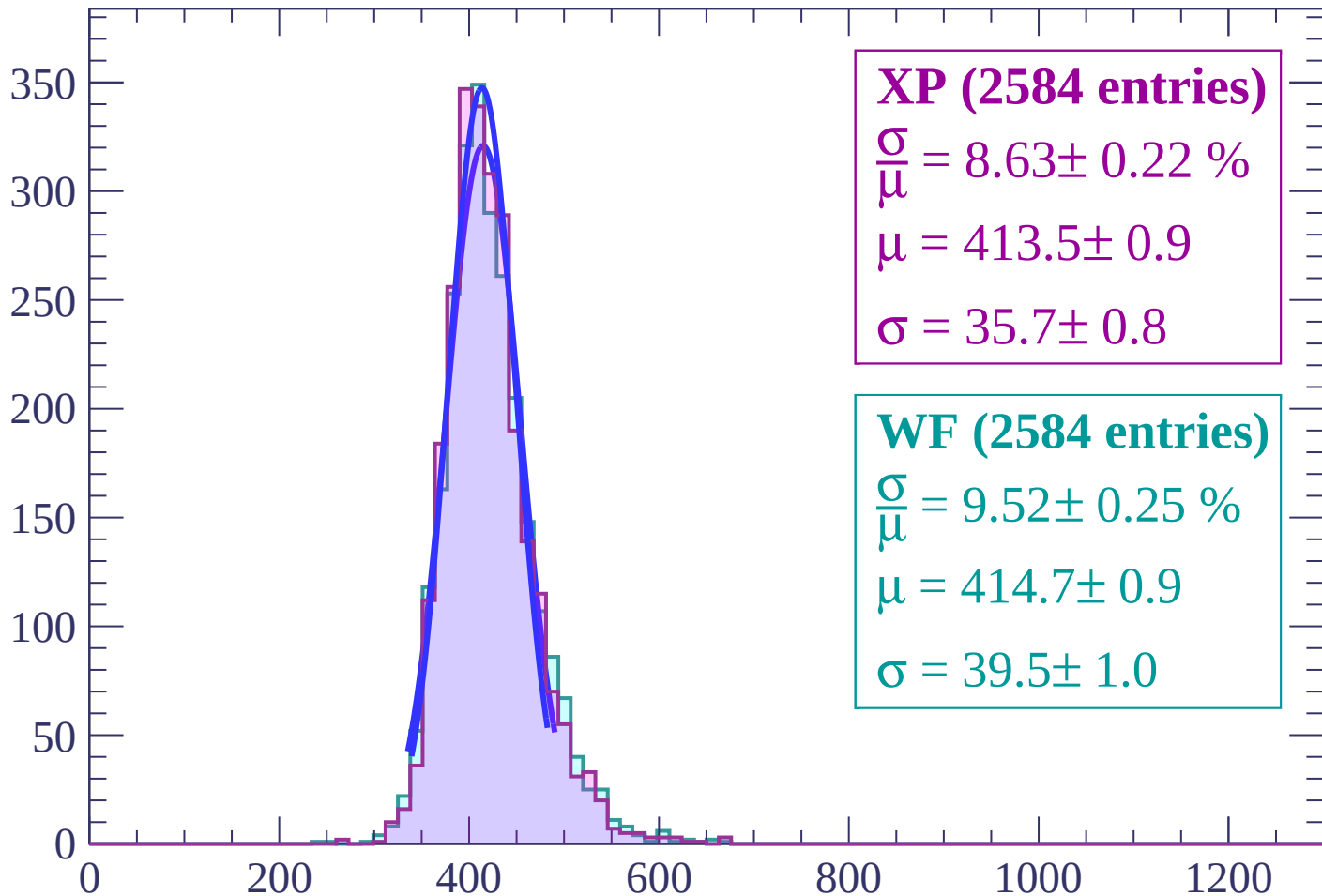
Energy loss | $18 < \theta < 20$

Count



Energy loss | $20 < \theta < 22$

Count



XP (2584 entries)

$\frac{\sigma}{\mu} = 8.63 \pm 0.22 \%$

$\mu = 413.5 \pm 0.9$

$\sigma = 35.7 \pm 0.8$

WF (2584 entries)

$\frac{\sigma}{\mu} = 9.52 \pm 0.25 \%$

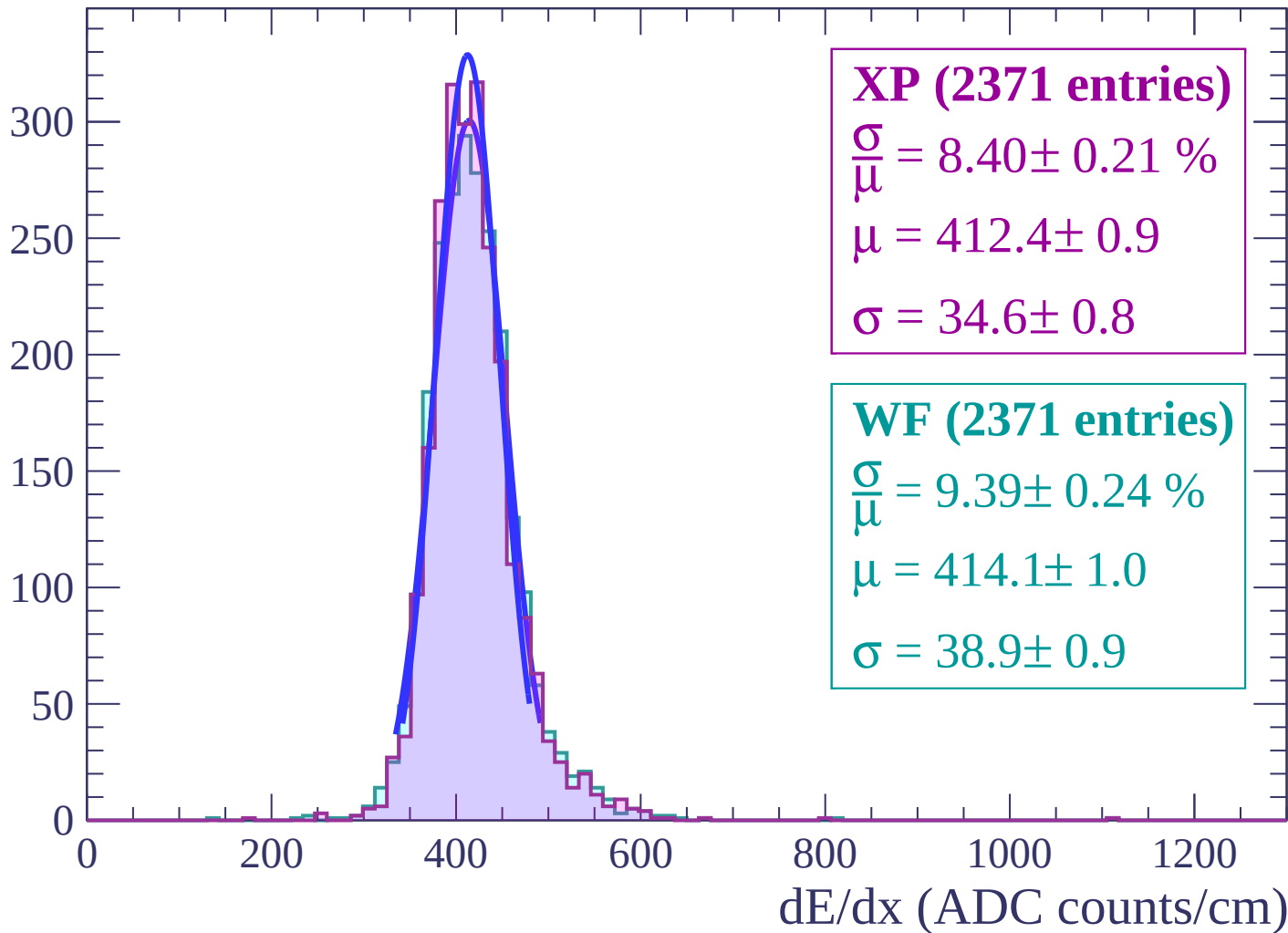
$\mu = 414.7 \pm 0.9$

$\sigma = 39.5 \pm 1.0$

dE/dx (ADC counts/cm)

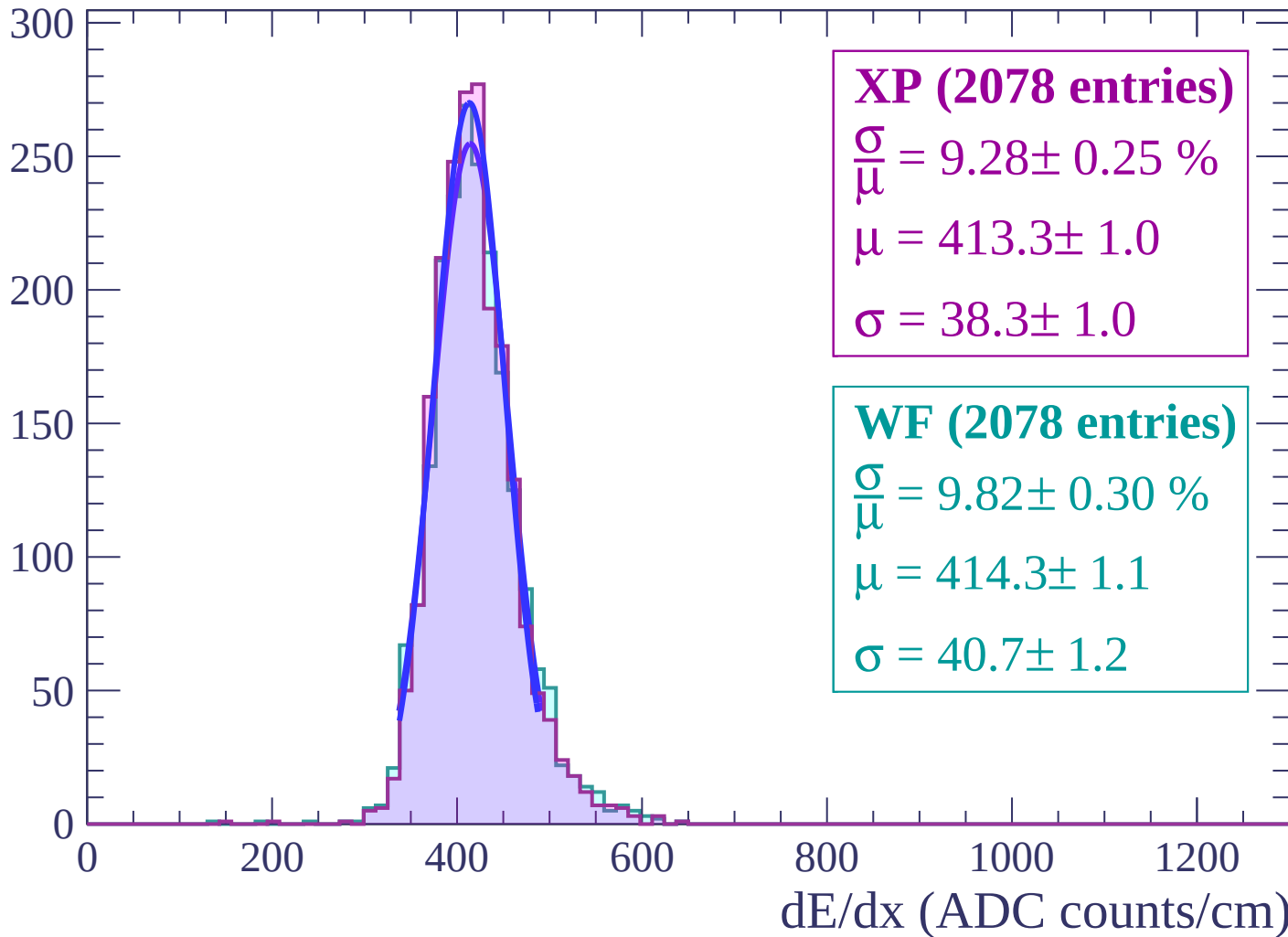
Energy loss | $22 < \theta < 24$

Count



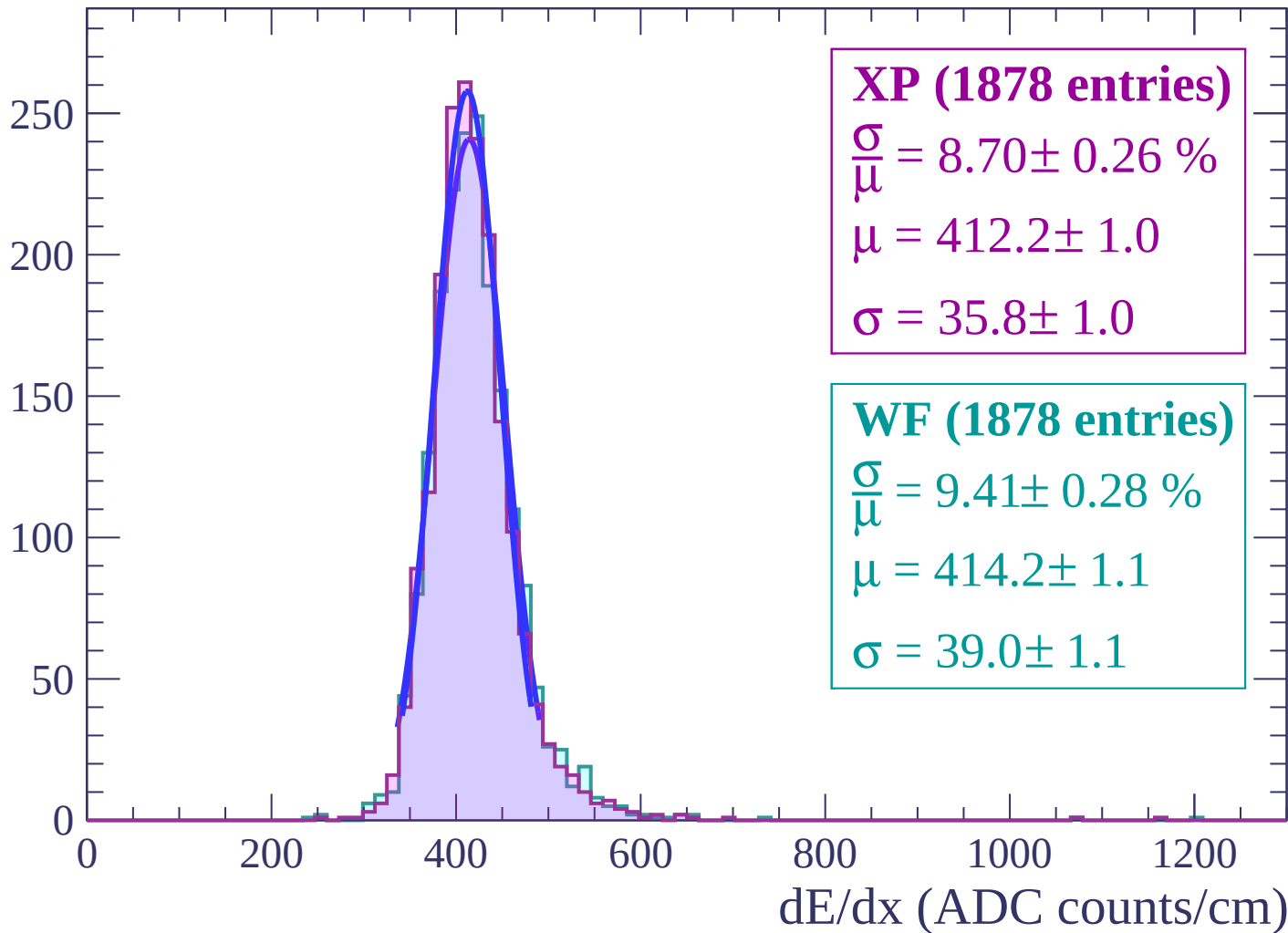
Energy loss | $24 < \theta < 26$

Count



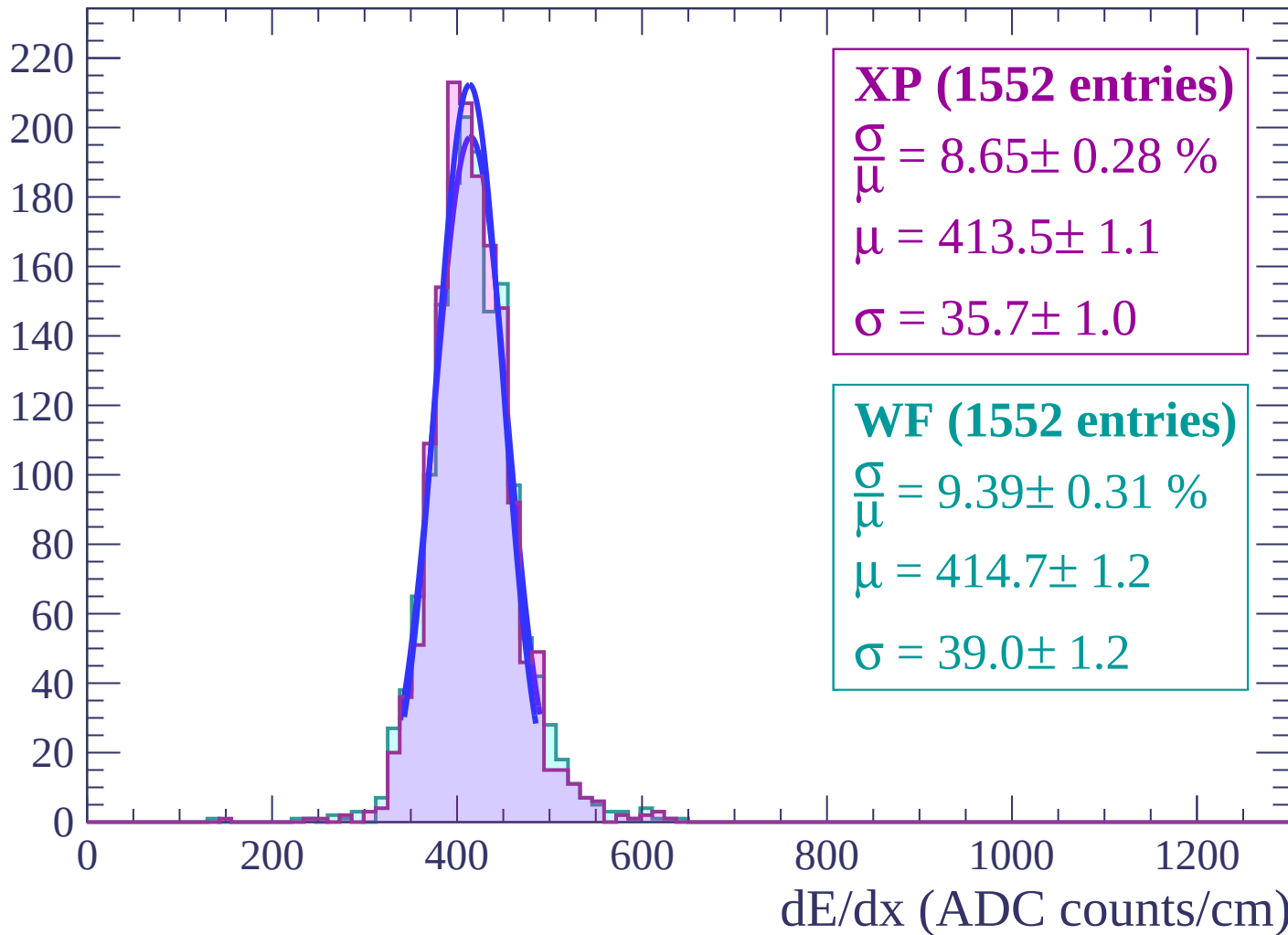
Energy loss | $26 < \theta < 28$

Count



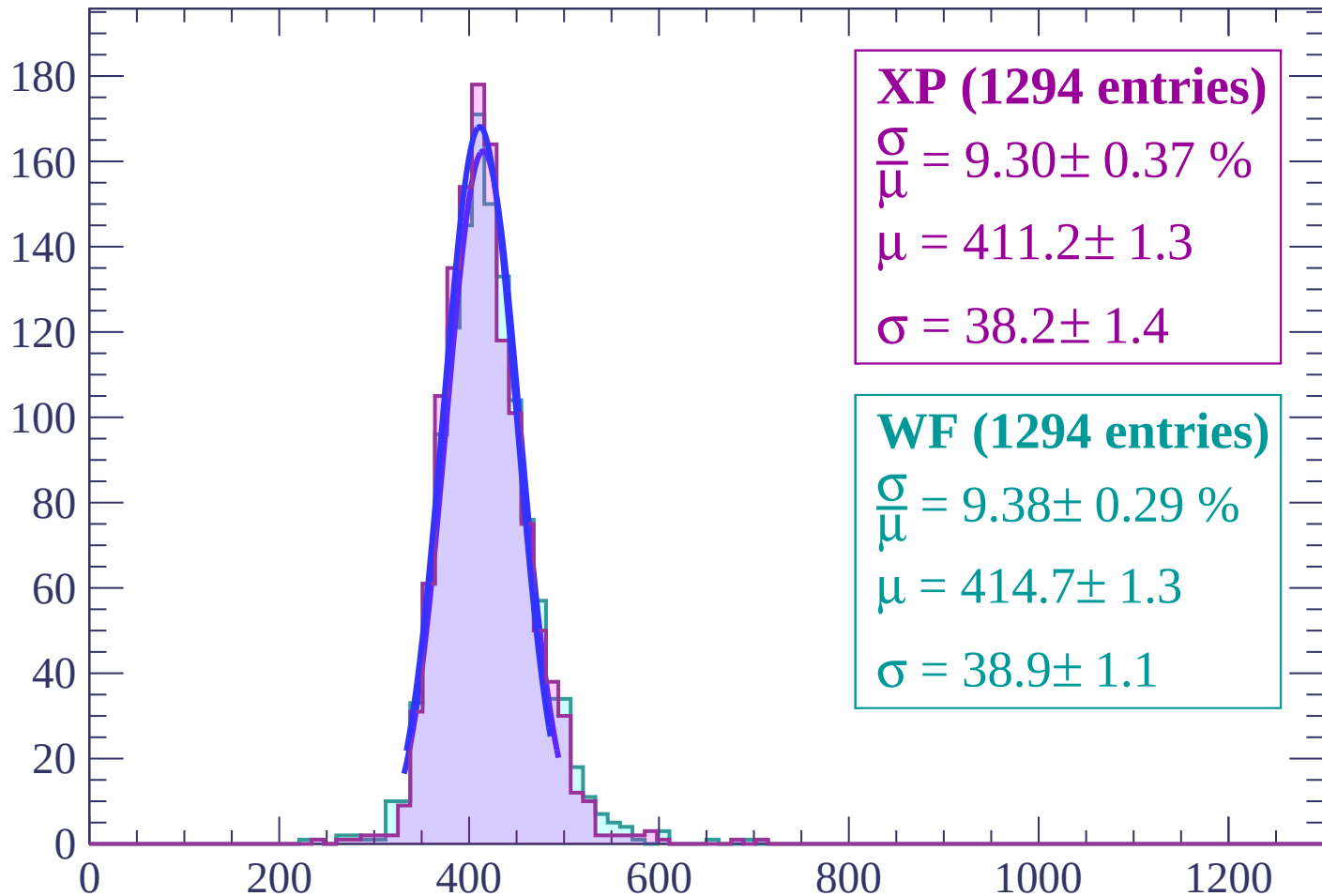
Energy loss | $28 < \theta < 30$

Count



Energy loss | $30 < \theta < 32$

Count



XP (1294 entries)

$\frac{\sigma}{\mu} = 9.30 \pm 0.37 \%$

$\mu = 411.2 \pm 1.3$

$\sigma = 38.2 \pm 1.4$

WF (1294 entries)

$\frac{\sigma}{\mu} = 9.38 \pm 0.29 \%$

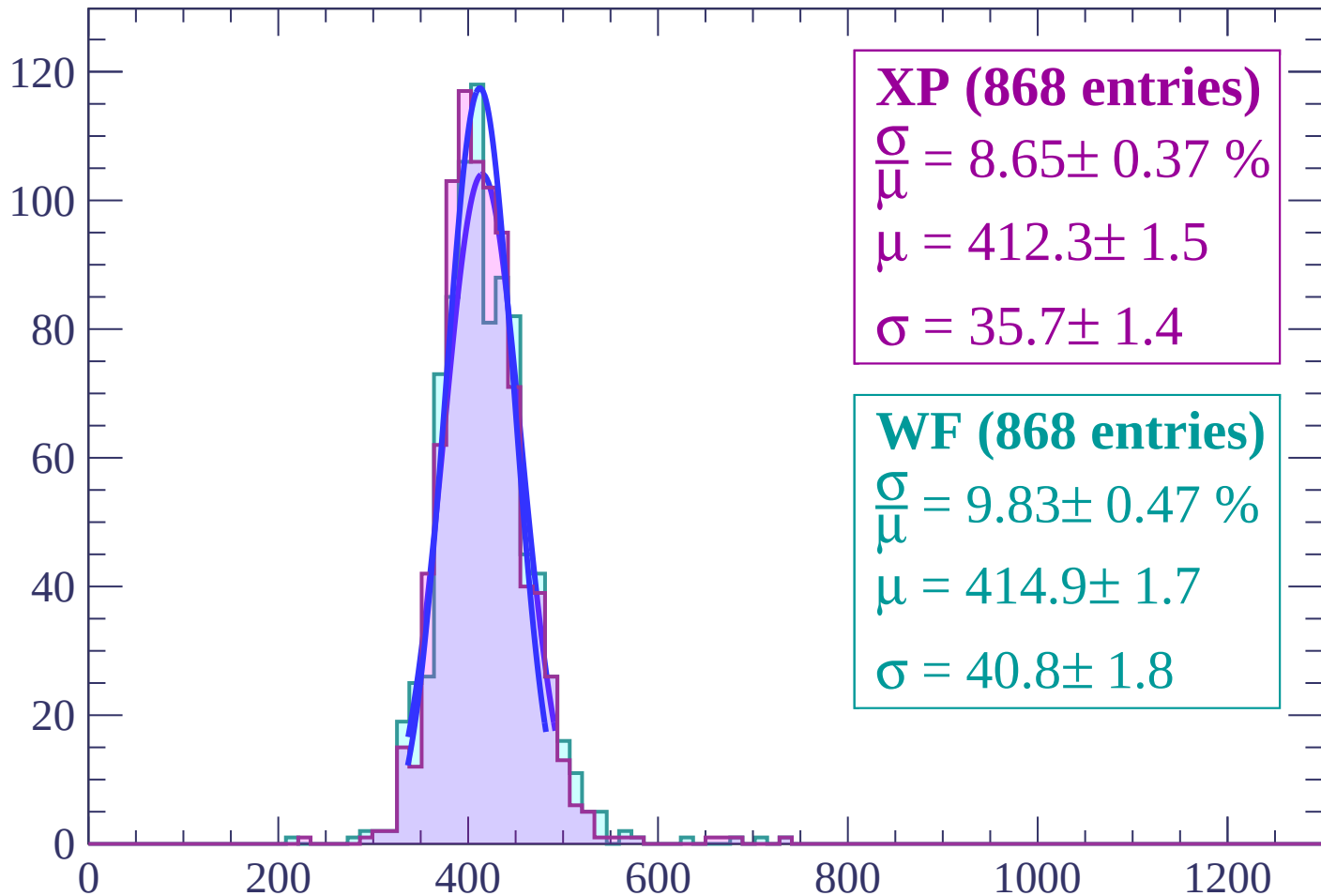
$\mu = 414.7 \pm 1.3$

$\sigma = 38.9 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $32 < \theta < 34$

Count



XP (868 entries)

$\frac{\sigma}{\mu} = 8.65 \pm 0.37 \%$

$\mu = 412.3 \pm 1.5$

$\sigma = 35.7 \pm 1.4$

WF (868 entries)

$\frac{\sigma}{\mu} = 9.83 \pm 0.47 \%$

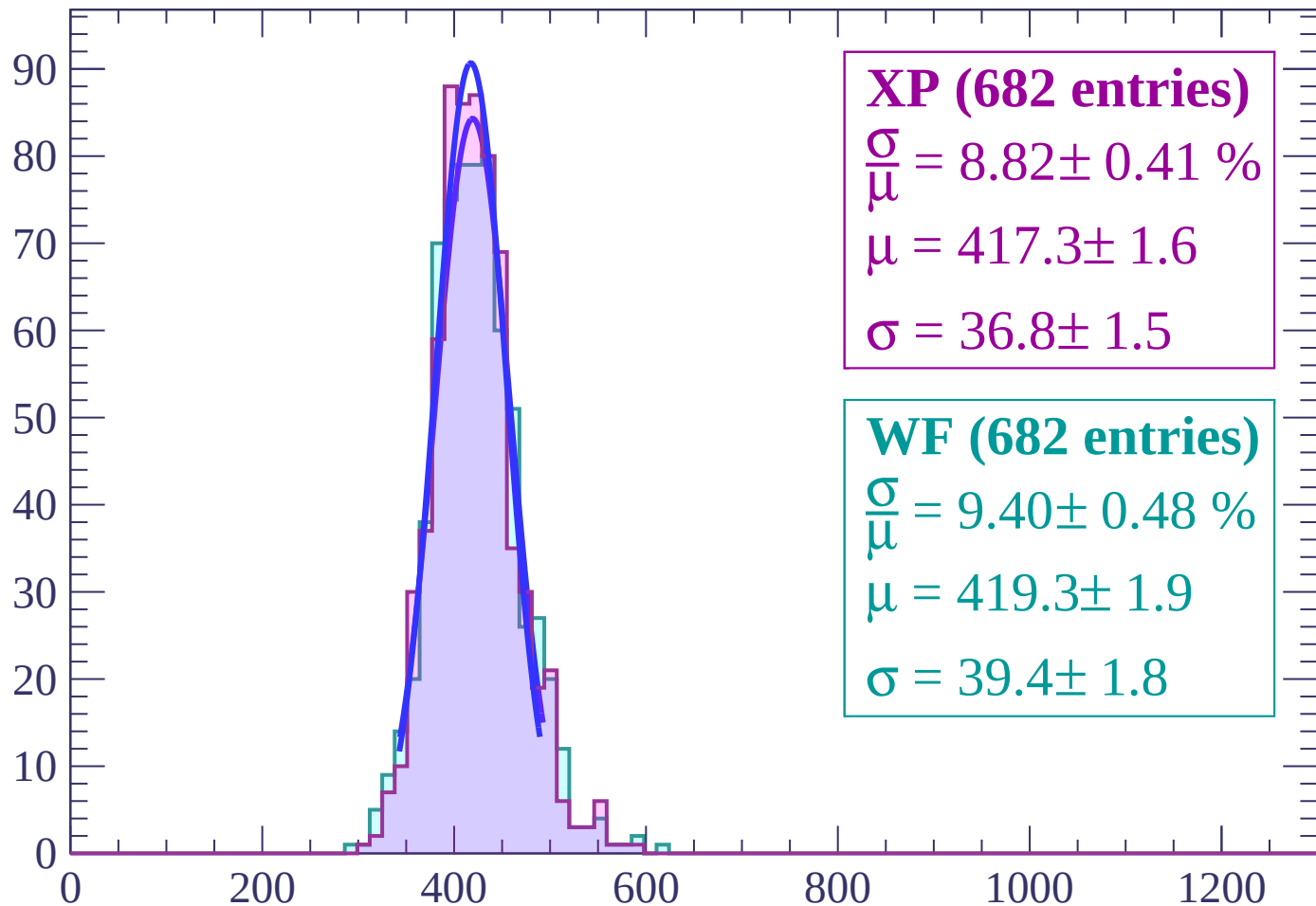
$\mu = 414.9 \pm 1.7$

$\sigma = 40.8 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $34 < \theta < 36$

Count



XP (682 entries)

$\frac{\sigma}{\mu} = 8.82 \pm 0.41 \%$

$\mu = 417.3 \pm 1.6$

$\sigma = 36.8 \pm 1.5$

WF (682 entries)

$\frac{\sigma}{\mu} = 9.40 \pm 0.48 \%$

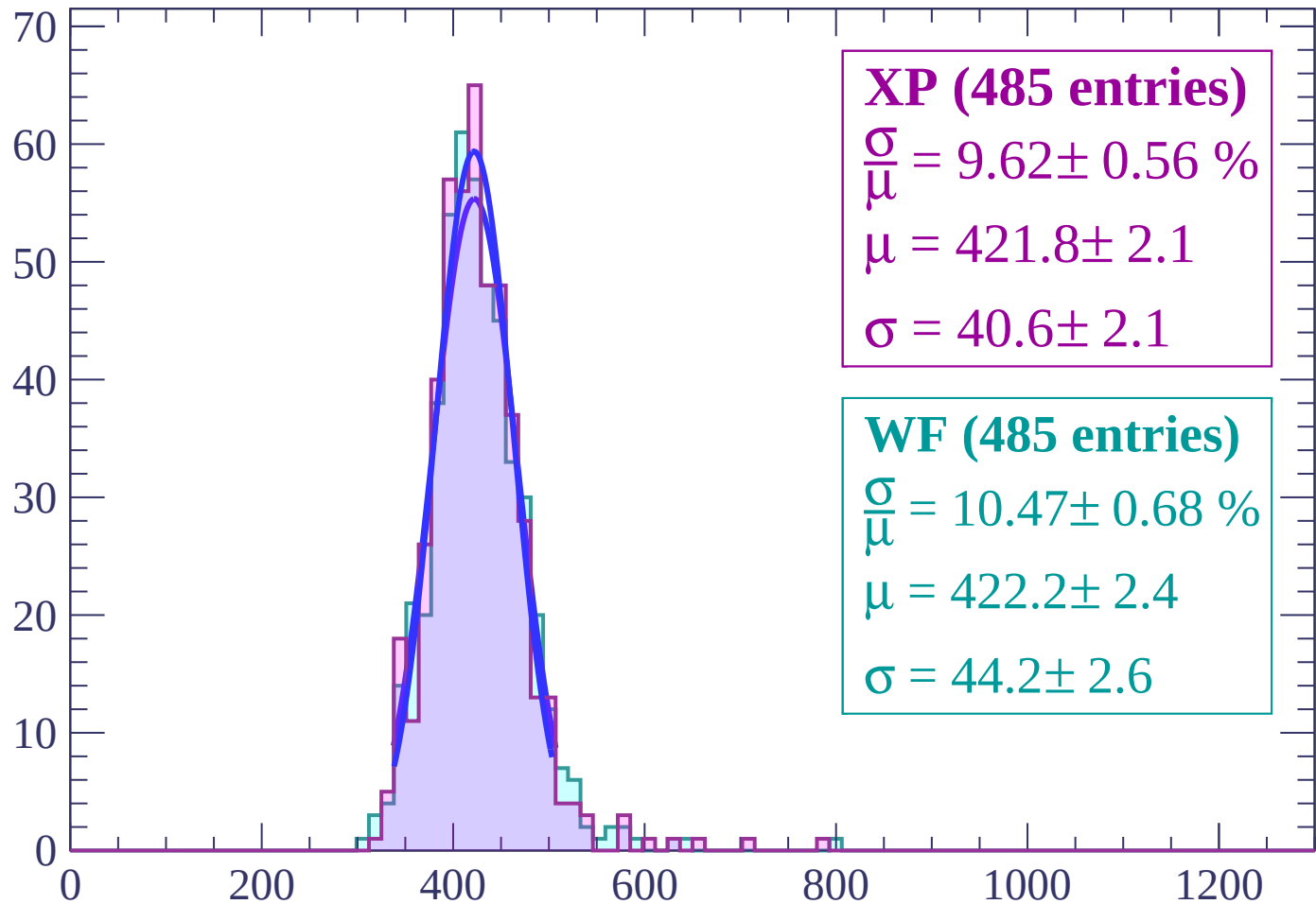
$\mu = 419.3 \pm 1.9$

$\sigma = 39.4 \pm 1.8$

dE/dx (ADC counts/cm)

Energy loss | $36 < \theta < 38$

Count



XP (485 entries)

$\frac{\sigma}{\mu} = 9.62 \pm 0.56 \%$

$\mu = 421.8 \pm 2.1$

$\sigma = 40.6 \pm 2.1$

WF (485 entries)

$\frac{\sigma}{\mu} = 10.47 \pm 0.68 \%$

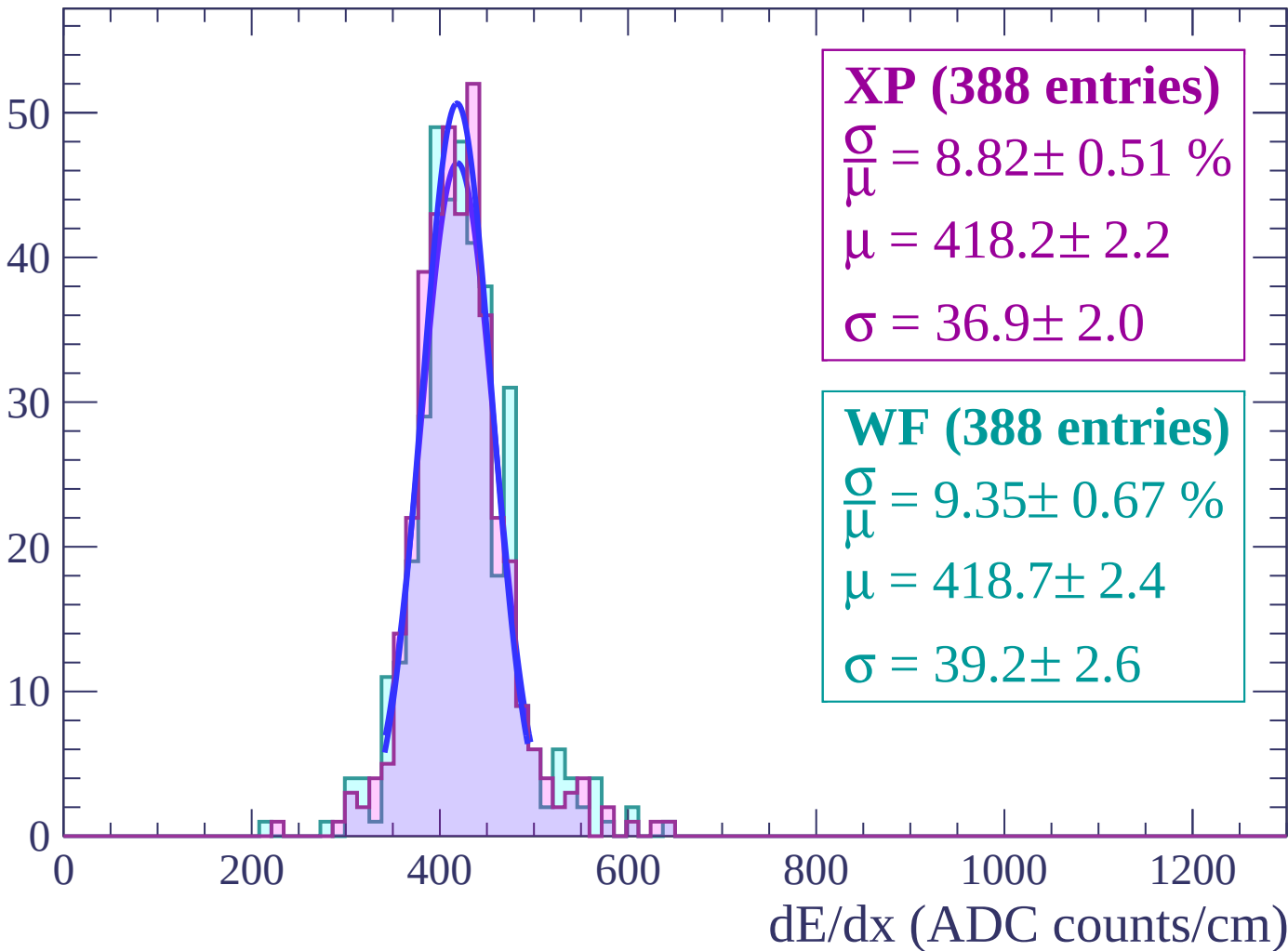
$\mu = 422.2 \pm 2.4$

$\sigma = 44.2 \pm 2.6$

dE/dx (ADC counts/cm)

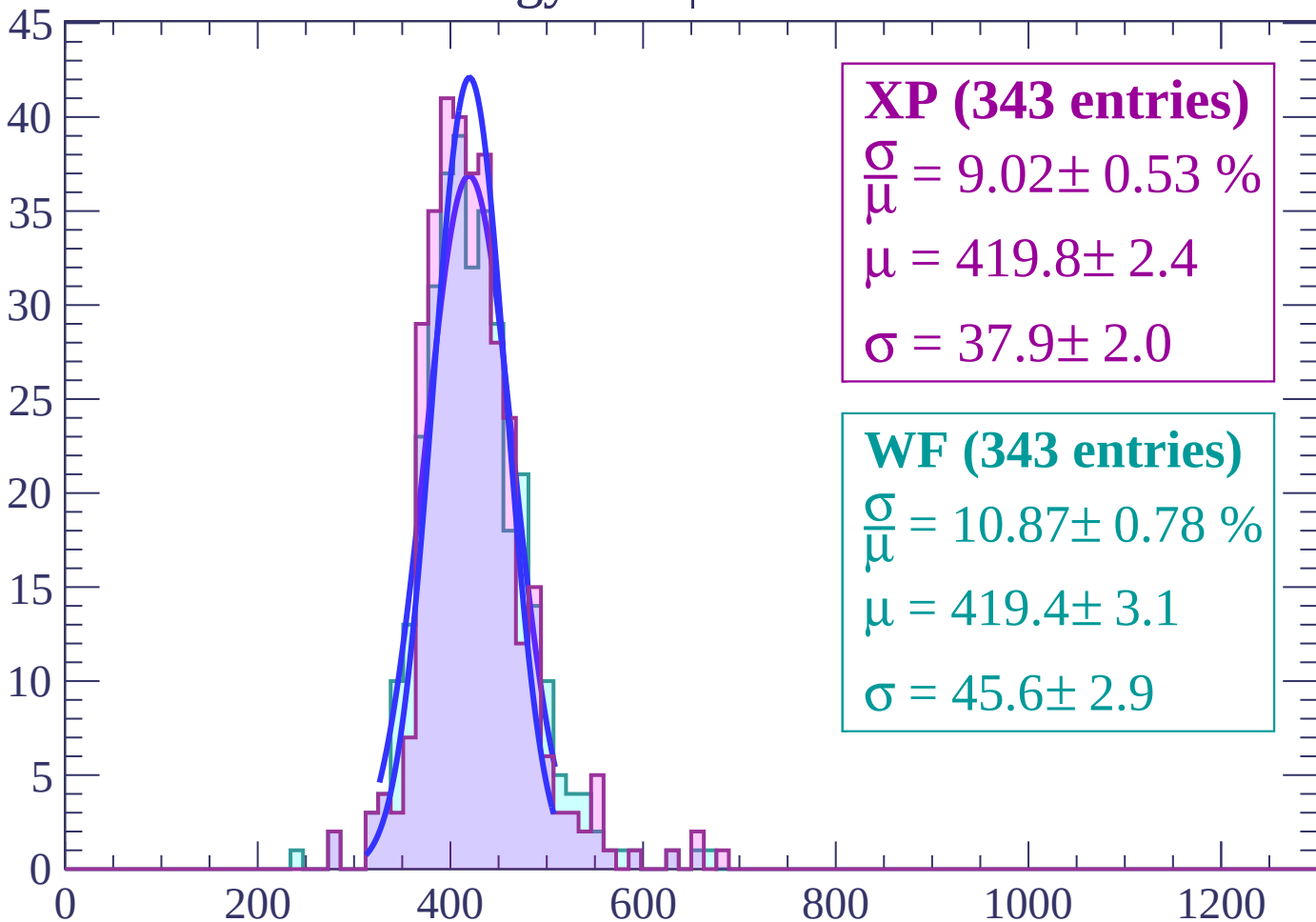
Energy loss | $38 < \theta < 40$

Count



Energy loss | $40 < \theta < 42$

Count



XP (343 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.53 \%$

$\mu = 419.8 \pm 2.4$

$\sigma = 37.9 \pm 2.0$

WF (343 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.78 \%$

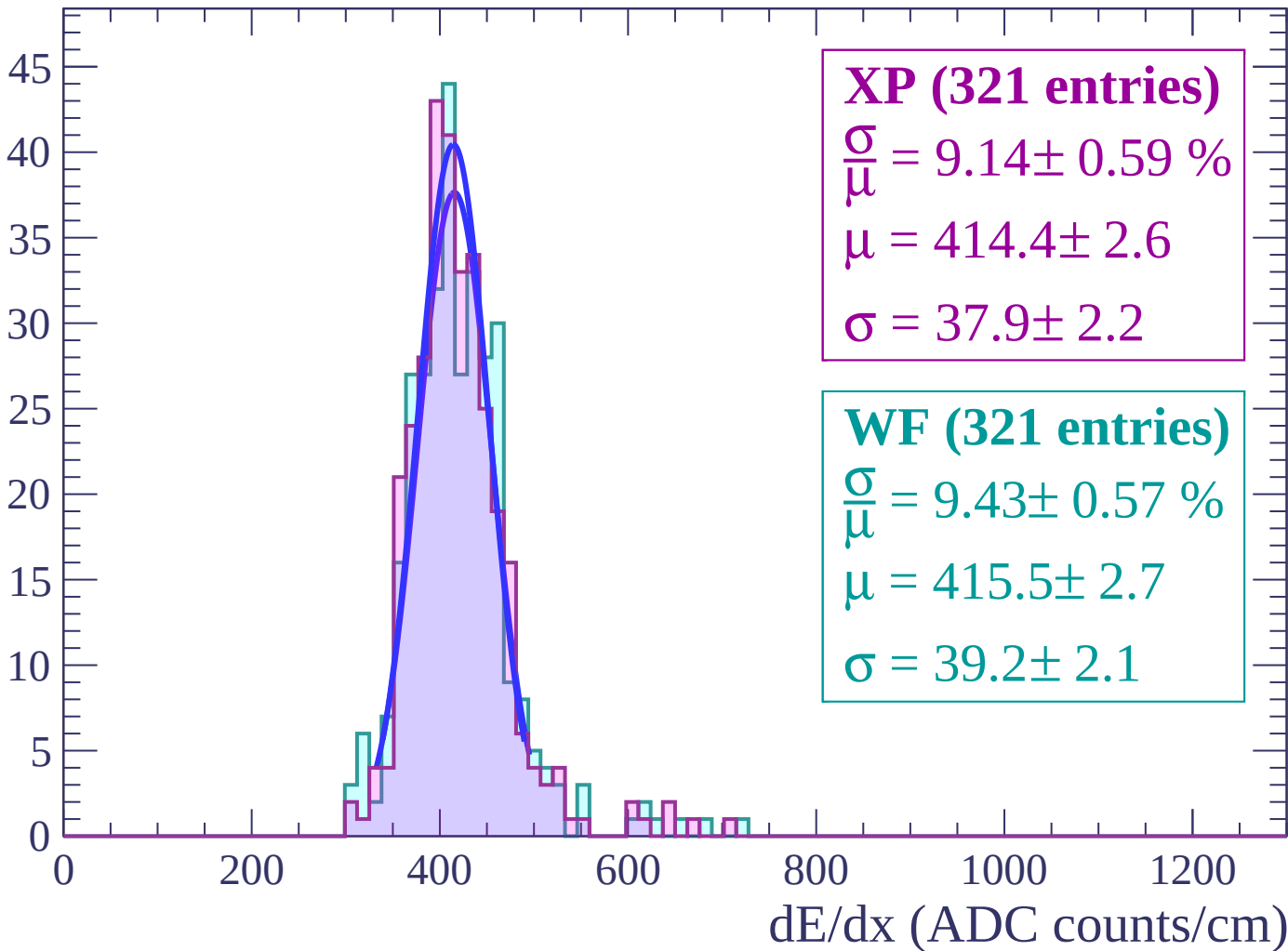
$\mu = 419.4 \pm 3.1$

$\sigma = 45.6 \pm 2.9$

dE/dx (ADC counts/cm)

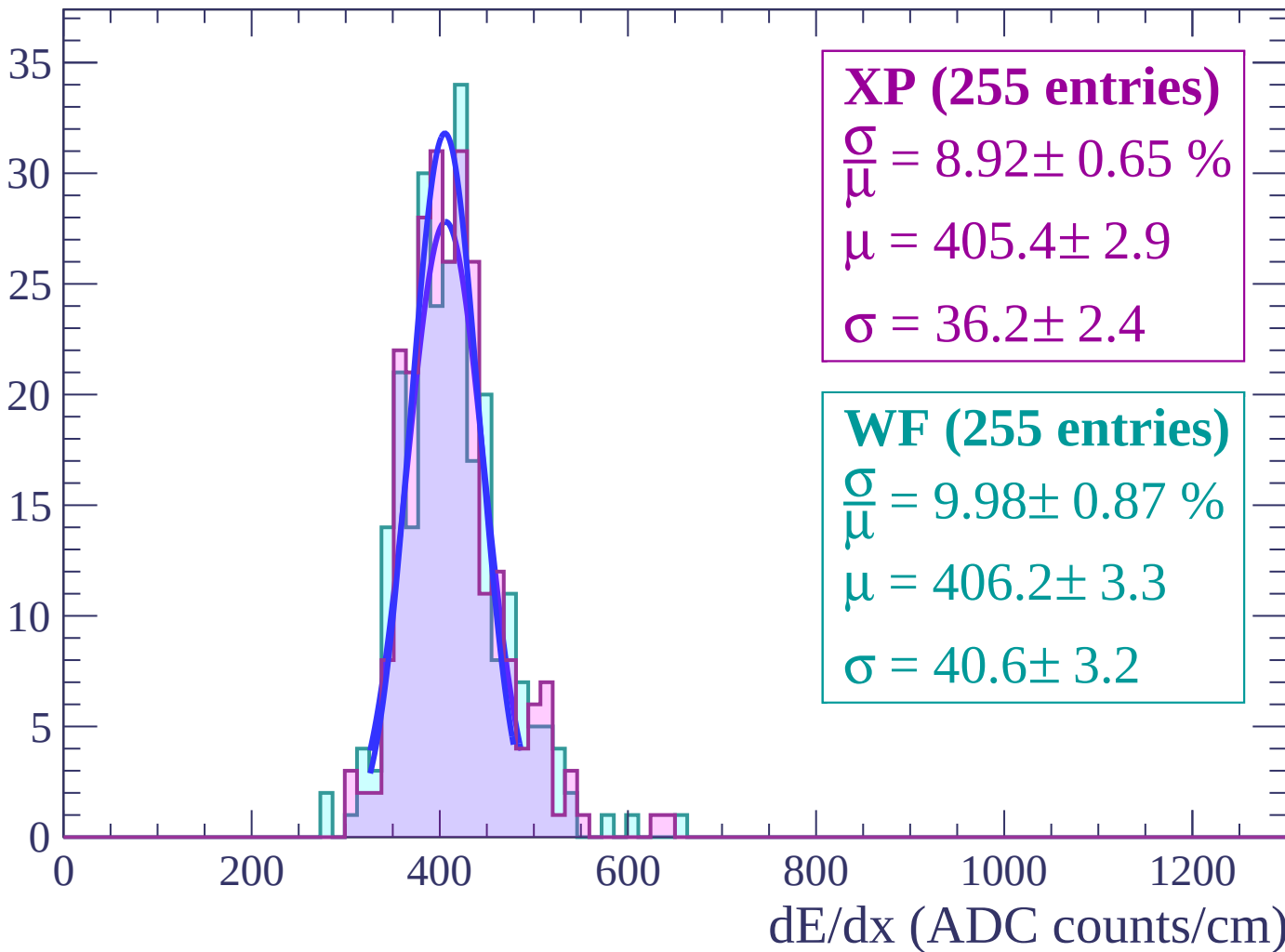
Energy loss | $42 < \theta < 44$

Count



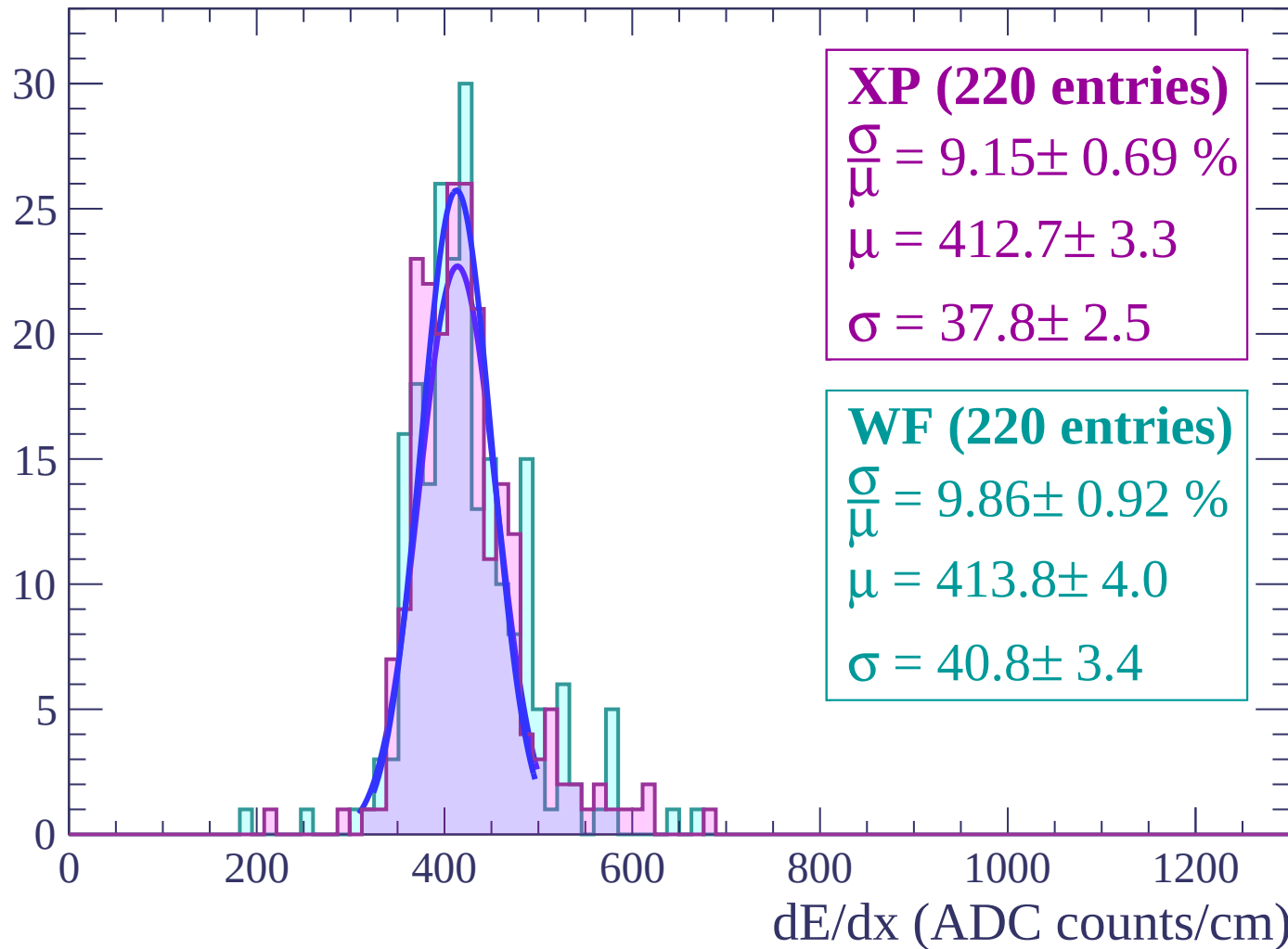
Energy loss | $44 < \theta < 46$

Count



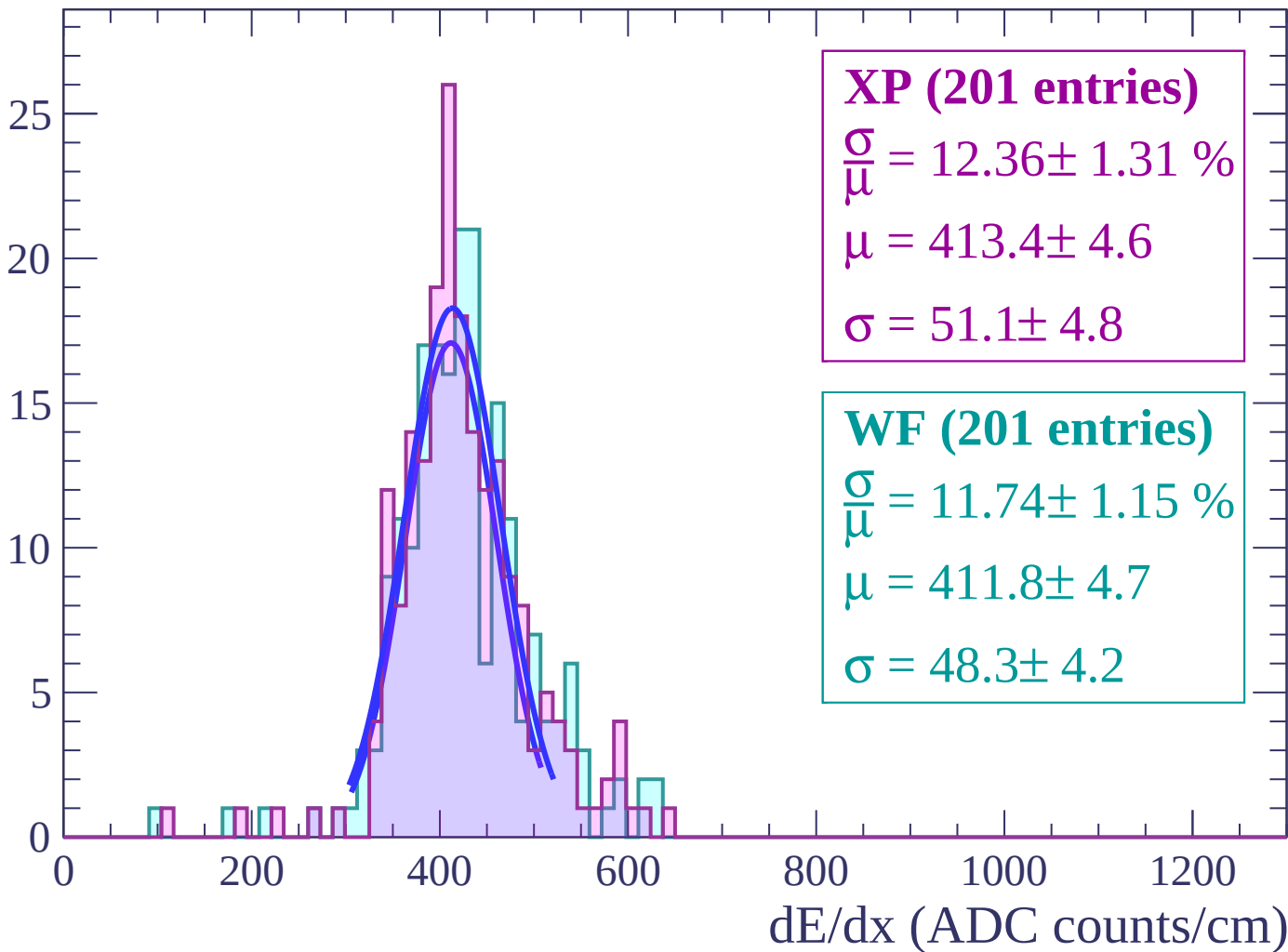
Energy loss | $46 < \theta < 48$

Count



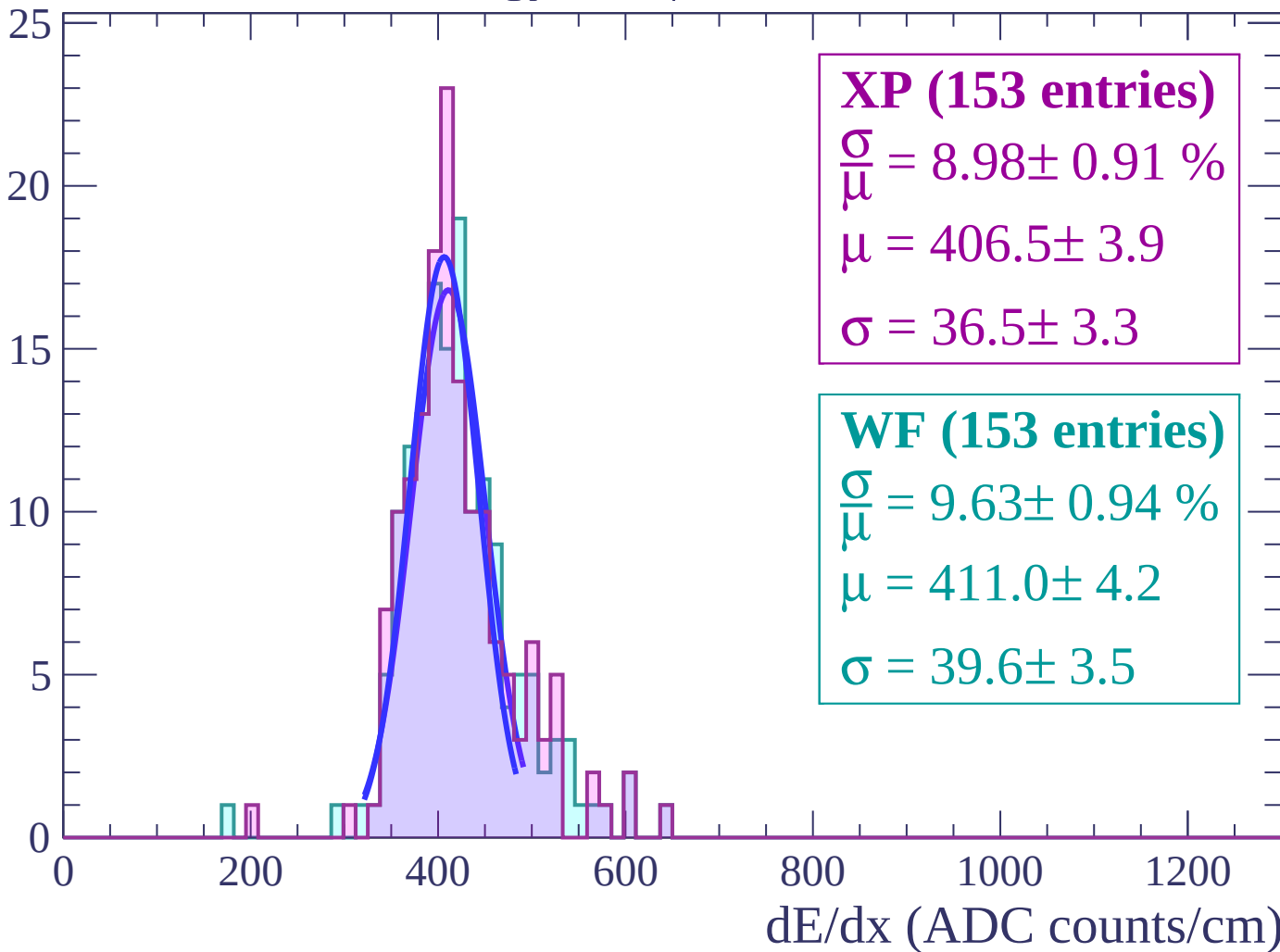
Energy loss | $48 < \theta < 50$

Count



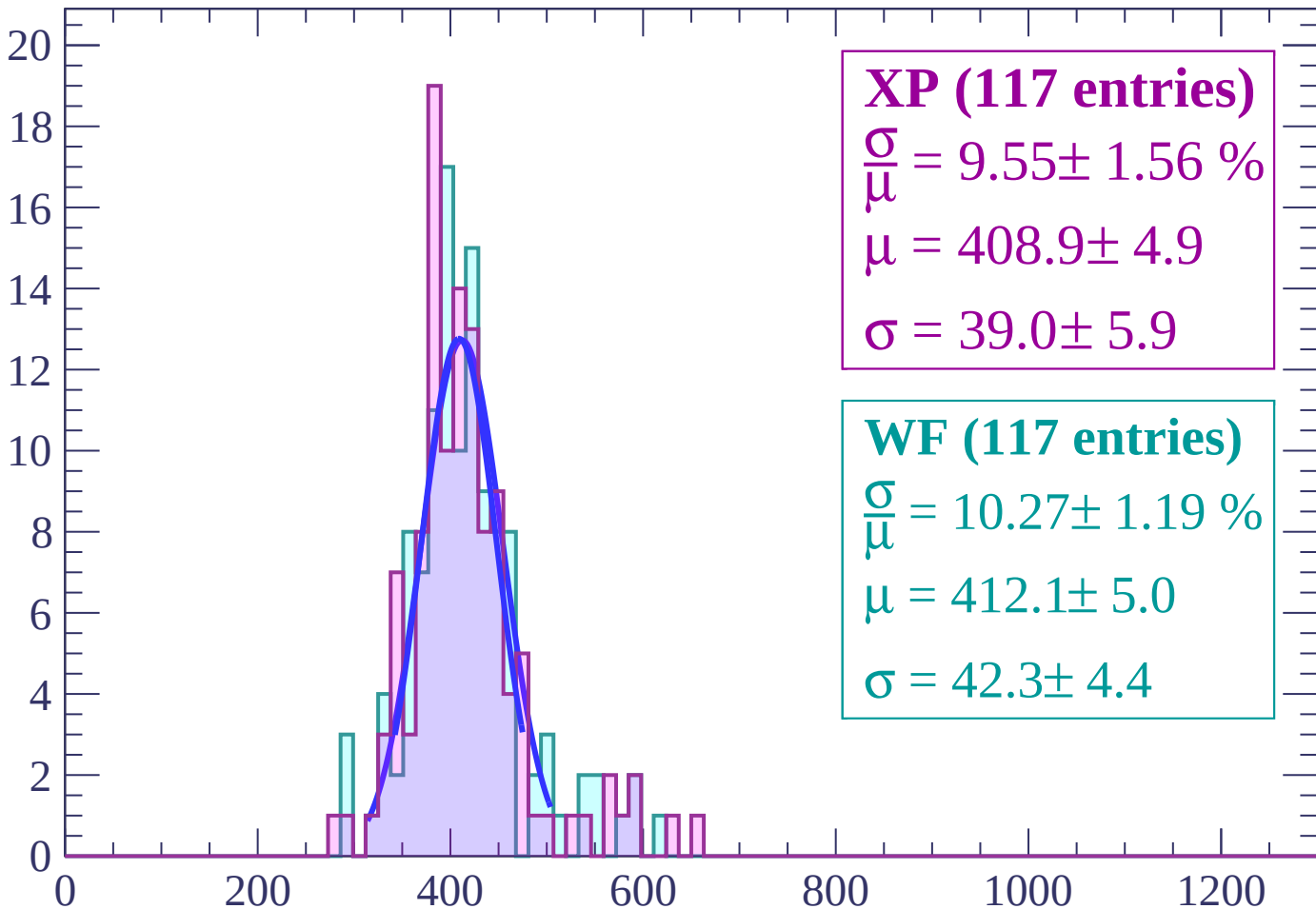
Energy loss | $50 < \theta < 52$

Count



Energy loss | $52 < \theta < 54$

Count



XP (117 entries)

$\frac{\sigma}{\mu} = 9.55 \pm 1.56 \%$

$\mu = 408.9 \pm 4.9$

$\sigma = 39.0 \pm 5.9$

WF (117 entries)

$\frac{\sigma}{\mu} = 10.27 \pm 1.19 \%$

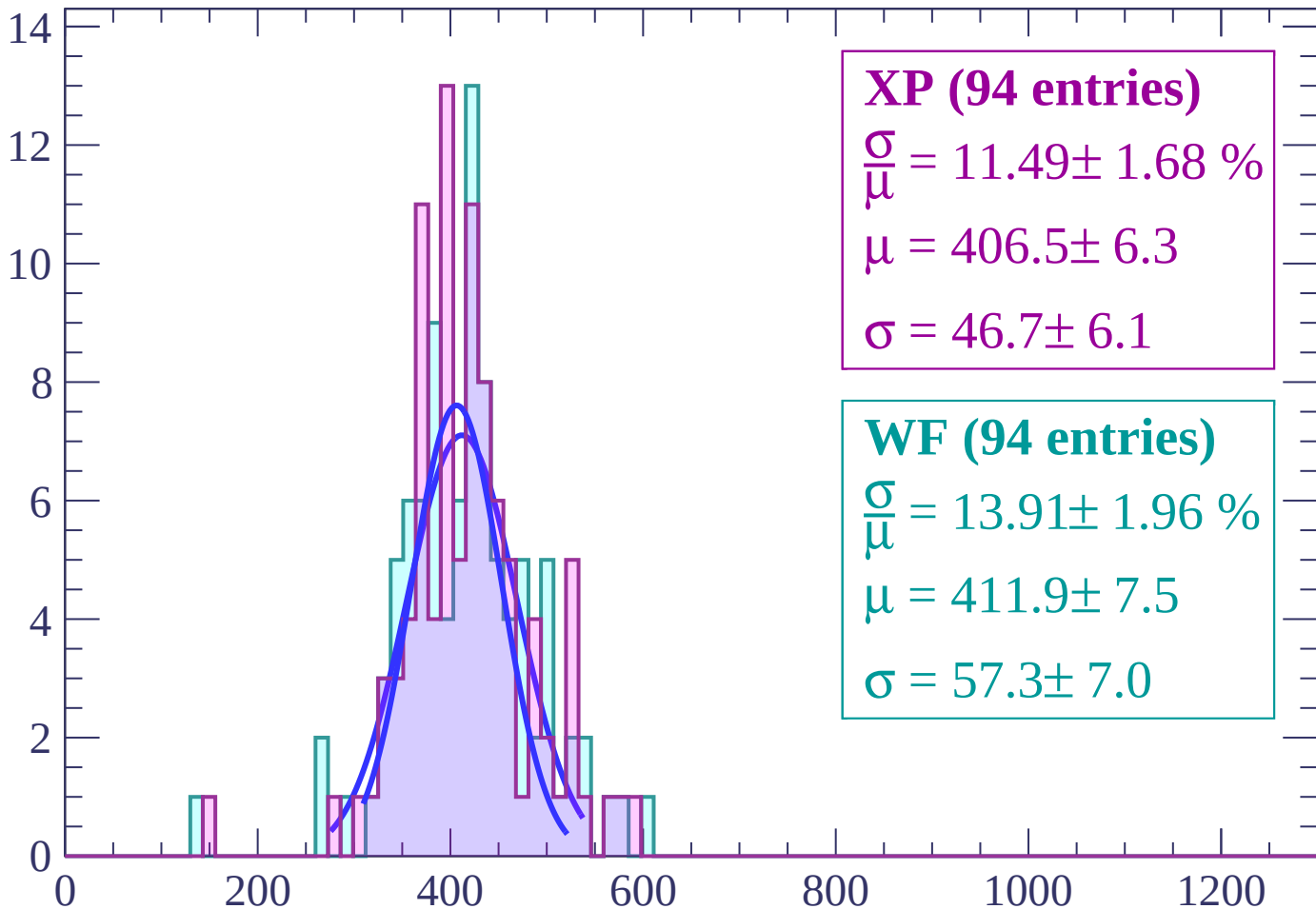
$\mu = 412.1 \pm 5.0$

$\sigma = 42.3 \pm 4.4$

dE/dx (ADC counts/cm)

Energy loss | $54 < \theta < 56$

Count



XP (94 entries)

$\frac{\sigma}{\mu} = 11.49 \pm 1.68 \%$

$\mu = 406.5 \pm 6.3$

$\sigma = 46.7 \pm 6.1$

WF (94 entries)

$\frac{\sigma}{\mu} = 13.91 \pm 1.96 \%$

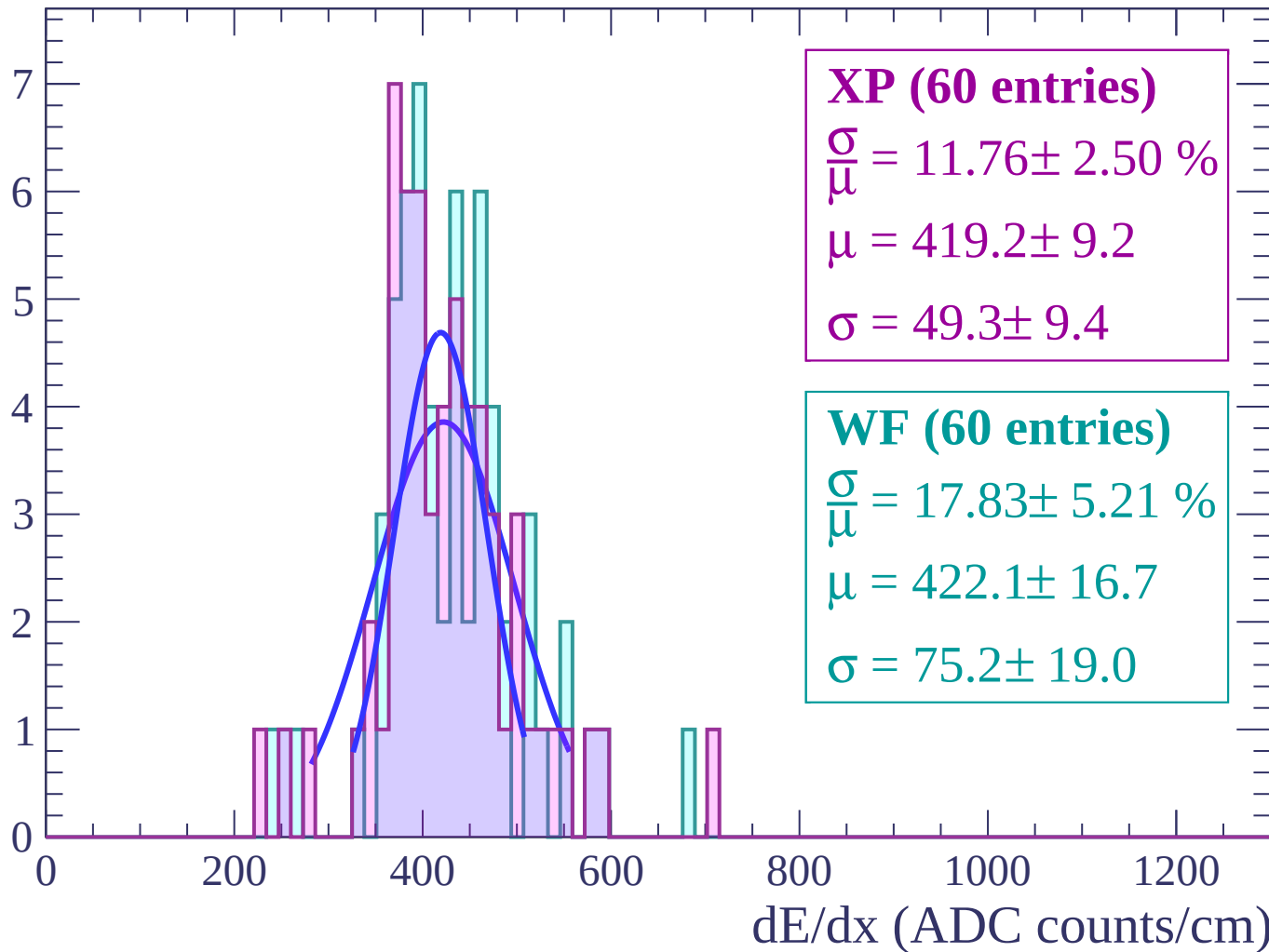
$\mu = 411.9 \pm 7.5$

$\sigma = 57.3 \pm 7.0$

dE/dx (ADC counts/cm)

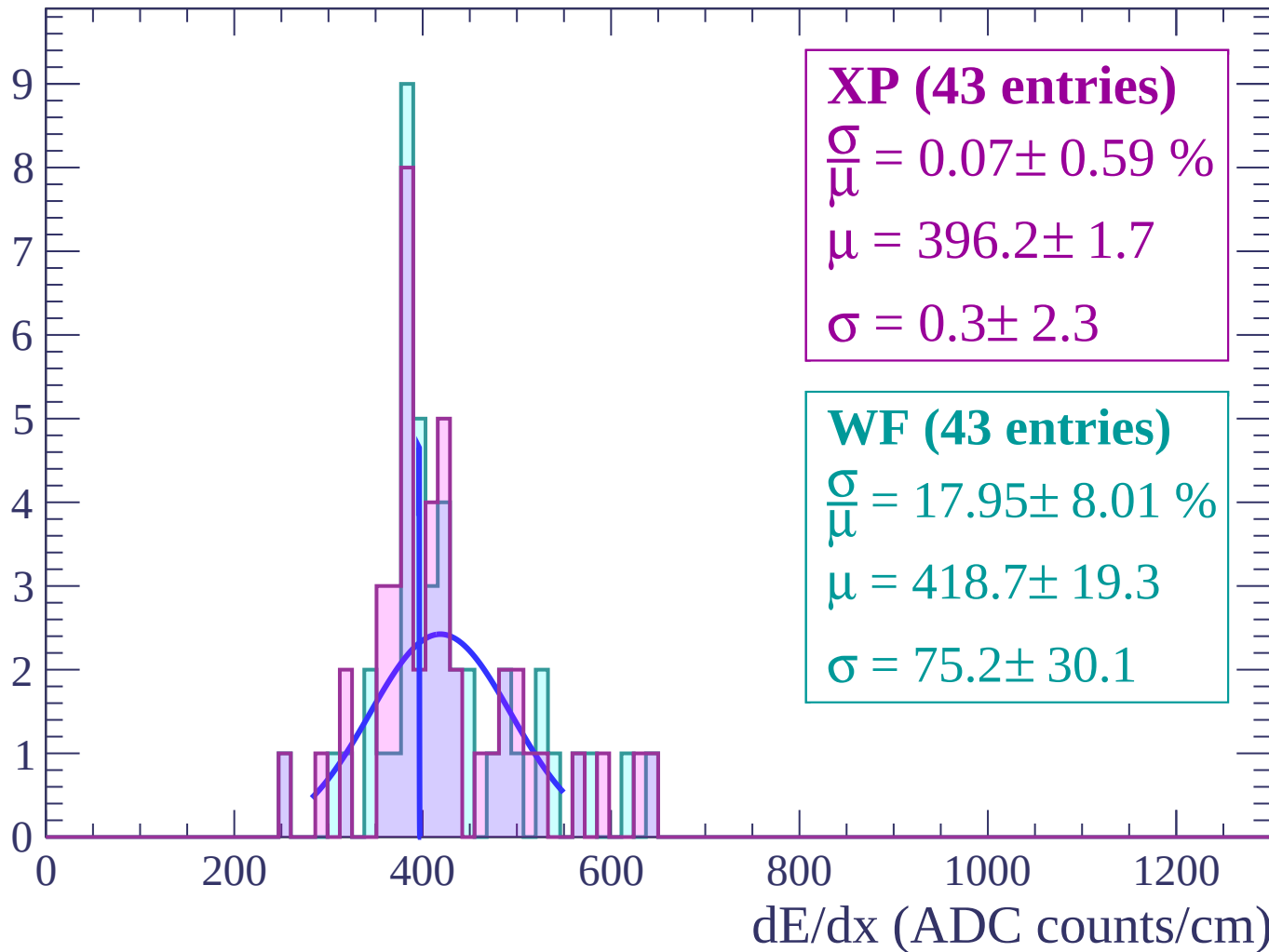
Energy loss | $56 < \theta < 58$

Count



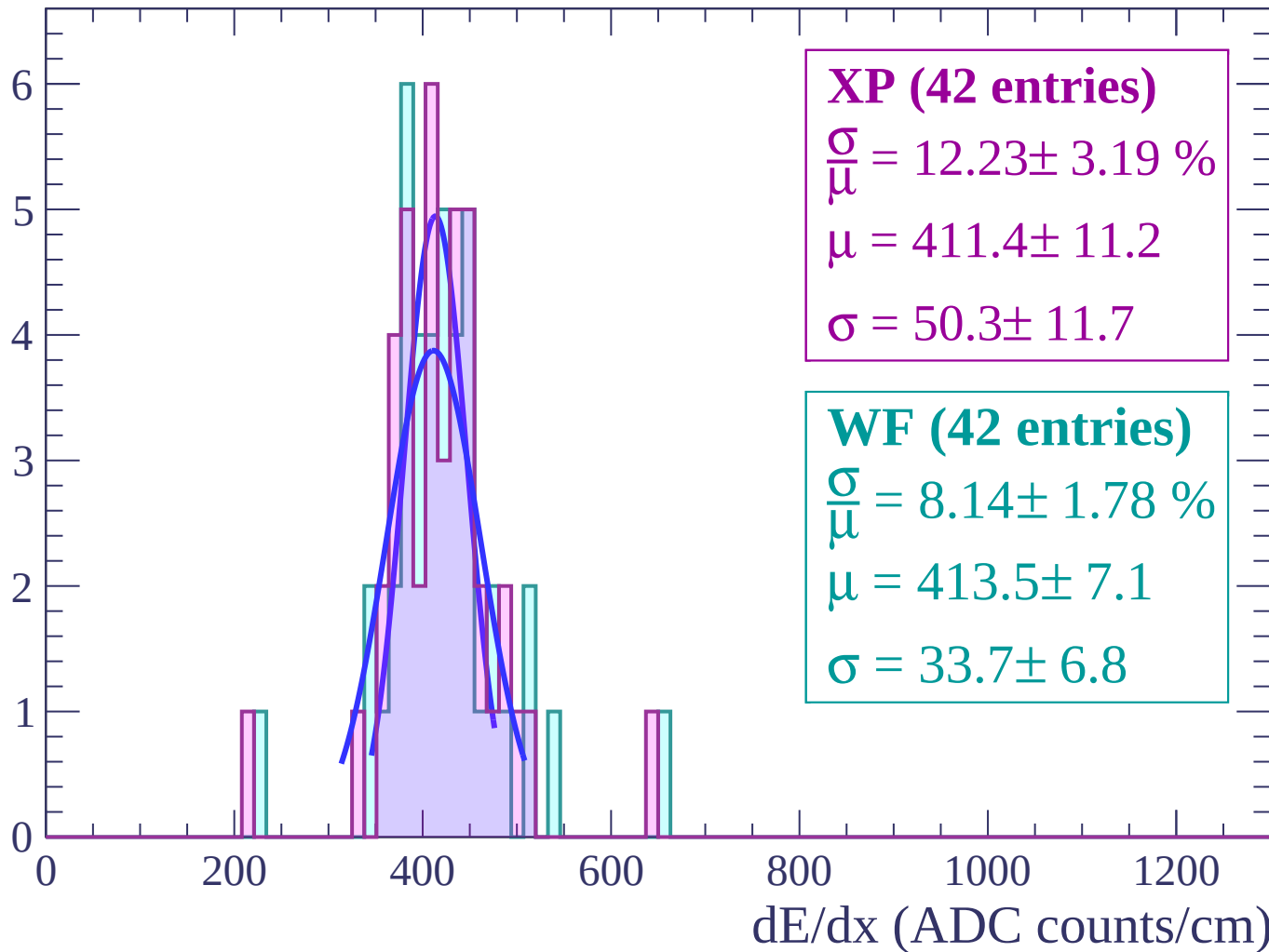
Energy loss | $58 < \theta < 60$

Count



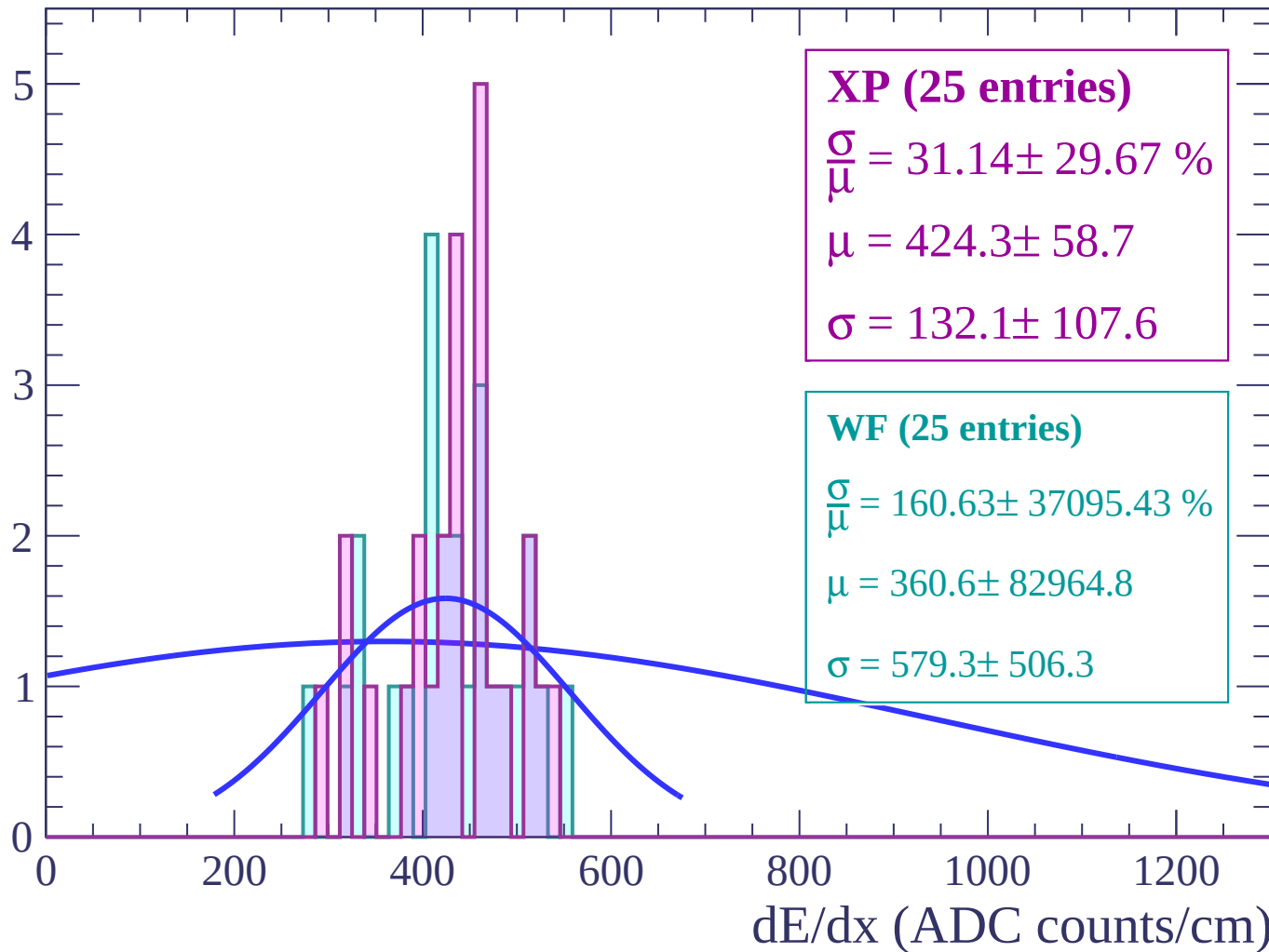
Energy loss | $60 < \theta < 62$

Count



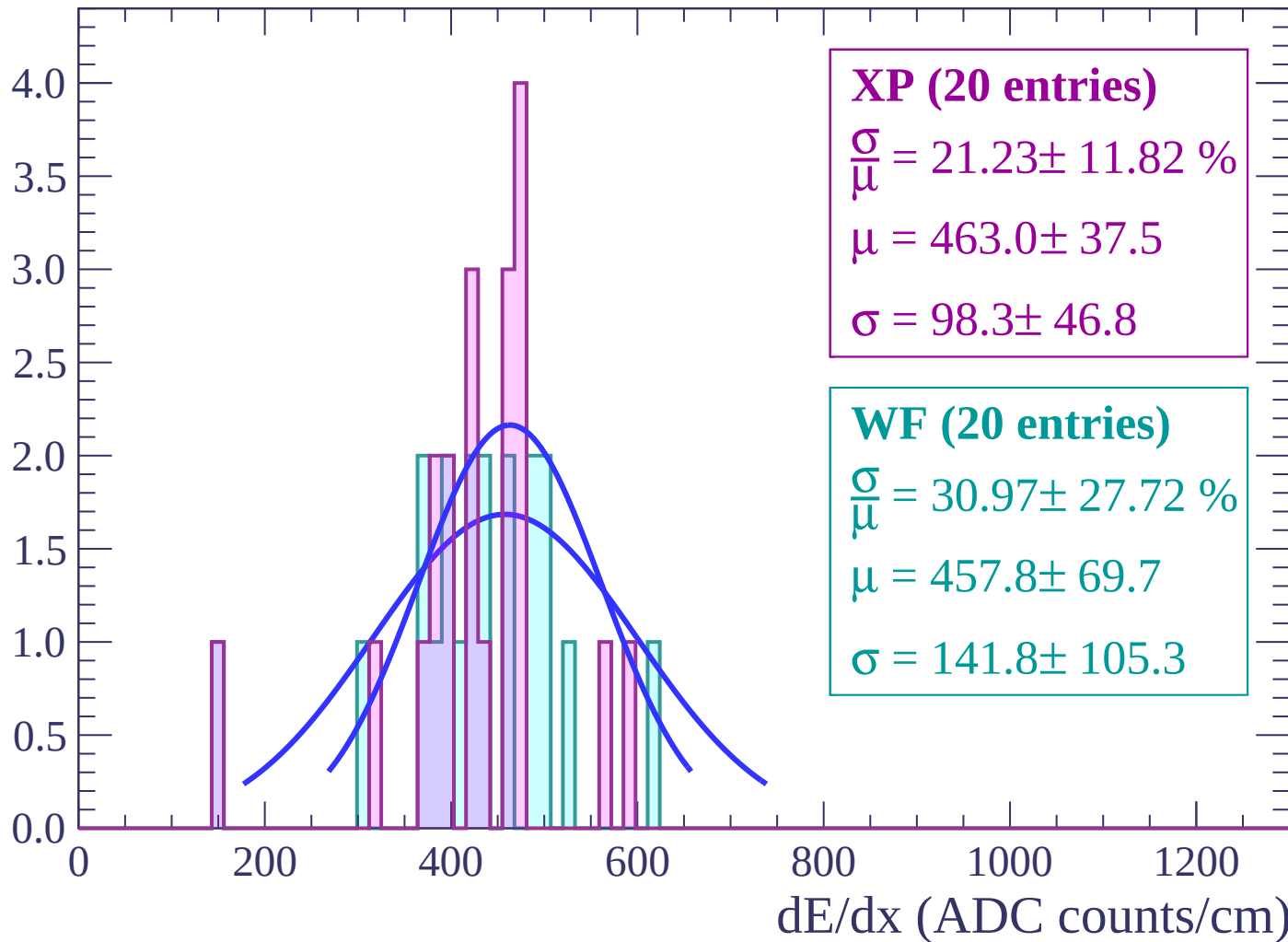
Energy loss | $62 < \theta < 64$

Count



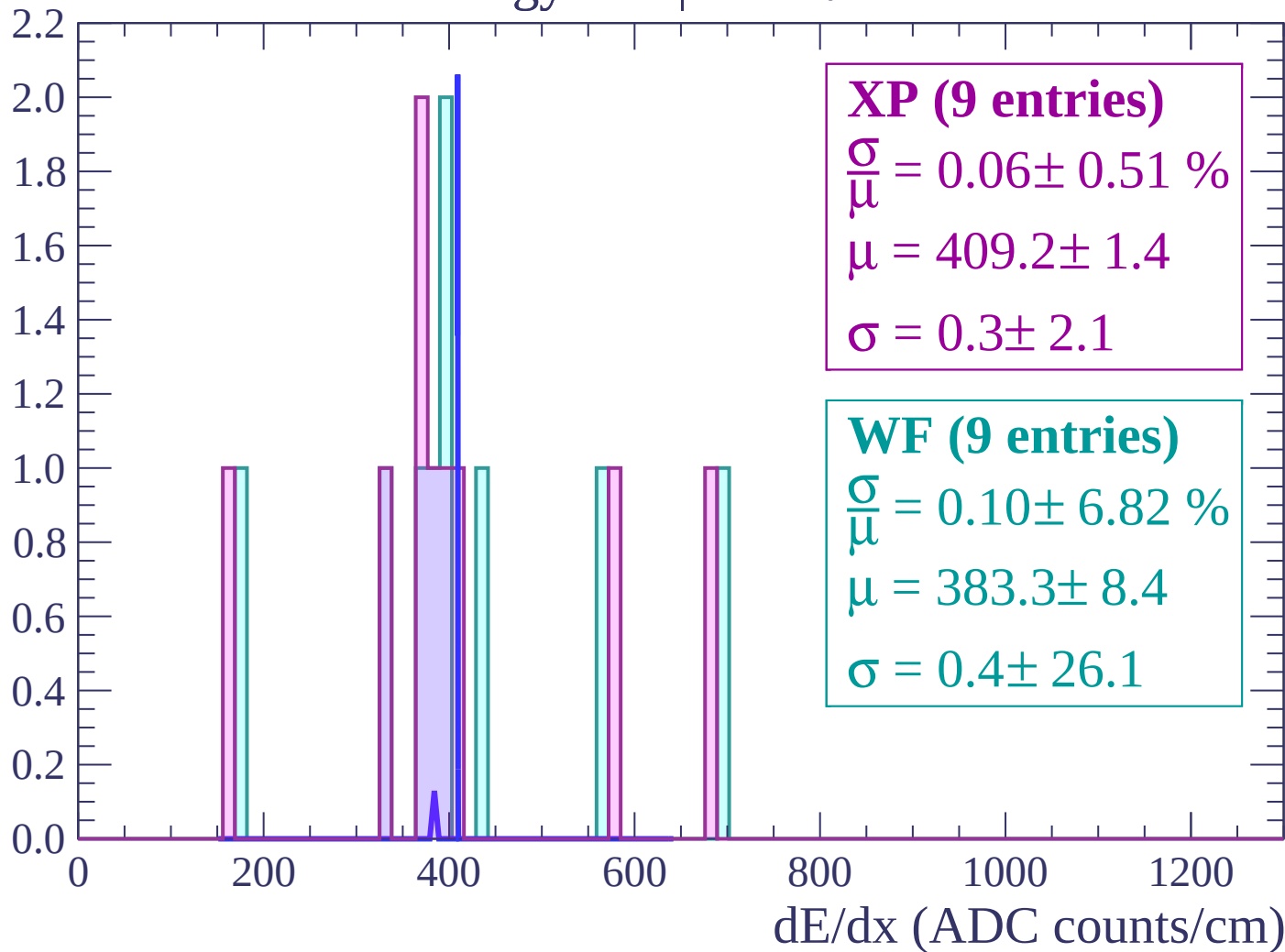
Energy loss | $64 < \theta < 66$

Count



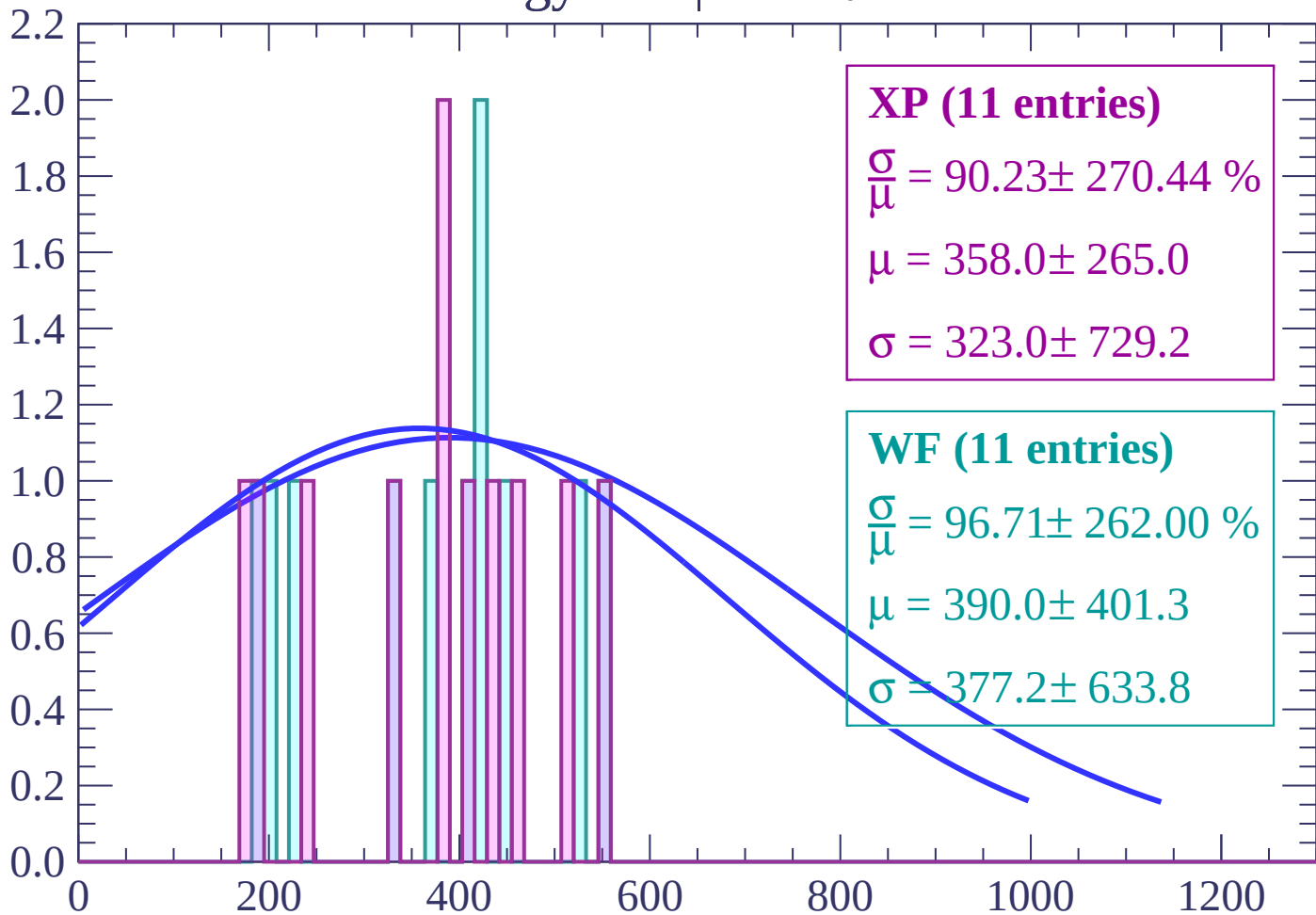
Energy loss | $66 < \theta < 68$

Count



Energy loss | $68 < \theta < 70$

Count



XP (11 entries)

$\frac{\sigma}{\mu} = 90.23 \pm 270.44 \%$

$\mu = 358.0 \pm 265.0$

$\sigma = 323.0 \pm 729.2$

WF (11 entries)

$\frac{\sigma}{\mu} = 96.71 \pm 262.00 \%$

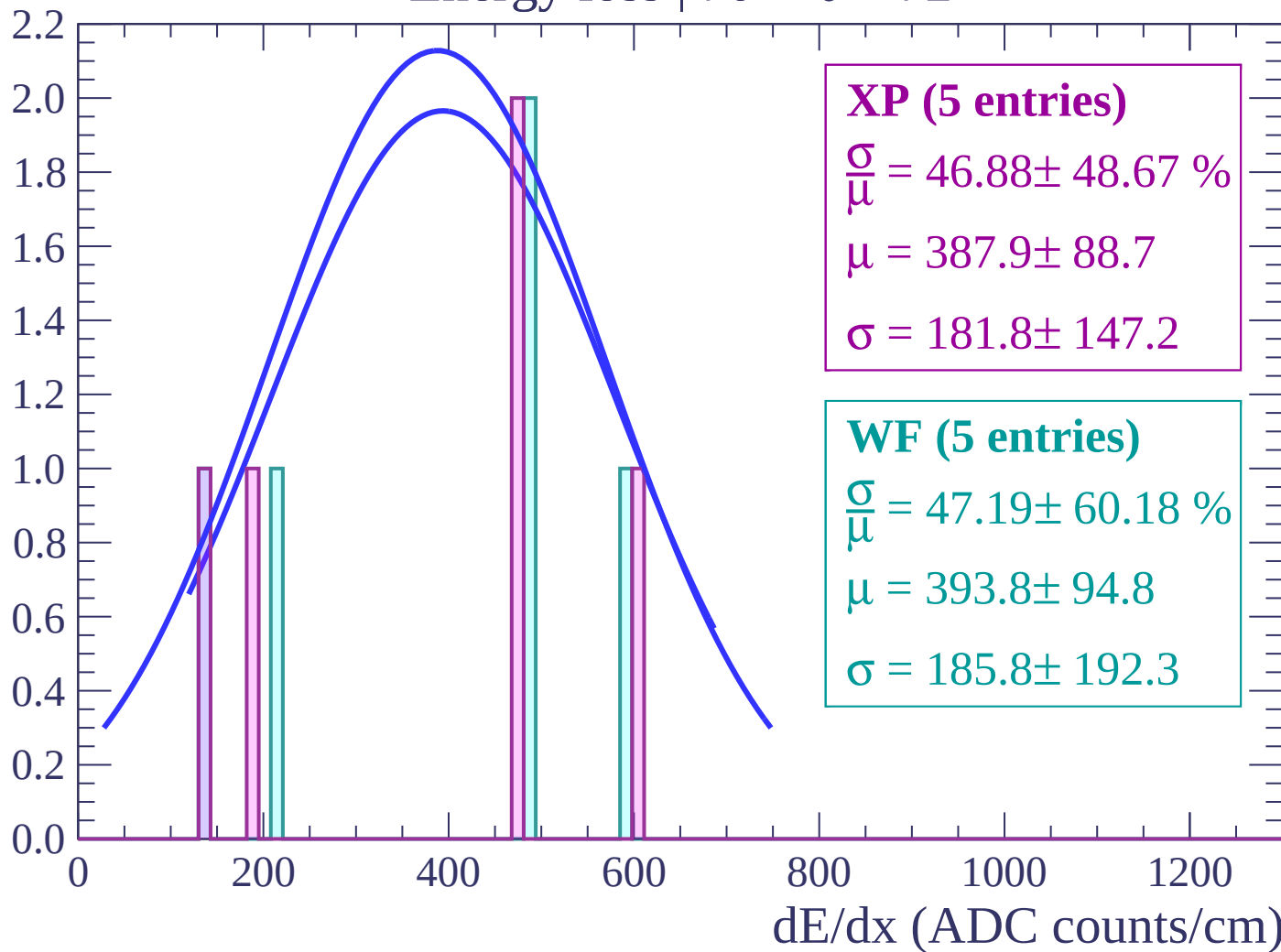
$\mu = 390.0 \pm 401.3$

$\sigma = 377.2 \pm 633.8$

dE/dx (ADC counts/cm)

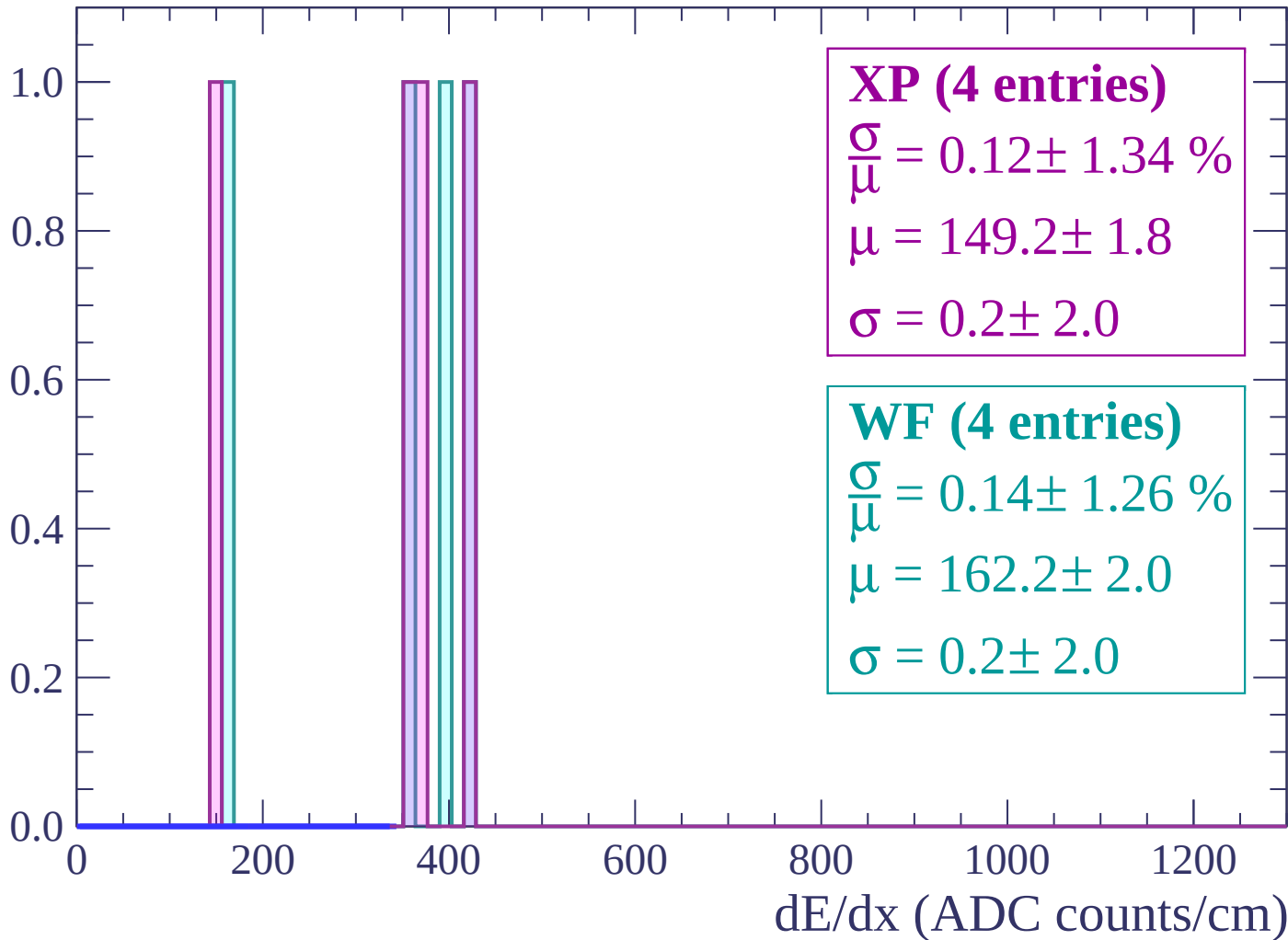
Energy loss | $70 < \theta < 72$

Count



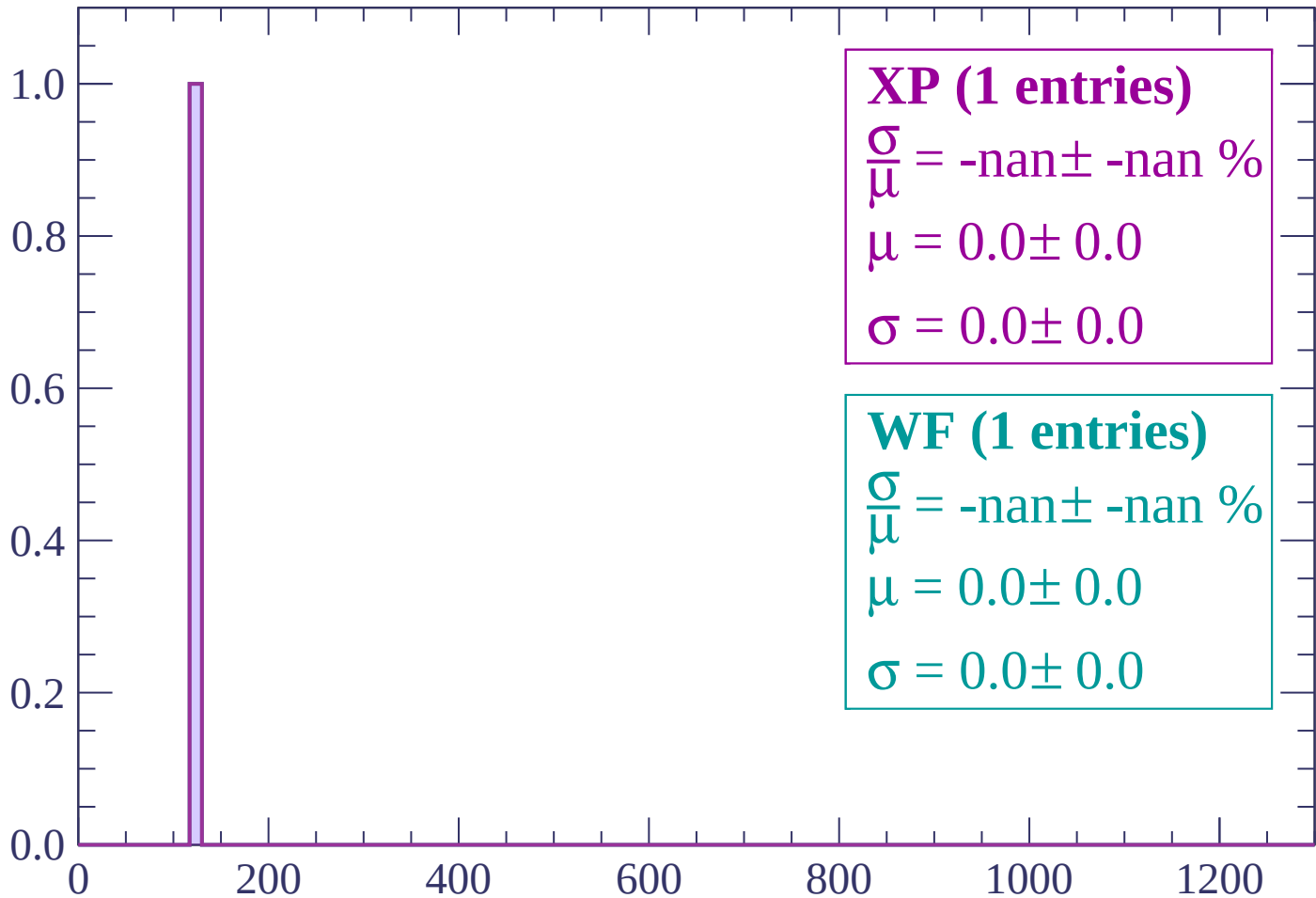
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $78 < \theta < 80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

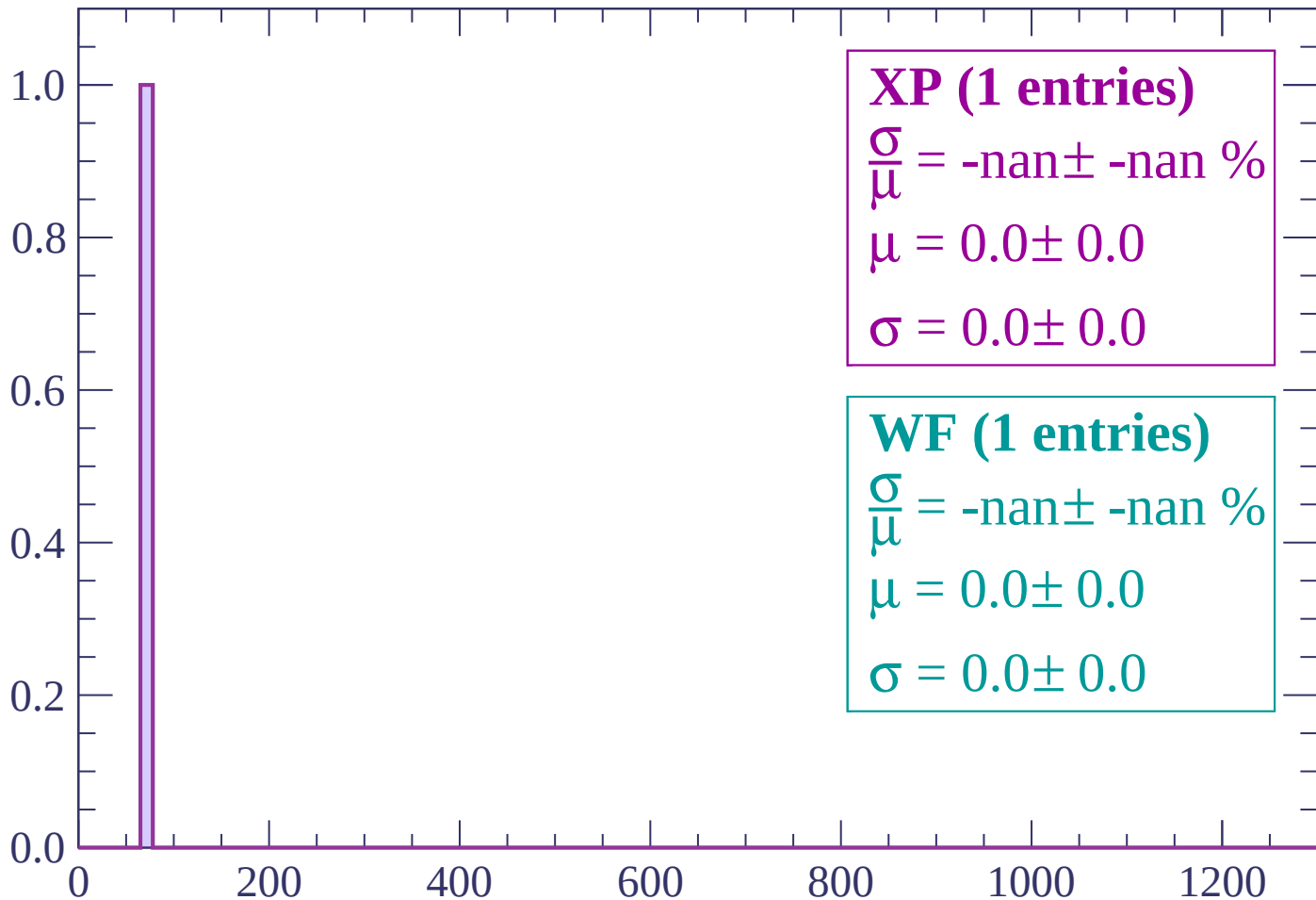
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $80 < \theta < 82$

Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

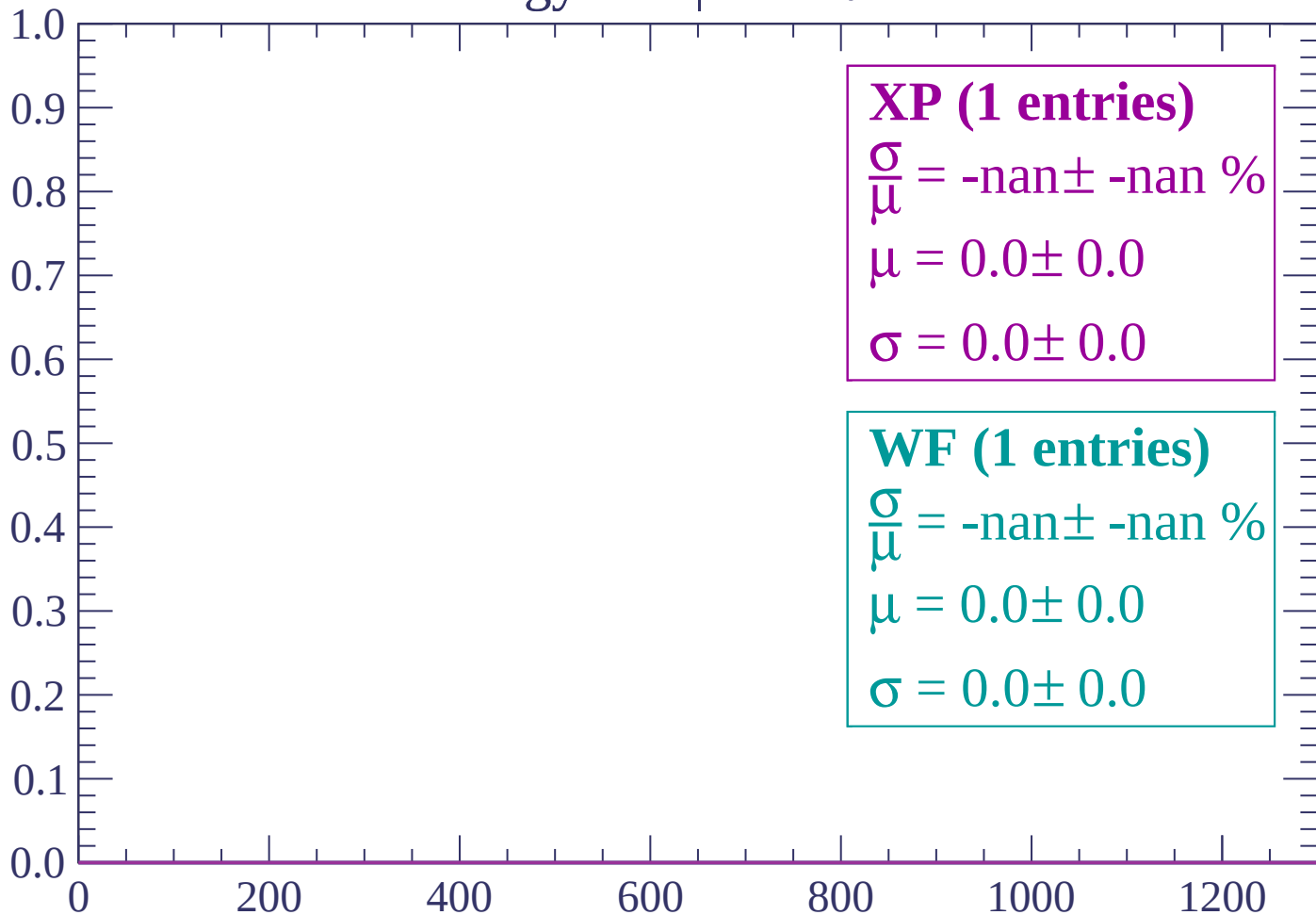
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

